Ram Calculus Applicability to the OpenAI Mathematics Collection

Manuscript-level Applicability and Contribution Catalogue

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Technical Report Assisted by AI Models

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1 Purpose and Interpretation

This catalogue screens 722 manuscripts in 372 result families, using the October 6, 2026 OpenAI release at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Manuscript identifiers M001–M722 follow the order of the pinned `CONTENTS.md`; family identifiers are the publisher's identifiers and are not a consecutive 1–372 numbering. A family may contain several distinct manuscripts. No family is silently substituted for a manuscript.

Applicability is NOT a validity, novelty, success, or solved-status score. L5 denotes a direct relation to the core problem; L4 a very relevant method or proof component; L3 significant but incomplete overlap; L2 a peripheral connection; L1 no concrete supported connection found in this screening. These initial scores reflect evidence available at the review date, not a claim that all source proofs or all corpus claims have been independently verified. The row-specific review-depth records identify how far inspection went. They are deliberately provisional where only a statement and proof structure were screened. Failure to locate a concrete link in this bounded search is not a theorem that no such link exists. Detailed studies may refine scores. Source-structure screening is distinct from a complete line-by-line audit. The initial detailed-report queue processed L5 first, then L4, in catalogue order; all 100 reports in that stage were delivered. On October 8, 2026, the user authorized the first 51 L3 manuscripts in the unchanged catalogue order, in 17 sequential batches of three. Their identifiers and batch boundaries are recorded in `l3-stage.json`. The other 88 L3 manuscripts and all L1–L2 manuscripts remain outside the detailed queue. Applicability ratings are unchanged by this expansion of the work scope.

Upstream revision notice (checked October 8, 2026). The publisher's current repository lists 719 manuscripts. Its October 7 history entry records the withdrawal of three papers: the split-abelian-eightfold Weil-class paper and the two dependent K3 correspondence/product papers. In this fixed catalogue these are M065, M060, and M059, respectively; all were rated L2 and none belongs to the selected L5/L4 detailed-study queue. The publisher also records proof repairs and dependency/citation updates elsewhere. See the dated upstream revision history. We retain the original 722 identifiers and screening rows for traceability, not as an endorsement of withdrawn claims. A result about a pinned version must not be silently presented as verification of a later revision.

2 Method and Evidence

The public collection was downloaded without changing confidential corpus files. All 9,019 acquired text-source artifacts have recorded SHA-256 hashes. Individual assessments identify source files, candidate target steps, corpus references, and untransferred obstacles. Corpus maps cover all requested directory groups by inventory and targeted theorem reading; they are not an assertion that every proof in the very large corpus was checked. The accompanying CSV and JSON preserve full source paths and more convenient machine-searchable evidence. A generic coordinate change, positive reference Gram matrix, or finite computation does not establish an arithmetic, topological, or infinite-dimensional theorem.

OpenAI explicitly states that verification stages differ across manuscripts. The presence of a Lean catalogue entry is not described here as a locally replayed proof. Public statements, imported hypotheses, local deductions, and proposed extensions must remain separate in every subsequent report. Confidential corpus contents were not uploaded or used as public queries.

3 Counts and Delivery State

L1	L2	L3	L4	L5	Total
312	171	139	47	53	722

The existence of an assessment row does not mean its detailed report has been delivered. The root `progress.json` records actual PDF and LaTeX delivery. Pending reports remain pending until both files have been written and checked.

4 Demonstrated Contribution Assessments

The additional **C1**–**C5** scale implements the requested assessment of progress, improvements, extensions and contributions made by applying Ram Calculus relative to each pinned OpenAI manuscript. It is distinct from **L1**–**L5 applicability**. A high L rating does not entail a high C rating. These are internal evidence-based judgments about the delivered reports, not independent peer-review verdicts, historical-priority findings, or certificates that the source headline theorems are correct.

Score	Meaning
C1	No contribution demonstrated.
C2	A little contribution, not significant.
C3	A notable or significant contribution.
C4	A substantial or high contribution.
C5	A very high contribution, with substantial and significant improvements, extensions, gap closures, or important new results.

Scores credit proved local results and explicitly conditional consequences within their actual scope. Proposed future work, imported source claims, mere terminology changes, and unmeasured speedups do not receive credit as new improvements. A substantial advance at a single important step can qualify for C4 without resolving the whole problem; a large improvement of an auxiliary constant alone is not automatically a high-impact result. The reasons below name both the contribution and its principal limitation. Standard mathematics used in a Ram formulation is not thereby claimed as an invention of Ram Calculus. Scores can change after substantive revisions or further independent verification.

A dash in the individual catalogue means *not assessed*, not C1. Uncompleted studies cannot yet be assigned an outcome score. The summaries count manuscripts, not independent result families; related reports are not independent research successes. No arithmetic mean of this ordinal scale is used.

4.1 Contribution Counts for Delivered Reports

C1	C2	C3	C4	C5	Assessed
0	55	182	2	0	239

All 239 delivered reports have been assessed. No selected reports remain pending. 483 unselected manuscripts are outside the present detailed-study queue.

4.2 Applicability and Contribution Cross-Tabulation

Applicability	C1	C2	C3	C4	C5	Unrated	Total
L1	0	0	0	0	0	312	312
L2	0	0	0	0	0	171	171
L3	0	28	111	0	0	0	139
L4	0	22	24	1	0	0	47
L5	0	5	47	1	0	0	53
All	0	55	182	2	0	483	722

4.3 Report-Level Reasons and Evidence

Report	Score	Demonstrated contribution and remaining boundary
M002	C2	Proved explicit rational rank-one kernel brackets, refinable weighted rational panels and a strictly smaller coefficient-tail majorant. Checked source-normalized central defect algebra and simultaneous point/Manin-factor covariance. Proved a conditional odd-integer interval bridge with a bounded point index. Improvements are modest analytic certification and bookkeeping; all-prime arithmetic inequalities, exact Sha order, source theta/unit prerequisites and saturation evidence remain imported. No historical novelty, measured runtime gain, real-curve certificate or global BSD proof implication. **Evidence:** \label{thm:bracket}; \label{thm:panels}; \label{cor:tail}; \label{prop:defect}; \label{lem:covariance}; \label{thm:odd}. **Report:** `TR-M002-Ram-Exact-BSD.tex`
M004	C2	Useful modest source-specific certification refinements: closed low-rank Mellin kernels, explicit exponential-fit log-panel budget, dyadic-only torsion-aware saturation reduction, and signed central determinant-defect propagation with a known-rank integrality consequence. Independent denominator data remain required for real-to-dyadic reconstruction. Classical analytic ingredients and deep integral source comparisons are imported; no universalBSD proof, significant arithmetic advance, measured speedup or historical priority claimed. **Evidence:** \label{thm:kernels}; \label{thm:panels}; \label{thm:index}; \label{cor:halving}; \label{thm:defect}; \label{cor:weaker-unit}; \label{thm:isolation}. **Report:** `TR-M004-Ram-Two-Primary-BSD.tex`
M005	C2	Source-specific exact finite optimality proof, independent certificates, localization, perturbation and a sharper conditional capacity loss; prior minimum discovery credited and no new global analytic theorem claimed **Evidence:** \label{thm:minimum}; On $\mathcal B$ the unique minimum of $G=N/(96j)$ is; \label{eq:optimal-square}; N(\delta,0)-96\gamma j(\delta,0)=2448(\delta-\delta_*)^2.; \label{prop:localization}; \label{prop:perturbation}; \label{prop:capacity}; \label{cor:bridge}; \label{app:review}; minimum originated in a prior local audit and is not claimed as a fresh; parts of that route. Its conclusions do not constitute a global. **Report:** `TR-M005-Ram-Quasi-Riemann.tex`
M006	C2	Source-specific Mellin isometry and first-residue/moment-cancellation refinement, odd-order separation saving one derivative and conditional transfer saving three, full-contraction depth certificate, and damped cube/finite residual reformulations. These are modest analytic improvements, not completed sextic-twist mean-square or QRH/RH certification; no M005 result reused or historical-priority claim. **Evidence:** \label{thm:unitary}; \label{thm:residue}; \label{prop:damped}; \label{thm:separation}; \label{cor:transfer}; \label{prop:depth}; \label{prop:residual}. **Report:** `TR-M006-Ram-Quasi-Riemann.tex`
M010	C2	Source-specific signed-moment and contact reconstruction, exact barrier replay, Bernstein exhaustion, and a sharper same-trial curvature certificate with an explicit perturbation budget; no new global irrationality proof, optimized trial discovery, historical novelty, or computational speedup claimed **Evidence:** \label{prop:contact}; \label{cor:cancel}; \label{prop:energy}; \label{lem:bernstein}; \label{lem:curvature}; \label{prop:curvature}; \label{thm:improved}; \label{eq:improved}; \label{prop:perturb}; \label{prop:bridge}; \label{sec:assessment}; \label{sec:review}. **Report:** `TR-M010-Ram-Catalan.tex`
M012	C3	Proved a source-specific conditional analytic-rank moment extension for every fixed 0<q<6/5 from the uniform k^-2 finite window and logarithmic cap; explicit small-block dyadic transfer permits a valid conditional finite-height modulus. Exact independent-block bias/error identity refines the source telescoping bound, with improvement only under an additional zero-bias premise. Ternary Gram and Fourier energy coordinates support error bookkeeping. Imported arithmetic estimates and companion density remain essential; no new global arithmetic proof or historical priority claim. **Evidence:** \label{thm:block}; \label{thm:dyadic}; \label{cor:fractional}; \label{ex:endpoint}; \label{cor:modulus}. **Report:** `TR-M012-Ram-Mean-Analytic-Rank.tex`

Report	Score	Demonstrated contribution and remaining boundary
M013	C2	Proved the parent-supplied optimized noncommuting resolvent transfer, and local energy-weighted tail, displacement-mass prefix, singleton calibration, polynomial restriction residual, and finite energy limitation certificates. These sharpen finite transfer and auditability but do not improve arithmetic rank-witness/forest estimates or establish a new Chowla theorem, exponent, or RH result. Classical finite methods and inherited source inputs are not credited as breakthroughs. C2 is independent of applicability L3. **Evidence:** \label{thm:transfer}; \label{thm:tail}; \label{thm:boundary}; \label{prop:calibration}; \label{prop:polynomial}; \label{prop:energy}; \label{sec:assembly}. **Report:** `TR-M013-Ram-Two-Point-Correlations.tex`
M021	C3	Proved source-specific truncated conditional-character Gram kernels give norm-one comparison with explicit residual and polynomial complexity for arbitrary big-prime subsets, without source near-orthogonality/Siegel-Walfisz at that step. Proved ideal construction consequence, orthogonal3^P to5^(P/2) error count, exact probe maximum, conditional reduced memory budget and local endpoint energy criterion. No global arithmetic or totient certification, historical novelty claim or unmeasured speedup credited. **Evidence:** \label{thm:cell}; \label{prop:major}; \label{thm:comparison}; \label{prop:error-sum}; \label{cor:ideal}; \label{prop:memory}; \label{prop:endpoint}. **Report:** `TR-M021-Ram-Totient-Fibers.tex`
M022	C3	Proved source-specific conditional cost and transport refinements: exact joint factorial Gram and line-class identities, explicit mark second-moment and covariance-weighted deweighting costs, a smaller sufficient cutoff at the source absolute-truncation step, and a finite-prefix plus missing-mass Wasserstein budget with an interior-test seminorm. Arithmetic comparison estimates remain imported. The qualitative l1 upgrade is a classical mass-conservation consequence, not a new prime-factor law; no new global arithmetic estimate or historical priority claim. **Evidence:** \label{prop:gram}; \label{prop:line}; \label{prop:cutoff}; \label{prop:mark}; \label{thm:deweight}; \label{lem:noescape}; \label{thm:transport}; \label{cor:powers}. **Report:** `TR-M022-Ram-Poisson-Dirichlet.tex`
M024	C2	Proved source-specific compressed positive-kernel mesh refinement, joint-TV operator conversion, same-site feature/mixed-row identities, constructive positive Bernstein lift, expected vertex-deletion estimate, finite adaptive sampling ledger, and credited parent exponent/lag refinements. These improve local approximation and auditability without improving the source arithmetic multiplicity, cancellation, all-cutoff theorem or number-theoretic rate. Classical/private/source foundations are inherited. C2 is independent of applicability L3. **Evidence:** \label{thm:mesh}; \label{lem:TV}; \label{prop:gram}; \label{thm:bernstein}; \label{prop:rows}; \label{lem:deletion}; \label{thm:sampling}; \label{cor:exponents}; \label{prop:lag}. **Report:** `TR-M024-Ram-Joint-Dickman.tex`
M034	C3	Explicit correlated local Holder moment with locally sharp constant and tail exponents below1/(8n); conditional short-interval range n*gamma<alpha-1/2, improved cusp remainder and polynomial radius range; exact Fourier/ideal probability energy normalization; dimension-deficit dominated-component entropy bookkeeping. Named arithmetic R/S and classification remain imported. No endpoint q^epsilon disproof, new classification, global equidistribution certification or historical priority claim. **Evidence:** \label{sec:fourier}; \label{prop:law}; \label{thm:moment}; \label{cor:tail}; \label{sec:kernel}; \label{thm:count}; \label{cor:cusp}; \label{cor:balls}; \label{thm:entropy}. **Report:** `TR-M034-Ram-Prime-Degree-Torus-Packets.tex`
M035	C3	Exponent-sensitive conductor period, p5 shift improvement, full Smith/Walsh information, exact scalar Euler normalization and L2/finite trace energy; conditional X49 profile cap retaining Delta=DF I^2. No global arithmetic or equidistribution certification. **Evidence:** \label{prop:scalar}; \label{cor:scalar-product}; \label{prop:smith}; \label{thm:period}; \label{prop:boundary}; \label{thm:L2}; \label{prop:finite}; \label{cor:cap}; \label{eq:branch}; \label{prop:moment}; \label{prop:entropy}. **Report:** `TR-M035-Ram-Quartic-Torus-Packets.tex`

Report	Score	Demonstrated contribution and remaining boundary
M039	C3	Proved source-specific largest-Smith and exact observation-index refinements of the source's existing elementary-divisor argument, with a strict manufactured example; proved a cancellation-sensitive matched-chart tail and a fixed algebraic graph certificate's residual precision threshold with graph-coordinate and product-formula costs. Classical multiplicity/defect estimates and the rank-five model are not new. Common all-place relations, branch-uniform nonidentity, auxiliary markings, global heights and counting finiteness remain independent inputs. No new quaternionic finiteness theorem, full source audit, historical-priority claim or adaptive-descent robustness theorem. **Evidence:** \label{thm:smith}; \label{thm:obs}; \label{prop:divdiff}; \label{prop:pairbudget}; \label{lem:residual}; \label{thm:robust}; \label{prop:tracegap}. **Report:** `TR-M039-Ram-Quaternionic-Division.tex`
M040	C2	Proved source-specific finite moment/jet certificates, explicit denominator and all-place coordinate costs, and transfer of the companion one-parameter cancellation-avoidance device to M040 with finite candidate and coefficient budgets. Arithmetic nonidentity, multiplicity, integral geometry and global counting remain imported or conditional; no historical novelty or stronger finiteness claim. **Evidence:** \label{prop:jet}; \label{prop:denominator}; \label{thm:circle}; \label{prop:momentmatrix}; \label{thm:avoidance}; \label{cor:avoidance}; \label{prop:gaugejet}; \label{thm:height}; \label{cor:admissible}; \label{prop:gap}. **Report:** `TR-M040-Ram-Elliptic-Squares.tex`
M041	C3	Proved full-packet unitriangular invariance of every full-row minor and used it to decouple short arithmetic from long analytic logarithm truncation. Balanced Gaussian-integer scaling and nonnegative aggregate exponent differences remove m and F0 from the leading LCM budget and retain selected-column savings. Exact distinct-Taylor-degree summation plus the source disk yields a sixfold leading collision constant with an explicit uniform remainder. A strengthened conditional finite/subsequence comparison and analytic determinant enclosure are proved. Classical partition algebra and moment coordinates are not claimed as new; center-uniform interpolation and the global exponent-two claim remain separate and uncertified. **Evidence:** \label{thm:unitriangular}; \label{thm:balanced}; \label{cor:decouple}; \label{thm:collision}; \label{thm:comparison}; \label{prop:enclosure}. **Report:** `TR-M041-Ram-Pi-Irrationality.tex`
M046	C3	Exact enlarged first quartic cell eta>=249/250 with continuous rational witnesses and conditional prime-band extension; unconditional dyadic Pell uniqueness removes an upper-tail logarithm. Classical CRT/Bernstein identities are not credited as new theorems. **Evidence:** \label{thm:cell}; \label{cor:band}; \label{thm:Pell}; \label{cor:Pell-tail}; \label{prop:CRT}; \label{prop:transfer}. **Report:** `TR-M046-Ram-Power-Free-Values.tex`
M047	C3	Significant realized local improvements relative to the source: exact all-length random-sieve variance with stable positive-measure ledger, finite-prime-set and nonunit-progression extensions, and exponential marked-window control extracted from the source's stronger cycle hypotheses. Classical and parent-supplied origins are credited. Actual-prime domination, inverse/stopping arithmetic and the global quadratic theorem remain imported; no historical novelty or global certification claim. **Evidence:** \label{lem:partition}; \label{prop:stopped}; \label{thm:variance}; \label{prop:stable}; \label{eq:nonunit}; \label{prop:remainder}; \label{eq:cocycle}; \label{prop:compression}; \label{thm:marked}; \label{eq:window-application}; \label{sec:assembly}. **Report:** `TR-M047-Ram-Jacobsthal.tex`
M048	C3	Proved an equivariant minimum-variance matching for the source prime-step polytope, with exact shared-residue covariance, zero-flux feasibility and optimality certificates, and a strict two-class reduction from r/p to zero versus permitted product coupling. Proved a uniform-integrability finite-block transfer preserving complete expected deletion and parameter order, weaker than the retained L2 interface. Load-sensitive clipping and loss-sensitive energy accounting are supporting refinements. Classical covariance and convexity are not claimed novel; no improved final broad energy or global arithmetic theorem is certified. **Evidence:** \label{thm:flux}; \label{thm:optimizer}; \label{prop:strict}; \label{prop:certificate}; \label{prop:clipping}; \label{prop:budget}; \label{prop:interval-gram}; \label{thm:UI}; \label{cor:limsup}. **Report:** `TR-M048-Ram-Inhomogeneous-Duffin-Schaeffer.tex`

Report	Score	Demonstrated contribution and remaining boundary
M049	C3	Notable source-specific local improvements: cube-lattice error sharpened to an explicit origin term plus rapid Schwartz remainder with exact Fourier/Riesz constants; distance-sensitive Perron summation reduces the absolute-cell-error loss from J^(3/2) to J^(1/2) under the stated uniform integrated input; sharp2/51 single-factor Gram margin independently proved after parent suggestion. Classical identities are not new credit. No global cancellation, effective uniform eta, historical novelty or normalization resolution; public rational theorem compatibility remains unresolved. **Evidence:** \label{thm:cubes}; \label{cor:cubes}; \label{thm:cells}; \label{cor:cells}; \label{prop:gram}. **Report:** `TR-M049-Ram-Cubic-Gauss-Sums.tex`
M052	C3	Notable source-specific local improvement: an admissible all-level truncated Poisson construction, including f0, gives exponential fixed-profile decay in maximum dimension instead of the source's flat-window polynomial bound. A parent-suggested rational taper is independently proved, and physical-norm perturbation and finite ordinary-density budgets are explicit. Classical origins, large profile/derivative/fourth-moment costs and fixed-k arithmetic order are retained. No new global arithmetic theorem, historical novelty claim, or automatic improvement of an actual-prime density constant is asserted. **Evidence:** \label{prop:deletion}; \label{lem:profile}; \label{lem:concentration}; \label{thm:poisson}; \label{cor:rate}; \label{prop:taper}; \label{prop:stability}; \label{prop:matrix}; \label{prop:density}; \label{prop:index}; \label{cor:assembly}; \label{sec:arithmetic}. **Report:** `TR-M052-Ram-Large-Prime-Gaps.tex`
M055	C3	Notable source-specific local improvements: exact principal Euler factorization with wider nonvanishing domain and isolated residual; constructive annular moment cancellation preserving principal residue; conditional shorter zero detector; exact smaller-line contour certificate and reflection-limited Mellin width. Classical coordinate identities and source fixed-base reduction not new credit. Theta/large-sieve/Kummer/TypeII/sieve inputs remain imported; no global Artin, RH or historical novelty certification. Parent suggested contour/endpoint choices, independently proved/tested by author. **Evidence:** \label{thm:euler}; \label{thm:weights}; \label{thm:detector}; \label{prop:contour}; \label{prop:least-line}; \label{thm:endpoint}. **Report:** `TR-M055-Ram-Primitive-Roots.tex`
M061	C3	Proved monotone multi-embedding batching reducing the source four-factor switch invocation bound to min(M, binomial(p,2)), with an exact matching-support model for the monotone optimum and the original nonzero polarization/Koszul contractions preserved. Proved explicit lattice-shell Gaussian tails and a finite wedge margin for the source sparse theta-gradient limit. These are notable local construction/certification improvements, not cover-degree, height, cycle-existence, rational CM-source extraction or global Hodge claims. Classical projector and private finite-span identities are supporting tools, not relabeled novelty. **Evidence:** \label{thm:batch}; \label{prop:matching}; \label{prop:composition}; \label{thm:tail}; \label{cor:wedge}; \label{prop:period}; \label{sec:certificates}. **Report:** `TR-M061-Ram-CM-Abelian-Hodge.tex`
M062	C3	Notable source-specific local gain: canonical functorial Clifford compression replaces an existential retraction and admits an exterior-degree-independent arithmetic denominator bound; exact polarized two-slot transport and an explicit D4 lattice/index certificate prevent normalization errors. A unit-plus-generator test certifies the specified vaw map with ambient defect constant7 and a strict known-denominator equality gap. Parent reflection-average suggestion and classical origins are credited. Algebraic seed cycles, coherent half sums, curve domination, two-square comparison and all-period specialization remain imported; no global Hodge theorem or historical novelty is claimed. **Evidence:** \label{thm:mask}; \label{cor:flags}; \label{prop:denominator}; \label{prop:slots}; \label{thm:generator}; \label{prop:defect}; \label{sec:lattice}; \label{sec:errors}; \label{sec:architecture}. **Report:** `TR-M062-Ram-K3-Kuga-Satake.tex`

Report	Score	Demonstrated contribution and remaining boundary
M063	C3	Notable local reduction of source condition (d) from all repeated determinant/CM lists to an exact projected integer-lattice basis using actual opposites and nonzero contractions, with finite CM-library localization. Explicit algebraic full-Weil-plane projector, minimal degree among polynomials in T^2, and one-cycle correlated two-column margin. No new cycle, companion CM theorem, global Hodge case, or geometric complexity bound. Classical finite generation and the source qualitative one-class-to-plane lemma receive no novelty credit. **Evidence:** \label{prop:hilbert}; \label{lem:subgroup}; \label{thm:lattice}; \label{thm:library}; \label{thm:projector}; \label{thm:plane}; \label{eq:balancedframe}; \label{eq:periodbudget}. **Report:** `TR-M063-Ram-Weil-Abelian-Powers.tex`
M064	C3	Source-specific one-point replacement of finite dual torsors, exact-order cyclic/lower-degree source reduction, and full-support Jacobian factor with bounded fixed-product sufficient mixed-power hypothesis. Local steps proved with rational whole-orbit and Tate normalization. Generic Fourier, known quadratic cases and test totals not novelty credit. Geometric Gale perfectness, global monodromy and companion Hodge inputs retained. **Evidence:** \label{thm:onepoint}; \label{thm:cyclic}; \label{thm:reduced}; \label{thm:support}; \label{prop:wedge}. **Report:** `TR-M064-Ram-Abelian-Covers-Gale.tex`
M066	C3	Notable distinct local improvement: replaces2^r signed-root polarization transports by a character-orthogonal list of length<2r with explicit coefficient/error control. The source proper-Hilbert proof yields a universal exceptional set, weaker nonmeagre/positive-outer-measure input, and one existential fixed Hilbert/denominator/degree format throughout a chart. Rational coefficient rank compression reduces a returned two-slot synthesis without nonsplit assumptions. Classical origins and internal attribution are recorded; M062 compression is not rescored. Initial algebraic-cycle existence, geometric background and rational descent costs remain conditional/imported; no global conjecture, effective cycle-search bound or novelty claim. **Evidence:** \label{thm:sparse}; \label{prop:signerror}; \label{eq:fieldfixture}; \label{prop:rank}; \label{thm:universal}; \label{cor:thickness}; \label{cor:curve}; \label{sec:certificates}; \label{sec:source}. **Report:** `TR-M066-Ram-Conditional-Kuga-Satake.tex`
M096	C3	Notable source-specific quantitative least-term transfer: finite positive-weight tail bounds, feasible annihilation of low-weight jet defects with a strict analytic two-column gain, and exact Euler/Fourier moment-block conditioning and leakage budgets with nonunitary metric costs. Classical identities and source qualitative projections receive no novelty credit. No new polarized-surface construction, effective global threshold, or maximal-Seshadri theorem. **Evidence:** \label{lem:gap}; \label{thm:tail}; \label{prop:cancel}; \label{thm:relative}; \label{eq:imperfect}; \label{ex:cancel}; \label{thm:blocks}; \label{prop:leak}; \label{eq:compressionbound}; \label{sec:boundary}. **Report:** `TR-M096-Ram-Seshadri-Surfaces.tex`
M097	C3	Notable source-specific local quantitative contribution: exact Laurent compression lifts with analytic error costs, anisotropic finite deficit-to-jet certificates, unequal-order exact Hermite staircase, and conditional degree rate. Classical ingredients are explicitly credited; no new global maximality theorem or uniform geometric certification. **Evidence:** \label{prop:remainder}; \label{thm:transfer}; \label{thm:filling}; \label{thm:hermite}; \label{thm:stencil}; \label{cor:effective}; \label{prop:hilbert}; \label{sec:boundaries}. **Report:** `TR-M097-Ram-Seshadri-Higher-Dimensions.tex`
M102	C3	Notable source-specific local extension: a computable planar residual-ball test, imperfect holomorphic splitting bound, geometric inexact correction policy with explicit pointwise domain gates, finite-prefix and side-sensitive composition stopping certificates, and quantitative finite Fourier stencil bounds preserving exact symplectic flux. Classical contraction/Cauchy/Jensen principles are attributed; dyadic optimality receives no improvement credit. Unknown geometric constants, uniform moving-centre coverage and global K3/Oka assembly remain source-level or imported obligations. No new global theorem, historical novelty or finite-test certification of algebraic/analytic geometry is claimed. ApplicabilityL3 is independent. **Evidence:** \label{thm:ball}; \label{prop:step}; \label{thm:inexact}; \label{prop:prefix}; \label{prop:optimal}; \label{thm:composition}; \label{ex:budget}; \label{prop:taylor}; \label{sec:flux}; \label{prop:fourier}; \label{sec:architecture}. **Report:** `TR-M102-Ram-K3-Oka.tex`

Report	Score	Demonstrated contribution and remaining boundary
M107	C3	Notable source-specific complete ambient LND kernel and degree-dependent specialization defect, including exact exceptional-divisor failure, with fully proved odd-degree/lower-auxiliary-term homogeneous rigidity and higher-root extraction extensions. Classical slice, saturation, and abc tools and source sharp examples receive no independent novelty credit. No full global cancellation verification, new geometric family, or all-LND classification is claimed. **Evidence:** \label{thm:kernel}; \label{thm:specialization}; \label{ex:specialization}; \label{prop:order}; \label{thm:odd}; \label{eq:sharp}; \label{prop:root}; \label{cor:rootodd}; \label{sec:boundary}. **Report:** `TR-M107-Ram-Affine-Cancellation.tex`
M109	C3	Notable source-specific local contribution: exact all-order obstruction ideals and terminating chart-data lifting certificate, plus binary-form rigidity for every homogeneous fiber shift below2. Classical ingredients credited; no new global stable-coordinate theorem or priority claim. **Evidence:** \label{thm:rees}; \label{cor:plinth}; \label{prop:transition}; \label{sec:gluing}; \label{thm:binary}; \label{sec:global}. **Report:** `TR-M109-Ram-Stable-Coordinate.tex`
M148	C3	Notable source-specific quantitative improvements: variable-contrast energy and an equilibrated residual interval for the normalized Laurent gain; weighted two-sided barrier exclusion; finiteN flux tail reduced16to8 using summable domination; full Laurent/area-slack budget with coefficient8-2pi and exterior map/derivative bounds. Exact square-panel hidden-mode fixture distinguishes a finite Ram moment solve from a continuum certificate. Classical coercivity, Prager-Synge, Cauchy-Schwarz and dominated convergence are credited; mere coordinate rewriting is not scored. Companion retained-test/actual-image construction, removability, trace-null and all-component obligations remain explicit. Contemporary survey still labels the implication open. No accepted global settlement or historical novelty claimed. ApplicabilityL3 is independent. **Evidence:** \label{prop:energy}; \label{thm:residual}; \label{prop:high-contrast}; \label{lem:filling}; \label{thm:slice}; \label{cor:exclude}; \label{thm:tail-flux}; \label{thm:defect}; \label{cor:exterior}; \label{sec:panel}; \label{sec:architecture}. **Report:** `TR-M148-Ram-Circle-Domain-Rigidity.tex`
M149	C3	The report proves a reduction of the finite stream boundary coefficient from 2pi to 2, coupled area-energy accounting, an exact distortion-capped energy optimum, and minimum-energy finite period repair. These are notable local improvements, but compatibility, smooth metric gluing, and all-component construction remain separate obligations, with no independent unrestricted uniformization proof or demonstrated end-to-end gain. **Evidence:** lem:variation; thm:stokes; thm:metric; thm:repair; thm:cluster; Costs and limitations. **Report:** `TR-M149-Ram-Koebe.tex`
M150	C3	The report proves a bounded-univalent quadratic counterexample to the earlier Ram pointwise tilted-variance premise, with variance T^2/12+O(T), and supplies tighter reciprocal-coefficient tails, a moment-matching obstruction to affine covariance, and explicit transport-defect certificates. These are notable corrections and local certification results, but no universal sharp upper certificate is realized, and the Brennan conclusion and loss-free endpoint estimate retain source or external dependencies. **Evidence:** thm:cusp; An exact critical-tilt symmetry; Why root moments cannot replace covariance; eq:tail; thm:defect; Loss-free endpoint bounds require another theorem; Advantages and costs. **Report:** `TR-M150-Ram-Brennan.tex`
M151	C3	Alongside the critical-tilt correction shared with M150, the report proves the nonzero-drift path identity, global coefficient-cover and residual certificate implications, and an exact Koebe obstruction eliminating unweighted block depths 1 through 13. These are notable local results, but the power-family and endpoint conclusions retain the geometric pairing dependency, and no successful all-class upper certificate or numerical strict gap is supplied. **Evidence:** thm:corner; thm:path; thm:certificate; prop:residual; prop:koebeblock; thm:coefficient; thm:parametertransfer; cor:endpoint; Costs and boundaries. **Report:** `TR-M151-Ram-Inverse-First.tex`

Report	Score	Demonstrated contribution and remaining boundary
M156	C3	Notable source-specific imperfect-join exterior-tail certificate combined with a finite two-sided wave-to-sign completion budget, exact energy-tail improvement, and proved physical lower-floor consequences and weak-limit obstruction. Zero-offset sparse rounding and classical Parseval, variance, convolution and compactness tools alone are not treated as major novelty. Companion packing/discrepancy inputs remain imported; no new global ultraflat family, effective all-length threshold, positive tower realization, or Banach-spectrum solution is established. **Evidence:** \label{thm:joins}; \label{prop:jump}; \label{thm:rounding}; \label{thm:completion}; \label{eq:energydefect}; \label{ex:soft}; \label{prop:grid}; \label{ex:five}; \label{thm:physical}; \label{prop:products}; \label{prop:weak}; \label{sec:limits}. **Report:** `TR-M156-Ram-Ultraflat-Littlewood.tex`
M157	C3	Notable local increment: fixed-geometry acute-angle taper improves the lower-envelope/projection remainder to N^-1/2 without enlarging the coefficient budget; sharp worst-case second-moment rescaling reduces the rounding defect. Classical log/product consequences secondary. No global source theorem certification, effective headline threshold or first-in-literature claim. **Evidence:** \label{thm:acute}; \label{prop:defect}; \label{cor:assembly}; \label{prop:logs}; \label{sec:limits}; \label{sec:literature}. **Report:** `TR-M157-Ram-Littlewood-Lower-Envelope.tex`
M158	C3	Notable local strengthening from one-sided source caps: exact normalized binary word products with summable cap excesses converge to a bounded positive-a.e density, with explicit total-variation/finite-Lp/relative-floor tails and a physical Morse-coordinate identification under declared classical gates. Exact autocorrelations, second-order grid cap certification and prefix-to-limit mass-scaled Gram margins provide a source-specific finite certificate. No fixed positive lower floor or factor independence required. Classical measure/matrix/frame tools credited; existing source merit bound and concentration corrections not counted as new. Full all-length sign construction, effective length thresholds and full-space Koopman cyclicity are not independently certified. No historical novelty/speedup/global Banach claim. L3 unchanged and independent. **Evidence:** \label{thm:product}; \label{cor:word}; \label{thm:physical}; \label{prop:grid}; \label{prop:gram}; \label{cor:tail-gram}; \label{prop:recursion}; \label{prop:cyclic}; \label{sec:corrections}; \label{sec:examples}. **Report:** `TR-M158-Ram-Littlewood-Maxima.tex`
M169	C3	The report uses rank-one unit-field Jacobians to make critical outer-scale certificates finite, proves a strictly larger Fourier-tail Dini domain with principal-value and restricted maximal control, and relaxes the measure-ratio summation requirement to every logarithmic exponent above one. These are notable proved extensions, but the all-L2 outer-scale consequence imports the source theorem, no all-L2 maximal estimate follows, and no substantial end-to-end improvement of the source's uniform oscillatory estimate is demonstrated. **Evidence:** lem:rank; thm:diff; cor:Wp; thm:Dini; Strict enlargement and logarithmic regularity; A restricted maximal-truncation conclusion; thm:kernel; thm:outerextension; Costs and unresolved implications. **Report:** `TR-M169-Ram-Hilbert.tex`
M181	C3	Notable source-specific exact cancellation of modulo-36 repeated-node jets with a closed geometric spectral counterterm, preserving the actual Schwartz realization, fixed columns and full residuals. The conditional bounds improve the original finite/tail coefficient errors and deleted-jet sign budgets while explicitly paying the inverse-coordinate cost. Classical Neumann, Schur, Fourier and Bernstein tools and finite tests alone are not credited as discoveries. No new extremizer, global universal-optimality certification, replacement arithmetic checker or demonstrated speedup is claimed. **Evidence:** \label{prop:aliasnorm}; \label{thm:cardinal}; \label{prop:density}; \label{thm:blocks}; \label{thm:improved}; \label{eq:originaltailbudget}; \label{cor:signbudget}; \label{eq:observable}; \label{sec:finite}; \label{sec:limits}. **Report:** `TR-M181-Ram-Triangular-Atomic.tex`
M182	C3	Notable source-specific conditional increment: actual sharp mixed auxiliaries from the fixed infinite spectral family, exact L1/all-inverse-moment Schwartz thresholds, weighted inverse and finite-window energy control without occupancy assumptions. Classical tools credited. No independent global optimality, screening or all-Schwartz uniqueness claim. **Evidence:** \label{thm:mixture}; \label{prop:analytic}; \label{thm:weighted}; \label{thm:finitewindow}; \label{cor:rate}; \label{sec:perturbation}. **Report:** `TR-M182-Ram-Triangular-Optimality.tex`

Report	Score	Demonstrated contribution and remaining boundary
M183	C3	Notable local quantitative extension: globally certified sign defects give an explicit all-packing density bound via an actual Gaussian Fourier repair and cumulative geometric leakage; source exact margins give directed finite-window shell budgets with linear first-contact and quadratic higher-node penalties. Strict shell residuals obstruct exact sharpness, and a coherent rational Bernstein fixture improves an enclosure while retaining shared errors. Classical tools and prior stability mechanisms credited. No new global witness or crystallization theorem, no historical novelty claim, and no source-margin speedup asserted. L3 unchanged and independent. **Evidence:** \label{lem:leakage}; \label{thm:repair}; \label{prop:defect}; \label{lem:shell}; \label{thm:contact}; \label{prop:slope}; \label{prop:slack}; \label{prop:coherent}. **Report:** `TR-M183-Ram-Planar-Packing.tex`
M203	C3	The report proves a sharper mass-coupled moment certificate with an exact separating square example, positive Hermite-tail and complete entropy certificates, quantitative curl-jet control, and exact chaos slack, and documents reconstructed finite certificate checks. These are notable certification improvements, but the strengthened entropy remainder retains upstream transport and scalar premises, does not improve the leading sharp constant or establish quantitative distance-to-equality stability, and does not independently prove the global entropy or simplex claims. **Evidence:** thm:coupled; An exact rational separation; thm:gramtail; thm:entropydata; thm:curl; thm:chaos; cor:strong; sec:audit; What remains costly or unavailable. **Report:** `TR-M203-Ram-Isotropic.tex`
M210	C2	Source-specific residual envelopes, asymmetric policy repairs, exact-zero margins, robust gain-bias witnesses and a sharp conditional stopping-discount bridge with complete bounded proofs; classical methods credited and no new universal discovery algorithm claimed **Evidence:** \label{thm:gaincertificate}; \label{lem:changes}; \label{thm:envelope}; \label{thm:policyrepair}; \label{prop:zero}; \label{thm:stopping}; \label{cor:rational}; \label{app:review}. **Report:** `TR-M210-Ram-Stochastic-Mean-Payoff.tex`
M217	C3	Notable source-specific local refinement: a uniform larger inset window, exact boundary dilution and central-loss costs coupled to a curvature correction give a sharp frontier within the declared scalar family. An explicit choice permits residual1e-13 while retaining central utility>1.1124e-6. Signed covector panels and chart-transfer bounds retain the assumptions and expose why no lower-degree substitution or exponent improvement follows. Classical LP/Bernstein/entropy tools are credited; source continuous-domain numerical inputs and tensor attainability remain imported. **Evidence:** \label{prop:signed}; \label{cor:panels}; \label{lem:inset}; \label{thm:robust}; \label{ex:tolerance}; \label{prop:frontier}; \label{prop:product}; \label{prop:charttransfer}; \label{prop:nearflat}; \label{sec:limits}. **Report:** `TR-M217-Ram-Matrix-Multiplication.tex`
M231	C3	Notable local source-specific improvement: smaller Casoratian z cutoff, sharp cleared-degree precision for the fixed denominator, and division-free structured generation/action with explicit model and dense-output costs. Classical tools credited; no global PIT certification or novelty claim. **Evidence:** \label{thm:order}; \label{thm:degree}; \label{thm:generation}; \label{thm:stream}; \label{thm:descent}; \label{sec:comparison}. **Report:** `TR-M231-Ram-Positive-Characteristic-PIT.tex`
M232	C3	Notable source-specific local improvement: O(d) integral-jet action and O(sd) known-formula detecting-column evaluation; minimal normalized positive rational diagonal integral gauge gamma_k=k!lcm(1,...,n)^k; sharp universal coordinate denominators and explicit rational numerator heights supporting division-free modular and enclosed-residual zero certificates. Classical ingredients credited; gauge minimality and sharpness are confined to their stated classes. No lower hitting dimension, no arbitrary black-box/full-output speedup, no positive-characteristic or inverse-gate extension, and no historical priority or new global derandomization claim. L3 unchanged and independent. **Evidence:** \label{prop:recurrence}; \label{prop:residual}; \label{thm:integral}; \label{thm:lattice}; \label{thm:apply}; \label{thm:height}; \label{cor:enclosure}. **Report:** `TR-M232-Ram-Rational-Hitting.tex`

Report	Score	Demonstrated contribution and remaining boundary
M238	C3	Source-specific notable finite increment: all17 exact profiles admit independent normalized odd tilts with common radius2^-15 and larger rowwise radii, preserving E_B exactly. Coupled quadratic variation, coefficientwise exact minima with J-positive endpoint reduction and base loss3|theta|/8 certify independent first integer Chebyshev edits up to3051. A further dependent residual budget retains cross terms and all source/global boundaries. Classical enclosure tools and original checker replay receive no novelty credit. **Evidence:** \label{lem:endpoint}; \label{prop:closure}; \label{prop:variation}; \label{thm:quadratic}; \label{thm:family}; \label{cor:integer}; \label{ex:edit}; \label{prop:budget}; \label{sec:semigroup}; \label{sec:limits}. **Report:** `TR-M238-Ram-Hellinger-Contraction.tex`
M277	C3	The report proves physical moment-to-jet error bounds, multiple-zero Bernstein-Rolle upper certificates, robust exact counts on exhaustively covered return intervals, and a quantitative cyclic inverse margin with explicit degeneration tests. These are useful local and finite-family certificate extensions, but neither a nontrivial new certified polynomial-field count nor a degree-uniform atlas, Hilbert-number upper bound, or terminating global counting algorithm is demonstrated. **Evidence:** thm:jets; prop:flow; eq:log-multiplier; thm:rolle; thm:exact; prop:cyclic-margin; prop:cubic; thm:cofinal; Verification boundaries. **Report:** `TR-M277-Ram-Limit-Cycles.tex`
M278	C2	Exact matching-multiplier interfaces, rational compact perturbation margins, and a reviewed conditional shape extension; no independent global theorem or established historical novelty **Evidence:** \label{thm:unit}; \label{cor:multiplier}; \label{prop:slopemargins}; \label{lem:rational}; \label{thm:threshold}; \label{lem:weakshape}; \label{thm:shape}; \label{cor:septic}; \label{eq:raysos}; \label{prop:septicsharp}; \label{app:evidence}. **Report:** `TR-M278-Ram-Quintic-Lienard.tex`
M279	C3	The report proves a 100-fold tightening of the ideal Hellinger loss coefficient, from 300delta^2 to less than 3delta^2, together with optimal zero-set-aware whitening, coordinate-invariant transfer certificates, and an explicit counterexample showing why fixed-power convergence alone loses cyclicity. These are notable realized local results, but the accumulated 4Kdelta^2 construction budget remains conditional on analytic realization, and no new smooth dynamical construction, independent global Banach proof, or implementation gain is established. **Evidence:** lem:sqrt; thm:three; cor:4; thm:whitening; thm:gram-transfer; prop:counterexample; thm:retention; sec:stage-boundary; sec:evidence. **Report:** `TR-M279-Ram-Banach.tex`
M280	C3	The report proves the sharp arbitrary-coupling information bound J(A) <= I(W;C)/n with equality examples, extends local orbit cancellation to centered L1 observables, improves the finite-block majorant, and demonstrates a signed-weight residual advantage with physical Gram-tail accounting. These are notable local quantitative contributions, but they do not construct the required saturated arrays and finite blocks, independently establish universal multiple mixing, or supply an effective general mixing rate, so the demonstrated contribution does not warrant C4. **Evidence:** thm:information; Sharpness and the scope of the improvement; thm:L1; prop:bessel; thm:signed; prop:gramtail; A finite example where signed weights help; sec:future; sec:verification. **Report:** `TR-M280-Ram-Mixing.tex`
M281	C3	The report proves optimal constant-one shortfall counting, sharp transfer-cone and return-graph certificates, and a weighted-defect factorization with uniform approximation-error bounds that avoid the potentially divergent unweighted mass cost. These useful local improvements and the projective-lift formulation produce no verified positive-entropy nonzero standard-map instance, effective parameter threshold or independently completed analytic closure, so they support C3 rather than a substantial demonstrated advance in the headline result. **Evidence:** thm:shortfall; prop:cone; prop:cancelword; thm:graph; thm:lift; thm:weighted; eq:weightedmass; eq:finitecaperror; sec:defects; Conclusion; app:evidence. **Report:** `TR-M281-Ram-Entropy.tex`

Report	Score	Demonstrated contribution and remaining boundary
M282	C3	The report proves exact circle preconditioning, a sharp centered rational primitive and a reciprocal-orbit-average flux inverse whose coefficient-disk corollary enlarges the elementary near-constant radius from 1/8 to 1/4, together with coercive, full-residual and quantitative support-collar certificates. These nontrivial local extensions improve admissible inverse conditions and error control, but do not establish uniform complex-parameter jets, the complete analytic construction or global ellipse rigidity, leaving the demonstrated effect at C3. **Evidence:** thm:circle; cor:precondition; thm:newton; thm:centered; thm:flux; cor:diskflux; thm:coercive; prop:gram; thm:collar; ex:adverse; sec:limits; sec:provenance. **Report:** `TR-M282-Ram-Collars.tex`
M283	C3	The report proves sharp finite complex-moment isolation for the line coalescent model, a reduced point-model moment test, stable analytic double division, and explicit odd-cycle inverse and sign-sensitive last-site error bounds. These are significant local verification refinements, but they do not supply effective elimination bounds, complete the global continuation and critical-wall arguments, or prove quantitative near-ellipse stability, and the companion flux inverse is an imported result rather than an additional independent contribution. **Evidence:** thm:finite-moments; thm:line-model; thm:point-model; thm:double-division; thm:odd-inverse; thm:modified-margin; prop:two-channel; thm:exit-budget; eq:last-finite-budget; sec:limits; app:evidence; app:analytic. **Report:** `TR-M283-Ram-Rigidity.tex`
M287	C3	Notable source-specific local quantitative contribution: normalized reflection-defect exclusion with actual angular tail, global elementary third-side phase margin, explicit triangle H1 mean/trace-to-amplitude transfer, and an exact rational irrational-angle fixture. Physical moment/Taylor residual certification and small-divisor counterexample delimit practical use. Classical Fourier/Poincare/trace/interpolation tools credited; no historical priority claim. The certificate requires actual YF, full edge and tail control and does not independently establish the global all-frequency billiard theorem or rates for the source's singular limits. Applicability L3 unchanged and independent. **Evidence:** \label{thm:amplitude}; \label{lem:phase}; \label{cor:floor}; \label{lem:trace}; \label{thm:physical}; \label{sec:example}; \label{prop:obstruction}; \label{sec:phaseapprox}. **Report:** `TR-M287-Ram-Triangular-Weak-Mixing.tex`
M288	C2	Explicit holonomy residual, twisted-annulus and anchored coefficient-gap bounds combine with a domain-correct finite-band full-residual certificate and derivative-tail budget. Classical ingredients are attributed, the geometric extension is explicitly source-dependent, and no independent global ergodicity proof, uniform infinite-mode gap or speedup is claimed. **Evidence:** \label{prop:finitehol}; \label{thm:annulus}; \label{thm:anchored}; \label{lem:matrix}; \label{thm:finiteband}; \label{cor:tail}; \label{thm:conditional}; \label{sec:ram}; \label{sec:tests}; \label{sec:assessment}. **Report:** `TR-M288-Ram-Triangles.tex`
M292	C3	Notable source-specific local improvement: optimized sparse-graph tensor packing exponent sqrt(d log(2+d)), enlarged summable ranks with explicit O(log A/sqrt A) selected-density residual, plus conditional retained-Gram square stability. Classical tools credited; no global mixing/pointwise theorem certification or overall convergence rate. **Evidence:** \label{thm:packing}; \label{cor:ranks}; \label{thm:dyadic}; \label{thm:square}; \label{cor:mass}; \label{sec:discussion}. **Report:** `TR-M292-Ram-Multiple-Ergodic-Averages.tex`
M293	C3	Notable source-specific sharp support-accuracy law 3c/(2R)+O(R^-2) for the exact three-moment signed profiles, with optimal atomic correction, variance-corrected smoothing, explicit variation/sampling and conditional block-oscillation errors. Secondary finite-test activation certificates are modest classical applications. No global fourfold convergence certification, H/P proof replacement, historical priority claim or repeated credit for M292 packing. **Evidence:** \label{thm:lower}; \label{lem:repair}; \label{thm:sharp-profile}; \label{prop:sampling}; \label{cor:oscillation}; \label{prop:saturation}; \label{prop:axes}. **Report:** `TR-M293-Ram-Fourfold-Ergodic-Averages.tex`

Report	Score	Demonstrated contribution and remaining boundary
M294	C2	Explicit arbitrary-signed-slope planting moments, a proved two-radius fourth-order summed error with its 17/3 leading-constant penalty, signed finite-horizon stability, and a bounded Bernoulli model-class proof. Classical methods and inherited M295 principles are credited. The global multilinear estimate is not independently certified; no global proof, novelty or speedup is claimed. **Evidence:** \label{prop:geometry}; \label{thm:richardson}; \label{cor:planttails}; \label{eq:richtransfer}; \label{prop:horizon}; \label{prop:bernoulli}; \label{sec:assessment}; \label{sec:evidence}. **Report:** `TR-M294-Ram-Distinct-Slopes.tex`
M295	C3	The report proves a delay-cost reduction from L to sqrt(L) and a smooth-row logarithmic-loss reduction from exponent 2 to 3/2 under unchanged local hypotheses, plus asymptotically optimal moment corrections, a robust oscillation criterion and second-order lattice-planting errors. These are notable realized quantitative gains, but the uniform oscillation closure is not independently established and the weak-mixing, endpoint Lp and general-system pointwise consequences remain conditional on that analytic premise, with no demonstrated higher-level effect sufficient for C4. **Evidence:** thm:robust; sec:profiles; eq:TVupper; eq:TVlower; prop:delay; prop:smooth; prop:plant; cor:all; cor:wm; thm:lp; Benefits, costs, and further research; sec:evidence. **Report:** `TR-M295-Ram-Triple.tex`
M301	C3	The report proves moment-free spectral-center stability and iteration error bounds, quantitative density and Fourier-tail estimates, a complete integral-coloring interpolation certificate, and an explicit 10^-7 clearance for the exact terminal spindle placement. The finite checks certify a four-chromatic seven-point terminal gadget rather than a non-five-colorable witness, while the Haar, palette and interface dependencies and Lean proof chain are not independently closed or replayed, so no new global six-color bound or demonstrated computational advantage is credited. **Evidence:** prop:density; thm:center; A controlled approximation module; prop:circle; thm:algebra; prop:clearance; sec:checks; Assessment and a focused research program; sec:provenance. **Report:** `TR-M301-Ram-Coloring.tex`
M303	C3	Notable source-specific local increment: exact half-open source keys have sharp multiplicity2, doubling both perturbation exponents and giving a conditional final-module c<gamma/80 budget. An independently proved integer optimizer for the actual half-digit/sphere colour budget gives leading log-length constant1/(8log2), explicit lower-order loss, and architecture-only optimality. Exact literal-event and mixed-base interfaces retain all dependencies and all-r quantifiers. Classical principles, finite fixtures and parent checks receive no invented novelty credit. **Evidence:** \label{lem:shell}; \label{prop:key}; \label{prop:prescription}; \label{thm:completion}; \label{cor:c}; \label{prop:CE}; \label{thm:mixed}; \label{prop:partition}; \label{thm:opt}; \label{prop:optimal-leading}; \label{sec:ram}; \label{sec:limits}. **Report:** `TR-M303-Ram-Van-der-Waerden.tex`
M308	C2	Useful but modest source-specific local contribution: exact optimum of all nonnegative single-deletion certificates, constructive integer sharpness in the four-uniform relaxation, and an explicit seven-vertex model violating a smaller planar constraint. Parity-aware defect and conditional incidence/support budgets, plus a scale-sensitive finite isotropy test, clarify limitations. These are limited consequences of classical counting, capacity allocation and finite linear algebra; they do not establish a new universal crossing bound, drawing realizability or an independent Harary-Hill proof. C2 is independent of unchanged applicability L3. **Evidence:** \label{thm:weighted}; \label{thm:integer}; \label{sec:examples}; \label{eq:burden}; \label{sec:stability}; \label{prop:residual}. **Report:** `TR-M308-Ram-Complete-Graph-Crossings.tex`
M309	C3	The report derives an exact nonnegative subspace and crossing-slack decomposition with equality restrictions, replication-weighted rank inequalities and a ceiling-sensitive odd correction, supported by complete local proofs and recorded exact finite checks. The detector and subspace theorem themselves are attributed source results, and the global crossing formula remains conditional in the report on the common-drawing geometric contract with formal dependencies unreplayed, so no improved universal crossing bound or extremal-drawing classification is demonstrated. **Evidence:** thm:defect; cor:slack; thm:weighted; prop:round; prop:transfer; thm:subspace; thm:detector; thm:conditional; sec:checks; sec:lean; Benefits, Limits, and Future Work; app:sources. **Report:** `TR-M309-Ram-Crossings.tex`

Report	Score	Demonstrated contribution and remaining boundary
M318	C3	The report demonstrates notable local contributions through an exact constraint-preserving rounding budget, explicit number-field tensor certificates, a smaller obstruction-color bound, and a proved derivation-to-tensor bridge recovering a seven-point obstruction. These go beyond restatement but do not establish a new Ramsey classification or historical novelty: the seven-point construction is attributed to an existing appendix, positive Ramsey consequences depend on the full sufficiency chain, and no effective forcing-dimension bound is delivered. **Evidence:** thm:necessity; prop:remainder; prop:rounding; prop:derivationbridge; thm:seven; thm:algebraic; prop:numberfield; ex:five; sec:audit. **Report:** `TR-M318-Ram-Ramsey.tex`
M319	C3	The report proves a quantitative deletion-stage identity, an accumulated deficit bound, and a nonuniform transitive-substitution estimate with a reduced sufficient uniform block size, alongside exact clone and weight-error certificates. These are useful local refinements rather than a new unrestricted existence proof: the universal second-neighborhood conclusion still depends on the source pruning argument, weighted existence imports that universal premise, and bounded graph tests do not establish either global claim. **Evidence:** lem:stage; cor:accum; thm:blowup; cor:threshold; thm:clones; prop:weighttransfer; prop:robust; Pruning: source reconstruction and proof obligations; Exact tests and their bounded meaning. **Report:** `TR-M319-Ram-Seymour.tex`
M320	C3	The report adds a proved unbounded cut-versus-spectral thinness separation, partition-specific packing and physical-metric error certificates, and residual and outer-threshold criteria that preserve packing under stated approximation bounds. These constitute notable local verification tools, but the universal thin-tree deduction remains conditional on the source hierarchy, no full response certificate is instantiated, and no improved universal thinness constant or effective construction is demonstrated. **Evidence:** prop:gap; prop:partcert; thm:domination; thm:metric; prop:vectorerror; thm:residual; prop:outer; sec:ledger. **Report:** `TR-M320-Ram-Thin-Trees.tex`
M321	C3	The report proves a whitening-free rational certificate for simultaneous cut control and packing, an explicit leverage-error budget, and a floor-loss refinement reducing the compressed-input transfer factor from 2 to at most 8/7 and the conditional thin-and-cheap bound from 24B to 96B/7. These are demonstrated local verification and reduction improvements, but the latter bounds assume a universal thin-tree routine, polynomial verification is measured in the supplied certificate length, and no implementation, speedup, or independent discharge of the hierarchy and signing dependencies is delivered. **Evidence:** thm:certificate; prop:leverage; prop:compression; thm:cheap; Advantages, limits and concrete next steps; Exact checks and their limits. **Report:** `TR-M321-Ram-Polynomial-Thin.tex`
M334	C3	The report demonstrates sharp local improvements through weighted residual and joint mass-variation bounds, a proved and attained three-strip boundary coefficient of 1 instead of 2, and sharper endpoint and integer-branch recurrence transfers. These warrant notable contribution credit but not a demonstrated halving-line power saving: sequence sharpness is not geometric realizability, the uniform all-ranks packing premise and a valid numerical scale remain unavailable, and the improved boundary coefficient does not change the required dimension threshold. **Evidence:** thm:weighted; thm:joint; thm:boundary; sec:tokens; thm:recurrence; cor:integer; sec:checks; sec:pros. **Report:** `TR-M334-Ram-Halving.tex`
M375	C2	Uniform finite positive-resolvent perturbation bound epsilon/(1-epsilon), paired semigroup control, and proved conditional weighted-path and L1-initial-density extensions; classical methods credited, all geometric/electrical inputs inherited, and no global FK-LQG proof, point-start result, novelty or speedup claimed. **Evidence:** \label{thm:perturb}; \label{eq:auxmass}; \label{eq:jointbudget}; \label{cor:semiperturb}; \label{thm:weightedpath}; \label{thm:tilt}; \label{prop:UI}; \label{prop:corner}; \label{cor:region}; \label{sec:tests}. **Report:** `TR-M375-Ram-FK-Walks.tex`

Report	Score	Demonstrated contribution and remaining boundary
M376	C3	The report proves a positive capped-kernel Hilbert-Schmidt error certificate with continuum mesh rate h^(1/113) under the stated ball bound, and quantitative inverse-to-heat bounds uniform for all times bounded away from zero, assembled into ordered spectral and heat-trace enclosures. These are meaningful analytic extensions beyond measure-aware relabeling, but certified quantum masses and ball constants are not supplied, no FK edge-count convergence rate or practical efficiency is demonstrated, and the probabilistic companion and all-level compactness inputs remain outside the local contribution. **Evidence:** prop:domain; thm:cap; eq:optimal-cap; thm:heat; thm:assembled; sec:checks; sec:future. **Report:** `TR-M376-Ram-FK-Spectrum.tex`
M377	C3	Retained-scale fixed-deterministic-mass L2 Green bound C sqrt(m) with quadratic tail, a simultaneous square-root modulus with every fixed logarithmic exponent above 1/2 and an L2-bounded coefficient, and a proved canonical Feller arbitrary-weak-start transfer under uniform source resolvent/exit inputs. Notable local conditional improvements; classical tools credited, imported global geometry and one-scale estimate not certified, no historical novelty or speedup claim, and no fresh credit for M375 inherited extensions. **Evidence:** \label{eq:B}; \label{thm:mean}; \label{thm:log}; \label{eq:ap}; \label{eq:log}; \label{eq:residualmod}; \label{prop:modbridge}; \label{thm:allstart}; \label{cor:random}; \label{sec:tests}. **Report:** `TR-M377-Ram-Tree-Walks.tex`
M386	C3	Notable source-specific local refinement: the actual corridor closed-neighborhood deletion majorant yields a Hilbert-Schmidt loss from finitely many square-summable columns even when susceptibility diverges. A fixed-parameter endpoint extension and finite Gram certificate preserve a quadratic omitted-energy cost while sharply separating the generally linear cross-block loss error. Explicit tree formulas and whole-operator residual gates make the limitations checkable. Classical operator tools and source global theorem receive no invented novelty or verification credit. **Evidence:** \label{lem:source}; \label{thm:bubble}; \label{cor:endpoint}; \label{prop:gram}; \label{prop:loss}; \label{prop:uniform}; \label{sec:tree}; \label{prop:obstruction}; \label{lem:schur}; \label{prop:neumann}; \label{cor:width}; \label{sec:ram}; \label{sec:limits}. **Report:** `TR-M386-Ram-Percolation-Nonuniqueness.tex`
M387	C3	Notable scoped source-specific quantitative improvement: the already-available uniform exponential moments replace the source fourth-moment clipping loss and retain every exponential rate strictly below the bounded-slab rate. Positive tail sums feed explicit physical-norm normalization bounds; source-disc/alignment and actual-law pair-product projections certify connected-observable errors. Classical tail integration, unitary dilation, Cauchy estimates and Hilbert projection are credited, and algebraic normalization identities and diagnostic fixtures are not treated as new global principles. The uniform RG, continuum, OS and gap construction remains imported; no historical-priority claim. C3 is independent of unchanged L3 applicability. **Evidence:** \label{thm:clipping}; \label{eq:mass-tail}; \label{eq:second-tail}; \label{thm:normerror}; \label{thm:cumulant}; \label{prop:composites}. **Report:** `TR-M387-Ram-O3-Continuum.tex`
M388	C2	Conditional extension of the source count to each fixed energy below twice the lowest mass and local finiteness on every compact subthreshold mass band, with explicit noisy-window, full-residual visibility and known-gap residue enclosures. Classical tools credited; continuum construction, numerical mass/residue, threshold control and historical novelty remain unestablished. **Evidence:** \label{thm:count}; \label{thm:support}; \label{prop:window}; \label{prop:visible}; \label{prop:residue}; \label{sec:checks}. **Report:** `TR-M388-Ram-O3-Pole.tex`
M389	C2	Explicit positive-resolvent residual-to-coupling and scaled-tail-mass enclosures, finite Cauchy moment recurrence, two-trace and sharp purity norm bounds, and robust seam-error propagation. Classical tools credited; endpoint asymptotics, RG, integrability and lattice bridges remain conditional. No global certification, historical novelty or speedup claimed. **Evidence:** \label{lem:scalar}; \label{thm:residual}; \label{cor:truncate}; \label{eq:ram-expansion}; \label{eq:recurrence}; \label{eq:boundedexample}; \label{thm:trace}; \label{thm:purity}; \label{cor:seam}; \label{sec:assembly}; \label{sec:tests}. **Report:** `TR-M389-Ram-O4-Mass.tex`

Report	Score	Demonstrated contribution and remaining boundary
M390	C3	The report proves a square-defect recursion coefficient of 13 in place of the source's relaxed 64, a positive finite-feature tail for the actual O(4) transfer kernel, trace and actual-power purity error certificates, and a compact-operator tail-resolvent count. These are useful model-specific local improvements, but no positive-coupling sharp-scale square test or practical mass computation is completed, uniform width control and the nonperturbative RG chain are not supplied by the certificates, and no new lattice mass-gap or continuum-existence theorem is established. **Evidence:** thm:13; thm:trace; thm:feature; sec:onesided; sec:intervals; thm:schur; sec:checks; sec:risks. **Report:** `TR-M390-Ram-O4-Mass.tex`
M394	C3	Two notable source-specific local increments: exact signed circular-hole defect with explicit O(sum rho_i²) enclosure replacing stated O(rho), including unequal radii, and average-loss kernel stability extended from1/64 tosqrt(5)-2 with quantitative terminal-density/depth bound on varying probability spaces. Classical projection/hypercircle/Fourier tools credited; no independent critical stiffness, lattice convergence, collision/tightness certification, optimal-threshold or historical-priority claim. **Evidence:** \label{thm:quadratic}; \label{cor:sector}; \label{thm:kernel}; \label{sec:kernel-example}; \label{prop:residual}. **Report:** `TR-M394-Ram-Critical-XY-Spin.tex`
M395	C3	Notable source-specific local refinement: the actual balanced quadratic Euler step admits a corrected phase clock whose half-log amplitude term removes the linear x error and yields an explicit7y^2 bound. A true-defect extension retains invariant-product positivity, segment branch costs and a separate phase-monotonicity gate. Exact affine-panel and shrinking-disk interfaces make its use checkable. Classical invariant/maximum-principle/Cauchy tools are credited; actual global RG, positive transversality and spin mass comparisons are not newly verified. **Evidence:** \label{sec:map}; \label{thm:phase}; \label{lem:euler}; \label{thm:clock}; \label{thm:robust}; \label{prop:panels}; \label{ex:step}; \label{sec:scaling}; \label{prop:disk}; \label{lem:secant}; \label{sec:ram}; \label{sec:gates}; \label{sec:limits}. **Report:** `TR-M395-Ram-XY-Essential-Singularity.tex`
M396	C3	Notable scoped source-specific refinement: an explicit finite charged-sector exclusion bound replaces a qualitative replica/soft-join contradiction, including its phase denominator, rational feasible threshold and O(Delta^1/4+kappa^1/8) fixed-charge consequence. Physical annular profile losses, heterogeneous allowed tilts and actual magnetic error accumulation make the source exponent obligations quantitative. The exact primary variance constant is supporting extraction. Classical Fourier approximation, projection, gauge, flow and coercive-residual identities are credited rather than claimed new. Uniform mixed limits, exact sector isolation, analytic entry and endpoint identification remain upstream; no new global BKT theorem or historical-priority claim. C3 is independent of unchanged L3. **Evidence:** \label{thm:charged}; \label{cor:explicit}; \label{eq:rational-fixture}; \label{eq:average-loss}; \label{thm:upper-budget}; \label{thm:lower-budget}. **Report:** `TR-M396-Ram-XY-Critical-Exponent.tex`
M397	C3	Notable local Gaussian-interface increments: two-sided quadratic HS complement certificate, necessary source first-order term with signed residual and explicit obstruction, and an area-weighted continuum penalty comparison/joint hard-band limit. Classical tools credited. Does not validate BKT/roughening/RG/field limits or recycle M394; fixed source unit-cell law is distinct. **Evidence:** \label{thm:det}; \label{thm:charged}; \label{thm:cellcompare}; \label{thm:joint}; \label{sec:example}; \label{sec:limits}. **Report:** `TR-M397-Ram-BKT-Universality.tex`
M398	C3	C3 is earned by a conditional source-specific Mellin module, not a new SK theorem. Exact cancellation in the squared logarithmic coefficients yields4^h energy growth; together with a polynomial-tail level law it proves a finite-variance unbiased moment specification with8^h replication, including the otherwise essential squared-mean term. The companion positive-kernel normalization, power-bias and explicit size-tail bounds provide a usable truncation/error certificate and expose its severe cost. A class-level geometric replication threshold is proved. Classical gamma integrals, total variation, Cauchy bounds and single-term unbiased estimation are credited. There is no finite expected-work, new limiting-law or efficient-sampler claim. **Evidence:** \label{sec:model}; \label{sec:architecture}; \label{sec:experiment}; \label{eq:continuation}; \label{lem:telescoping}; \label{lem:quadrature}; \label{thm:positive}; \label{prop:cutoff}; \label{cor:cost}; \label{ex:cutoff}; \label{thm:energy}; \label{thm:unbiased}; \label{prop:threshold}; \label{thm:cf}; \label{cor:normalization}; \label{sec:marginal}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M398-Ram-SK-Limiting-Law.tex`

Report	Score	Demonstrated contribution and remaining boundary
M399	C3	Notable source-specific local increments: optimal fixed-n clock allocation removes the source's log^2 projection loss and changes its named off-diagonal log^8 budget tolog^4; sharp one-sided unsmoothing yields all-center/fixed-width anti-concentration and explicit finite-window variance enclosures. Existing augmentation and classical entropy/Stieltjes/log-concavity tools are credited. No global SK exponent, limiting law, prefactor or uniform changing-temperature theorem is newly certified. **Evidence:** thm:projection; eq:pair-saving; thm:unsmoothing; cor:SK-concentration; thm:finite-window; sec:evidence. **Report:** `TR-M399-Ram-SK-Fluctuation-Scale.tex`
M403	C3	Notable local increment: ordered finite-prefix first/second response formulas with explicit all-omitted-scale curvature enclosure, checked against the exact nearest-neighbor normalization; sharp common-tilt modulus and correct singleton-sensitive joint-source error certificate. Classical methods credited; no independent RG/radial/conformal validation or finite-to-infinite shortcut. **Evidence:** \label{thm:slope}; \label{thm:curvature}; \label{thm:certificate}; \label{thm:tilt}; \label{prop:partition}; \label{sec:examples}; \label{sec:assessment}; \label{sec:verification}. **Report:** `TR-M403-Ram-Ising-Conformal-Universality.tex`
M406	C3	C3 is earned by the source-specific relative boundary-dissipation certificate and its actual Schur/law interfaces, not by renaming classical Woodbury algebra. A complete correlated boundary error of relative norm tau<1 gives sharp hyperbolic tau^2/(1-tau), pseudohyperbolic tau/(2-tau), covariance and retained-logdet control without an unweighted resolvent-norm factor. A weighted whole-error certificate, a zero-height branching gate, and a good-event W2 budget connect it to the source's separated mean/fluctuation stability estimates. Exact worked blocks exhibit a genuine gain over the ordinary inverse-norm sufficient condition. The full relative metric, inverse-margin and probabilistic coupling hypotheses remain inputs; no new disorder threshold, law construction, speedup or universality proof is claimed. **Evidence:** \label{sec:model}; \label{sec:map}; \label{sec:recursion}; \label{ex:tail}; \label{lem:compression}; \label{thm:relative}; \label{eq:bilinear}; \label{eq:covariance-order}; \label{eq:logdet}; \label{ex:sharp}; \label{sec:block}; \label{eq:boundary-metric}; \label{lem:branch}; \label{eq:disk-error}; \label{ex:gain}; \label{prop:rows}; \label{ex:rows}; \label{eq:law-budget}; \label{prop:two-mode}; \label{eq:two-budget}; \label{sec:private}; \label{sec:evidence}. **Report:** `TR-M406-Ram-Anderson-Regular-Graphs.tex`
M411	C3	C3 notable local contribution: proves a spin-contrast, first-moment approximation uniform over all finite depths and masses, a finite-leaf-alphabet infimum bound, and an assembled omitted-degree/physical/normalized-moment evaluation certificate. Classical coupling, centering, Poisson and Taylor tools are credited; no historical novelty or independent variational equality, optimizer or finite-branching algorithm is claimed. **Evidence:** thm:messages; cor:alphabet; prop:stoploss; thm:cap; thm:assembled; thm:moment; prop:accumulate; sec:example. **Report:** `TR-M411-Ram-Diluted-Spin-Glasses.tex`
M412	C3	Zero-hierarchy-coefficient-safe amplitude moment enclosures, additive finite-recursion errors, and a finite-trial lower/supporting-field upper bracket for the full variational functional, plus explicit finite-third-moment disorder replacement. Classical tools credited; source thermodynamic identification, hard constraints, optimized numerical certificates and speedup remain unestablished. **Evidence:** \label{thm:amplitude}; \label{prop:chain}; \label{thm:bracket}; \label{thm:replacement}; \label{cor:disorder}. **Report:** `TR-M412-Ram-Ising-Perceptron.tex`
M413	C3	The report supplies exact local Ram moment and weak-form identities, boundary-aware residual estimates, and independently checked tighter profile and shooting error bounds. These are notable certificate improvements, but do not narrow the physical exponent interval, complete the original driver, or independently establish global profile selection and microscopic universality. **Evidence:** Exact Ram Moment and Jet Reformulation; Weak Ram Transform and Boundary Realization; Independent audit tightening, with its proof mechanism; Finite inward propagation and endpoint signs; Arithmetic, Replay, and Independently Executed Checks. **Report:** `TR-M413-Ram-Jamming.tex`

Report	Score	Demonstrated contribution and remaining boundary
M414	C3	Notable scoped increment: conditional quadratic-growth extension with explicit dimension obstruction, entropy-aware spectral-block certificate and mass-relative radial-grid enclosure. Classical tools credited; source angular/Gaussian/Haar theorem remains imported, not independently certified. **Evidence:** \label{thm:quadratic}; \label{lem:budget}; \label{prop:dual}; \label{thm:block}; \label{thm:grid}; \label{prop:quadraticexact}; \label{prop:aspect}; \label{sec:scope}; \label{sec:verification}. **Report:** `TR-M414-Ram-Spherical-Perceptron.tex`
M415	C2	Additive positive-Gaussian recursion residual budgets, a hierarchy-independent terminal Gaussian L1 range-factor bound, an exact finite-entropy criterion and endpoint-cap cost, and bounded-Lipschitz trial cutoff/finite-cover consequences are proved. Classical mechanisms and inherited source derivatives credited. No independent global pressure proof, historical novelty, uniqueness, hard-constraint result, or measured speedup claimed; L4 applicability remains separate. **Evidence:** \label{thm:residual}; \label{prop:center}; \label{thm:terminal}; \label{thm:entropy}; \label{prop:endpoint}; \label{thm:cutoff}; \label{prop:interval}; \label{thm:cover}; \label{sec:tests}. **Report:** `TR-M415-Ram-Spherical-Perceptron.tex`
M419	C2	Source-specific boundary-correct Green/Ram mapping and fully proved restricted residual, flux, finite parameter, smooth-transport, regularized Newton and drift-tail certificates. Classical arguments under explicit hypotheses, with no established source gap closure, historical novelty, global SLE improvement or computed full-drift certification. L4 applicability is separate from realized C2 contribution. **Evidence:** \label{thm:rough}; \label{thm:energy}; \label{thm:flux}; \label{thm:parameter}; \label{thm:shape}; \label{thm:newton}; \label{thm:drift}; \label{sec:assessment}. **Report:** `TR-M419-Ram-Massive-SLE.tex`
M423	C3	C3 is earned by the source-specific weakening from full isotropy to finite rotational symmetry in the signed annular module, the endpoint9/4 forcing bootstrap and its three-rate Banach-message interface, with explicit angular and color defect costs. The slow-frequency counterexample and full-space residual budget delimit what a finite certificate can establish. Optimized near/far exponents and finite-integral/all-future-tail enclosures are useful secondary consequences. Classical Fourier and semigroup tools are credited. Whole-chamber regularity, discrete domination, arm and calibration inputs remain inherited; no new Cardy proof, stochastic input or computed amplitude. **Evidence:** \label{sec:model}; \label{sec:source}; \label{sec:chamber}; \label{eq:mode-energy}; \label{thm:filter}; \label{eq:two-filter}; \label{ex:slow}; \label{thm:endpoint}; \label{eq:filter-defect}; \label{prop:forcing}; \label{cor:shape}; \label{thm:three}; \label{eq:defect-cost}; \label{thm:rate}; \label{eq:integral-certificate}; \label{eq:amplitude-interval}; \label{sec:ram}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M423-Ram-Voronoi-Pivotal-Amplitude.tex`
M425	C2	C2: explicit finite-Q endpoint, pointwise residual/quadrature and physical Kohn error certificates are useful, but closely quantify source/classical Neumann and shape-derivative methods. No forced C3, historical novelty, thermodynamic uniformity or nonlinear GFF verification is claimed; L3 unchanged. **Evidence:** thm:boundary; thm:residual; cor:neumann; eq:rational-budget; prop:ratio; prop:kohn; sec:architecture. **Report:** `TR-M425-Ram-Balanced-Six-Vertex.tex`
M426	C3	Notable source-specific local increment: descendant-budget forest weights replace the source depth-loss scale by an exponential-in-M detector guarantee; an interval-preserving spectral residual and conditional iterated-limit near-linear rectangle modulus are proved. Classical algebra is credited; source geometric/color, localization and CLE identification inputs are not independently certified. **Evidence:** \label{thm:forest}; \label{lem:chains}; \label{thm:detector}; \label{thm:tail}; \label{lem:recursion}; \label{cor:modulus}; \label{sec:scope}; \label{sec:verification}. **Report:** `TR-M426-Ram-Half-Plane-Double-Dimers.tex`

Report	Score	Demonstrated contribution and remaining boundary
M427	C3	C3 for the source-specific odd-sector convolution-row certificate, factorized warming cost and actual moving-entrance-density residual budget, combined with conditional even/odd two-rate relaxation. The exact Gibbs example and domain/leakage guards make this a usable quantitative interface rather than a covariance relabeling. Classical semigroup and Gram tools are credited; the source already supplies the essential two-copy tail and strong-semigroup passage. No new probabilistic input, uniform random gap, critical exponent or independent universality proof is claimed. **Evidence:** \label{eq:true-pairing}; \label{eq:V}; \label{thm:odd}; \label{eq:rows}; \label{eq:better-warming}; \label{cor:passage}; \label{thm:residual}; \label{eq:moving}; \label{prop:three}; \label{eq:weighted-leak}; \label{thm:rates}; \label{eq:two-rate}; \label{prop:trace}; \label{sec:example}; \label{eq:two-diff}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M427-Ram-Critical-SK-Quench.tex`
M428	C3	The report derives a domain-correct Ram moment realization, dual-energy trace and residual certificates, an explicit conditional angular tail bound, and a resolvent-only heat-trace approximation with quantified error. These are notable analytic and certification contributions, but the microscopic correctors, effective coefficient, changing-space limit and disorder universality remain imported obligations, with no demonstrated computational speedup. **Evidence:** Exact Ram Moment Systems for the Target; Local Proof: The Infinite Source Lives in Dual Energy; Local Proof: Compression and Error Certificates; A quantitative curve certificate from repeated resolvents. **Report:** `TR-M428-Ram-SK.tex`
M429	C2	C2: source-specific quadratic leading entropy budget, explicit binary/affine gap certificates and coherent likelihood residual accounting are useful but modest quantitative refinements using classical tools. No new full-phase mechanism, historical novelty or independently verified global SK theorem is claimed; L3 unchanged. **Evidence:** lem:envelope; prop:entropy; thm:gap; eq:fixture; lem:likelihood; prop:physical; sec:architecture; sec:obstructions. **Report:** `TR-M429-Ram-High-Temperature-SK-Gap.tex`
M430	C2	Metric-correct finite moment formulation and conditional complementary-space, heat, spin-edge and block-error certificates. Classical mechanisms credited. No evaluated limiting SK rate, improved cutoff regime/window or computational speedup. **Evidence:** \label{thm:tail}; \label{thm:semigroup}; \label{prop:edge}; \label{prop:blocks}; \label{thm:time}. **Report:** `TR-M430-Ram-SK-Cutoff.tex`
M431	C3	Notable conditional increment: retaining both nonlinear cavity factors gives the explicit ratio constant exp(1/2)/E sech G with n^(1/3+epsilon) exceptional-block control. A separate actual-cell TV certificate and bounded-even-tilt corollary preserve the realized-start mechanism. Classical identities credited; critical global static inputs remain inherited. **Evidence:** \label{lem:even}; \label{prop:twofactor}; \label{thm:constant}; \label{thm:cells}; \label{sec:perturb}; \label{cor:tilt}; \label{sec:scope}; \label{sec:verification}. **Report:** `TR-M431-Ram-Critical-SK-Slowing.tex`
M433	C3	C3 for the source-specific exact Gaussian reverse-entropy cost and optimized simultaneous coverage budget, with a law-preserving terminal-field error bound and positive normalized scalar-panel certificate. The expected cost improves from O_beta(n sqrt(delta)) to overlap-dependent O_beta(n delta); the source forward-entropy input then yields n^(1-f/4) instead of n^(1-f/8) at its delta choice. Classical tools are credited. No global mixing exponent, adaptive proposal replacement or independent Parisi/mixing proof is supplied. **Evidence:** \label{thm:IBP}; \label{eq:C}; \label{eq:linear}; \label{lem:reverse}; \label{thm:coverage}; \label{eq:coverage}; \label{eq:optimizer}; \label{cor:exponents}; \label{thm:terminal}; \label{prop:panels}; \label{eq:panel-structure}; \label{eq:gaussian-tail}; \label{eq:moment-panel}; \label{sec:mixing}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M433-Ram-Low-Temperature-SK.tex`
M434	C2	C2: explicit ratio-five capped-profile extraction, sharp moment truncation with optimized warm integral, and uniform defect/physical-clock accounting are useful but limited deterministic refinements of classical methods. No new critical profile, exponent, random uniformity theorem or historical novelty is demonstrated. L3 applicability unchanged. **Evidence:** thm:layers; cor:five; lem:truncate; thm:envelope; thm:mixing; eq:primitive; prop:pencil; sec:architecture; sec:evidence. **Report:** `TR-M434-Ram-Critical-SK-Mixing.tex`

Report	Score	Demonstrated contribution and remaining boundary
M437	C3	The report gives exact moment-to-Bernstein certificate transfer, a sharper attained scalar bound, and conditional extensions to finite-mean offspring, observed random edge strengths, explicit finite-depth and quenched rates. These are notable local and conditional contributions, but the central pairwise certificate traversals remain unreplayed, historical novelty is not established, and neither a stronger unconditional threshold nor a speedup is demonstrated. **Evidence:** Direct Ram correspondence, not a renamed algorithm; Independent scalar certificates and a sharper lower bound; Conditional all-merges extension and explicit rates; Quenched and survival-conditioned consequences. **Report:** `TR-M437-Ram-Potts.tex`
M438	C3	Unconditional symmetric-posterior local estimates improve the quadratic-loss constant7 to5 and remove channel smoothing from the third-moment estimate. Under the explicitly retained preserved-class companion premise MS, merge-by-merge cubic-potential subadditivity removes the uniform offspring bound and proves the nonlinear L3-capacity criterion for all locally finite deterministic trees with known heterogeneous strengths 0<lambda_e<=ell<1, with explicit capacity/advantage bounds and sharper terminal-noise loss. Classical capacity recursions, inherited source results and exact Ram coordinate transfer are credited, not counted as novelty. No independent full companion certificate replay, unconditional global Potts theorem, historical priority, finite-to-infinite certification by samples, or measured speedup is claimed. C3 contribution is separate from L4 applicability. Essential F1 correction explicitly fixes earlier-dead-leaf labels at zero in the finite residual proposition; the unchanged proof is valid under that boundary condition. One exact regression added, not credited as a new contribution; score unchanged. **Evidence:** \label{thm:local}; \label{lem:corrected}; \label{thm:extension}; \label{cor:quantitative}; \label{eq:improvedloss}; \label{sec:tests}. **Report:** `TR-M438-Ram-Potts-Tree-Capacity.tex`
M439	C3	The report proves a stratum-aware fixed-denominator completion with a Blackwell sandwich, exact improvement ledger, finite-alphabet extension and a sharp spread budget, supported by exact finite witnesses and internal targeted review. Full threshold certificates, global recursion improvement, historical novelty and measured runtime gains remain unestablished, so the demonstrated result is a notable local contribution rather than a completed global advance. **Evidence:** thm:completion; eq:saving; prop:q; thm:spreadcost; cor:roundbudget; prop:poisson27; sec:verification. **Report:** `TR-M439-Ram-Three-State.tex`
M442	C2	Proved countable-path integral residual and inexact-Picard bounds, finite-prior-field shift and decoder sensitivity, conditional influence-tail truncation/convergence, a product-continuity counterexample, and finite readout TV/sublinear-slope guarantees. Classical mechanisms and inherited factorial inverse are credited. No new global FUSF theorem, unconditional spatial rate, finitary upgrade, historical novelty, or measured speedup; L4 applicability remains separate. **Evidence:** \label{thm:residual}; \label{cor:budget}; \label{thm:inexact}; \label{prop:parameter}; \label{prop:counterexample}; \label{thm:tails}; \label{cor:tail-convergence}; \label{thm:readout}; \label{cor:slopes}; \label{sec:tests}. **Report:** `TR-M442-Ram-FUSF-Factor-IID.tex`
M443	C3	Notable local increment: uses the source existing exponential tails to improve its gap/collision/boundary second-moment transfer losses to full failure exponent with logarithmic factors, and proves a conditional boundary/contact/bulk Green-to-Wick finite-law certificate. Classical tools and all global probabilistic premises remain explicit; no credit for restating source concentration. **Evidence:** \label{thm:rare-modulus}; \label{cor:source-errors}; \label{thm:green-defect}; \label{prop:wick-error}; \label{thm:law}; \label{sec:comparison}; \label{sec:verification}. **Report:** `TR-M443-Ram-Lipschitz-Height-GFF.tex`
M444	C3	Notable source-specific local increment: exact restricted-field boundary-displacement energy including annular and zero-mode costs, observable Gaussian likelihood/TV, certified positive series and sharp distinct mark/buffer limits. Conditional normalization uncertainty retains the genuinely unknown angular tail. Classical tools are credited; no independent global GFF or historical-priority claim and no repeated credit for M443. **Evidence:** \label{thm:energy}; \label{thm:entropy}; \label{prop:tail}; \label{thm:motion}; \label{cor:product-tail}; \label{prop:normalization}; \label{sec:scope}; \label{sec:verification}. **Report:** `TR-M444-Ram-Two-Arc-Lipschitz-GFF.tex`

Report	Score	Demonstrated contribution and remaining boundary
M445	C3	C3 for the source-specific two-endpoint Green estimate and Frostman boundary energy improvement, from the optimized displayed source epsilon^1/4 logarithmic bound to epsilon^1/2 logarithmic bound at exponent1/2, together with a conforming energy/probabilistic covariance budget and conditional-law perturbation control. Classical Beurling, hypercircle, energy and tail tools are credited. No stationary stiffness, uniform stochastic rate, stronger overall localization exponent or independent GFF/SLE theorem is supplied. **Evidence:** \label{lem:two-end}; \label{eq:two-stop}; \label{thm:strip}; \label{eq:half}; \label{ex:flat}; \label{eq:flat-exact}; \label{prop:flux}; \label{thm:three-budget}; \label{eq:variance-interval}; \label{ex:triangle}; \label{eq:local-rate}; \label{lem:TV}; \label{cor:records}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M445-Ram-Real-Lipschitz-Surfaces.tex`
M446	C2	C2: explicit slit-ellipse translation exponents, reflected Cauchy-defect bounds and a finite color/shielding error ledger refine specific source interfaces; exact Gaussian-shift constants and physical norms support the comparison. The methods are classical and upstream uniform constants/rates remain unevaluated, so the demonstrated increment is modest. No global joint-limit or novelty claim; L3 unchanged. **Evidence:** thm:ellipse; cor:moves; prop:cauchy; thm:laplace; prop:colors; thm:screen-budget; cor:flat; prop:shift; lem:weak; sec:architecture; sec:evidence. **Report:** `TR-M446-Ram-Ashkin-Teller-Currents.tex`
M447	C3	Notable local quantitative increment: ordered mean-preserving spectral bins with exact signed remainders and certified quadratic gap-dependent variational enclosure; sharper fixed-cap constrained-block loss and explicit two-spin/supporting-plane error under the actual source cascade law. Classical tools credited; global Haar/cavity pressure identification not independently proved. **Evidence:** \label{thm:bracket}; \label{lem:remainders}; \label{thm:quadratic}; \label{thm:block}; \label{cor:tangent}; \label{sec:comparison}; \label{sec:verification}. **Report:** `TR-M447-Ram-Orthogonally-Invariant-Ising.tex`
M451	C3	C3 for the source-specific exact sign-count A-packet envelope and half-range, asymmetric strict-negative count-tail interval, and depth-independent original-law leaf stability yielding explicit all-count message/exponent rationalization constants. This strengthens finite certificate error control without changing root/site laws or ordered exponents. Classical tools are credited; no numerical threshold, practical runtime improvement or independent variational completeness theorem is supplied. **Evidence:** \label{prop:range}; \label{eq:RA}; \label{thm:tail}; \label{eq:tailinterval}; \label{thm:leaf}; \label{eq:abel}; \label{prop:exponent}; \label{eq:variancebudget}; \label{thm:allcount}; \label{eq:Cmsg}; \label{eq:Cexp}; \label{cor:margin}; \label{ex:root}; \label{sec:logic}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M451-Ram-Random-3SAT-Threshold.tex`
M452	C3	C3 for the proved exact finite-depth physical optimizer, its closed information and truncation formulas, and separate quantitative infinite-row/finite-boundary tail budgets. Classical least squares, Green recurrences, martingale inequalities and causal contraction are credited. These are notable local refinements, not historical novelty claims or an independent proof of the global factor threshold. **Evidence:** lem:gram; thm:optimum; cor:depthtail; eq:weightregret; prop:attenuation; thm:infinite-tail; prop:finite-tail; prop:causal; sec:gates; sec:evidence. **Report:** `TR-M452-Ram-Free-Ising-Factor-IID.tex`
M455	C3	Notable local increment: direct singular-value clipping sharpens the source finite-core log-determinant error from O(L xi^(1/5)) to O(L xi^(1/3)); explicit near-unit integer-grid transfer and conditional count/planar LDP preserve actual source laws. Classical tools credited; global contour, pressure and lattice exponent remain source inputs. **Evidence:** \label{thm:det}; \label{thm:grid}; \label{cor:grid}; \label{thm:ldp}; \label{sec:comparison}; \label{sec:verification}. **Report:** `TR-M455-Ram-Cylinder-Loop-Nesting.tex`

Report	Score	Demonstrated contribution and remaining boundary
M457	C3	C3 for the Jordan-degree polynomial-moment weakening of the source signed trace coefficient lemma, explicit discarded-word remainder, algebraic anchored-defect sufficient regime, and its grouped two-gap and cancellation-safe finite-window interfaces. Classical tools and the source's existing Jordan/cancellation mechanisms are credited. Rates require actual whole-operator/tail certificates; no new polygon exponent, amplitude computation or independent global proof. **Evidence:** \label{thm:moment}; \label{eq:remainder}; \label{eq:epsilon}; \label{ex:sharp}; \label{par:UI}; \label{prop:words}; \label{eq:algebraic-tail}; \label{ex:algebraic}; \label{thm:window}; \label{eq:data-posterior}; \label{eq:Vwindow}; \label{thm:two-gap}; \label{eq:grouped}; \label{sec:Ram}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M457-Ram-Signed-Cylinder-Polygons.tex`
M458	C3	C3 for the proved source-specific conditional narrowed logarithmic bridge window, explicit auxiliary-to-actual-height conversion, and smaller polynomial-error recovery cutoff. Positive Fourier norm and residual comparisons are supporting classical certificates. Imported cylinder, polygon, signed-flux and actual connector geometry remain explicit; no historical novelty or independent global exponent certification is claimed. Applicability remains L3 independently. **Evidence:** thm:moments; prop:pruning; thm:recovery; thm:window; cor:height; prop:paired; prop:residual; sec:parameters; sec:architecture; sec:evidence. **Report:** `TR-M458-Ram-Cylinder-Bridge-Lengths.tex`
M459	C3	Notable source-specific uniform bounded-error denominator refinement, average-node displacement stability of its Jensen certificate, and explicit analytic truncation budgets; no global polygon/one-arc improvement claimed. **Evidence:** thm:bounded; thm:stability; prop:truncation; rem:increment; sec:architecture; sec:limits. **Report:** `TR-M459-Ram-Marked-Polygon-Correlations.tex`
M460	C3	The report establishes an exact effective-Gram variational formulation, dimension-uniform alternating scalar certificates and metric-relative residual controls, and sharpens the conditional two-arm sewing loss with explicit recovery hypotheses. These are notable local verification and representation gains, but the source transfer identity, contour estimates and spatial hypotheses remain unverified and no new critical exponent or historical novelty is claimed. **Evidence:** sec:audit; sec:moments; thm:effective; thm:alternating; prop:constants; thm:certificate; thm:sewing; cor:armgain. **Report:** `TR-M460-Ram-Disk-Transfer.tex`
M461	C3	C3 for source-specific original-pole-law positive boundary error budgets, an explicit rational common kernel envelope, and relative preservation of every fixed integer length moment in A(h+1)log(h+1) corridors at cA>2j/3+1/4, with hypothesis-level sharpness. Classical cofactor/quotient/Jensen tools and source mechanisms are credited. No new critical exponent, computed physical amplitude, effective asymptotic constants or global proof certification. **Evidence:** \label{prop:quotient}; \label{lem:Christoffel}; \label{thm:positive-ratio}; \label{prop:envelope}; \label{eq:winfty-moments}; \label{thm:ledger}; \label{prop:shape}; \label{thm:insertions}; \label{thm:changes}; \label{thm:corridor}; \label{thm:sharp}; \label{sec:complex}; \label{sec:Ram}; \label{sec:evidence}. **Report:** `TR-M461-Ram-Polynomial-Vacuum-Bridges.tex`
M463	C3	A proved slope-controlled hard-edge weight-class extension, exact normalization-aware sensitivities, and source-specific primal-dual/endpoint certificates. No new lattice realization, independently certified global lattice theorem, uniform matrix conditioning or measured computational speedup. **Evidence:** \label{thm:extension}; \label{thm:dual}; \label{prop:tilt}; \label{prop:threshold}; \label{prop:bridge}. **Report:** `TR-M463-Ram-Honeycomb-Strip-Mass.tex`
M464	C3	C3 for explicit all-width translated-minor positivity, source-specific leading-pole residual bounds with proper full-jet brackets, all-width denominator nonvanishing, and quantified endpoint corner/marked-ratio cutoff losses. Classical tools are credited; actual current, uniform ensemble asymptotic, marked-pair geometry and exact-height bridge inputs remain imported. No global proof or historical novelty claim. Applicability remains L3 independently. **Evidence:** thm:margin; thm:residual; prop:jet; cor:box; lem:corner; prop:corner-tail; prop:ratio; sec:ram; sec:architecture; sec:scope. **Report:** `TR-M464-Ram-Uniform-Polygon-Bridge-Moments.tex`

Report	Score	Demonstrated contribution and remaining boundary
M465	C3	Notable source-specific initial-arch budget refinement and explicit finite-height/unequal-cap amplitude enclosures, with a classical full-state overlap certificate for the middle remainder. No global honeycomb exponent or uniformN theorem claimed. **Evidence:** prop:marks; lem:tail; thm:height; prop:unequal; prop:overlap; sec:limits; sec:evidence. **Report:** `TR-M465-Ram-Cap-Selected-Amplitudes.tex`
M473	C2	C2: modest explicit carrier-multiplicity gain for heterogeneous/longer sweep decay, exact worst-edge matching bias budget after a smoothed prefix, and smaller exception-count compression sum. Classical tools and source mechanisms are credited; no new uniform mixing theorem, effective universal P, or complexity improvement. **Evidence:** \label{lem:moment}; \label{thm:heterogeneous}; \label{sec:carriers}; \label{prop:bias}; \label{thm:ledger}; \label{prop:count}; \label{prop:metric}; \label{sec:architecture}; \label{sec:evidence}. **Report:** `TR-M473-Ram-Random-Subspace-Sweep-Tests.tex`
M476	C3	C3 for a representation-free explicit binary-line contraction deficit and a fixed-slit Green transfer with explicit exponent and improved small-radius barrier order, supplemented by a finite reciprocal-dimension remainder. Classical methods are credited; simultaneous conditioned moments, signed densities and source constants remain imported. No global mixing-order improvement, historical novelty or independent Lean certification. Applicability remains L3. **Evidence:** thm:gap; cor:radius; lem:green; thm:transfer; eq:asymptotic; cor:budget; prop:remainder; ex:pole; sec:architecture; sec:ram; sec:scope. **Report:** `TR-M476-Ram-Conditional-Sweep-Analytic-Transfer.tex`
M477	C3	C3 for the complete open robust exponent range p>pc, the sharp single-permutation TV obstruction and parity classification, and the exact source positive column law with explicit nonlinear local certificates. Classical BC, compactness and covariance tools credited. The compact complement gap is not numerically evaluated; source global recursion/mixing and external full proofs remain imported. Applicability remains L3. **Evidence:** thm:robust; prop:obstruction; thm:local; thm:parity; thm:column; eq:bernstein-poly; sec:recursion; sec:ram; sec:evidence. **Report:** `TR-M477-Ram-Four-Row-Permanent.tex`
M510	C2	Proved an explicit signed Morrey-source H^-1 budget, a conditional full-space mixed inverse bound for the actual nonsymmetric linearization, and branch-preserving charged-area support, sensitivity and finite-profile bounds. Correct source speed-gradient product is smooth at zero slope. Classical residual/Cordes/contraction mechanisms are credited, not claimed novel. No geometric candidate, global/neutral Penrose proof, original-data rigidity, historical novelty or measured speedup is certified; L4 applicability is separate. **Evidence:** \label{eq:smooth-source-derivative}; \label{thm:morrey}; \label{cor:frozen}; \label{thm:block}; \label{thm:branch}; \label{prop:grid}; \label{sec:assessment}; \label{sec:validation}. **Report:** `TR-M510-Ram-Charged-Penrose.tex`
M511	C2	Source-specific restricted results: a rank-one nondivergence residual inverse under explicit Dirichlet-domain and smallness hypotheses; quadratic-gradient H1-to-Hminus1 obstruction; complete first/second coupled residual jet specification with zero-slope-safe invariant remainder; a bounded-domain charged flux-conversion contraction/error certificate; explicit Morrey, filling, magnetic-graph and branch-budget statements with proofs. Classical ingredients are credited. Exact symbolic and conditional arithmetic checks are not a certified solve of the assigned geometry. No independent neutral-input, global Penrose, exhaustion, singular-multiplier, rigidity, or historical novelty certification. L4 applicability is separate from C2 contribution. **Evidence:** \label{sec:audit}; \label{sec:corpus}; \label{thm:rankone}; \label{prop:surrogate}; \label{thm:h1}; \label{prop:morrey}; \label{sec:derivatives}; \label{thm:ball}; \label{thm:conversion}; \label{eq:conversionforms}; \label{sec:handoff}; \label{prop:filling}; \label{prop:magnetic}; \label{prop:branch}; \label{sec:tests}; \label{sec:limits}; \label{sec:assessment}. **Report:** `TR-M511-Ram-Higher-Dimensional-Charged-Penrose.tex`

Report	Score	Demonstrated contribution and remaining boundary
M512	C2	Limited source-specific local contribution: complete coupled variation with actual conormal boundary residual; Fredholm bordered inverse; metric/envelope bounds; exact charged-period loss; weighted TT least-norm identity with divergence/end defect. Classical mechanisms credited. No global Penrose or original-data rigidity proof, uniform AdS family theorem, historical novelty, speedup or finite-check guarantee claimed. L4 applicability is distinct. **Evidence:** \label{prop:full}; \label{prop:border}; \label{prop:ram}; \label{prop:area}; \label{prop:envelope}; \label{prop:periods}; \label{prop:tt}; \label{eq:tt-defect}; \label{prop:radii}; \label{sec:pros-cons}; \label{sec:evidence}. **Report:** `TR-M512-Ram-Spacetime-Penrose-Composite.tex`
M513	C2	Proved source-specific smooth trace variations including the off-root normalization term, a unit residual bound for the rescaled height and conditional Z sensitivity, drift-correct weak conversion, a conditional full-space lower-order/Cordes inverse gate, corrected total-angular-flux defects, and physical-branch parameter/error certificates. Classical maximum-principle, Cordes, Green, Lax-Milgram and contraction mechanisms are attributed. No geometric candidate, coupled exterior continuum certification, global Penrose proof, original-data rigidity, historical novelty or measured speedup is demonstrated. L4 applicability is separate. **Evidence:** \label{prop:variation}; \label{prop:weak}; \label{thm:height}; \label{cor:Z}; \label{thm:inverse}; \label{thm:ball}; \label{prop:flux}; \label{prop:mixed}; \label{thm:branch}; \label{prop:box}; \label{sec:future}; \label{sec:assessment}; \label{sec:validation}. **Report:** `TR-M513-Ram-Kerr-Newman-Penrose.tex`
M514	C4	The report proves scalar existence and uniqueness for every finite amplitude on the spherical exterior, an actual nonzero resonant quadrupole logarithm, and weighted residual-to-mass and explicit geometry-to-defect certificates, removing a substantive local smallness restriction rather than merely improving a numerical factor. Arbitrary-amplitude horizon geometry, the corrected-total-angular-momentum formulation, historical novelty, and independent numerical PDE validation are not established, and the global source conclusions retain explicitly audited geometric dependencies. **Evidence:** Scalar Solvability Without Smallness; thm:all-amplitudes; thm:unique; thm:log; A mass-sensitive weighted space; Explicit Barrier and Finite-Parameter Certificate; Verification and Review Record. **Report:** `TR-M514-Ram-Electromagnetic-Tails.tex`
M515	C3	Explicit source-specific saturating Legendre dual and nonlinear Bregman flux certificate, fixed-geometry O(1/D) full-cut approximation by combining source pole barrier and coarea, timelike four-flux bounds and static residual obstruction with stated inverse constant. Complete restricted proofs; independent review corrected one exterior admissibility condition. No independently certified global Penrose or rigidity theorem, numerical geometry solve, speedup, or external novelty claim. **Evidence:** \label{thm:dual}; \label{thm:gap}; \label{thm:rate}; \label{prop:fourflux}; \label{thm:staticcert}. **Report:** `TR-M515-Ram-Anti-de-Sitter-Penrose.tex`
M516	C2	C2: explicit source-specific aligned scalar/current/boundary defect costs, retained height derivative, and conformal all-charge nonlinear tail bounds. These are modest refinements of classical identities and already sophisticated source mechanisms; actual state enclosures and global geometry remain unverified. **Evidence:** \label{thm:defects}; \label{cor:aligned}; \label{prop:mass}; \label{thm:mass-budget}; \label{sec:panels}; \label{prop:annulus}; \label{sec:architecture}; \label{sec:enclosure}; \label{sec:evidence}. **Report:** `TR-M516-Ram-Maximal-Hyperbolic-Penrose.tex`
M517	C3	The report proves an exact inexact-potential TT residual identity, explicit nonlinear branch and mass-error bounds, and a differentiated dipole obstruction certificate that make a notable source-specific local verification improvement. It does not independently verify the complete quartic geometry or nonlinear Penrose theorem, compute a certified numerical seed or quartic constant, or establish historical novelty. **Evidence:** Exact inexact-potential TT identity; Explicit residual-to-branch and mass bounds; Residual form of the dipole obstruction; Quartic symmetry and the near-kernel boundary; thm:residual-square; thm:branch; eq:deficit-error; prop:elimination; thm:dipole-stability. **Report:** `TR-M517-Ram-Local-Penrose.tex`

Report	Score	Demonstrated contribution and remaining boundary
M518	C3	Notable local removal of the source's enclosing-area loss via a proper contracting end flow, and exact source-weighted convex penalty/ADM residual budgets. Classical tools credited; all global PDE and geometric obligations retained. **Evidence:** lem:contract; thm:lossless; cor:source; prop:cubic; prop:bracket; cor:massbudget; prop:residual; sec:scope; sec:verification. **Report:** `TR-M518-Ram-Spacetime-Penrose-Enclosing-Area.tex`
M519	C3	C3: source-specific doubling of the displayed fixed-time mesh area-loss exponent from alpha/(2n) to alpha/n, with a one-sided density proof and nonuniform budget, plus a sharp coupled mass-window/residual certificate. Classical whole-exterior primal-dual and harmonic-tail tools support implementation boundaries but are not claimed novel. Global weak-flow construction, detachment, positive mass and long-time geometry remain imported. **Evidence:** \label{lem:swept}; \label{thm:mesh}; \label{thm:window}; \label{cor:window-data}; \label{thm:capacity}; \label{prop:tail}; \label{cor:combined}; \label{sec:no-rate}; \label{sec:map}; \label{sec:evidence}. **Report:** `TR-M519-Ram-Conformal-Flow-Minimizing-Frontiers.tex`
M520	C3	Notable source-specific fixed-gauge refinement: sharp physical horizon Robin residual and mode-complement certificates, independent weighted mass control, and exact tensor/escaping-tail counterexamples prevent invalid promotions. Classical analytic ingredients are credited; global source rigidity and quantitative near-equality geometry remain unproved. **Evidence:** prop:scalar; ex:dipole; lem:coercivity; thm:weak; thm:modes; cor:tail; thm:mass; prop:escape; prop:perturb; sec:limits; sec:verification. **Report:** `TR-M520-Ram-Spacetime-Penrose-Rigidity.tex`
M521	C2	Complete restricted interface/exponential identity, surface Lp flux-potential budget with critical obstruction, exact scalar-elimination differential, and conditional nonmaximal polynomial tail. Classical degree, regularity, and square completion credited as inherited; no global Penrose extension, novelty priority, continuum certificate from finite tests, or speedup claimed. L4 relevance is distinct. **Evidence:** \label{prop:schur}; \label{prop:interface}; \label{prop:Lpflux}; \label{prop:critical}; \label{prop:polytail}; \label{sec:outlook}; \label{sec:assessment}. **Report:** `TR-M521-Ram-Boundary-Graph-Deformations.tex`
M523	C3	The report proves source-specific physical residual, residual-to-Ward-form, fixed-reference and exact-range certificates, with explicit weighted reset radii and a conditional bounded-density readout extension that constitute a notable local verification contribution. These results do not certify adaptive all-scale probability bounds or the headline spectral theorem, and no historical novelty, source-scale efficiency or numerical speedup is established. **Evidence:** Physical Resolvent Residual Certificates; Residual-to-Form and Committed-State Tests; thm:form; thm:commit; A Certified Full-Range Clearing Test; thm:range; Explicit Weighted Reset and Form Bounds; thm:reset; prop:resetform; Sharp Classical Cayley Readout; Conditional Infinite-Volume Extension. **Report:** `TR-M523-Ram-Anderson-AC.tex`
M524	C3	Proved source-specific passive closure and residual certificates, an absorbed-response error bound, and a centered conditional-law decrement with a finite error ledger notably strengthen the pinned finite-port comparisons. The hierarchy mismatch-versus-centered-error estimate, uniform fractional-moment bootstrap and headline localization theorem remain unverified, and no historical novelty or computational speedup is claimed. **Evidence:** thm:quadratic; thm:feedback; thm:feedback-residual; thm:residual; thm:centered; cor:centered-ledger; The Conditional Infinite-Volume Bridge; Realized contribution assessment. **Report:** `TR-M524-Ram-Anderson-Localization.tex`
M525	C3	C3 for quantitative global physical-potential Lp reconstruction from the source Young/coercivity budget, all-set truncated-defect channel drift with rank/gap costs, and ordered Cayley residual stability. These are notable source-specific local/conditional improvements using classical tools; no independent global classification, global near-extremizer theorem, historical priority or measured speedup is claimed. Applicability remains L3. **Evidence:** thm:reconstruction; thm:drift; cor:energy; thm:cayley; cor:riccati-input; prop:gram; sec:examples; sec:source-map; sec:ram; sec:limits. **Report:** `TR-M525-Ram-Matrix-Lieb-Thirring-Equality.tex`

Report	Score	Demonstrated contribution and remaining boundary
M526	C3	The report proves a positive-part extension of the pinned integrated trace comparison, characterizes equality by commutation, and derives explicit commutator and finite-verification error bounds. These are notable local refinements but do not independently prove the full Lieb-Thirring theorem, establish global spatial stability, realize an indefinite integrated matrix, or demonstrate historical novelty or measured speedups. **Evidence:** A positive-part trace extension; thm:positive-part; Equality and quantitative noncommutativity; thm:equality; thm:comm-gap; Continuum and numerical certificates; thm:cutoffs; prop:deficit-error; Verification, comparisons, and limits; Contribution assessment. **Report:** `TR-M526-Ram-Matrix-Lieb-Thirring.tex`
M527	C3	C3 for the sharp spectral-width quadratic common-cutoff Gram bound and exact first-list trace/moment transfer, with angular mass coupling and full-form residual extension. These meaningfully refine the source's existing generic finite exhaustion estimate. Classical tools are credited; no new global constant, independent connecting-path proof, unseen-spectrum certification, historical novelty or measured speedup. Applicability remains L3. **Evidence:** lem:ims; thm:quadratic; ex:sharp; cor:exhaustion; thm:trace; cor:moments; prop:residual; ex:profile; ex:wells; sec:source-map; sec:ram; sec:evidence. **Report:** `TR-M527-Ram-Lieb-Thirring-Constants.tex`
M528	C3	C3: source-specific improvement of the local screening budget from the displayed0.02 gain to1/11 or1/5 under stricter offset ceilings and admissible new cuts; direct unsmoothed-field transfer retains the two actual laws and removes the inverse-master-width loss, giving conditional1/77 and1/35 rates. Whole-space TF and smearing certificates support those interfaces. Classical tools and source duality are not counted as historical novelty; global ionization/radius inputs remain imported. **Evidence:** \label{thm:rates}; \label{prop:optimal}; \label{prop:pointwise}; \label{lem:moment}; \label{thm:transfer}; \label{thm:dual}; \label{sec:fixture}; \label{prop:smear}; \label{sec:global}; \label{sec:evidence}. **Report:** `TR-M528-Ram-Coulomb-Excess-Charge.tex`
M529	C2	Source-specific annular Green/lift equivalence, monotone full-residual and exact torsion error bounds, sharp threshold-aware first variation, nonradial charge/tail and coefficient transfer, sharp restricted barrier amplitude, finite-window Mellin boundary identities and finite-price sector enclosure. Complete restricted proofs and bounded executable checks. Classical mechanisms acknowledged. No independently verified quantum-to-profile limit, global ionization improvement, certified numerical q, Lean kernel run, novelty claim, or measured speedup. L4 applicability remains separate. **Evidence:** \label{thm:annular}; \label{thm:energy}; \label{thm:torsion}; \label{thm:variation}; \label{prop:mean}; \label{prop:flux}; \label{prop:barrier}; \label{prop:mellin}; \label{prop:noline}; \label{thm:prices}; \label{sec:assessment}. **Report:** `TR-M529-Ram-Coulomb-Ionization-Energies.tex`
M530	C3	Source-specific conditional analytic synthesis: explicit boundary-correct Mellin/angular Green reconstruction, nonlinear annular residual and contraction bounds, whole-cylinder residual rigidity, and uniform conditional-density/exceptional-event/far-tail/packet error budget yielding physical radius intervals. Independent audit clarified trace, core and composition hypotheses. No independently certified quantum limit, finite-charge rate, actual quantum computation, speedup or historical novelty claim. **Evidence:** \label{thm:contraction}; \label{thm:corridor}; \label{thm:tail}; \label{thm:radius}; \label{sec:composite}; \label{thm:rigidity}. **Report:** `TR-M530-Ram-Outer-Electron-Radii.tex`
M533	C3	Notable conditional source-specific local refinement: continuum center concentration sharpens differentiated beam and packet exponent budgets, including a sharp fold model; exact positive triangular and separate phase/Hamiltonian certificates make the finite-window physical assumptions explicit. Classical tools credited; actual Kerr concentration and all global evolution/inextendibility inputs remain unproved. **Evidence:** lem:gaussian; thm:beam; sec:jets; prop:fold; ex:fold; thm:budget; prop:phase; thm:triangular; prop:root; sec:limits; sec:verification. **Report:** `TR-M533-Ram-Near-Kerr-Packets.tex`

Report	Score	Demonstrated contribution and remaining boundary
M534	C3	C3 for loss-reduced physical positive-constraint localization: unit-mass filter removes L1 normalization cost, logarithmic absolute-value modulus avoids the first square-root loss, with cb channel/time-leakage and linear summed-error transfer. Classical tools and source general-alpha summation are credited; no new area law, whole-proof certification, historical novelty or measured efficiency. Applicability remains L3. **Evidence:** prop:filter; cor:positive; thm:modulus; cor:local; prop:time; thm:ground; cor:radii; ex:schur; ex:root; ex:bell; ex:volume; sec:source-map; sec:ram; sec:evidence. **Report:** `TR-M534-Ram-Area-Law-Global-Gap.tex`
M535	C2	Complete restricted exact encoded-range/leakage and finite-model proofs, sharper whole-group nuclear-tail and coefficient budgets, conditional nonuniform Gaussian slot allocation, sharp column normalization, and physical residual accounting. Classical ingredients are credited and no priority claim made. Full M535 read is not global proof certification; companion collar/area-law chain and uniform geometry remain unverified global dependencies. L4 applicability is distinct from realized C2; no algorithm, measured speedup, positivity restoration or finite-test guarantee of the global theorem. **Evidence:** \label{prop:frame}; \label{prop:model}; \label{lem:tail}; \label{thm:truncate}; \label{thm:slots}; \label{prop:column}; \label{prop:residual}; \label{sec:limits}; \label{sec:assessment}. **Report:** `TR-M535-Ram-PEPS-Ground-States.tex`
M536	C3	Proved shared-displacement support domination and configuration extensions, source-specific phase remainder and cover refinements, and an exact Fourier-panel certificate establish notable local contributions with full proofs and bounded checks. The missing production verification assets prevent independent authentication of the source exclusion, and no global exclusion gain, historical novelty, or measured speedup is claimed. **Evidence:** thm:gram; thm:laplacian; prop:six-row; thm:joint; thm:bernstein; prop:arc; eq:normal-op-exact; prop:closed-distance; prop:rect-counter; sec:assessment. **Report:** `TR-M536-Ram-MUB-Six.tex`
M537	C3	The proved contrast-Gram constraints, sharper bounds on named moments, and bounded-elimination and residual certificates constitute a notable source-specific local refinement. No supplied function is removed by the order-two rule and the full pair and mixed witness systems and residual identity remain unreplayed, so neither headline theorem nor a computational speedup is independently established. **Evidence:** Order-Two Compression and Contrast Positivity; Sharper Bounds on Named Original Moments; Lossless Elimination and Rational Dual Witnesses; Residual-Aware Physical Certificates; Cubic Defect Identity; Verification Methods and Evidentiary Boundary. **Report:** `TR-M537-Ram-Hadamard-Fourier.tex`
M543	C3	Exact retained-seed restarts and the sharp purity inverse strengthen conditional periodic-gap certificates, while physical-Gram rounding and weighted-moment proofs tighten and clarify finite certification. Complete thermal generation and large trial contractions remain unreplayed, and neither historical novelty nor independent verification of the source theorem is claimed. **Evidence:** Sharp inversion of purity and a conditional gap refinement; Seven exactly certified restarts; Physical-Gram rounding and trace-error certificates; Positive weighted-moment certificates. **Report:** `TR-M543-Ram-Periodic-Haldane.tex`
M544	C3	Proves a sharper seed-preserving conditional boundary bootstrap at every required odd length and certifies physical-metric thermal schedules plus a strictly improved normalized cubic filter with executed finite checks. Periodic spectral and long-chain variational premises remain unverified independently here, and neither a global theorem proof, historical novelty nor a measured runtime speedup is claimed. **Evidence:** Physical-Metric Horner Certificates; thm:residual; thm:schedule; eq:reviewfloor; Projected Polynomial Moment Filters; prop:leading; Sharper Conditional Boundary Bootstrap; thm:bootstrap; cor:newgap; Executed Verification and Honest Scope; Contribution Assessment. **Report:** `TR-M544-Ram-Boundary-Haldane.tex`
M545	C3	Proved source-specific pair-energy finite-flux transfer budget on redundant wedge ranges, extended slow-charge exponent to fixed p0<1/3, optimized annular rate to 1/10 at p0=1/4,p=3/2, and quantified coercivity/relative-bound/spectral-transport closure. Exact replay of all 23 appendix four-body blocks and 13 three-body margins. Analytic locality, pinning, loss and imported global gap inputs remain inherited; no effective threshold, independent kernel build, unconditional new many-body theorem or historical novelty claimed. L4 applicability is separate. **Evidence:** \label{thm:transfer}; \label{thm:exponents}; \label{cor:rate}; \label{prop:coercivity}; \label{thm:relative}; \label{thm:stability}. **Report:** `TR-M545-Ram-Spherical-Laughlin-Stability.tex`

Report	Score	Demonstrated contribution and remaining boundary
M546	C3	Exact rational quotient certificates, explicit pair-energy error and support-aware allowance budgets, and a proved spherical particle-addition projector yield notable source-specific local gains with bounded exact checks. Finite-flux thresholds and the full headline gap remain unverified, while the modest allowance improvement and classical coordinate machinery do not establish historical novelty or measured speedups. **Evidence:** thm:quotient; thm:budget; prop:allowance; prop:support; eq:sharpeffective; thm:addition; sec:checks; sec:independent; sec:assessment. **Report:** `TR-M546-Ram-Laughlin-Gap.tex`
M547	C2	C2: explicit source-normalized odd-parity residual/tail constants and conditional whole-block recovery rates sharpen the source cutoff interfaces; matrix-tail and physical-Gram budgets quantify its eventual finite-list projection argument. These are modest applications of credited classical cutoff/layer-cake/Gram methods, not a new BFSS threshold-state proof. Actual tail envelopes, global profiles, index and massive gap remain inputs. **Evidence:** \label{lem:primitive}; \label{thm:odd}; \label{cor:mixed}; \label{prop:recovery}; \label{cor:rates}; \label{prop:matrix-tail}; \label{prop:gram}; \label{sec:examples}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M547-Ram-BFSS-Threshold-State.tex`
M548	C3	Notable source-specific analytic refinement: exact pair budgeting and sharp physical color averaging increase the confinement coefficient, with explicit fermionic constants, a weaker sufficient rotation threshold, quantitative positive gap and a symmetry-preserving perturbation region. Classical tools credited; source sector/reduction inputs remain named and no finite Ritz or fixture promotion to a global BFSS theorem is made. **Evidence:** lem:pair; lem:orbit; thm:confinement; prop:constants; cor:threshold; prop:gap; prop:split; thm:perturb; ex:coupling; sec:comparison. **Report:** `TR-M548-Ram-BFSS-Positive-Sectors.tex`
M550	C3	C3 for the combined explicit signed all-order ideal heat/Bose remainder, anisotropic/fugacity uniformity and certified residual-tail improvement, plus exact directional capacity-error transfer on the source test space. Classical coefficients, energy duality and source identities are credited; no global interacting theorem, third physical coefficient, historical novelty or measured speedup is claimed. L3 remains applicability. **Evidence:** thm:heat; thm:bose; cor:acceleration; thm:capacity; cor:averaged; prop:dual; prop:physical; ex:anisotropy; ex:capacity; sec:normalization; sec:source-map; sec:ram; sec:evidence. **Report:** `TR-M550-Ram-Bloch-Lattice-Correction.tex`
M553	C3	The report proves a source-specific minor-residual entanglement witness with finite error budgets, justifies a positive quotient trace metric through projective realness, and sharpens the conditional secret-bit gap with a measurable retained-mass and imbalance-sensitive bound. No numerical full-spark margin is certified and the finite computations and targeted reviews do not independently validate the complete all-copy and all-protocol headline theorem or establish historical novelty. **Evidence:** thm:real; cor:trace-metric; eq:lambda; thm:minor; thm:frame; cor:m553-residual; prop:budget; prop:ram; thm:retained; cor:sharp-gap. **Report:** `TR-M553-Ram-Entanglement-Key.tex`
M555	C2	C2: exact positive-root weighted coefficient and grid-error budgets, the shifted interval, smaller-root logarithmic degree overhead, and finite Newton support interface refine this source family. These use credited classical approximation and coordinate tools; no new power threshold, independent parity lower-bound proof, historical novelty or measured computational speedup is established. **Evidence:** \label{thm:exact}; \label{cor:degree}; \label{thm:root-allocation}; \label{prop:residual}; \label{eq:ram-metric}; \label{thm:interface}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M555-Ram-QAC0-Parity.tex`
M557	C3	The report proves a physical two-particle Coulomb compression bound of order m^(7/2)/D in place of the source's m^5/D bound and a positive angular quadrature with cubic rather than quartic spatial node count, together with explicit complement certificates. These are notable local refinements using stated source analytic hypotheses, not independent verification of the global hardness theorem, historical novelty, or a measured runtime improvement. **Evidence:** thm:pair; thm:compression; thm:quadrature; thm:block; prop:matrix-NO; cor:simulation. **Report:** `TR-M557-Ram-Coulomb-Binary.tex`

Report	Score	Demonstrated contribution and remaining boundary
M558	C3	Optimized Coulomb dual certificates with explicit site-count-independent separation and neutral counterterm robustness give notable local gains, including an exact radial hopping formula and quadratic oriented-exchange sensitivity. Actual source-orbital interval data, a complete nuclear implementation and independent verification of the global continuum hardness theorem remain outside the demonstrated contribution. **Evidence:** Whole-Plane Resolvent and Physical Normalization; An exact radial hopping formula; Coulomb Dual Certificates and Uniform Separation; Neutral Occupancy and Counterterm Robustness; Oriented Denominators and Exchange Error; Verification, Limits, and Research Directions; prop:bessel; thm:dual; thm:packing; thm:counterterm; thm:exchange. **Report:** `TR-M558-Ram-Coulomb-Unit.tex`
M559	C3	Notable source-specific increment: exact retained thermal concavity slack with a sharp uniform coefficient, conditional near-optimal ensemble and identity-partner linear-entropy bounds, and positive-polynomial scalar value/optimizer certificates with rigorous tails. Classical tools credited; zero-block entropy and global coding inputs remain explicit, not supplied by finite PSD checks. **Evidence:** thm:thermal; cor:branch; thm:ensemble; cor:product; prop:curvature; thm:series; prop:bernstein; ex:certificate; sec:comparison. **Report:** `TR-M559-Ram-Amplitude-Damping.tex`
M561	C3	Conditional enlargement of the excluded Kohn-Sham multiplication-potential class beyond global L3/2+Linfinity, with strict sparse-spike example; exact physical angle residual, weighted IMS and finite/complement gap, Pauli/density defect certificates. Source Coulomb channel construction, nonnumerical charge and global proof remain inherited, not independently certified. No novelty or runtime claim. **Evidence:** \label{prop:angle-defect}; \label{prop:ims}; \label{thm:schur}; \label{prop:pauli-defect}; \label{thm:extension}; \label{prop:spikes}. **Report:** `TR-M561-Ram-Kohn-Sham-Representation.tex`
M564	C3	C3 for a demonstrated source-specific square-root to linear helper-probability readout under actual excluded classical first-chaos structure, with exact leakage and sharpness controls. Earlier coloring losses and global synthesis/seed/thermodynamic proofs remain imported; classical chaos, Gram and telescoping tools are not claimed new. L3 stays applicability. **Evidence:** thm:helper; lem:conditional; lem:row; cor:synthesis; prop:sharp; prop:leak; prop:telescoping; cor:twoinputs; prop:gram; sec:budget; sec:architecture; sec:imports; sec:ram; sec:evidence. **Report:** `TR-M564-Ram-QAOA-SK.tex`
M565	C3	The report proves local jump-sensitive curvature and weighted-mass identities with quantitative residual and gap tests, supported by exact coercivity certificates and a sharp one-half value-stability bound. The inherited full-support theorem still depends on the source analytic inputs, and neither global historical novelty nor numerical PDE performance is established. **Evidence:** Exact improved integrated coercivity; Curvature as a measure and quantitative mass; Error-stable integration and finite moment certificates; Sharp value stability; Exact optimum in a restricted coefficient cone. **Report:** `TR-M565-Ram-SK-Full-Support.tex`
M567	C3	Source-relative proved finite-period Poisson covariance, square-root sparse encoder repetition schedule, logarithmic field cutoff and retained full SK recurrence yield explicit improved slot/qubit/query/gate upper bounds. Exact feedback residual/Gram identities preserve original physical operators. Classical mechanisms credited; contracting-block, coherent table existence and global synthesis construction remain inherited. No historical novelty, optimality, measured speedup or efficient oracle-table construction claim. ApplicabilityL4 remains distinct. **Evidence:** \label{prop:residual}; \label{thm:tail}; \label{thm:encoder}; \label{prop:cutoff}; \label{lem:SK}; \label{thm:ledger}; \label{sec:future}. **Report:** `TR-M567-Ram-Boolean-Oracle-Synthesis.tex`
M570	C3	Proved leakage-sensitive compression and quadratic Schur gap, full-margin equivariant homotopy, quantitative injection, and finite-free-factor scalarization for every actual unnormalized trace $0<a<1$ with exact row and inverse-cost bounds. Global group construction and regular-norm realization remain inherited; no priority claim. **Evidence:** \label{thm:block}; \label{thm:gap}; \label{thm:injection}; \label{prop:scalar}; \label{thm:balanced}; \label{prop:homotopy}. **Report:** `TR-M570-Ram-Kadison-Kaplansky.tex`

Report	Score	Demonstrated contribution and remaining boundary
M577	C2	C2: explicit branch-gap and canonical-corner budgets, the unitarity-capped Fourier constant40768/3, and matrix-level discarded-tail bounds refine this source construction. The methods are credited classical tools; no new topological principle, optimal near-inclusion distance or independent global counterexample theorem is established. **Evidence:** \label{prop:branch}; \label{lem:intertwiner}; \label{thm:corner}; \label{prop:fourier}; \label{prop:cb}; \label{eq:reordering-tail}; \label{prop:implementation}; \label{sec:examples}; \label{sec:evidence}. **Report:** `TR-M577-Ram-Near-Inclusions.tex`
M580	C3	The report proves a finite nontracial recovery certificate with leakage and kernel-error budgets, a polynomial-size exact phase family, and explicit bandwidth and graph-norm control. These are meaningful local extensions using classical ingredients, not a new sharp cutoff constant, an effective ergodic rate, a measured speedup, or independent verification of global bicentralizer triviality. **Evidence:** Finite-phase recovery with an explicit error budget; Polynomial-size exact phase families; Explicit filters, bandwidth, and finite-time control; Why scalar centralizer cannot simply be dropped; Verification and contribution boundary. **Report:** `TR-M580-Ram-Modular-Recovery.tex`
M587	C3	Exact cancellation of the nested exceptional drift yields a finite normalizer on quantified cylinders and a sharp Hp-to-L2 coordinate-truncation power, with a matching rare-cylinder obstruction and finite-table commutator certificate. These are notable local source-specific results; weighted classification and global hyperinvariant/transitive-commutant claims remain imported. Classical tools and source prior bounds are credited. **Evidence:** lem:cancel; thm:normalizer; thm:sharp; lem:escape; thm:tables; sec:noncompact; sec:comparison. **Report:** `TR-M587-Ram-Backward-Intertwiners.tex`
M594	C3	Exact multiplicity-aware rooted certificates, signed gap-free insertion-cost margins, and half-radius deterministic descent provide notable source-specific local verification and quantitative gains. These classical constructions do not independently establish the arbitrary-MASA theorem, certify global measurable coloring, improve the final paving exponent, or demonstrate a runtime speedup. **Evidence:** thm:cyclic; thm:zero; cor:nobuffer; thm:sensitivity; cor:selection; lem:halfradius; thm:descent; thm:bernstein. **Report:** `TR-M594-Ram-MASA-Paving.tex`
M595	C3	Exact color-sign cycle and forest formulas, sharp half-radius mixed-polynomial locality, explicit finite-ball cost certificates, and a proved buffer-32 assignment step provide notable source-specific local and conditional quantitative gains. The improved global constant inherits the source's analytic and structural inputs and does not independently establish arbitrary-MASA paving, improve the quadratic exponent, prove historical novelty, or demonstrate runtime acceleration. **Evidence:** thm:cycle; thm:forest; thm:half; lem:bernstein; thm:cost; cor:selection; thm:buffer; sec:carry; thm:global-conditional; prop:pencil. **Report:** `TR-M595-Ram-Quadratic-Paving.tex`
M610	C3	C3 for a sharp, source-specific coefficient-survival and boundary-defect interface through maximal-index restriction, with supported exact repair and ambient-filling controls; also a reserved-relation-plane kernel bonus. Classical building/PBW/chain tools are credited. No global Quillen certification or new uniform source constant is claimed. Applicability remains L3. **Evidence:** thm:opposite; prop:sharp; thm:restriction; prop:repair; sec:examples; prop:kernel; eq:kernel-bonus; sec:architecture; sec:ram; sec:literature; sec:evidence. **Report:** `TR-M610-Ram-Quillen-Homology.tex`
M630	C3	Source-specific exact Faber-metric defect identity, quantitative necessary near-extremality estimates, and two-tail finite-section certificate with exact complementary Schur contribution. The universal complete inequality and main amplification mechanism remain inherited; no new universal constant, approximate singular-pair implementation, global certification, novelty, or speedup claim. **Evidence:** \label{thm:defect}; \label{cor:near}; \label{prop:tails}; \label{thm:cutoff}; \label{prop:schur}. **Report:** `TR-M630-Ram-Complete-Crouzeix.tex`

Report	Score	Demonstrated contribution and remaining boundary
M631	C3	Proved conditional synthesis specific to the source's metric SDP and common positive boundary density: model-error-aware Stein repair, self-normalized dual residual lower bound, endpoint leakage, physical and certified-face distinctions, and positive Cesaro/nodal approximations with full bias and aliasing tails. Exact optimal nilpotent fixtures include singular Stein slack. Universal complete constant two, optimal similarity existence and common exact density remain inherited. No global certification, universal improvement, novelty, runtime or all-faces rationality claim. **Evidence:** \label{thm:repair}; \label{thm:dual}; \label{prop:leakage}; \label{prop:facerepair}; \label{prop:rational}; \label{thm:fejer}; \label{cor:nodes}; \label{sec:examples}. **Report:** `TR-M631-Ram-Crouzeix-Positive-Representation.tex`
M638	C3	C3: arbitrary-scale opposite-crossing budget, provably strict representation-dependent improvement with a realizable endpoint law below the trivial radius cap, and linked-parameter residual certificate. Norm-weighted endpoint transfer sharpens this source route, with same companion constant credited. No new global renorming theorem or optimal constant is claimed. **Evidence:** \label{thm:scale}; \label{prop:optimizer}; \label{prop:sign}; \label{ex:law}; \label{ex:cusp}; \label{thm:average}; \label{cor:modulus}; \label{prop:interval}; \label{sec:boundary}; \label{sec:evidence}. **Report:** `TR-M638-Ram-Midpoint-Lenses.tex`
M639	C3	Shared subtree energy accounting improves the direct correction constant from sixteen to twelve; initialized adaptive witnesses yield D^2>=1+k/12. A distinct fixed-zero-root norm retains its actual radius to improve the conditional cubic depth constant. These are notable specific proof refinements, not a stronger global headline than the source midpoint result or a new classification. **Evidence:** thm:twelve; thm:adaptive; cor:direct; cor:finite; thm:cubic; prop:cubic-error; sec:examples. **Report:** `TR-M639-Ram-Diamond-Distortion.tex`
M641	C3	Notable source-specific local improvement: true quadratic-quotient stopping K*<52/17 instead7, with RMS and approximate-head consequences; weighted child-probability quota extension and joint terminal budget giving25 instead252 in dyadic predictable energy. Classical tools credited, minor free-root identification not the score basis. No global renorming or geometric-optimality claim. Applicability remainsL3. **Evidence:** lem:stopping; thm:main; eq:actual-decomposition; cor:head; cor:lens; lem:weighted; thm:weighted; eq:joint; prop:free; sec:ram; sec:map; sec:limits; sec:evidence. **Report:** `TR-M641-Ram-Tree-Potentials.tex`
M642	C3	C3: proved source-specific full-excess bounds uniformly dominate all source exponential/square parameter choices; extend the actual conditional construction to every Gamma shape; replace second-moment weak-child argument by an integrable density-weighted proof. Classical tools credited; no optimal modulus or historical priority claim. **Evidence:** \label{thm:gamma}; \label{prop:domination}; \label{prop:lograte}; \label{lem:weak}; \label{thm:noauc}; \label{ex:pareto}; \label{prop:bernoulli}; \label{prop:panels}; \label{cor:residual}; \label{ex:exit}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M642-Ram-Independent-Products-L1.tex`
M643	C3	Retaining the actual accumulated exit-energy exponent yields power e in the unchanged source schedule, and an explicit power-Q extension for schedules p_h^h<=Q. A fully proved additive local-tightness defect theorem weakens the four-field hypothesis; its exact case improves the source logarithmic root detector to a polynomial certificate. These are notable local quantitative refinements, not generic metric relabeling, a new global classification, or historical priority claims. **Evidence:** thm:component; lem:envelope; thm:outer; cor:e; thm:slack; cor:polynomial; ex:schedule. **Report:** `TR-M643-Ram-Recursive-Potentials.tex`
M644	C3	Notable source-specific quantitative increment: optimal18 for the actual cubic/quartic inequality with sharpness, retained3B+tE cost, exact geometric-law comparison14+4sqrt2, and law-preserving capped/residual bounds. Classical tools and source construction credited. No optimal cotype, exact extension constant, historical novelty, or finite-test global certification claim; applicability remainsL3. **Evidence:** thm:flat; eq:sos; lem:initial; thm:retained; thm:geometric; cor:cotype; prop:residual; prop:capped; sec:examples; sec:ram; sec:map; sec:limits; sec:evidence. **Report:** `TR-M644-Ram-Markov-Cotype-L1.tex`

Report	Score	Demonstrated contribution and remaining boundary
M645	C3	C3: source-specific uniform finite-input bound440->115 proved with complete six-error, initialization and infinite-block scheduling analysis, including architecture-limited optimality; exact physical periodic residual cost complements it. Geometric/small-mode dependencies retained; classical smoothing/least-norm tools credited, no new global immersion certification or historical novelty claim. **Evidence:** \label{thm:energy}; \label{ex:loop}; \label{prop:ledger}; \label{thm:115}; \label{cor:166}; \label{prop:optimal}; \label{eq:memory}; \label{eq:tail}; \label{sec:boundary}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M645-Ram-Isometric-Four-Space.tex`
M646	C3	The source short-pulse mechanism is strengthened to an explicit residual-sensitive finite inequality with physical moving-boundary energy, differentiated coefficients and terminal/initial defects. A fresh width-dependent curvature estimate proves the conditional regime tau*w->0 and tau^2*w->infinity; power widths1<a<2 admit a balanced relative tau^-1/2 budget with all finite-seminorm costs retained. This is a notable source-specific local extension, not a new cap theorem, global smooth nonimmersion certification, historical novelty claim or generic finite-jet substitution. **Evidence:** thm:finite; lem:forcing; thm:schedule; ex:schedule; ex:resonance; prop:orientation. **Report:** `TR-M646-Ram-Local-Immersion-Obstruction.tex`
M648	C3	Notable local extension beyond the source: radial ellipticity admits nonmonotone auxiliary profiles; normalized nonlinear height comparison and an optimal two-variable residual scalar penalty yield a finite inexact shortening budget with an explicit third-derivative negative control. Classical tools are credited. No global rational-inessentiality proof, historical novelty, numerical solver, or global finite-test certification is claimed; applicability remains L3. **Evidence:** thm:profile; lem:chord; thm:comparison; thm:residual-scalar; eq:two-squares; thm:pass; eq:multipass; eq:oscillation; sec:ram; sec:map; sec:literature; sec:limits; sec:evidence. **Report:** `TR-M648-Ram-Scalar-Curvature-Inessentiality.tex`
M649	C3	C3: actual graph preprocessing reserve100n+203->4n+4/3 by coupled pressure/drift completion; whole-domain slope-independent C1-residual extension with explicit cost and actual-graph C0 negative control. Classical algebra/elliptic/Bernstein tools credited; source stretching/partition global inputs retained. **Evidence:** \label{prop:existence}; \label{thm:reserve}; \label{thm:residual}; \label{eq:bernstein}; \label{ex:affine}; \label{ex:oscillatory}; \label{sec:physical}; \label{eq:scale}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M649-Ram-Spectral-Scalar-Width.tex`
M654	C3	Notable conditional local refinement: the source sufficient jet budget (d+1)nu>30 is replaced by 5(d+1)nu>1, with a complete domain-safe product-tail proof and all downstream e^-s geometric comparisons. Variable degree/radius-gap exhaustion is separately proved. Classical tools credited; no global geometric certification, historical novelty, or measured speedup claim. **Evidence:** thm:shadow; cor:finite; lem:sum; thm:jet; ex:budget; thm:exhaustion. **Report:** `TR-M654-Ram-Kahler-Uniformization.tex`
M657	C3	Notable local quantitative increment: exact midpoint-metric energy including topological defect, sharp4 square-root density bound with actual closed-form sharpness, physical cone-angle transfer and enlarged exact-form correction certificate. Annular truncation is separately bounded. Classical uniqueness and tools, the source inverse and global deformation claim are not credited as new; applicability remains L3. **Evidence:** thm:midpoint; eq:topological; eq:hellinger; thm:metric; ex:torus; ex:cohomology; thm:radius; prop:annulus; sec:ram; sec:architecture; sec:comparison; sec:literature; sec:evidence. **Report:** `TR-M657-Ram-Hypersymplectic-Deformation.tex`
M660	C3	C3: source-specific joint retention of both positive Schur path remainders through the exact weighted cancellation, with relative matrix bad-set and ratio-free density/allocated-reserve bounds. Classical tools credited; no new global classification or uniform geometric stability. **Evidence:** \label{prop:paths}; \label{eq:retained}; \label{thm:budget}; \label{thm:density}; \label{cor:global}; \label{ex:curve}; \label{ex:tail}; \label{ex:resonance}; \label{prop:volterra}; \label{eq:detcert}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M660-Ram-Metric-Blaschke.tex`

Report	Score	Demonstrated contribution and remaining boundary
M663	C3	Notable source-specific local increment: larger Gaussian localization window with a matching abstract atomic scale obstruction, reserved signed margin, and domain-correct finite contact certificate separating cumulative payoff and initial-source errors. Classical tools credited. No independent global regularity certification or historical novelty claim. **Evidence:** thm:window; prop:atomic; cor:margin; prop:tails; thm:finite; prop:initial; cor:combined. **Report:** `TR-M663-Ram-Varifold-Regularity.tex`
M669	C3	Exact boundary margins yield notable source-specific improvements through a shorter terminal rigidity argument and quantitative spectral, gradient, and moment-balance defect bounds. These local gains retain the source analytic and geometric dependencies, do not weaken global theorem assumptions, and do not independently establish a broader geometric class or metric near-rigidity. **Evidence:** Source-specific matrix coercivity; Potential zero set and anisotropic residual budget; A residual-aware norm-gradient theorem; Potential-driven terminal rigidity and spectral concentration; An explicit moment-balance error. **Report:** `TR-M669-Ram-Einstein-Zero-Plane.tex`
M670	C3	A notable local quantitative certificate: exact arbitrary-potential square identity with quadratic full residual and no inverse-Laplacian constant, physical orthogonality, retained initial-tail bounds and an explicit existing-time bootstrap. Classical sharp almost-Schur and global topology/classification are not credited as new. Applicability remains L3. **Evidence:** thm:identity; prop:orthogonal; thm:certificate; thm:bootstrap; ex:bootstrap; sec:examples; sec:source; sec:ram; sec:status; sec:verification. **Report:** `TR-M670-Ram-Einstein-Energy-Gap.tex`
M671	C3	Exact boundary-sensitive certificates give a proved two-well terminal closure, an additional quadratic gradient defect, and conditional residual and spectral-concentration bounds. These are notable local improvements, but source differential and global geometric dependencies remain assumptions and no approximate-metric classification or historical priority is established. **Evidence:** thm:quadratic; prop:two-well; cor:terminal; thm:residual; cor:strict; cor:concentration; prop:imbalance. **Report:** `TR-M671-Ram-Einstein-Positive.tex`
M673	C2	C2: modest source-specific calibration of the affine logarithmic loss and a harmonic-jet profile refinement with explicit coefficient and missing-source costs. Conditional norm and regular-zero certificates retain critical-set and nonlinear-gauge limitations; classical tools, no new global nodal theorem. **Evidence:** \label{prop:atom}; \label{cor:logloss}; \label{thm:jet}; \label{prop:finite}; \label{eq:gauge}; \label{prop:norm}; \label{thm:regular}; \label{sec:global}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M673-Ram-Nodal-Length.tex`
M674	C3	Proved source-relative joint-jet loss improvements (3D 115h+226, 4D 206h+403), a fixed-metric parameter-uniform positive-factor estimate replacing P1=P2+8 by P2+1, and conditional all-dimension m>=3 localized metric realization. Full retained-residual and sign-margin/spectral-gap implications delimit certification. Classical identities credited. Global profile/wave/simplicity/fixed-metric sequence remain inherited; no historical novelty, global certification, all-orders uniformity or measured speedup claim. **Evidence:** \label{thm:jet}; \label{eq:q3}; \label{eq:q4}; \label{prop:remainder}; \label{thm:factor}; \label{thm:localmetric}; \label{prop:margin}; \label{prop:spectral}. **Report:** `TR-M674-Ram-Smooth-Nodal-Counterexamples.tex`
M675	C3	Source-specific joint first-jet volume improves noncancellation from exponent 500 to 398; homogeneous full-jet quotient reduces a sufficient C^R correction residual budget to 841+443R. Weighted resolvent/sign-margin criterion and exact witness transfer are proved with explicit dependencies. Constants and gamma are not improved, and the global fixed-metric construction and formalization validity remain inherited/unverified here. No historical priority or numerical interval claim. **Evidence:** \label{prop:bank}; \label{prop:volume}; \label{thm:398}; \label{thm:quotient}; \label{cor:budgets}; \label{thm:resolvent}. **Report:** `TR-M675-Ram-Power-Law-Nodal.tex`

Report	Score	Demonstrated contribution and remaining boundary
M676	C3	Notable source-specific constant-order reaction refinement using the actual multiplicity-weighted matrix and its large spectral gap, with a proved conditional q^-1 barrier budget and causal residual/error controls. Classical tools credited; no independent global Ricci-flow construction or historical priority claim. **Evidence:** thm:reaction; lem:block; prop:uncertainty; thm:barrier; cor:schedule; thm:causal. **Report:** `TR-M676-Ram-Ricci-Flow-Blowup.tex`
M677	C2	Proved source-specific reflexive reduction B=LAL from ALA=A, actual weighted quadratic Ricci map, third-order collar predictor and second-order compatibility obstruction, inexact-to-exact branch and boundary-preserving residual transfer, full parameter derivatives under a Bochner W1,1 hypothesis, and physical tail solver-error budgets. Classical tools credited. Bounded continuum L and geometric collar comparison inherited; concentration/static inequality/path uniformity and global same-manifold Ricci-flow extension not independently certified. No historical novelty, finite certification or speedup claim. **Evidence:** \label{prop:range}; \label{lem:nonlinear}; \label{thm:predictor}; \label{cor:compat2}; \label{thm:residual}; \label{cor:match}; \label{prop:time}; \label{prop:tail}. **Report:** `TR-M677-Ram-Ricci-Flow-Extension.tex`
M678	C2	Modest source-specific improvement in auditability: full augmented residual criteria, an infinite-complement trace Schur residual, fixed-domain finite-rank update bounds, a uniform nonlinear/holomorphic criterion and conditional observable/integrable-weight extensions are derived explicitly. No validated geometric inverse constants, stronger tree inequality, speedup or independent global proof is obtained. Classical/private source results and finite checks are not counted as new geometric contributions. **Evidence:** Locally proved operator residual criteria; Finite-rank changes that are genuinely finite-rank; Complex lengths and nonlinear radii; A source-specific approximation tolerance; The final contradiction and a limited generalization. **Report:** `TR-M678-Ram-Ricci-Flat-Trees.tex`
M679	C3	Notable source-specific conditional improvement: original-square matching is contractive below 0.36, yielding unique selected-branch parameters and residual/perturbation amplification at most 25/16; recomputed bounds sharpen R0 and enlarge the finite shooting radius by 50 percent. Uniform derivatives, all tail d-dependence and norm conversions are proved and independently checked. C3, not C4: this remains a narrow local finite/matching advance conditional on the imported ledger and analytic enclosures; ancient inversion, compact gluing and the headline global theorem are not newly established. L5 applicability is separate; no historical novelty or measured speedup claimed. **Evidence:** New finite-box refinement and enlargement; New result: contractive finite-to-tail matching; Ancient, parabolic, and compact dependencies; Bounded execution contract. **Report:** `TR-M679-Ram-Calabi-Singularity.tex`
M680	C4	Substantial proved source-relative extension: a rational invariant box gives smooth nonquadratic classical affine-maximal graphs in every n>=10; strictly increasing normalized trajectories identify the lower stationary corner and regular-variation indices; a global Berwald-Blaschke metric formula and positive limiting logarithmic derivatives prove affine completeness, explicitly not asserted in the supplied dimension-ten source. Additional exact jets, quantitative contraction, weighted residual and finite-face certificates are proved with their domains and errors retained. The ansatz, reduction and n=10 qualitative construction are inherited, and the stationary polynomial is classical. C4 reflects the demonstrated global/geometric extension, not L4 applicability or historical priority. C5, blanket source validity, lower-dimensional rigidity, uniform Hessian bounds, rates and positive finite asymptotic coefficients are not claimed. **Evidence:** \label{thm:local}; \label{cor:radius}; \label{prop:jets}; \label{thm:weighted}; \label{thm:box}; \label{thm:dimensions}; \label{thm:rv}; \label{thm:complete}. **Report:** `TR-M680-Ram-Affine-Maximal-Graph.tex`
M681	C3	Notable source-specific local gain: complete admissible dimension/direction range for the retained tube identity, explicit Hessian-weighted logarithmic constants and a precise endpoint obstruction. Classical algebra and source global Bernstein/centering/regularity results are not credited as new. Applicability remains L3. **Evidence:** lem:identities; thm:range; thm:log; sec:examples; prop:box; prop:tests; sec:ram; sec:architecture; sec:literature; sec:evidence. **Report:** `TR-M681-Ram-Affine-Bernstein.tex`

Report	Score	Demonstrated contribution and remaining boundary
M682	C2	Signed fourth-moment and panel bounds, a sharper switch-safe curvature constant, and exact quadratic interpolation produce executed local robustness improvements. No new geometric conclusion, historical novelty, speedup, or formal global verification is claimed; the geometric chain is imported. **Evidence:** sec:moments; sec:curvature; sec:certificates; sec:execution; sec:assessment. **Report:** `TR-M682-Ram-Cubic-Torus-Isoperimetry.tex`
M683	C3	Proved conditional enlargement of fixed-radius preliminary contour data from B^-4 to B^-p for alpha+1<p<=4, 2.1<alpha<3, retaining full weighted augmented domains; sharp drift-division disk margin; minimal radical-dimensional compact observation completion; full Schur residual transport. Inherited global uniqueness, finite-jet, action-jump and geometric propagation not independently established. No historical novelty or measured speedup claim. **Evidence:** \label{thm:normalization}; \label{thm:completion}; \label{thm:schur}; \label{thm:tolerance}; \label{cor:original-comparison}. **Report:** `TR-M683-Ram-Tangent-Flow-Uniqueness.tex`
M684	C2	C2: modest source-specific explicit acceleration and singular-measure budget, retained finite-viscosity defects, and conditional physical coefficient-tail/Lp controls. The source already proves the core cancellation; classical inequalities and local fixtures do not establish the global Alexandrov characterization. **Evidence:** \label{thm:budget}; \label{eq:error}; \label{eq:index}; \label{prop:tail}; \label{cor:lp}; \label{ex:slow}; \label{sec:singular}; \label{prop:measurement}; \label{sec:fixture}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M684-Ram-Distributional-Curvature.tex`
M685	C2	Explicit curvature-dependent resolvent constants, positive-parameter drift/forcing and graph errors, and a conditional finite entropy-to-chord budget are proved. These are useful modest quantitative extractions using classical tools; source geometric hypotheses, qualitative regularity and global theorem are not improved. **Evidence:** lem:resolvent; thm:shift; cor:perturb; prop:graph; thm:chord; cor:entropy. **Report:** `TR-M685-Ram-Weak-Hessian-Geodesics.tex`
M686	C2	Modest proved source-specific quantitative refinements: symmetric-width commutator conditioning, coupled dyadic Taylor/tail minimax and physical inexact-update budgets. Classical mechanisms and source all-scale geometry remain imported; no global or historical novelty credit. Applicability remains L3. **Evidence:** thm:width; cor:uncertain; prop:recovery; thm:scale; cor:resolution; prop:residual; cor:metric-loss; sec:examples; sec:ram; sec:route; sec:evidence. **Report:** `TR-M686-Ram-RCD-Coordinates.tex`
M687	C2	C2: modest exact source-specific momentum Riesz defect, normalized measurement and whole-interval boundary/jump residual certificate, with an actual noncommuting local neck model. The source already proves the endpoint minimization and uniform all-geodesic estimate; no improved global construction parameter or new manifold theorem is established. **Evidence:** \label{thm:defect}; \label{cor:probability}; \label{prop:margin}; \label{thm:residual}; \label{sec:negative}; \label{ex:matrix}; \label{sec:global}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M687-Ram-Three-Manifold.tex`
M688	C3	Exact regularized-log Fourier spectrum gives geometric later-density interference and a sharper sufficient separation threshold; separated nongeometric degrees transfer under finite exponential growth; all-fixed-order weighted phase derivatives; strengthened beta Gram inequality controls the actual mixed Xi block and yields an explicit conservative uniform curvature sign threshold from full Frobenius bounds. Source density existence, analytic input constants, completeness and coordinate obstruction remain inherited/conditional as stated. No historical priority, optimized numerical metric, runtime, finite continuum certificate or generic inverse claim. **Evidence:** \label{thm:interference}; \label{thm:degrees}; \label{prop:bernstein}; \label{thm:gram}; \label{thm:threshold}. **Report:** `TR-M688-Ram-Kahler-Threefold.tex`
M690	C3	A rank-resolved physical Jacobi pole with explicit bounded remainder, sharp square-root contamination scale and conditional endpoint/residual enclosures materially refine the source scalar trial interface. Complete local proofs and variable-curvature/negative examples are supplied; global MTW support and convexity remain imported, and classical methods are credited. **Evidence:** thm:compat; thm:pole; cor:contamination; prop:data; prop:energy; ex:variable. **Report:** `TR-M690-Ram-Weak-MTW.tex`

Report	Score	Demonstrated contribution and remaining boundary
M691	C3	The full center mixed block yields an attained least admissible pole matrix and a sharp determinant minimum strictly stronger than the source retained-Schur estimate; controlled annihilation/form errors give an explicit finite contradiction budget. Complete local proofs, endpoints and counterexamples are supplied. Classical identities are credited and global Holder/conjugacy inputs remain imported. **Evidence:** thm:least; thm:det; sec:comparison; thm:robust; sec:jetscope; prop:quadratic; ex:budget. **Report:** `TR-M691-Ram-Holder-Transport.tex`
M692	C3	Proved source-relative analytic refinements: a global history-independent round scalar baseline yields O_C1(j^-6) actual-to-adaptive pulse/full-period error with exponentially small entrance discrepancy, while the adaptive ideal remains O(j^-2) from the fixed word; a noncommuting PSD angular-leg theorem gives the sharp Loewner entrance bound for SPD matrix Theta; nonuniform damping and physical endpoint/determinant/Jacobian margins yield a conditional exact-control certificate. These are not a new global counterexample or a numerically constructed full source word. The source already proves scalar-baseline integrated damping, C1 control, exact tuning and cycle growth. C3 reflects demonstrated analytic gains, not applicability L4, historical novelty, or blanket certification. **Evidence:** \label{thm:adaptive}; \label{thm:leg}; \label{thm:weighted}; \label{prop:normalization}; \label{thm:inverse}. **Report:** `TR-M692-Ram-Harmonic-Dimension-Comparison.tex`
M693	C3	Proved source-specific conditional slow-time extension 1/2<rho<1 and 1/2<beta<1 retaining full trace tail, Ricci Schur margin, all crossing controls and infinite compatibility; explicit original O(T^3/4) and modified O(T^2/3) every-radius logarithmic growth budgets. Quantitative two-residual, finite-horizon and nonuniform-product graph estimates, plus a conditional finite odd-order crossing lemma. Angular program and global source counterexample remain inherited; no numeric effective radius, independent global validation, historical novelty or measured speedup claimed. **Evidence:** \label{thm:graph}; \label{cor:products}; \label{lem:green}; \label{thm:slow}; \label{thm:growth}; \label{cor:threshold}; \label{prop:cumulative}. **Report:** `TR-M693-Ram-Three-Dimensional-Harmonic.tex`
M694	C2	C2: modest source-specific off-geometry defect, exact tangent conditioning and weak primitive error/jump/entry certificate. The source already proves the signed cancellation; no uniform singular-tail/occupation certificate or global Vlasov-Maxwell theorem is established. **Evidence:** \label{thm:defect}; \label{cor:pointwise}; \label{prop:metric}; \label{ex:cone}; \label{thm:weak}; \label{cor:labels}; \label{sec:negative}; \label{prop:closure}; \label{sec:global}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M694-Ram-Vlasov-Maxwell.tex`
M695	C2	Source-specific weighted weak-observation truncation bound in the quotient by exact collision invariants; physical paired-collision positive cubature using classical convexity; explicit even-observation, Holder, angular and stress constants; optimized compact small-mass2/3modulus retaining source tail deletions and nonuniform constants. Useful proved local refinements, not new nonuniqueness, global2/3law, historical novelty, inverse recovery, solver speedup or finite continuum certificate. Source collision kernel, sheet bounds and global continuum construction remain inherited as stated. **Evidence:** \label{thm:tail}; \label{thm:cubature}; \label{prop:holder}; \label{prop:angular}; \label{eq:stress}; \label{thm:smallmass}. **Report:** `TR-M695-Ram-Local-Conservation-Boltzmann.tex`
M696	C3	Sharp source-relative global axial concentration exponent 2-q, proved by event-preserving exponential shell summation and a matching atomic-sheet lower box. Global axis-input and mismatch powers, simultaneous off-axis narrow-sheet estimate, conditional two-gain error budget, retained moment/gain tails, and essential-sup moving-atom counterexample add usable analytic scope. Source already proves local powers, qualitative moduli, parameter tails, five-gain rank and global nonuniqueness; none counted as new. C3 is independent of applicability L4; no historical priority or blanket/global proof certification. **Evidence:** \label{thm:axial}; \label{thm:axis}; \label{thm:offaxis}; \label{prop:two}; \label{thm:jets}; \label{thm:discontinuity}. **Report:** `TR-M696-Ram-Periodic-Boltzmann.tex`
M697	C2	Modest proved source-specific finite refinements: occupation-weighted triangular charge budget, exact adapted Poisson-binomial discard tail, and cancellation-preserving signed-measure certificates. Classical mechanisms and source global stability/exposure/closure are not new contributions; no kinetic-rate or historical-novelty credit. Applicability remains L3. **Evidence:** thm:charges; cor:weights; thm:discard; prop:moments; prop:pairerror; ex:exclusion; ex:tail; ex:tv; ex:finite-difference; sec:interfaces; sec:comparison; sec:evidence. **Report:** `TR-M697-Ram-Boltzmann-Grad.tex`

Report	Score	Demonstrated contribution and remaining boundary
M698	C3	Grand-canonical factorial domination improves the source fixed-moment centering transfer to the full eta exponent and the sharp abstract o(epsilon) discrepancy scale. Explicit tree sums give initial characteristic-function and excluded-volume remainder budgets. Causal Banach-scale truncation supplies a conditional collision-covariance error enclosure. Classical tools and source Gaussian representation are credited; no new global kinetic CLT or historical-priority claim. **Evidence:** thm:transfer; ex:centering; thm:initial; prop:bias; prop:dyson; prop:cov-error. **Report:** `TR-M698-Ram-Hard-Sphere-Fluctuations.tex`
M699	C2	Explicit local singular-value margins and derivative certificates, ordered sphere-moment/displacement constants, scale-dependent defects and seed-error bounds modestly strengthen the source audit. Global continuation and angular coercivity remain analytic premises; no new global inverse theorem, numerical reconstruction, universal stability rate or major historical novelty is claimed. L5 relevance is separate. **Evidence:** thm:jetcertificate; thm:displacement; sec:defects; prop:weak; prop:seed; sec:assessment. **Report:** `TR-M699-Ram-Metric-Unitary-Boundary.tex`
M700	C2	Explicit affine-defect and sharp signed harmonic-quadratic certificates, an executed finite-moment obstruction, and a proved conditional weighted-density bootstrap tradeoff are realized local results. The tradeoff quantifies density loss, regularity, dimension and radius profile but does not identify a new metric class satisfying its premises. Standard Taylor, Gram and interpolation machinery and source results are not credited as new global advances. The inherited L5 relevance is distinct from the C2 contribution; uniform angular coercivity and global uniqueness are not independently certified. **Evidence:** sec:moments; sec:quadratic; sec:angular; sec:tradeoff; sec:execution; sec:assessment. **Report:** `TR-M700-Ram-Calderon-Boundary.tex`
M701	C3	C3: boundary-parameter product reduces at most8 leaves to at most4 and gives the sharp uniform product interval with quadratic rather than the source's convenient cubic contrast budget. Exact least-Frobenius correction and physical weak-flux bounds support the conditional realization; no independent full-DN nonuniqueness proof. **Evidence:** \label{thm:four-leaf}; \label{thm:uniform}; \label{cor:scalarization}; \label{prop:minimal}; \label{eq:reserve}; \label{ex:rank}; \label{prop:hypercircle}; \label{sec:localization}; \label{sec:branch}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M701-Ram-Scalar-Conductivity.tex`
M702	C2	Modest source-specific quantitative pure-translation cutoff and conditional harmonic interaction tolerances, with physical fixed-crack energy certificates and negative controls. The Dini condition does not enlarge the absolute-minimizer class because linear growth is automatic. Classical projection/Harnack/hypercircle tools and imported global free-discontinuity classification are not credited as new theorems. Applicability remains L3. **Evidence:** thm:cutoff; thm:approx; prop:flux; prop:automatic; thm:dini; sec:fixture; sec:route; sec:evidence. **Report:** `TR-M702-Ram-Mumford-Shah.tex`
M703	C3	Exact target coercivity improved to 1/18 with full-norm residual reserve 1/10000, boundary lower bound improved beyond 123/500, and a source-specific sharp Kato/Bochner cutoff extension plus strict variational budget proved. C3 is local contribution, not a new dcrit theorem or independent full analytic verification; L5 relevance is separate. **Evidence:** Sharper finite margins with unchanged tables; A sharper Hessian-zero integrability theorem; A quantitative variational consequence; Residual-aware Ram certification and its limits. **Report:** `TR-M703-Ram-Bernoulli-Critical-Dimension.tex`
M705	C2	Modest explicit source saturation interface: classical dimension-free and sharp quartic constants, one contracted actual trace moment, dirty-energy and physical reconstruction budgets, exact quartic moments with remainder and negative controls. The source already has Bregman/mean recovery and calibrated coordinate cancellation. No new global diffeomorphic approximation theorem or historical novelty is claimed. Applicability remains L3. **Evidence:** lem:constants; thm:moment; cor:dirty; prop:physical; sec:examples; sec:controls; sec:route; sec:ram; sec:evidence. **Report:** `TR-M705-Ram-Diffeomorphic-Approximation.tex`

Report	Score	Demonstrated contribution and remaining boundary
M706	C3	Proves a conditional physical-L2 spectral-gap and Schur-deflated Gram-tail gradient bound, with robust Gram-Weyl, direct Hilbert-Schmidt tail and deflation-error extensions; no new global theorem, historical priority, or continuum numerical benchmark claimed **Evidence:** \label{thm:gradientcertificate}; \label{prop:robust}; \label{eq:directtail}; \label{eq:deflationerror}. **Report:** `TR-M706-Ram-Hot-Spots.tex`
M707	C2	C2: retains the exact nonnegative midpoint and unequal-weight endpoint allowance in the source interval comparison, with general even decreasing kernels, clearance gains and finite-panel errors. The midpoint sign and classical Taylor/duality tools are credited; no uniform continuum gain or independent global Henon-Lane-Emden proof. **Evidence:** \label{thm:retained}; \label{cor:clearance}; \label{ex:separated}; \label{ex:negative}; \label{prop:moments}; \label{prop:errors}; \label{prop:closing}; \label{sec:route}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M707-Ram-Henon-Lane-Emden.tex`
M708	C2	Source-specific exact tail-image improvement and residual, modulation and nonlinear budgets, with a parent-supplied independently reviewed conditional time-one certificate; no instantiated PDE observation bound, new global theorem, demonstrated speedup or established historical novelty **Evidence:** \label{prop:degreeimprovement}; \label{sec:ray}; \label{thm:timeone}; \label{eq:explicitM}; \label{sec:sequence}; \label{sec:assessment}. **Report:** `TR-M708-Ram-Defocusing-Torus-Blowup.tex`
M709	C3	Proved terminal normal completion cancels the leading physical divergence defect and improves residual h^(N-1) to h^N without another planar transport solve; N=2 yields the leading actual displacement-divergence convergence, versus minimal unmodified N=3 (source chooses N>=4). Exact derivative/support/direction costs and polynomial example. Proved four explicit null directions suffice for the scalar-matrix inference with a deterministic rank/error margin; added conditional C1-tensor finite-type residual closure. Global frames, physical inverse/transfer, supported left-inverse continuation, holomorphic gauge and arbitrary-smooth uniqueness remain source inputs, not new contributions. No historical priority or DN stability claim. **Evidence:** \label{thm:completion}; \label{sec:example}; \label{thm:four}; \label{eq:rankmargin}; \label{prop:finiteerror}. **Report:** `TR-M709-Ram-Isotropic-Elasticity.tex`
M710	C3	Complete source-specific optimizer classification by two measurable parity weights, including extreme optimizers, forced half-half central splitting and unique full symmetry; explicit K=4s branch conditioning; exact cyclic cell realization with quadratic rather than source-linear cost error; TV separation and full-slack residual bounds. Source smooth counterexample and classical cyclic approximability remain inherited. No arbitrary-density extension, historical priority, numerical uniform support certificate or speedup claimed. **Evidence:** \label{prop:center}; \label{thm:classification}; \label{cor:extreme}; \label{prop:tv}; \label{cor:singular}; \label{thm:cyclic}; \label{thm:quadratic}. **Report:** `TR-M710-Ram-Coulomb-Monge.tex`
M711	C3	Proved source-relative reduction of the stationary moment derivative constant162 to(2+sqrt(3/2))^2, preserving zero energies and configuration-dependent finite thresholds. Directional target-width interpolation with independent slice scales gives an explicit all-target constant. Physical mass-residual and conditional verified graph-margin/slab certificates include an attaining two-cell example. Sharp one-third exponent, source midpoint construction and exact three-atom obstruction remain inherited. C3 independent of applicabilityL4; no historical priority or blanket proof certification. **Evidence:** \label{thm:moment}; \label{thm:directional}; \label{cor:global}; \label{prop:mass}; \label{thm:inverse}; \label{lem:slabs}; \label{cor:localmap}. **Report:** `TR-M711-Ram-Brenier-Stability.tex`

Report	Score	Demonstrated contribution and remaining boundary
M712	C3	Proved exact optimal finite-window coefficient (2sqrt(2))^-1 coth(P/sqrt(2)) for the assigned positive limiting cone using a Wronskian single-crossing comparison; source previously has a full-line constant and existence of a sufficient finite window. Proved additive curvature and mean-curvature mass budget by preserving even Green-source and nonnegative homogeneous contributions, explicit finite-window curvature loss, and strict-slack small-defect transfer. Boundary-aware Green, finite moments, reaction-flux identity and signed residual/stress budgets retain actual domains and errors. The inherited compactness and positive-mode classification are not counted as new; global finite density, reach, fourth-order growth, radial contradiction and dimension-eight rigidity are not independently certified. Effective defect thresholds and historical priority are not claimed. **Evidence:** \label{thm:window}; \label{cor:window}; \label{thm:jointlimit}; \label{eq:curvaturetail}; \label{prop:hypercircle}; \label{prop:residual}. **Report:** `TR-M712-Ram-De-Giorgi-Eight.tex`
M713	C3	The complete four-shear optimization lowers the source uniform full-box excursion factor from 2 to (1+sqrt(2))/2, sharp within the declared alternating class. A rational 5/4 schedule preserves mixed-height fluid cancellations and improves chart/observer buffers. Exact force-cost and finite-tube analysis expose costs and limits. Classical algebra and variational methods are credited; no new global undecidability, unforced regularity, speedup or historical-priority claim. **Evidence:** thm:anisotropic; thm:uniform; cor:rational; lem:layers; thm:compiler; prop:cost; prop:tube. **Report:** `TR-M713-Ram-Solenoidal-Shear.tex`
M714	C3	For the unchanged nondegenerate source gate mechanism, two-sided force norms establish the exact finite-order Lp clock criterion and all-order superexponential growth characterization, with an explicit integer O(n log n) log-duration schedule and tail bounds. A mode-resolved pressure calculation retains the vertical-average component and proves oscillatory gains without a false Linfinity projection claim. Endpoint and history-error obstructions delimit the improvement. Classical tools and source coding are credited; no global NS/halting certification, historical novelty, uniform faster execution or measured speedup claimed. **Evidence:** prop:pressure; lem:nondegenerate; thm:clock; cor:allorders; thm:integerclock; cor:threshold; prop:endpoints; prop:gate. **Report:** `TR-M714-Ram-Fixed-Particle-Test.tex`
M715	C2	C2: exact startup forcing-action and pressure budgets, a three-regime fixed-phase derivative-cost bound, and finite phase-aligned error/latency certificates for the source scalar-clock family. Classical mechanisms are credited; no full-action optimum, new universality theorem, arbitrary-forcing regularity theorem or independent compiler certification. **Evidence:** \label{sec:route}; \label{prop:small}; \label{thm:action}; \label{eq:gaugecost}; \label{cor:cost}; \label{thm:clock}; \label{cor:startup}; \label{thm:phase}; \label{cor:observer}; \label{prop:spatial}; \label{sec:ram}; \label{sec:evidence}. **Report:** `TR-M715-Ram-Stationary-Forcing-Clock.tex`
M716	C3	Notable source-specific local increment: sharp positive-thickness and recording-coefficient feasibility for the nonreciprocal three-shear sheet map, exact two-coordinate normal/partial-path accounting, and a clock travel-cost obstruction excluding L1 forcing for fixed nonzero eventually periodic templates while preserving p>1 constructions. Classical tools credited; M713 four-shear optimization and global compiler/halting claims not credited again. Applicability remains L3. **Evidence:** thm:envelope; thm:thickness; cor:minimax; sec:two; prop:collar; thm:clockendpoint; prop:power; sec:ram; sec:negative; sec:route; sec:evidence. **Report:** `TR-M716-Ram-Solenoidal-Forcing.tex`
M717	C3	Exact trace/determinant classification extends the source simultaneous straight-path normal-compensation method from positive diagonal branches to admissible planar affine maps, with sharp quadratic determinant thickness and retained convex observer paths. The original diagonal normal bound is sharpened. Certified collars, oblique geometry, coefficient uncertainty and finite causal error budgets are proved with explicit boundaries. These are source-specific local increments using credited classical tools, not a new global fluid computation theorem or a repeat of companion shear/clock results. **Evidence:** prop:diagonal; thm:affine; thm:routing; prop:detbudget; prop:box; thm:causal; prop:word. **Report:** `TR-M717-Ram-Prefix-Instructions.tex`

Report	Score	Demonstrated contribution and remaining boundary
M718	C2	C2: a less aggressive computable scale recurrence for the source addressed-curl architecture, with preserved all-order decay, exact precision/signal comparisons, physical Lp/tail budgets and finite core-error certificates. This is a modest architecture-specific parameter and error refinement; no new universality class, uniform detector robustness or runtime improvement is established. **Evidence:** \label{sec:route}; \label{thm:criterion}; \label{thm:adaptive}; \label{prop:comparison}; \label{ex:scale}; \label{prop:Lp}; \label{prop:tail}; \label{thm:plateau}; \label{cor:detector}; \label{ex:false}; \label{prop:template}; \label{sec:ram}; \label{sec:limits}; \label{sec:evidence}. **Report:** `TR-M718-Ram-Rapidly-Vanishing-Forcing.tex`
M719	C3	Proved source-specific exponential stopped-tube mass-loss bounds, including loading and conditional stage survival; logarithmic tolerance radius; normalized logarithmically inflated critical spatial expansion with complete derivative/effectivity proof, removing the source's extra4^n geometric factor in the sufficient schedule. Continuous L1 and equal-mass observation certificates and compact disjoint-mass refinement support the analysis. Classical stochastic/positivity tools, imported reversible compiler and source halting equivalence are inherited. No optimality, historical priority, runtime speedup, arbitrary-force regularity or finite infinite-time certificate. **Evidence:** \label{thm:tube}; \label{cor:geometric}; \label{thm:critical}; \label{prop:L1}; \label{cor:robust}; \label{prop:compact}. **Report:** `TR-M719-Ram-Navier-Stokes-Computation.tex`
M720	C3	Notable source-specific local improvement: direct second variation and exact planar area constraint reduce the source transported-bump majorant from cubic to quadratic exponential in logical time and give a sharper single-burst diffusion budget. Compact Euclidean full-convection forcing asymptotics and physical potential error controls complement this. Classical mechanisms credited; no global compiler/halting/detector theorem, prior report result or historical novelty claimed. Applicability remains L3. **Evidence:** thm:variation; lem:area; thm:bump; cor:burst; sec:parameters; thm:clock-cost; prop:residual-error; sec:negative; sec:architecture; sec:ram; sec:evidence. **Report:** `TR-M720-Ram-Scalar-Potentials.tex`
M721	C3	Notable local extension from positive diagonal boxes to arbitrary rational SL(3) maps on full parallelepipeds, with constructive shape paths, true support separation and effective material collars. Classical linear algebra/curl tools credited; source coding, fixed-observer and all-time statements not newly established. **Evidence:** lem:factor; lem:separate; lem:curl; thm:routing; sec:examples; sec:comparison; sec:evidence. **Report:** `TR-M721-Ram-Box-Transport.tex`

`contributions.json` records each source hash, assessment date, report location, and evidence sections; catalogue CSV/JSON include the scores. Completed reports remain unchanged. Future abstracts end with their contribution assessment, and this catalogue is refreshed after each batch.

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5 Individual Assessments

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
001/001	Milne's rationality conjecture for abelian varieties	L2	–	Finite period matrices can check independence and equality of proposed balanced Weil classes after exact rational membership is supplied. This is a peripheral verification route for the transported classes, not Milne rationality itself. **Boundary:** Integral-model transport, weak admissibility and rational all-prime specialization are not supplied. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
002/002	Exact Birch-Swinnerton-Dyer Formula from Low Selmer Corank	L3	C2	The central L-value/derivative, height Gram determinant and normalization side of the exact BSD formula have a concrete certified log-moment and regulator counterpart. These are significant analytic and numerical parts; the manuscript's integral determinant-length argument is different. **Boundary:** Low-Selmer converse, theta congruence, integral lattice saturation and equality of arithmetic orders still require the source proof. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
003/002	The Selmer converse for elliptic curves at every prime	L2	–	Once a particular elliptic curve and claimed rank-one point are available, central-jet and height certificates can test the final nonvanishing and independence consequences. They do not replace the tame deformation versus cuspidal extension length comparison driving the converse. **Boundary:** Every-prime deformation lengths, theta cusp lifting, Galois extension extraction and transfer remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
004/002	The two-primary Birch-Swinnerton-Dyer formula in Selmer corank at most one	L3	C2	Certified central derivatives, regulator determinants and explicit normalization/saturation accounting bear directly on substantial factors in the two-primary BSD identity. The corpus's saturation warning is especially pertinent because losing an index changes its square, including powers of two. **Boundary:** Siegel-unit cochain integrality, split local measures, real-component factors and two-primary alternation are not proved by these tools. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
005/003	The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane Re s> 7/8	L4	C2	Very relevant to two actual proof steps: exact log-coordinate Mellin/Fourier inversion in the common-signal continuation criterion, and rational Bernstein certification of the compensated endpoint polynomial after its positive denominator is cleared. A subsequent independent bounded audit establishes the exact local minimum (-4473+35sqrt(16354))/12696, including all 39 Bernstein coefficients and the source algebra. This does not accept the 7/8 theorem as established. **Boundary:** Completed theta reflection, compensated low/high arithmetic estimates, short-prime moment bounds and uniform loss/height bookkeeping remain unaudited; no stronger zero-free line follows from the improved finite margin. **Review:** Selected proof audit: common-signal criterion, principal normalization and endpoint proof read; source endpoint reduction and square identity checked exactly; parent optimal certificate replayed without writing inputs; all 39 Bernstein coefficients recovered by independent rational evaluation-matrix inversion, coverage checked, and exact quadratic-surd minimum proved. The rest of the arithmetic proof is not audited.
006/003	The Quasi-Riemann Hypothesis (alternate 11/12 proof)	L3	C2	The weighted Mellin/Poisson energy formulation and exact separation of short-cube, long-cube and transfer terms fit the corpus's transform, norm and remainder bookkeeping. Applicability is to significant analytic reductions, not a derivation of the completed sextic-twist mean square. **Boundary:** Theta reflection, quadratic large-sieve estimates, arithmetic dualization and uniform recursive scale descent require independent verification. **Review:** Main statement, proof roadmap, canonical-energy proposition, completed sums, short/long cube reduction and transfer statement read; not a full reflection or zero-free proof audit.
007/003	Uniform exclusion of Landau-Siegel zeros	L2	–	Finite polynomial coordinates can represent the anisotropic interpolation/evaluation matrix and check determinant row operations. This is narrower than the proof's essential arithmetic mechanism: inert-prime Frobenius congruences yield divisibility of the integer norm. **Boundary:** Prime bias from a hypothetical exceptional zero, uniform inert-prime log mass, norm integrality and divisibility lower bound are not supplied. **Review:** Statement, interpolation lemma/proof, weighted determinant construction and competing determinant bounds read; arithmetic inputs not independently established.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
008/004	Hilbert's tenth problem over the rational numbers	L1	–	No concrete corpus mechanism was identified for the existential integer test inside rational arithmetic. Finite analytic certification of elliptic data does not implement the compactness, odd-pole and index-multiplication argument. **Boundary:** Rational definability, elementary-extension transfer, five-point height constraint and prime-pattern construction remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
009/004	A pointwise 2-converse for elliptic curves with rational two-torsion	L2	–	The finite address/forest matrices and contraction coefficients admit exact coordinate and graph-algebra checks. Such bookkeeping is a peripheral aid to the missing-vertex nonvanishing argument, not its arithmetic supply. **Boundary:** Integral cyclotomic and ring-class interpolation, auxiliary twists and pointwise missing-vertex nonvanishing are outside the matched tools. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
010/005	Catalan's constant is irrational	L4	C2	The signed mixed-moment determinant, polynomial row/contact identities and rigorous one-variable barrier maximization provide substantial, concrete matches to finite moment representation and certified polynomial enclosures. Bernstein subdivision can independently bound derivative numerators after denominator signs are fixed, complementing the source's Descartes/root-bracket and rational-log certificate. **Boundary:** The determinant is signed/mixed, not automatically a positive Gram matrix. Its arithmetic nonvanishing, odd-prime and 2-adic denominator bounds, and hence irrationality, are not certified by the transform or this screening. **Review:** Statement and proof structure read; selected foundations and energy proofs plus full finite certificate read; corpus finite-moment and Bernstein proofs read; certificate arithmetic not replayed and arithmetic-prime proof not audited.
011/006	Goldfeld's analytic density conjecture and the 2-converse for elliptic curves	L2	–	Finite binary-cube identities and local Selmer matrices can be checked exactly, and central-jet enclosures can test selected twist examples. Neither operation provides a density theorem or the uniform missing-vertex arithmetic nonvanishing needed here. **Boundary:** Prime graph supply, Heegner nonvanishing, every-curve 2-converse and passage to Goldfeld density remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
012/006	The mean analytic rank of quadratic twists of elliptic curves	L3	C3	The paper's log-weighted second moments and block-mollifier comparison can use exact moment coordinates with explicit truncation/mean-square error accounting. This reaches significant analytic parts, especially distinguishing a finite model from the rank-weighted tail required for convergence of the mean. **Boundary:** The requisite arithmetic moment estimates and uniform rank-tail bound are not consequences of finite covariance identities; the density companion is not treated as established. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
013/007	Ordinary two-point correlations of multiplicative functions	L3	C2	Centered adjacency powers, high traces and transfer to weighted correlations are concrete finite spectral/moment operations. The corpus can organize these and enforce retention of boundary and cross terms; it cannot make the trace small merely by representing it. **Boundary:** Arithmetic rank-witness exceptional-word bounds, forest enumeration and uniform every-cutoff cancellation must still be proved. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
014/008	The Deligne-Drinfeld conjecture	L1	–	The core construction uses characteristic-two pentagon degeneration, deletion operators and categorical holonomy to establish Lie freeness. No matched corpus theorem supplies these algebraic/categorical implications; finite matrix representation alone is not a concrete substitute. **Boundary:** Odd depth-one generators, integral dimension control and the freeness induction remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
015/009	Reconstruction of Function Fields from Mod-l Milnor K-Theory	L1	–	Recovering valuations, residual curve fields and the common scalar from mod-l Milnor symbols is not a finite inverse problem of the kind treated by the corpus. No concrete applicability beyond an analogy in the word reconstruction was found. **Boundary:** Pencil support, valuation recognition, compatible curve-field recovery and projective field reconstruction remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
016/009	Reconstruction from Milnor K-theory modulo the characteristic	L1	–	The essential p-independent intersection and logarithmic-derivation calculation recover projective lines in characteristic p. Real/complex local-moment inversion does not supply the Steinberg-symbol or Frobenius-sensitive field reconstruction. **Boundary:** Mod-p symbol relations, exponent comparison, common scalar and recovered field structure remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
017/009	The Bogomolov-Pop reconstruction theorem	L1	–	The pro-l abelian-central Kummer argument depends on valuation and residual tests followed by curve-field recovery. Finite spectral or moment reconstruction does not address that anabelian recognition problem. **Boundary:** Uniform pencil bound, recovery of rational subfields and the Pop reconstruction input are outside the identified corpus mechanisms. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
018/010	Unrestricted pro-modularity at the prime two	L1	–	Potentially automorphic support and diagonal local admissibility in a two-adic trace-deformation family have no concrete counterpart in the reviewed operator/moment results. **Boundary:** Regular-point construction, completed Hecke support and propagation across deformation components remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
019/010	The Dimension of the Two-Adic Hecke Algebra at Odd Level	L1	–	The dimension-four claim rests on cuspidal density, Eisenstein-component control and a three-dimensional adjoint-Selmer tangent bound. Numerical matrix rank or finite-jet dimension does not establish these deformation-theoretic dimensions. **Boundary:** No matched result supplies component density, Newton–Thorne-type Selmer input or the passage from relative to arithmetic dimension. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
020/010	Fontaine-Mazur modularity at the prime 2	L1	–	The characteristic-two patching/connectedness and classical descent route is not covered by the local transform corpus. Its curve and tangent terminology concerns Galois deformation spaces, not analytic inversion. **Boundary:** Controlled-field Selmer loss, propagation on patched modules, potential pro-modularity and classical modularity descent remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
021/011	Weighted dilation graphs, smooth shifted primes and totient fibers	L3	C3	Weighted dilation adjacency, ideal kernels and frequencywise transference have a concrete finite spectral/energy representation. These tools can expose weights, norms and discarded terms in significant analytic stages before prime extraction. **Boundary:** Shifted arithmetic correlations, Type-II savings, candidate prime mass and totient-fiber extraction are not generated by the representation. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
022/011	The Poisson-Dirichlet law for prime predecessors	L3	C3	Marked determinant expansions, signed memory and factorial-moment bookkeeping fit finite joint-moment coordinates and norm/error ledgers. This is meaningful for the analytic passage from marked observables to a candidate limiting law. **Boundary:** Uniform major-arc/correlation estimates, size-biased no-escape and the full Poisson–Dirichlet convergence require independent arithmetic/probabilistic arguments. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
023/011	Prime Predecessors with an Even Number of Prime Factors	L2	–	The long-prime polynomial and bilinear weighted expressions can be represented exactly in finite moment coordinates, offering coefficient and error checks. The even-parity conclusion is controlled by a much more specialized prime-extraction mechanism. **Boundary:** Parity-twisted shifted correlation, congruence-uniform Type-II savings and balanced semiprime sieve extraction are not supplied. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
024/012	The joint Dickman law for consecutive integers	L3	C2	The smoothing kernels, mixed amplification and second moments of latent rows fit the corpus's kernel-moment and joint-covariance machinery. Finite integral-feature approximation also has an explicit remainder analogue, making several analytic steps concrete matches. **Boundary:** Vanishing arithmetic energy, endpoint uniformity, control of all samples and deduction of the joint Dickman law are not implied by moment representation. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
025/013	The additive indecomposability of the primes	L2	–	Finite-field character arrays and binary-tree histories admit exact finite polynomial/graph coefficient checks. This is peripheral to the higher-order translated-character cancellation needed for additive indecomposability. **Boundary:** Arbitrary-center character supply, overlap estimates and the final additive contradiction are not supplied by generic spectral calculus. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
026/014	Global Arthur Enhancements of Cuspidal Excursion Parameters	L2	–	Finite trace matrices and prescribed test-function pairings can be represented and checked without changing their physical pairing. This assists bookkeeping in the enhancement criterion, but not the global geometric existence or support arguments. **Boundary:** The conditional finite-level decomposition, shtuka support, open-orbit enhancement and simultaneous lower/boundary estimates remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
027/014	Rationality of the Canonical Unramified Arthur Filtration	L2	–	The finite bilinear-measure/trace part permits exact coordinate checks and explicit rank tests once the rational Frobenius model is given. That does not establish the canonical rational descent of the whole Arthur filtration. **Boundary:** Orbit categories, compact spectral rational descent and rational structure on Hecke support are not supplied. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
028/014	Ramanujan-Arthur Decompositions of Cuspidal Functions at Full Finite Level	L2	–	Polynomial spectral filtering is a real operational overlap, provided the full spectral measure and continuous contribution are retained. It can help audit filter identities, but the corpus's finite graph calculus is not a theorem on the full automorphic spectrum. **Boundary:** Rotation-shtuka trace lower bounds, extreme-slide upper bounds, ordered Bernstein weights and full finite-level decomposition remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
029/014	Temperedness at ramified places for globally generic exceptional groups	L1	–	Purity of adjoint local monodromy and Shahidi-coefficient regularity, not numerical spectral positivity, drive the ramified temperedness argument. No concrete corpus theorem supplies those representation-theoretic inputs. **Boundary:** Positive-exponent contradiction and use of the unramified companion require a separate audit. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
030/014	The Restricted Geometric Langlands Equivalence in Positive Characteristic	L1	–	The missing-support contradiction uses two transverse Hecke correspondences and a Levi/Higgs nearby-cycle slice. A finite cochain or residual solver does not supply this geometric support theorem. **Boundary:** Existence/coherence of the restricted Langlands functor, local slices, nonzero nearby cycles and coefficient descent remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
031/014	Constructible tame Hecke eigensheaves in positive characteristic	L1	–	The tame eigensheaf construction requires inverse-monodromy node gluing, enhanced torus descent and nonvanishing nearby cycles. None is provided by the reviewed finite reconstruction or analytic residual results. **Boundary:** Perverse extension, full fusion coherence and specialization to positive characteristic remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
032/014	Tame Hecke Eigensheaves with Several Marked Points	L1	–	The several-marked-point problem adds simultaneous nodal compatibility and finite-quotient descent to eigensheaf existence. Finite coefficient coordinates do not establish these sheaf-theoretic compatibilities. **Boundary:** Multipoint lifting, enhanced torus descent, coherence and nonzero simultaneous specialization remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
033/014	Frobenius Structures on Tame Hecke Eigensheaves	L1	–	The Frobenius structure is an equivariant descent problem on an arithmetic tower with a compact generator. It is not certified by knowing a finite Frobenius matrix or solving a local analytic equation. **Boundary:** Finite-power equivariance, gluing across the tower and preservation of eigenstructure under nodal specialization remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
034/015	Equidistribution of Prime-Degree Torus Packets with Arbitrary Local Type	L3	C3	The arbitrary-order ideal sums and local Fourier normalization have a concrete weighted-transform/energy interface, with explicit measures and removed Euler factors. This can support significant analytic bookkeeping in the local packet-mass and nonescape bounds. **Boundary:** Uniform arithmetic packet estimates, the small-ball exponent and entropy on every dominated component still require their own proofs; homogeneous measure classification is not supplied. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
035/015	Equidistribution of primitive quartic torus packets for arbitrary orders	L3	C3	Local Fourier conductor weights and additive packet-distribution estimates match transform normalization and weighted-energy bookkeeping. The exterior-multiplicity/cubic-resolvent structure must be kept in the coefficients rather than discarded under a generic norm estimate. **Boundary:** Cubic-resolvent multiplicities, unramified sieve and exclusion of homogeneous obstructions remain specialized inputs. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
036/015	Equidistribution of Primitive Sextic Torus Packets	L2	–	Conditional plaque masses and product-rule probability tables can be checked under a single declared law with all cross terms retained. This is a limited aid to the entropy bookkeeping, not a proof of active-root propagation. **Boundary:** Arithmetic packet bounds, long-diagonal entropy, root activity and the Haar classification step are not covered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
037/016	The abelian Zilber-Pink conjecture	L2	–	Height Gram matrices and rational period/projector checks offer concrete tests of the normalized Neron–Tate geometry at a fixed stage. They do not provide the multiscale geometric and arithmetic contradiction used for finiteness. **Boundary:** Ax-type tangent positivity, high-scale Vojta–Remond input, bounded arithmetic scales and the product-cut argument remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
038/016	The E x CM component of Zilber-Pink for curves in A2	L2	–	Period-jet and finite-span error accounting can assist the local period calculations after a family and exact classes are fixed. The proof's uniform boundary-length and height-to-Galois-orbit passage is not supplied by those calculations. **Boundary:** Uniform curve-height bounds, essential threefold periods, Galois-orbit growth and unlikely-intersection finiteness remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
039/016	Quaternionic division points on curves in the Siegel threefold	L3	C3	The two-argument interpolation in a period differential system is a specific fit for variable-coefficient analytic jets and correctly remainder-bounded period matrices. These address a significant analytic stage of the quaternionic-point counting argument. **Boundary:** Common arithmetic relations at all relevant places, auxiliary-marking heights and the counting-to-finiteness argument remain independent. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
040/016	Elliptic squares and Zilber-Pink for curves in A2	L3	C2	The shared small-parameter interpolation and period equations can use exact finite jets with analytic remainder bounds. The corpus's warning that a small residual is not exact equality is pertinent to the paper's separate local-equation/nonidentity stages. **Boundary:** Geometric monodromy, finite correspondences, bad-disk exclusions, integral height frames and all-point nonidentity/counting estimates are not supplied. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
041/017	The irrationality exponent of pi is 2	L3	C3	Separated-weight interpolation is explicitly a finite coefficient-packet surjectivity problem, and the later analytic determinant can be checked with exact polynomial coordinates and remainder-controlled jets. The match reaches substantial linear-algebra/analytic steps, but not the uniform interpolation theorem itself. **Boundary:** Weights chosen before arbitrary point locations, all-large-degree surjectivity, arithmetic determinant lower bounds and the irrationality exponent are not consequences of a finite rank computation. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
042/018	The Margulis-Platonov conjecture over global function fields	L1	–	The function-field Margulis–Platonov proof uses power-coset approximation, norm tori, odd division quotients and character-killing recurrence. No local transform theorem establishes that arithmetic normal-subgroup classification. **Boundary:** Simultaneous approximation, additive recurrence over finite quotient palettes and degree induction remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
043/018	The Margulis-Platonov conjecture over number fields	L1	–	The number-field argument depends on actual arithmetic cosets, unitary induction and exceptional D4/E6 structures. General finite-group coordinate transport does not furnish the required global normal-subgroup result. **Boundary:** Unitary and exceptional-type reductions, torsors, character elimination and arithmetic approximation remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
044/019	Etale covers with a prescribed exterior sheet	L2	–	The component-cycle/character bookkeeping on a specified finite graph can be audited through exact cochain maps and separating period rows. This is a peripheral finite subproblem of the prescribed-sheet cover construction. **Boundary:** Existence of the connected etale cover, disk/exterior analytic geometry and stabilizer-quotient realization do not follow from a graph certificate. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
045/019	The p-adic section conjecture	L1	–	Branch-disk patching, arbitrary collar roots and a completed universal tower form the core section-conjecture argument. The real/complex moment tools do not give these p-adic covers or the section-to-point implication. **Boundary:** Tame and wild root realization, tower completion and the additional global descent hypothesis remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
046/020	Squarefree values of quartics and power-free values of polynomials	L3	C3	The exact parameter certificate is a concrete finite rational-inequality problem, compatible with polynomial coefficient checks and rational Bernstein enclosures once positive denominators and powers are handled. This is a substantial, isolated verification stage in the determinant-method tail bound. **Boundary:** Ideal factorization, uniform determinant cuts, Hilbert bounds, exceptional-curve estimates and the large-prime tail are not supplied by the finite parameter certificate. **Review:** Statement/proof structure and complete embedded rational certificate read; certificate not executed here; arithmetic determinant and sieve proof not audited.
047/021	A quadratic bound for Jacobsthal's function	L3	C3	The comparison of the reference prime tree with actual classes includes probability kernels, regeneration occupation and variance terms that can be represented with a common moment law and explicit approximation errors. These are significant analytic/probabilistic parts of the proof. **Boundary:** The inverse modular-hyperbola estimate, sieve-delay control and uniform transfer to all prime-class paths require independent arithmetic work. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
048/022	The Weak Inhomogeneous Duffin-Schaeffer Conjecture	L3	C3	Weighted interval rows and the finite second-moment comparison with a deletion-free model have a concrete covariance/energy counterpart. Retaining common-law cross terms and the full expected deletion loss is essential to the positive-measure step. **Boundary:** Close-pair inverse arithmetic, prime-factor constraints, sparse prescriptions and the full limsup statement are not supplied by the moment identity. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
049/023	An unconditional first moment for cubic Gauss sums	L3	C3	Corrected dispersion explicitly expands a square into diagonal, mixed model and exceptional-frequency pieces. Exact weighted energy and Mellin normalization tools are useful for this substantial analytic stage and help prevent losing the cubic-frequency correction. **Boundary:** Cubic large-sieve savings, SW uniformity, exceptional character moments and the unconditional prime first-moment asymptotic remain separate claims. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
050/024	An asymptotic formula for the number of totients	L2	–	The finite layer/limit bookkeeping in extracting the totient main-term coefficient admits exact coefficient and weighted-moment checks. The decisive shift-product and collision estimates are arithmetic, so this is peripheral applicability. **Boundary:** Prime-prefix supply, collision uniqueness, uniform limiting main term and the totient asymptotic remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
051/025	Short Egyptian fractions	L2	–	Uniform divisor moments and finite residue-weight computations can be organized in exact joint-moment coordinates. This assists the quantitative estimates, but not the exceptional-set descent converting a dense supply into every integer. **Boundary:** Uniform divisor estimates, exceptional descent, prescribed-denominator construction and exact final padding are not provided. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
052/026	Positive lower density of large prime gaps	L3	C3	The adjacent-dimension quadratic forms w and v, cumulative integration T, and square of a signed divisor sum are specific finite-feature/moment objects. Exact coefficient assembly and mean-square error bounds are relevant to constructing and checking the low marked-mass family. **Boundary:** Uniform arithmetic mixed-moment asymptotics, singular-series control and passage from weighted detection to ordinary positive index density are not supplied. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
053/027	Integral points on character varieties of curves	L1	–	The integral-point construction relies on surface diagrams, exact Jordan boundary data, finite-group characters and Lang–Weil irreducibility. Numerical or finite-moment reconstruction does not establish integrality or Zariski density here. **Boundary:** Rank reduction, arithmetic determinant roots, descent and dense integral specialization remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
054/028	Bounded-Step Walks on Gaussian Primes	L1	–	The bounded Gaussian-prime walk claim rests on a common sampling schedule, residue avoidance and a quantitative block-entropy contradiction. No specific reviewed corpus result supplies the required entropy cost versus coverage estimate. **Boundary:** Uniform residue zero-avoidance, self-avoidance/coverage transfer and finiteness of every bounded-step component remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
055/029	Primitive roots for every admissible integer base	L3	C3	The paper has its own theta/Poisson/Mellin zero-detection stage, so the exact logarithmic transform, weight conventions and energy-error decomposition apply to a substantial intermediate analytic argument. This is not reliance on the new 7/8 manuscript as established input. **Boundary:** Cubic theta reflection, uniform Kummer splitting, rough-prime marked Type-II bounds and the primitive-root sieve conclusion remain unaudited arithmetic inputs. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
056/029	Simultaneous primitive roots: a conditional lower bound for prime bases	L2	–	The conditional theorem's marked families and finite block moments permit coefficient and covariance checks, conditional on its explicitly assumed estimates. The reviewed corpus does not discharge those analytic-number-theory assumptions. **Boundary:** Hecke, marked/block and density hypotheses, progression nonsquares and Kummer bad-order removal must remain explicit; no unconditional lower bound is inferred. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
057/030	Modularity of elliptic curves over imaginary quadratic fields	L1	–	Simultaneous residual specialization of cyclic-cover systems with prescribed local types is an automorphy-lifting construction, not a transform inversion problem. No concrete matching theorem was identified. **Boundary:** Character orders 1,p,3p,3, common integral lattice, monodromy, residual compatibility and automorphy descent remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
058/031	Open Homomorphisms of Global Solvably Closed Galois Groups	L1	–	Kummer/Frobenius character detection and residue-characteristic matching reconstruct a field from an open Galois-group map. Finite linear representations alone do not supply this anabelian theorem. **Boundary:** Uniform character-detection bounds, faithful representation comparison and unique field identification remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
059/032	The rational Hodge conjecture for products of K3 surfaces	L2	–	After actual cycles are constructed, finite period matrices and rational symmetry blocks can check the exact Kuga–Satake tensor and mixed-product span. This is a meaningful but peripheral verification layer because the manuscript's construction is largely Floer/mirror geometric. **Boundary:** Controlled Lagrangian realization, disk-curvature removal, ordinary Fukaya comparison and algebraic cycle existence are not supplied. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
060/032	Algebraicity of Kuga-Satake Correspondences for K3 Surfaces	L2	–	The prescribed cup-product kernel and exact full Clifford class can be checked against rational periods if actual algebraic candidates and an injective test lattice are supplied. Small period residuals alone cannot establish the Kuga–Satake correspondence. **Boundary:** Immersed topology, energy/arity-limit disk obstruction removal, ordinary Fukaya realization and semiregular polarized deformation remain outside the matched results. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
061/032	The rational Hodge conjecture for CM abelian varieties	L3	C3	The CM character tensors, theta-period differential and finite rational span tests have a substantive counterpart in the Hodge corpus's period-certification and rational-symmetry machinery. This can check exact tensor targets and nonzero finite period derivatives once the geometric inputs exist. **Boundary:** The projective-surface construction, ball-quotient Frobenius slopes forcing CM types and existence of algebraic cycle supplies remain independent. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
062/032	Algebraic Kuga-Satake correspondences and Hodge conjectures on a K3 quadratic locus	L3	C3	The exact full Clifford tensor criterion and finite correspondence/period comparisons are concrete targets for rational block and lattice-gap certificates. They matter substantially because realizing an isomorphic summand is weaker than realizing the specified Kuga–Satake map. **Boundary:** Abelian-double-cover construction, coherent half-sums, curve domination, K3 pencil correspondences and all-period cycle existence remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
063/032	Weil classes and Hodge classes on abelian powers	L3	C3	The separation of pairing tensors from top exterior character classes, followed by rational block assembly on powers, matches the corpus's exact symmetry-projector and finite-span tests. These can verify which rational Hodge subspace proposed cycle supplies actually fill. **Boundary:** CM cycle supply, the seed's Hochschild obstruction, spin-character construction and period descent are not supplied by a span certificate. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
064/032	Abelian covers, Gale correspondences, and the Hodge conjecture for powers	L3	C3	Finite group averaging here is an actual algebraic quotient/correspondence, not merely a change of coordinates. The corpus's rational symmetry blocks and finite period-span budget directly support checks of character determinants, complete Galois orbits and the assembled rational cohomology supply. **Boundary:** Full marked variation, rank-one monodromy, connected irreducibility and the geometric Gale correspondences still require the source arguments. **Review:** Statement/proof structure plus finite quotient, bundle/blowup supply and start of Fermat-supply proofs read; rational-projector and period gates read in corpus; not a full monodromy or Hodge-conjecture audit.
065/032	Algebraicity of Weil classes on split abelian eightfolds	L2	–	Exact period targets and relative-weight identities can be checked at the finite-class level. The local disk discussion, however, involves compactness, nodal gluing and virtual integration far beyond a finite moment reconstruction, so applicability is peripheral. **Boundary:** Uniform disk area/gluing estimates, virtual boundaries, obstruction removal, category limits, mirror comparison and coherent algebraic lifting remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
066/032	A Conditional Reduction for Algebraic Kuga-Satake Correspondences	L3	C3	The conditional reduction explicitly requires equality with the full Clifford multiplication map, including polarization signs and both tensor slots. Rational exact matrix/block and period-lattice checks are a concrete, substantial aid to that identification and its specialization. **Boundary:** Existence of the nonzero algebraic correspondence or surface wedge is a premise; the corpus does not turn the conditional reduction into unconditional algebraicity. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
067/033	Orbifold and logarithmic Iitaka subadditivity	L1	–	The proof's relative root-Hodge line, reduced-boundary comparison and adjoint positivity are global birational/Hodge-theoretic constructions. No concrete finite-period computation or local operator problem was identified to which the corpus theorem applies. **Boundary:** General-type addition, root comparison and orbifold/logarithmic subadditivity remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
068/033	Logarithmic Kodaira dimension and whole-fiber variation	L1	–	Detecting the parameter field and proving whole-fiber birational constancy, then descending markings, requires a global variation theorem. Finite transform injectivity does not establish that constancy or descent. **Boundary:** Root-Hodge construction, constancy estimates, whole-fiber normalization and the numerical extension remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
069/033	The reverse logarithmic Kodaira inequality and additivity	L2	–	The coefficient lattice of the Hodge vector and finite period/rank comparisons admit an exact rational-period audit after the marked variation is given. This is a limited check within the boundary-rank analysis, not the reverse Kodaira inequality. **Boundary:** Boundary Higgs-tail ranks, relative section-space equality, period quotient geometry and nonvanishing removing the twist remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
070/033	Projective Hodge lines and ordinary Iitaka subadditivity	L2	–	Fixed rational period coordinates and boundary rank-loss matrices can be checked with the corpus's finite-span error ledger. Such a rank certificate is peripheral to constructing the projective period quotient and deriving section growth. **Boundary:** Minimal VHS, tangential limit geometry, adjoint bigness, root-cover growth and integral Chern–Thom construction remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
071/033	B-semiampleness for compact log-smooth Kahler fibrations	L1	–	B-semiampleness here depends on product decomposition, period compactification and exact extension/descent of a norm line. A transform metric or finite norm identity does not supply that line-bundle descent. **Boundary:** Calabi–Yau/torus/symplectic decomposition, exclusion of flat twists and finite analytic norm descent remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
072/034	Log abundance for compact Kahler spaces under logarithmic Iitaka subadditivity	L1	–	This is a conditional birational reduction under logarithmic Iitaka input, followed by boundary generation and simple-space nonvanishing. No reviewed local operator result supplies those hypotheses or global implications. **Boundary:** The assumed subadditivity, required MMP, signed rigidity and meromorphic nonvanishing must remain explicit. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
073/034	Uniform indices for semi-log-canonical log Calabi-Yau pairs	L2	–	The scalar/diagonal jet evaluation matrices used in Hodge-rank calibration can be assembled exactly on a fixed finite polynomial space. This is a peripheral finite verification task; unisolvence in the corpus does not prove the manuscript's uniform rank bound. **Boundary:** Normalized-volume/Seshadri calibration, removal of that auxiliary bound, uniform character exponent and conductor-residue descent remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
074/034	Conditional good minimal models for compact Kahler fourfolds	L1	–	The fourfold good-model statement is expressly conditional on orbifold Iitaka, MMP and nonnegative-Kodaira abundance inputs. The reviewed analytic residual tools do not discharge these premises or the empty-divisor argument. **Boundary:** Global minimal-metric reduction, boundary completion and each stated conditional input remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
075/034	Log abundance in characteristic zero	L2	–	The moving-jet and finite Frobenius coefficient identities offer a limited exact-algebra audit target. This does not identify the geometric jet spaces with arbitrary polynomial spaces or prove the global vanishing/nonvanishing implications. **Boundary:** Geometric exclusions, Frobenius fixed-data supply, Hodge lifting, signed descent, minimal models and canonical-ring finite generation remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
076/034	Minimal metrics and interior injectivity for nef adjoints	L1	–	Zero Lelong numbers for minimal adjoint metrics and interior injectivity concern singular global metric envelopes and cohomology. The available local PDE residual bounds have no matched coercive equation, boundary realization or uniform inverse here. **Boundary:** Metric minimality, Lelong-number vanishing, injectivity and the Euler-characteristic abundance consequence remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
077/034	Fourfold nonvanishing by minimal metrics and moving jets	L2	–	The two-slot finite jet maps and finite-input compatibility can be audited coefficientwise, once the geometric spaces and centers have been constructed. This is a peripheral check within the nonvanishing proof. **Boundary:** Moving-center bounds, full nef-dimension geometry, signed representatives and the finite-Frobenius contradiction are not supplied. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
078/034	Lifting sections from the reduced support of an adjoint	L2	–	Residue maps and boundary-graph compatibility at a fixed finite level can be checked as exact cochain/period identities. This does not establish section lifting through all infinitesimal thickenings. **Boundary:** Cyclic-cover geometry, positive-frame Euler argument, filtered-Hodge lifting and asymptotic section growth remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
079/034	Schnell fiber spaces and good canonical models	L1	–	The Schnell-fibration argument is a numerical-geometric reduction using good canonical models and a companion abundance input. No concrete moment, finite operator or analytic-jet task is central to the stated deduction. **Boundary:** Fiber Kodaira-zero section equality, pseudo-effectivity versus base ampleness and the abundance input remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
080/034	Uniform log Iitaka fibrations and bounded moduli denominators	L2	–	Semilinear product-block and pluriweight matrices permit finite exact coefficient checks for fixed degeneration data. That is a limited bookkeeping match, not the dimension-uniform denominator theorem. **Boundary:** Curve MMP, degeneration weight bounds, moduli denominator control and recovery of the field from all section ratios remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
081/034	Uniform Pluricanonical Iitaka Fibrations	L2	–	The diagonal scalar-jet and rational-transport identities can be checked on specified finite data. The corpus cannot promote such checks to the uniform pluricanonical threshold, which depends on geometric control of arbitrarily large indices. **Boundary:** Curve-chain geometry, the uncountably-many-birational-maps contradiction and inductive uniform denominator bounds remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
082/034	Relative denominators and effective systems for log Calabi-Yau fibrations	L1	–	The exact-pullback canonical bundle formula, whole-fiber dlt residues and torsion on rationally connected fibers are global algebraic-geometric inputs, not finite moment reconstruction problems. **Boundary:** Relative denominators, effective base linear systems and torsion uniformity in dimension at most four remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
083/034	Arithmetic Stein-degree bounds for log Calabi-Yau pairs	L1	–	Arithmetic Stein-degree bounds are obtained from geometric boundedness, complements and finite constants fields along boundary strata. No domain-correct operator/moment model for this arithmetic degree bound was identified. **Boundary:** Fano boundedness, boundary induction and control of constants-field extensions remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
084/034	Uniform effective log Iitaka fibrations for fourfolds	L2	–	Fixed scalar-jet matrices and the many-factor rational transports have a finite coefficient-checking interface. This is peripheral to proving a uniform index bound across all fourfold data. **Boundary:** Generic-chain constructions, uniform index completion, geometric Frobenius supply and relative denominator inputs remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
085/034	Abundance after nonvanishing for compact Kahler fourfolds	L2	–	At a fixed cover/thickening, residue and scalar-action compatibility can be checked using exact finite complexes and period rows that separate the relevant classes. The corpus does not supply compatible sections at every order. **Boundary:** dlt adjunction, root-neighborhood splitting, analytic extension, all-order Hodge lifting and semiampleness remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
086/035	Log abundance in numerical dimension one for threefolds in positive characteristic	L1	–	The key issue is torsion on a full nonreduced boundary after separating and isolating the canonical cycle in characteristic p. Finite characteristic-zero jets or numerical rank tests do not establish this torsion. **Boundary:** Coefficient-one isolation, nonreduced boundary torsion and reduction of the abundance obstruction remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
087/035	Abundance in numerical dimension one for terminal threefolds in positive characteristic	L1	–	Periodic cohomology rows on Cartier thickenings and strong tangent semistability in characteristic p drive the proof. Their use of jets is Frobenius/Cartier geometric, not the real/complex Taylor-moment calculus of the reviewed sources. **Boundary:** Prepared nef-cycle normal-line torsion, polar bounds, p-power periodicity and Cartier descent degree contradiction remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
088/036	Numerical semiampleness of nef adjoint classes on compact Kahler manifolds	L1	–	Numerical semiampleness is obtained by algebraic reduction, flat-twist descent and mixed Hodge–Riemann intersection arguments. No specific finite analytic operator calculation was identified that supplies these global steps. **Boundary:** Descent of the nef/flat class, projective adjoint-base construction, cycle restrictions and removal of the negative part remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
089/036	Numerical Semiampleness of Nef Adjoint Divisors	L2	–	The finite two-slot jet kernels can be represented exactly with all mixed coefficients retained. This is a peripheral audit of the contradiction's finite data, not a proof of the geometric rescaling/curve-degree bounds. **Boundary:** Calabi–Yau reduction, bounded-volume versus large-curve estimates, Frobenius contradiction and index-cover descent remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
090/036	Minimal models and Mori fibre spaces for generalized log canonical Q-pairs	L1	–	The local phase theorem is a valuative/birational statement involving cones, Cartier selection, exact rounding and integral adjoint jumps. It is not a finite spectral phase or operator-stability problem. **Boundary:** Independent degree scales, bounded-Fano jump control, trait lifting and the global minimal-model/Mori-fiber conclusion remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
091/036	Minimal models in numerical dimension one	L1	–	Nefness from two surface tests and termination of a fixed-model sequence of K-negative operations are birational/numerical arguments. No matching finite operator or transform calculation was identified. **Boundary:** Positive perturbation control, surface tests and finite contractions/flips in numerical dimension one remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
092/037	The ordinary-double-point gap in every dimension	L2	–	Fixed canonical-jet and saturated-semigroup coefficient data can be checked exactly, with truncation clearly separated from all-degree claims. This is peripheral to the normalized-volume gap and equality classification. **Boundary:** K-semistable cone reduction, small stabilizers, two-filtration bounds, slice bootstrap and equality for the original germ remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
093/037	The normalized-volume gap in dimension four	L2	–	The finite Hilbert-series/weight calculations at a specified cone have a coefficient-checking interface. A finite spectral or moment representation does not establish the pole-frequency gap or its geometric implications. **Boundary:** Uniform stabilizer/discrepancy jet bounds, the Hilbert pole-frequency estimate and terminal-threefold section argument yielding the value 162 remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
094/038	Fujita's freeness conjecture	L2	–	The finite exponential minimization can be audited with bounded analytic jets and rigorous local enclosures after its domain and tails are specified. This is only a peripheral analytic inequality within the freeness proof. **Boundary:** The ordinary-order alternative, removal of positive-dimensional centers, multiplier-ideal isolation and Nadel lifting to the n+1 bound remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
095/039	Nagata's conjecture for plane curves	L2	–	Theta-basis collision jets give a definite finite coefficient/rank-checking task after degeneration data are fixed. Moment unisolvence does not establish the required uniform interpolation exponent for all point configurations. **Boundary:** Incidence symmetrization, degeneration along the cubic, normal-bundle control and the global Nagata inequality remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
096/039	Maximal Seshadri constants on arbitrary polarized surfaces	L3	C3	The source's jet test is literally a full-column-rank finite matrix, and its transfer uses persistence of a nonzero maximal minor under holomorphic scaling. Exact coefficients and the corpus's singular-value-versus-perturbation bound provide a domain-correct quantitative version of this substantial proof step. **Boundary:** Construction of the exponent polygon, lattice compression/interpolation in every required order and deduction of maximal nef thresholds remain independent. **Review:** Statement/proof structure and selected finite-jet persistence/initial-transfer proof read; corpus finite perturbation theorem read; all-order geometric interpolation not audited.
097/039	Maximal Multipoint Seshadri Constants in Higher Dimensions	L3	C3	Monomial coefficient packets, graded-support multiplication and finite jet surjectivity are substantive exact algebraic interfaces. The finite perturbation theorem applies to the adjoint of a full-row-rank jet map, once the analytic-limit remainder is bounded in a common norm. **Boundary:** Complete-intersection flag construction, two-intercept reshaping, interior lattice saturation and a uniform all-large-r threshold require separate geometric proof. **Review:** Actual manuscript introduction/proof structure and selected least-term/finite-jet transfer proofs read; bundled surface companion not substituted for this source; not an all-degree saturation audit.
098/039	Maximal multipoint Seshadri constants in positive characteristic	L2	–	The Hasse-jet evaluation maps have an exact finite coefficient/rank representation, but ordinary factorial Taylor formulas in the analytic corpus cannot be imported in characteristic p. Applicability is limited to characteristic-correct algebraic matrix checks. **Boundary:** Positive-characteristic transfer, flag reshaping, lattice filling and the all-large-r positivity bound remain uncovered; no division by nonunits such as m! is allowed. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
099/040	Bloch's conjecture for surfaces with p_g=0	L1	–	The argument must prove a universal diagonal relation in Chow, using det-fixed Quot obstruction theory and cosection support. A homological period or finite cochain equality is insufficient for rational equivalence of zero-cycles. **Boundary:** Virtual-class construction, nonzero universal coefficient including K^2=9 and the integral Bloch conclusion remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
100/041	Projective-space bases of Lagrangian fibrations	L1	–	Finite cotangent stacks, quotient-stratum Euler obstructions and deformation to a second Lagrangian system drive the base-classification proof. No reviewed operator theorem establishes these stack/geometric steps. **Boundary:** Nonquotient-stratum exclusion, b2-specific deformation arguments and elimination of quotient stabilizers by abelian covers remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
101/041	The strong hyperkahler SYZ conjecture	L2	–	Metric-estimate subproblems could use residual and inverse bounds only after the exact Ricci-flat/special-Lagrangian equation, gauges and function spaces are fixed. This is a peripheral local verification interface, not a direct match to the global SYZ construction. **Boundary:** Degenerating torus-scale uniformity, flat-cylinder special Lagrangian torus existence, rotation and semiampleness are not supplied. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
102/042	Every complex K3 surface is Oka	L3	C3	The open-overlap gluing step actually uses bounded splitting operators and a supremum-norm contraction for the nonlinear transition defect. The Banach-ball and error-composition framework is a concrete match, with base/fiber margin losses retained, to a significant analytic part of the Oka argument. **Boundary:** Construction of the local operation L on every K3, uniform buffered splitting constants, degeneration/sewing and globalization in every source dimension still require their own arguments. **Review:** Statement/proof structure and buffered-overlap splitting plus nonlinear gluing proof excerpt read; corpus full Banach-ball hypotheses read; global K3/Oka proof not audited.
103/043	P=W in composite rank for fixed determinant	L2	–	The common cohomology operator and double-commutator identities can be audited on a specified finite complex while preserving grading and all class-separating period rows. That is a peripheral finite algebra check within P=W. **Boundary:** Betti curious hard Lefschetz, logarithmic family restriction, specialization and bounds identifying the full perverse and weight filtrations remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
104/044	The equivariant cohomological Hikita conjecture for arbitrary quivers	L2	–	The explicit Laurent coefficient ideals and multipartition Hilbert counts permit exact finite-degree algebra checks. They must retain fixed-scheme nilpotents; a reduced numerical rank test would miss the stated equivariant scheme structure. **Boundary:** Stable-pair test completeness, flavor/deformation loop removal and equality of the full cocharacter fixed scheme with the cohomology presentation remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
105/046	A projective fourfold with large fundamental group and non-Stein universal cover	L1	–	The non-Stein universal-cover construction relies on arithmetic arrangements, a local hyperbolic-path criterion and projective realization. Finite graph reconstruction does not establish the topology or holomorphic-convexity obstruction. **Boundary:** Cyclic image of surface fundamental groups, arrangement realization and the global non-Stein property remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
106/046	A surface counterexample to Shafarevich holomorphic convexity	L1	–	The weighted pro-2 Golod–Shafarevich quotient and paired-puncture surface construction require profinite group and global complex geometry. No concrete finite moment/operator result supplies the infinite-quotient or noncompact-hull argument. **Boundary:** Infinitude of the pro-2 quotient, family pinch/quotient realization and infinite fundamental-group image of the rational-fiber graph remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
107/047	An explicit failure of complex affine-space cancellation	L3	C3	The rigidity proof uses locally nilpotent derivations, homogeneous components and localization in polynomial charts. The corpus has an exact characteristic-zero local-slice analysis/synthesis and a separate saturation test, so it supports substantive invariant-ring/localization checks rather than just generic coordinates. **Boundary:** Lifting the hypothetical affine-space action, the polynomial-abc rigidity obstruction and the cylinder/noncancellation conclusion still need the source proof; a localized slice alone cannot settle the exceptional locus. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
108/048	A singular normal affine surface with free tangent sheaf	L2	–	The finite vector-field jet-gluing equations and residue/trace coefficients are accessible to exact coefficient and rank checks. The analytic contraction and algebraization are separate geometric issues, so the match remains peripheral. **Boundary:** Curve construction, analytic contraction, passage to an algebraic normal singularity and freeness of the completed tangent module remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
109/049	A stable coordinate that is not a coordinate in four variables	L3	C3	This rigidity argument concretely computes localized LND invariants on the a and d charts and then intersects the charts to remove denominators. That is closely aligned with the corpus's local-slice and saturation distinction and supports a substantial portion of the invariant-theoretic obstruction. **Boundary:** The graded-degeneration lift, implication from being a coordinate, and stable automorphism construction are separate; no arbitrary LND can be assumed to possess the required slice globally. **Review:** Statement/proof structure and full rigidity/localization proof read; corpus local-slice and saturation proofs read; construction and degeneration lift not fully audited.
110/049	An explicit noncoordinate polynomial with affine three-space zero fibre	L2	–	The quotient isomorphism, Bezout identity, explicit inverse and critical-point calculation are finite polynomial identities and can be represented exactly. This covers many checkable expressions, but the available Ram contribution is generic coefficient/jet algebra; it is not a specialized new method for the core embedding problem. **Boundary:** The global ring-isomorphism and Jacobian obstruction remain the source's algebraic proof; finite moment coordinates add no theorem that upgrades a local or approximate identity to a global coordinate result. **Review:** Full mathematical source, including lifting lemma, main proof and explicit-coordinate appendix, read; symbolic identities not independently executed; not a validation claim for the preprint.
111/050	Ample rank-two bundles on the quadric surface without Griffiths-positive metrics	L2	–	At the finite stage the two generator matrices, positive quadratic-form normalization and sampled power-pullback moments can be checked exactly with a shared measure. This is a peripheral consistency aid to the metric-limit obstruction. **Boundary:** Global ampleness, extraction of the invariant limit measure and incompatibility of expanding AB with a common fixed measure require independent proof. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
112/051	Canonical ampleness of compact hyperbolic Kahler manifolds	L2	–	The frame-variation and determinant-Hessian calculations have a local analytic-jet interface, with full coefficient derivatives and controlled smoothing errors. That does not establish a positive global curvature/intersection bound from finite samples. **Boundary:** Extremal-disc regularity, bounded global potentials, nefness and the uniform volume-to-ampleness argument remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
113/052	Universal-cover splitting for compact Kahler manifolds	L1	–	Global splitting is obtained from Hartogs boundary crossing and continuation of path rectangles, without assuming a parallel metric. A local block-diagonal operator or product-coordinate chart is not a matched proof of that global continuation. **Boundary:** Finite-grid nonexpansion, boundary crossing, simply connected globalization and the canonical nef/big case remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
114/052	Integrability of split tangent bundles on rationally connected manifolds	L2	–	The formal diagonal transport equation d_v F=P2(F)d_u F admits coefficientwise jet-recursion checks with the full nonlinear derivative retained. This is a peripheral local step before the rational-geometric argument. **Boundary:** Positive normal-bundle bounds, rationality of coordinate ratios, removal of base points and algebraic P1 x P1 boxes forcing bracket vanishing remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
115/053	A high-arity counterexample to Pixton completeness in Chow	L2	–	The finite Frobenius-algebra detector and symplectic tensor contractions can be checked as exact multiplication tensors and coefficient identities. This is a genuine finite subproblem, but it does not demonstrate a relation in the Chow ring. **Boundary:** Exclusion from the full Pixton span at huge arity, Quot relation construction, compactification and support induction proving Chow-zero remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
116/054	Irrational cubic fourfolds with geometric K3 categories	L2	–	Specified transcendental-lattice and correspondence matrices can be checked exactly, with rational span and integral discriminant information kept distinct. This is a peripheral audit of finite data, not a birational irrationality criterion. **Boundary:** Gromov–Witten factorization, divisorial birational cocycles, conic-bundle correction and uniform link/discriminant bounds remain uncovered; approximate periods cannot certify integral lattice equivalence. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
117/055	A Gepner stability condition on every smooth quintic threefold	L2	–	The numerical central charge and periodic functor action admit finite exact matrix/sign checks on the stated lattice. These are peripheral to constructing the heart and slicing, which require control of every relevant object. **Boundary:** Double-tilt aisle inclusions, support inequality, discrete phase grid and finite Harder–Narasimhan filtrations are not supplied. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
118/055	Prescribed large-volume charges on threefolds with trivial canonical bundle	L2	–	Finite charge matrices and proposed quadratic-support forms can be checked exactly for specified classes. The manuscript's uniform apex inequality ranges over objects and twists, so finite positivity tests cannot stand in for its main argument. **Boundary:** Cohomological restriction bounds, all-object/all-twist apex control, wall propagation and openness of the required heart cone remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
119/056	Termination of generalized log canonical flips on compact Kahler fourfolds	L1	–	The termination proof depends on terminal junctions, low-discrepancy places, branch topology and a descending difficulty. Its surface intersection matrices encode geometry not covered by a generic moment or spectral positivity theorem. **Boundary:** Uniform extraction of places, branch inequalities, difficulty descent and the klt-to-lc completion remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
120/056	Termination of generalized-canonical flips on compact Kahler fourfolds	L1	–	Strict discrepancy increase, blowup witnesses and loss of surface classes supply the generalized-canonical termination mechanism. No concrete reviewed operator/moment theorem gives that finite descent. **Boundary:** The equality-one discrepancy case, terminal smooth flipped surfaces, witness persistence and finiteness of descending boundary data remain uncovered. **Review:** Source statement/introduction and proof-section structure screened; selected corpus statements and boundaries checked (map-mediated references identified in evidence_files); not a full proof audit.
121/056	Termination for projective log canonical fourfolds with rational boundary	L1	–	The claimed termination of every projective log-canonical fourfold MMP uses cycle ranks, branch topology and discrepancy descent. No concrete Ram result located supplies these birational termination invariants. **Boundary:** Local operator inversion or polynomial coordinates do not bound flip sequences. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
122/056	Finite ordinary minimal model programs on compact Kahler fourfolds	L1	–	This version claims termination of every maximal ordinary Kaehler fourfold program, using global Weil Q-factoriality and real-boundary arguments. No matching Ram termination or cone theorem was located. **Boundary:** The all-program quantifier is stronger than the existence statement in 123; neither is obtained by finite matrix closure. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
123/056	Finite ordinary minimal model programs on compact Kahler fourfolds	L1	–	The September Kaehler manuscript constructs a terminating ordinary program through scaling and limiting terminal models. The screened corpus supplies no concrete step in this birational existence argument. **Boundary:** Do not identify existence of one program with termination of all programs in 122. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
124/057	The abelianity conjecture for special compact Kahler manifolds	L1	–	The target is virtual abelianity of the entire fundamental group of a special compact Kaehler manifold, through tensor, spectral-cover and Albanese arguments. No concrete Ram bridge was found. **Boundary:** Finite spectral coordinates do not control the global fundamental group or specialness. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
125/057	Two-step monodromy of special quasi-projective varieties	L1	–	The quasi-projective statement concerns complex linear images of the fundamental group, with a nilpotency-class bound and mixed-Hodge/unipotent structure. No relevant Ram group-theoretic or Hodge-theoretic lemma was located. **Boundary:** This is not the entire-group virtual-abelianity assertion of 124. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
126/057	A conditional abelianity theorem for special fourfold pairs with a half-weight divisor	L1	–	The conditional orbifold reduction builds a higher-dimensional manifold to handle order-two root data. The local corpus does not supply the required specialness, covering or orbifold descent arguments. **Boundary:** Finite cohomology transport is not an orbifold-to-manifold reduction theorem. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
127/058	Semialgebraic universal covers of normal projective varieties	L1	–	The semialgebraic universal-cover classification relies on global quotient/window constructions and the product structure of the cover. No matching Ram classification result was found. **Boundary:** A local coordinate representation cannot establish global covering-space or algebraic structure. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
128/058	Symmetry of semialgebraic bounded domains with compact quotient	L1	–	The bounded semialgebraic affine-domain result uses boundary scaling to force a symmetric domain under a compact quotient. No concrete Ram boundary-rigidity input was located. **Boundary:** Generic analytic transforms do not provide boundary scaling limits or symmetry. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
129/059	Ambiently homeomorphic isolated hypersurfaces of multiplicities two and three	L1	–	The isolated weighted-homogeneous singularity argument uses integral Seifert lattices, tensor constructions and local/global lattice comparison to distinguish multiplicities. No directly applicable Ram singularity theorem was found. **Boundary:** A finite bilinear-form representation alone does not establish link topology or integral lattice equivalence. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
130/059	Ambiently homeomorphic isolated hypersurface germs in C4 with multiplicities four and five	L1	–	The surface-germ construction with different multiplicities depends on cone-tube topology, Bockstein data and integral lattice fillings. No specific Ram topological construction was located. **Boundary:** The integral and topological constraints are not supplied by moment or Gram coordinates. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
131/060	Global Spherical Shells on Minimal Surfaces of Class VII	L1	–	The class-VII global-spherical-shell argument proceeds through weighted Dolbeault analysis, cyclic covers and residues producing curves. No Ram theorem found addresses that curve-production step. **Boundary:** Boundary-aware PDE bookkeeping would not establish the global shell or surface classification. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
132/062	Contact Fano manifolds and the LeBrun-Salamon conjecture	L1	–	The contact-Fano classification uses families of rational curves, smoothing and tensor-jet geometry to reach an adjoint variety. No concrete Ram contact-geometric input was found. **Boundary:** The word jet here does not identify a finite Ram operator calculation with the required geometric argument. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
133/063	The generalized Mukai conjecture	L1	–	The generalized Mukai inequality is approached through quantum divisors, descendants and spectral/trace constraints. No applicable Ram theorem supplies the extremal-ray or equality classification. **Boundary:** Finite trace identities do not imply the claimed Fano classification or Picard-number bound. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
134/064	Topological triviality of mu-constant families of surface singularities	L1	–	The mu-constant surface-family argument needs semistable/logarithmic boundary control and an Euler-gradient trivialization. No matching Ram equisingularity mechanism was located. **Boundary:** A local differential representation does not prove a global right-trivialization of the family. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
135/065	Virasoro Constraints under Projectivization	L1	–	Virasoro for projectivizations of arbitrary bundles is derived through ancestor theory, master-space localization and reconstruction of blocks. No concrete Ram localization or Gromov-Witten theorem was found. **Boundary:** Generic operator or spectral terminology is insufficient for relevance to virtual-cycle identities. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
136/065	Virasoro Constraints for Projective Complete Intersections	L1	–	The complete-intersection Virasoro claim includes primitive and odd sectors, handled through cap detection and sewing. The corpus screen found no tool for those enumerative identities. **Boundary:** Do not substitute ordinary finite tensor algebra for the primitive/odd reconstruction argument. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
137/066	Bounded klt complements for Fano contractions	L1	–	Bounded klt complements are approached through normalized volumes and rank-growth/trace-lifting constructions. No Ram boundedness or complement theorem was located. **Boundary:** Finite rank calculations alone do not give uniform algebraic-geometric bounds. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
138/066	Uniform Cartier sections for Fano type contractions	L1	–	The uniform Cartier-threshold argument uses signed phases, cutoff fields and log-canonical slices. No specific Ram result supplies the uniform Cartier index or threshold control. **Boundary:** The appearance of phases and fields is not an integral-transform applicability bridge. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
139/067	The Campana-Peternell conjecture in dimension six	L1	–	The six-dimensional Campana-Peternell proof outline uses rational curves, characteristic classes and arithmetic comparisons. No concrete Ram nef-tangent classification input was found. **Boundary:** Exact finite arithmetic could check an isolated identity but does not establish the geometric reduction. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
140/068	Anticanonical nonvanishing from smooth semipositivity	L1	–	The smoothly semipositive anticanonical nonvanishing argument uses metric/finite-volume structure and induction to produce sections. No Ram result located provides this section-existence mechanism. **Boundary:** Positivity of an analytic model is not algebraic effectivity. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
141/068	Invariant anticanonical indices and conversion of twisted differentials	L1	–	This anticanonical variant relies on an invariant Euler polynomial and conversion of a fixed pseudoeffective error. No concrete Ram theorem addresses that conversion. **Boundary:** Finite polynomial representation does not yield a nonzero invariant section. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
142/068	Bounded anticanonical metrics on klt pairs and torus quotients	L1	–	The klt bounded-metric version uses nonzero Euler characteristic and stable torus-quotient structure. No matching Ram theorem was found for the required geometric descent. **Boundary:** Operator conditioning does not replace the quotient or nonvanishing argument. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
143/068	Cohomological transfer and equivariant anticanonical sections	L1	–	The arbitrary-linearization version transfers invariant sections through harmonic/cohomological fields. No Ram transfer theorem was found with the necessary equivariant algebraic hypotheses. **Boundary:** Finite observable transport is not invariant-section descent. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
144/068	Metric descent and rank-preserving contractions	L1	–	The integrable-metric version depends on corrected anticanonical base descent and rank preservation. No concrete Ram lemma was located for this positivity-to-effectivity step. **Boundary:** An integrable analytic representation does not prove the algebraic descent statement. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
145/068	Integrable metrics and effectivity with controlled boundary	L1	–	The strict-curvature integrable-boundary argument uses a fixed-error passage to effectivity. No Ram theorem in the inspected material supplies that passage. **Boundary:** Curvature strictness and the existence of a global section remain distinct obligations. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
146/068	Exact orders and invariant anticanonical linear systems	L1	–	The zero-valuation variant requires attainment on a curvature-null face and finite generation of invariant sections. No concrete Ram result establishes those hypotheses. **Boundary:** A limiting zero residual is not attainment, finite generation or nonvanishing. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
147/069	Global quantum geometric Langlands at irrational level	L1	–	The actual target is an irrational-level DG-category equivalence with localization-Whittaker, collision and central-torsor compatibility. The finite Hitchin/intertwiner themes in the Ram routing material do not identify a usable step in that categorical construction. **Boundary:** Finite Fourier-Mukai surrogates or matrix intertwining do not produce this equivalence for all global forms. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
148/071	Removable Boundaries and Rigidity of Circle Domains	L3	C3	The rigidity proof explicitly imports the companion's retained-test model construction and must exclude nonpoint omitted components before applying removability. The Ram circle-domain report supplies a concrete all-component compactness and image-identification framework for this substantial intermediate step, although rigidity itself is a different target. **Boundary:** The corpus does not supply the retained-test selection estimate, Sobolev removability or the final flux equality. No finite cycle bank or checked disk list covers arbitrary connectivity automatically. **Review:** manuscript: Full introduction, including the precise removability-to-rigidity theorem and its dependency on the companion; proof-file headings screened. Corresponding corpus compactness/image-identification statements and proofs and period/noncollapse section read.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
149/071	Koebe's Circle-Domain Conjecture	L5	C3	Direct core-problem match: both the manuscript and TR-item-343 target circle-domain uniformization of arbitrary sphere domains, including arbitrary connectivity. The manuscript's retained-test/period construction and component identification map concretely to the corpus's period, inverse-containment, noncollapse and all-component compactness obligations. **Boundary:** L5 denotes the same core problem and a concrete domain-correct mapping, not a proof supplied by Ram. The corpus leaves universal construction/compactness hypotheses open. The manuscript's transfer, barrier and fusion arguments have not been fully verified. **Review:** manuscript: Full introduction and 02-transfer read; remaining proof architecture screened by headings. Corpus target, periods/noncollapse, image-identification theorem/proof, all-component compactness theorem/proof and conditional assembly read.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
150/072	Brennan's conjecture and sharp inverse-square integral means	L5	C3	Direct match to the Brennan p=-2 endpoint of the integral-means problem in TR-item-077. The manuscript's reciprocal-derivative transfer operator, weighted affine law and logarithmic-derivative moments are concrete objects for the corpus's normalized log-partition, dyadic-scale and analytic-error framework. **Boundary:** The manuscript ranges over all normalized schlicht maps; the corpus pins the bounded-univalent spectrum, which is a subclass and must not be silently exchanged at positive exponents. Ram's conditional cumulant/variance criteria do not supply the critical-point transport estimate or validate the claimed endpoint. **Review:** manuscript: Introduction read in full; selected weighted-law and moment proof passages and the full integrability section read; other proof sections structurally screened. TR-item-077 sections 1-12 read, including fixed-function and full-tail cautions. No complete transport-proof audit.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
151/072	A strict inverse-first-power bound for univalent functions	L5	C3	Direct match to the interior p=-1 value of the very same bounded-univalent integral-means spectrum in TR-item-077. The manuscript claims B_b(-1)<1/4, contrary to the quadratic value discussed in that report, and makes the uniform reciprocal-derivative operator growth and equality case explicit. **Boundary:** Direct relevance does not mean agreement with the corpus's conjectural formula. The claimed strict inequality would change that formula, not follow from it. The corpus does not provide strict transport or a positive exponent gap; neither the disproof claim nor historical novelty was certified. **Review:** manuscript: Full introduction and full harmonic equality-barrier/final theorem proof read; operator/law/pairing/transport architecture screened, not all its proofs. TR-item-077 sections 1-12 read.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
152/073	The Falconer distance conjecture in all dimensions	L1	–	The all-dimensional Falconer argument requires angular projection gains, oscillatory distance-shell bounds and a multiscale recursion. No inspected Ram theorem supplies these incidence or dimensional gains. **Boundary:** Merely rewriting a shell integral as a transform is not concrete progress on the distance-set estimate. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
153/074	The Kakeya maximal conjecture in three dimensions	L1	–	The three-dimensional Kakeya maximal argument uses weighted critical configurations, planar filling and two-frame entropy projections. No matching Ram geometric estimate was located. **Boundary:** The planar Lipschitz-field problem in TR-item-355 is a different operator and dimension. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
154/074	Every four-dimensional Kakeya set has full Hausdorff dimension	L1	–	The four-dimensional Hausdorff-dimension claim relies on isotropic/horizontal improvement, fitted algebraic sheets and unbounded-narrowness iteration. No Ram result was found with these scale and incidence quantifiers. **Boundary:** Finite polynomial fits alone do not prove the Kakeya gain. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
155/075	Almost-everywhere Fourier convergence in L log L	L1	–	Almost-everywhere Fourier convergence in L log L is reduced to entropy compression, thermal labels and endpoint summation. No Ram theorem found supplies the endpoint maximal estimate. **Boundary:** Finite Fourier reconstruction or Parseval identities do not control full-sequence almost-everywhere convergence at this endpoint. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
156/076	Ultraflat real Littlewood polynomials	L3	C3	The claimed two-sided ultraflat real-sign polynomials directly address a substantial flat-polynomial module in the Ram Banach-spectrum program: normalized moduli, logarithmic products and correlation control. This is a concrete polynomial/spectral connection, not just a Fourier title match. **Boundary:** Real coefficients of both signs are not the positive coefficients of ordinary rank-one towers. Ultraflat factors alone neither realize a probability-preserving Koopman operator nor establish cyclic Haar spectrum. The corpus's positive-coefficient obstruction is not an objection to signed Littlewood polynomials. **Review:** manuscript: Main ultraflatness theorem excerpt and construction/rounding proof headings screened; no full oscillatory or rounding audit. Corpus block/completion passages and the positive-coefficient, log-product and moving-window corrections read.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
157/076	Nearly minimal maxima and positive minima of Littlewood polynomials	L3	C3	The nearly minimal upper modulus together with a fixed positive lower modulus provides nonvanishing flat-factor control relevant to the Ram spectral-product module. Its quantitative content is distinct from the two-sided (1+/-epsilon) assertion of 156. **Boundary:** A fixed lower bound and an asymptotically optimal upper bound do not automatically give the corpus's summable logarithmic deviations, uniform integrability or cyclicity. Signed polynomials are not positive tower polynomials. **Review:** manuscript: Main theorem excerpt and sampling/correction/rounding structure screened; not a full proof review. Corpus flatness obstruction, replacement product criterion and physical cyclicity statements/proofs read.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
158/076	Asymptotically minimal maxima of real Littlewood polynomials	L3	C3	The upper-modulus theorem gives autocorrelation/merit-factor consequences and an explicit Morse-shift spectral application, making its connection to the Ram Banach-spectrum program concrete. The source carefully asserts a nonzero absolutely continuous component, not the full simple Lebesgue target. **Boundary:** Upper flatness alone does not control deep zeros. Simple full Koopman spectrum plus absolute continuity on one cyclic subspace is not simple Lebesgue spectrum on the whole mean-zero space. No transfer of the companion's stronger dynamical claim is assumed. **Review:** manuscript: Main theorem excerpt and proof headings screened; full merit-morse.tex read. Relevant corpus completion, log-product and moving-window correction passages read; main polynomial construction not fully audited.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
159/077	Elliptic capacity propagation and Fourier restriction to the sphere	L1	–	The spherical restriction proof requires elliptic packet capacity propagation and exclusion of calibrated entropy paths. No concrete Ram result supplies that geometric propagation exponent. **Boundary:** Re-encoding a packet frame or its Gram matrix does not yield the required p>3 bound. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
160/077	Diagonal Fourier extension for positively curved surfaces in three dimensions	L1	–	The curved-surface extension result imports a packet-capacity theorem and performs density, sparse-ball and global exponent transfers. No Ram theorem found replaces those quantitative inputs. **Boundary:** A Fourier transform identity does not establish uniform extension bounds over the stated phase class. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
161/078	Bochner-Riesz multipliers in three dimensions	L1	–	The three-dimensional Bochner-Riesz endpoint route depends on packet uncertainty and repayment estimates with frequency summation. No inspected Ram theorem gives that oscillatory saving. **Boundary:** Finite spectral computation does not control all frequency scales or every positive summability order. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
162/079	Critical local smoothing for the three-dimensional wave equation	L1	–	The three-dimensional wave local-smoothing proof uses cylinder entropy, cone geometry and physical-frequency reassembly. No applicable Ram cone estimate was located. **Boundary:** Local moment calculus does not remove the local-smoothing derivative loss. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
163/080	Endpoint convergence for the planar Schrodinger equation	L1	–	The two-dimensional Schrodinger endpoint argument requires a Gaussian-first maximal bound at H^(1/3), all-real-time control and one common full-measure set. No concrete Ram theorem covers those quantifiers. **Boundary:** Fixed-band approximations and dense-time tests do not establish the claimed endpoint or common exceptional set. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
164/080	Endpoint pointwise convergence for the Schrodinger equation in higher dimensions	L1	–	The higher-dimensional Schrodinger endpoint uses dimension-dependent angular/frame estimates at H^(n/(2(n+1))). No Ram endpoint gain was located. **Boundary:** The higher-dimensional all-scale maximal estimate is not furnished by the two-dimensional or finite-frame identities. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
165/081	Riesz transforms and uniform rectifiability in higher codimension	L1	–	The Riesz-transform/uniform-rectifiability argument relies on reflectionless-measure rigidity, fractional energy and packing. No relevant Ram rigidity theorem was found. **Boundary:** Generic singular-integral representations do not imply quantitative rectifiability in higher codimension. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
166/082	Annular variation of the triangular Hilbert transform at the symmetric point	L1	–	This is a bilinear triangular Hilbert r-variation estimate with disjoint-annulus control and matrix-energy dissipation. No Ram result located applies to that entangled multilinear form. **Boundary:** It is not the linear frozen-direction Hilbert transform in TR-item-355 or manuscript 169. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
167/082	An L3 bound for the dyadic triangular Hilbert form	L1	–	The dyadic triangular form claims an absolute L3 trilinear bound via matrix-curvature splitting. No matching Ram dyadic entangled-form estimate was found. **Boundary:** A positive matrix representation is not an estimate for the full triangular form. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
168/082	The maximal triangular Hilbert transform at the symmetric point	L1	–	The hard two-endpoint maximal triangular estimate uses masked matrix heat energies and jump control. No concrete Ram maximal entangled-form theorem was located. **Boundary:** Neither the dyadic absolute bound nor the planar linear directional report supplies this maximal assertion. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
169/083	A uniform Hilbert transform estimate for Lipschitz directions	L5	C3	Exact core operator/model match to TR-item-355: the planar Hilbert transform samples frozen straight lines in a Lipschitz direction field at an outer scale proportional to the reciprocal Lipschitz seminorm. The manuscript's uniform inner-cutoff and band-reassembly claims directly meet the unresolved quantifiers isolated in that report. **Boundary:** The Ram positive results cover restricted difference/Sobolev domains and frequency-dependent bounds, not uniform all-L2 control. The manuscript puts the supremum outside the norm and gives principal values on Schwartz inputs plus an L2 extension; do not upgrade it to a maximal-truncation theorem or almost-everywhere principal values for every L2 function. **Review:** manuscript: Full introduction and first 260 lines of analytic reductions read, including operator, endpoint-change and restricted-to-strong arguments; remaining hierarchy proof structurally screened. Corpus target, pullback/band/difference estimates and global-gap criteria read. Full all-scale proof not audited.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
170/084	The geometric case of the Erdos similarity conjecture	L1	–	The affine-avoidance construction for fixed geometric ratios uses periodic random routing and high-measure sets. No Ram similarity-avoidance construction was located. **Boundary:** Transforming indicators does not produce the required simultaneous affine exclusions. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
171/084	The dyadic case of the Erdos similarity conjecture	L1	–	The earlier dyadic-ratio avoidance statement uses a restricted geometric sequence and its own routing construction. No concrete Ram step was found. **Boundary:** This restricted-ratio statement is not the all-fixed-ratio extension in 170. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
172/085	An Endpoint Gradient Bound for the Centered Disk Maximal Operator	L1	–	The centered disk maximal-function W^(1,1) argument needs signed finite-radius labels, prices and cancellation to control the gradient. No matching Ram endpoint maximal estimate was found. **Boundary:** Exact moments or affine cancellation identities alone do not control the supremum over radii. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
173/086	An L3 bound for the trilinear Hilbert transform	L1	–	The slopes-1,2,3 trilinear Hilbert transform uses quadratic charts, U3 detection and nonlinear counting. No Ram estimate matching this multilinear operator was found. **Boundary:** Do not confuse it with either the triangular planar form or Stein's linear Lipschitz-direction transform. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
174/087	The symmetric Mahler conjecture and its equality cases	L1	–	The symmetric Mahler bound and Hanner equality classification depend on holomorphic-lens and simplex-feasibility geometry. No concrete Ram volume-product proof step was located. **Boundary:** The corpus's spectral Mahler problem is different; the simplex isotropic bound has different extremizers and constants. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
175/087	The Mahler Conjecture for General Convex Bodies	L1	–	The nonsymmetric Mahler argument uses biased projections and a Gaussian-entropy comparison. No inspected Ram theorem supplies its sharp volume-product or equality argument. **Boundary:** The related implication from sharp isotropic constants does not make moment-coordinate identities a proof of this independent route. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
176/087	Symplectic Balls in Symmetric Polar Products	L1	–	The symplectic-ball construction in a polar product aims at width four and then symmetric Mahler. No Ram holomorphic or symplectic realization theorem was found. **Boundary:** Finite analytic coordinates do not establish a symplectic embedding or a sharp capacity. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
177/088	Petty's projection-volume conjecture in dimensions at least four	L1	–	Petty projection-volume minimality is pursued through spherical gauge variations and a sign estimate. No Ram theorem located provides this projection-body variational inequality. **Boundary:** A general moment or optimization certificate is not an all-body projection inequality. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
178/088	A product counterexample to the simplex maximum for projection-body volume	L1	–	The product-of-simplices counterexample uses an explicit finite combinatorial/projection-volume calculation. No Ram-specific operator, error certificate or geometric result is needed for the displayed route. **Boundary:** Elementary exact arithmetic alone is not sufficient to raise applicability. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
179/089	Planar Graph Metrics Embed into L1 with Constant Distortion	L1	–	The planar-graph L1 embedding theorem needs globally compatible cut/partition fields and anchor interpolation. No matching Ram metric-embedding theorem was located. **Boundary:** Graph spectral filters do not furnish L1 cut decompositions with uniform distortion. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
180/089	L1 Embeddings of Graphs of Bounded Treewidth	L1	–	The bounded-treewidth embedding uses cut-law transport and reservations across a tree decomposition, uniformly over depth and edge ratios. No concrete Ram embedding lemma was found. **Boundary:** Finite-state representation alone does not give depth-independent distortion. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
181/090	An atomic certificate for triangular-lattice universal optimality	L3	C3	A significant part of the atomic Gaussian certificate is an actual l1 finite/tail operator problem plus whole-parameter polynomial signs. Ram's bounded-inverse and exact-enclosure methods apply to those explicitly identified analytic certificate steps, independently of generic Gram representations. **Boundary:** The corpus does not supply the atomic columns, numerical bounds, Fourier minorant or global energy comparison. Its conditional Neumann principle must be instantiated with the manuscript's true l1 realization and complete tails. Numerical certificate programs were not executed. **Review:** manuscript: Main statement and complete section structure screened; full 'The full infinite correction' section read, including operator norms and residual propagation. Corpus infinite-inverse proposition/proof and Bernstein conversion/hull/subdivision proofs read; sign and certificate tables not replayed.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
182/090	Universal optimality of the triangular lattice	L3	C3	The earlier triangular-lattice proof has a distinct exact-interpolation module: genuine spectral integrals replace a finite quadrature block, followed by a bounded inverse on paired l1 lists. This matches Ram's explicit operator-realization, perturbation and complete-tail methodology for a substantial construction step. **Boundary:** The sine-cardinal construction, uniform kernel bounds, Schwartz realization and all-configuration/renormalized-energy transfer remain manuscript-specific. Finite invertibility alone does not imply full-space invertibility or uniqueness among all radial Schwartz functions. **Review:** manuscript: Actual introduction in 01-uniform-gaussian-theorem.tex and proof headings screened; full 04-exact-interpolation.tex read. Corpus inverse and Bernstein sections read. Gaussian/energy-transfer proofs and finite certificates not fully audited or run.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
183/090	A sharp Fourier certificate for planar circle packing	L3	C3	The sharp packing certificate is built from a concrete l1 interpolation operator with two Schur eliminations and exact whole-interval sign bounds. Ram's complete-tail inverse and Bernstein enclosure machinery applies to this significant analytic certification module. **Boundary:** Ram does not construct the sharp auxiliary function or certify its Fourier/global signs from a finite Gram matrix. The packing bound and periodic equality classification still require the manuscript's normalization and geometric argument. No interval program was replayed. **Review:** manuscript: Main theorem/introduction excerpt and proof headings screened; full interpolation.tex read. Corpus infinite-inverse and Bernstein sections read. Certificate/sign tables, Fourier construction and normalization not fully audited.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
184/090	Triangular minimality for planar Coulomb renormalized energy	L2	–	There is a concrete peripheral application to the finite scalar verification: polynomial/interval enclosures can audit the compact-rectangle inequalities after the manuscript has enclosed its infinite-mode functions. The Voronoi dual-potential construction and whole-plane energy reduction are not supplied by Ram. **Boundary:** The Bernstein mechanism is an alternative enclosure tool for suitable polynomial bounds, not a certificate for the raw infinite Green series. It does not establish matching traces, the sector inequality, periodic approximation or the renormalized-energy minimum. **Review:** manuscript: Main theorem and full proof headings screened; first 180 lines of 06-computation.tex read, including outward arithmetic, derivative jets and table initialization. Corpus Bernstein proof read; geometric and infinite-mode proofs not fully audited.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
185/091	The logarithmic Brunn-Minkowski conjecture	L1	–	The logarithmic Brunn-Minkowski claim needs an even whole-space variance/divergence inequality in moment coordinates. No inspected Ram theorem supplies the sharp convex-geometric inequality. **Boundary:** The use of moment coordinates or a symmetric tensor does not establish the universal logarithmic inequality. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
186/092	A single-lattice covering bound of order n log n	L1	–	The single-lattice covering proof uses folded Gaussian hole estimates and a branch/shear construction to reach order n log n. No concrete Ram covering result was located. **Boundary:** One lattice covering is stronger than an arbitrary finite union of translated lattices; generic Gaussian moments do not give it. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading. The actual theorem and proof overview in overview.tex were separately read after the first extraction selected the main-file preamble.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
187/092	Translative covering densities of order n log n	L1	–	The translative covering lower bound is a random-cap and Poisson-pattern obstruction for selected symmetric bodies. No Ram uniform covering-obstruction estimate was found. **Boundary:** This lower-bound construction is distinct from the single-lattice upper bound in 186. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
188/093	A dimension-free logarithmic Sobolev inequality for subgaussian log-concave measures	L1	–	The dimension-free log-Sobolev argument uses regularization, diffusion eigenfunctions and a covariance-information estimate. No Ram theorem located supplies the dimension-free entropy-energy constant. **Boundary:** A finite matrix or posterior-moment identity does not establish log-Sobolev control for every subgaussian log-concave law. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
189/094	Subpolynomial dimension reduction in Lp	L1	–	The nonlinear same-exponent Lp reduction uses normalized increments, signed Poisson sampling and row projection. No applicable Ram dimension-reduction theorem was found. **Boundary:** Generic finite-rank reconstruction cannot give subpolynomial embedding dimension at fixed distortion. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
190/095	Hyperbolicity Cones Without Semidefinite Lifts	L1	–	The no-semidefinite-lift result uses local non-SOS obstructions for an explicitly hyperbolic cone. No inspected Ram certificate proves impossibility of every finite lift. **Boundary:** Failure of a particular Gram representation or rational certificate is not nonexistence of all real affine semidefinite lifts. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
191/095	A nonspectrahedral hyperbolicity cone	L1	–	The actual theorem rules out every finite homogeneous symmetric pencil in the original coordinates for its explicit hyperbolicity cone. No matching Ram obstruction theorem was located. **Boundary:** Original-coordinate nonspectrahedrality differs from nonexistence of lifts. The read source states degree 20, whereas the screening abstract says 16; the source statement governed this screen. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
192/095	An Exact Semidefinite Lift of a Nonspectrahedral Hyperbolicity Cone	L1	–	The construction gives a size-100 pencil with 307 auxiliary variables and explicitly covers singular-X boundary points. No concrete Ram construction of this lift was found. **Boundary:** A generic Schur complement valid only for invertible X would miss the theorem's boundary assertion; this does not contradict 191. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
193/096	The Gaussian propeller bound in every dimension	L2	–	The Gaussian propeller proof contains explicit one-variable polynomial enclosures of exponential and Gaussian integrals used in its numerical inequalities. Ram panel positivity can independently check these small scalar comparisons, but does not supply the centroid competition or elimination of extra cells. **Boundary:** This is scalar-certificate applicability only. Moment coordinates do not prove the sharp 9/(8*pi) partition bound, classify extremizers or establish the kernel-clustering reduction. **Review:** manuscript: Main theorem and proof headings screened; first 135 lines of certificates.tex read. Corpus Bernstein identities/enclosure limits read. No full partition proof or numerical replay.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
194/097	The Euclidean Steinitz-Bergstrom theorem	L1	–	The prescribed-order signing and Steinitz reordering bounds require simultaneous control of every prefix via Gaussian survival and variance filtering. No matching Ram discrepancy theorem was located. **Boundary:** Finite covariance representation does not produce one signing valid for all prefixes. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
195/098	A doubling Hilbert subset with no finite-dimensional bi-Lipschitz embedding	L1	–	The Hilbert-subset counterexample uses sheet geometry, derivatives and rectangle-energy packing to obstruct every finite-dimensional bi-Lipschitz target. No concrete Ram obstruction was found. **Boundary:** Finite-measurement dimension obstructions concern linear information, not arbitrary nonlinear bi-Lipschitz embeddings. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
196/099	Edit Distance in l1: Matching Bounds up to Constants in the Exponent	L1	–	The edit-distance distortion theorem combines recursive incompatible displacements with an upper embedding at the matching exponential scale. No Ram theorem located supplies either distortion bound. **Boundary:** Fourier or histogram coordinates alone do not establish the edit-distance comparison constants. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
197/099	Finite-Circle Obstructions, Binary Codes, and Histogram Embeddings for Edit Distance	L1	–	This version has two finite-circle lower constructions, binary conversions and a deterministic histogram upper embedding including the empty word. No applicable Ram metric lemma was found. **Boundary:** The construction and alphabet-uniform distortion analysis are not consequences of a transform representation. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
198/099	Tree Constructions for the l1 Distortion of Binary Edit Distance	L1	–	The tree constructions prove binary lower bounds at every sufficiently large length cap and a prescribed-length binary conversion. No Ram tree-metric obstruction matches the argument. **Boundary:** This manuscript supplies lower constructions, not the companion's full upper embedding theorem. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
199/100	Finite angular cylinder covers below the half-area bound	L1	–	The explicit finite angular cylinder cover uses triangular bases and a quantified fourth-order remainder to beat the half-area bound. No specific Ram covering construction or needed new error theorem was located. **Boundary:** Representing its elementary polynomial remainder in another basis would not itself be a substantive application. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
200/100	Finite cylinder approximation of ruled sets	L1	–	The ruled-set approximation depends on trace-zero differentials, eigenvalue constraints, square-zero shear and finite oriented tiles over arbitrary compact labels. No matching Ram geometric covering lemma was found. **Boundary:** A local Jacobian or moment calculation does not prove coverage or the total perpendicular-area limit. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
201/100	Slope-field perturbations of the two-cylinder covering	L1	–	The slope-field perturbation uses rational/nilpotent regimes and singular cutoffs to turn an integral cost improvement into finite cylinder covers. No Ram construction supplies that mechanism. **Boundary:** Generic local field approximation would still require the coverage and area arguments. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
202/100	Finite triangular approximation of radial sweeps	L1	–	The radial-sweep approximation uses aligned velocities and tagged partitions to obtain triangular-base covers and convergence of their cost. No concrete Ram covering result was located. **Boundary:** A quadrature identity does not establish that the finite cylinders cover every segment. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
203/101	A sharp entropy bound and the simplex inequality for isotropic constants	L5	C3	Exact core-problem match: TR-item-344 and the manuscript use the same affine-invariant det(Cov)/volume^2 functional and the same dimension-specific simplex constant over every convex body. The actual cone entropy-to-covariance calculation also gives a concrete link to the corpus's augmented moments and realizability checks. **Boundary:** The Ram report does not prove simplex extremality. Its finite-body/family certificates do not establish the manuscript's global entropy inequality, noncommuting chaos estimate or equality rigidity. L5 is applicability, not a correctness or novelty endorsement. **Review:** manuscript: Full introduction and geometry proof read; selected long certificate passages read, including central polynomial/error arithmetic and Bernstein verification; other proof files structurally screened. Corpus target, affine moments, error, family-certificate and adverse-case passages read. Numerical tables not replayed and entropy/rigidity proofs not fully audited.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
204/102	The Unique Games Theorem	L1	–	The Unique Games theorem needs explicit affine constraints, shortcode decoding and a deterministic 3SAT reduction with fixed-alphabet soundness. No Ram hardness-reduction theorem was found. **Boundary:** Finite moment or matrix checks do not supply the universal completeness/soundness proof. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
205/102	A Direct Proof of Optimal Max-Cut Hardness	L1	–	The direct optimal Max-Cut hardness result needs a gap reduction, affine games and noisy Fourier decoding. The corpus's finite Folkman/Max-Cut certificates do not apply to that approximation-hardness proof. **Boundary:** Solving or certifying one finite cut instance is different from proving the Goemans-Williamson NP-hardness threshold. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
206/102	The Factor-Two Hardness Threshold for Vertex Cover	L1	–	The factor-two Vertex Cover hardness proof uses conditional-gradient juntas and pair decoding. No inspected Ram theorem provides this soundness-to-cover reduction. **Boundary:** General Boolean moment identities do not give the required junta and decoding estimates. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
207/102	Constant-factor hardness of Min-UnCut	L1	–	The Min-UnCut all-constant hardness proof depends on masked face correlations, lists and an affine outer verifier. No matching Ram gap-reduction lemma was located. **Boundary:** A finite cut certificate is not a uniform NP-hardness construction. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
208/102	Constant-factor hardness of directed feedback vertex set	L1	–	The directed feedback-vertex hardness argument uses rank spreading, pivotality/juntas and list decoding. No concrete Ram source supplies the necessary directed soundness argument. **Boundary:** Finite graph reconstruction or spectral filtering does not establish all-constant approximation hardness. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
209/103	Exact derandomization of logarithmic space: L = RL = BPL	L1	–	The logspace derandomization claim requires a stream-compression compiler with simultaneous time and space bounds for acceptance probabilities. No Ram result found gives those resource bounds. **Boundary:** An exact finite transition-matrix computation is not a logarithmic-space derandomization algorithm. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
210/104	Turn-Based Stochastic Mean-Payoff Games in Deterministic Quasipolynomial Time	L4	C2	Very relevant stochastic-game match. The manuscript explicitly reduces simple stochastic reachability to its expected-liminf mean-payoff model and then uses a quantified discounted Bellman problem. TR-item-312 supplies concrete domain-correct value, residual, stopping and rational-precision checks for this large algorithmic interface. **Boundary:** The corpus pins reachability, not arbitrary expected-liminf rewards, so the reduction and discount bridge must be retained. Its polynomial-duration subclass does not establish the manuscript's all-instance quasipolynomial labeling algorithm, nor does a small Bellman residual identify a nonstopping value without additional structure. **Review:** manuscript: Full introduction and discount.tex read, including gain/bias, threshold gap and behavioral-strategy proof. Labeling/application proof structure screened, not audited in full. Corpus residual-envelope, duration and counterexample sections read.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
211/104	Mean-payoff parity games in quasipolynomial time	L1	–	The mean-payoff parity result adds dominion recursion and unrestricted binary priorities to the mean-payoff subroutine. No Ram theorem found addresses parity or the combined recursion. **Boundary:** Stochastic reachability residual certificates do not control the parity winning condition. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
212/104	Deterministic quasipolynomial-time mean-payoff games	L1	–	The deterministic mean-payoff algorithm relies on label boxes, exact potentials and a quasipolynomial recursion for binary rewards. No concrete Ram result supplies that recursion. **Boundary:** The inspected reachability-game certificate theory concerns different payoff semantics and does not imply this all-instance time bound. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
213/104	Randomized quasipolynomial-time mean-payoff games	L1	–	The randomized version needs an adaptive orientation argument, worst-tape runtime and verified winning-region witnesses. No Ram theorem found provides the probabilistic recursion or its success bound. **Boundary:** Existence of efficiently checkable game certificates is not an algorithm for finding these witnesses in quasipolynomial time. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
214/105	Perfect completeness for 2-to-1 games	L1	–	Perfect-completeness 2-to-1 games are constructed through affine lists, trees of function spaces and explicit constraints. No applicable Ram PCP or soundness theorem was located. **Boundary:** Exact finite algebra does not furnish a perfect-completeness reduction. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

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215/106	Hardness of finding large independent sets in three-colorable graphs	L1	–	The three-colorable independent-set hardness proof needs distributionally aligned lists and a decoding chain. No concrete Ram theorem supplies the gap amplification. **Boundary:** A graph Gram matrix does not prove hardness of finding large independent sets. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
216/107	An Upper Bound of 9/4 for the Matrix Multiplication Exponent	L1	–	The 9/4 complex matrix exponent is obtained through tensor degenerations and shared-leg character constructions. No inspected Ram theorem gives such a tensor extraction or asymptotic rank bound. **Boundary:** Fast representation, work/span accounting or a basis change is not an improvement in the arithmetic exponent. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
217/107	Complex Matrix Multiplication Below 2.258 and Rectangular Bounds	L3	C3	The comparison bound contains a substantial explicit rational polynomial certificate over a continuous chart, including an approximately harmonic residual and boundary positivity. Ram's whole-panel enclosure methods apply to that identified verification task, not to matrix-multiplication tensors merely because they are algebraic. **Boundary:** The corpus does not provide tensor degenerations, label-tree attainability, arithmetic-progression comparison or the asymptotic exponent. Polynomial certificates are useful only after those reductions and all nonpolynomial remainder bounds have been established. **Review:** manuscript: Main statement/introduction excerpt and proof headings screened; first 160 lines of tex/certificate.tex read, including the precise certificate theorem and finite chart recurrence. Corpus Bernstein proof read. Full comparison, boundary and tensor arguments not audited; certificate code not run.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
218/107	Staggered extraction for exact matrix multiplication over every field	L1	–	The all-field matrix exponent uses integer tensors, staggered multidepth extraction and entropy/hash arguments. No matching Ram extraction theorem was found. **Boundary:** Characteristic-uniform tensor identities, not generic matrix algebra, determine the claimed exponent. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
219/108	A cubic lower bound for border determinantal complexity of the permanent	L1	–	The cubic border determinantal-complexity lower bound uses coefficient selectors, adjugates and conductor/polar constraints stable under degeneration. No relevant Ram algebraic-complexity obstruction was located. **Boundary:** Finite PIT or determinant evaluation does not yield a border-representation lower bound. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
220/109	Integer multiplication below n log n	L1	–	The integer-multiplication claim is in a fixed finite-tape bit model and depends on butterfly layout and a rank-saving motif. No Ram source located supplies the required tape-level schedule or bound. **Boundary:** An FFT identity or arithmetic-operation saving does not prove sub-n-log-n time on that machine model. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
221/110	Squared-logarithmic randomized k-server on arbitrary metrics	L1	–	The k-server ratio is obtained through allocation partitions, epochs, rounding and compactness. No concrete Ram online-competitive-analysis result was found. **Boundary:** Existence of an online policy with a horizon-uniform ratio is not a transform reconstruction theorem. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
222/110	Uniform computation of the squared-logarithmic k-server bound	L1	–	This finite rational k-server implementation compiles a policy into bounded per-request computation with a large additive constant. No matching Ram uniform decision-table or competitive guarantee was located. **Boundary:** The implementation claim is separate from the nonconstructive policy existence in 221; generic finite-state encoding proves neither. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
223/111	One Sample Suffices for Matroid Prophet Inequalities against an Almighty Adversary	L1	–	The one-sample matroid prophet inequality faces an almighty adversary and uses reverse exposure, safety and rank accounting. No Ram theorem matches this information pattern. **Boundary:** Conditional simulation or covariance identities do not establish adversarial online selection guarantees. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
224/112	Beyond the Square-Root Exponent for Depth-Three Boolean Circuits	L1	–	The explicit P-language depth-three circuit lower bound uses restrictions, sparse partitions and finite-field structure. No inspected Ram theorem proves a Boolean circuit-size lower bound. **Boundary:** Polynomial or finite-field coordinates are not a lower-bound argument. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
225/113	A Fully Polynomial Randomized Approximation Scheme for Perfect Matchings in General Graphs	L1	–	The general perfect-matching FPRAS depends on a logical hole bound and bottleneck-tree mixing. No Ram mixing or approximate-counting theorem was located for this chain. **Boundary:** Representing the transition matrix does not prove conductance, mixing or polynomial bit complexity. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
226/113	Entropy and Face Dimension of the Perfect-Matching Polytope	L1	–	The perfect-matching entropy inequality needs a face-uniform covariance/trace argument and a sharp Bethe-entropy coefficient. No inspected Ram log-partition theorem supplies this combinatorial inequality. **Boundary:** General covariance identities do not establish the sharp entropy deficit or its counting-factor consequence. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
227/114	An FPRAS for Common Integer Polymatroid Bases with Binary Capacities	L1	–	The integer-polymatroid intersection FPRAS requires signature transport and annealing with binary capacities and exact rank oracles. No concrete Ram mixing bound was found. **Boundary:** Local moment closure does not control oracle complexity or sampling on the full integer base set. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
228/114	Approximate counting of common bases of two matroids	L1	–	The two-matroid FPRAS uses paired ground sets, Lorentzian signatures and transport to obtain a sampling/counting algorithm. No applicable Ram theorem was located for this combinatorial state space. **Boundary:** Formal finite operators do not imply the necessary mixing or independence-oracle bounds. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
229/115	Exact Uniform Sampling of Contingency Tables with Arbitrary Margins	L1	–	Exact uniform contingency sampling uses a near-uniform sampler followed by an exact correction and dense-block conductance. No Ram source found establishes those mixing and correction costs. **Boundary:** Unbiased Monte Carlo estimation is not exact uniform sampling with polynomial expected runtime. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
230/115	An FPRAS for Cell-Bounded Contingency Tables	L1	–	The cell-bounded contingency FPRAS uses factorial lifts, port weights, annealing and trace chains in a binary bit model. No concrete Ram conductance or counting input was located. **Boundary:** An exact state-space representation does not establish polynomial dependence on the encoded margins and bounds. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
231/116	Uniform Matrix Hitting Points in Every Positive Characteristic	L3	C3	A substantial part of the positive-characteristic hitting construction is exact evaluation in a finite coefficient ring and faithful multiplication-matrix descent. TR-item-317's characteristic-safe jets, quotient-algebra semantics and correct transported products apply concretely to that step. **Boundary:** The Ram report's general target is commutative circuit PIT, not this noncommutative singleton construction. It supplies neither multiplicative-shift word independence nor the uniform order bound. Ordinary factorial-divided derivatives and real Gram positivity cannot be imported in characteristic p. **Review:** manuscript: Main theorem and proof headings screened; full construction.tex through the main algorithm/corollary read. Corpus finite-algebra/product/dimension sections read. Shift-independence and order-bound proofs not fully audited.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
232/116	One Rational Matrix Hitting Point for Noncommutative Formulas	L3	C3	The explicit integration operators J_i(a)=integral_0^z a(t)/(t+i) dt become finite rational matrices on a rigorously chosen Taylor quotient. This is a concrete and substantial local-operator/jet module matching Ram exact coefficient algebra, with the manuscript's multiplicity bound supplying the nontrivial truncation size. **Boundary:** The corpus does not provide the faithful iterated-integral encoding or Moura bound for this black-box noncommutative class. A generic Ram transform is not itself a polynomial-size hitting point, and no conclusion for arbitrary shared commutative circuits follows. **Review:** manuscript: Full introduction, construction.tex and multiplicity.tex read. Corpus finite-algebra, product, dimension and structured-span passages read. Iterated-integral independence and formula-system proofs only structurally screened.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
233/116	Polynomial Hitting Lists for Noncommutative Rational Formulas	L2	–	The rational-formula hitting list has a concrete peripheral connection to exact algebra/domain bookkeeping: every inverse operand must remain invertible and every evaluation must preserve ordered products. The Ram PIT report explicitly isolates these unit/denominator limitations, but its division-free algorithms do not perform this construction. **Boundary:** No Ram result inspected supplies the division-algebra deformation, all-small-subspace expansion or polynomial list size at unrestricted inverse depth and constant height. The support/degree promise for division-free jets cannot silently be reused. **Review:** manuscript: Main theorem/introduction excerpt and encoding/division/expansion/generator headings screened. Relevant corpus quotient/unit and denominator warnings read. No detailed rational-formula proof audit.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
234/117	Constant-factor hardness of uniform sparsest cut	L1	–	Uniform sparsest-cut hardness relies on explicit bit tests, pencil construction and score-path compilation. No Ram gap-reduction result matches the proof. **Boundary:** Finite spectral or cut computations do not establish all-constant NP-hardness. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
235/117	Near-square-root logarithmic integrality gaps for uniform sparsest cut	L1	–	The uniform-demand Goemans-Linial integrality gap needs a negative-type metric, rounded charts and cut lower bounds. No applicable Ram metric construction was found. **Boundary:** A Gram representation certifies only part of negative type, not the metric, demand normalization or integrality gap. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
236/118	Additive hardness and unbounded configuration gaps in bin packing	L1	–	The bin-packing result uses cover gadgets and two-tree constructions to force unbounded additive configuration-LP gaps. No matching Ram combinatorial construction was located. **Boundary:** Generic optimization certificates do not produce the required family of LP gaps or additive hardness reductions. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
237/119	Sharp binary-information contraction on the discrete cube	L1	–	The binary-information contraction argument uses an optimized soft channel, three-point entropy comparisons, joining capacities and thinning. No inspected Ram theorem supplies that sharp channel inequality. **Boundary:** The Boolean-influence and log-partition reports concern different functionals and quantifiers. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
238/119	Hellinger contraction with arbitrary Boolean output bias	L3	C3	A significant part of the Hellinger contraction proof is explicitly reduced to seventeen polynomial profiles and whole-rectangle scalar sign certificates. The exact Bernstein conversion/subdivision in the Ram corpus maps directly to these stated checks and their continuum coverage. **Boundary:** This is applicability to a substantial scalar verification module, not a Ram proof of arbitrary-bias Hellinger contraction. Interval endpoint normalization, transcendental remainders and the induction reduction remain separate; listed certificate success was not rerun. **Review:** manuscript: Main statement and proof headings screened; scalar.tex lines 1-230 and profile/sign-verification passages including the Bernstein formula and leaf rules read. Corpus conversion/hull/subdivision proofs read. Full contraction argument and certificate programs not audited.; corpus: Exact sections named in corpus_refs read as described in manuscript-depth notes; corpus maps also consulted. These primary-source checks were scoped, not exhaustive.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
239/120	Almost-Linear-Time Maximum-Cardinality Matching in General Graphs	L1	–	The almost-linear general-matching algorithm depends on phase prices, factor-critical structure, routing and a cycle backend, with runtime bounded on every tape. No concrete Ram combinatorial algorithm supplies these components. **Boundary:** Graph operator compression does not imply an almost-linear exact maximum-matching algorithm. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.
240/121	An Almost-Linear Approximation Scheme for Edit Distance	L1	–	The approximate edit-distance algorithm needs cost cones, entropy-regret control and an exact sampling/evaluation schedule. No Ram method found supplies the expected almost-linear bound. **Boundary:** A histogram or transform representation does not establish the adaptive runtime or approximation guarantee. **Review:** manuscript: Main theorem excerpt (up to the first two theorem environments where detected), or introductory passage, and selected section/subsection headings of the proof files were read. This is statement-and-structure screening, not full proof reading.; corpus: All three corpus maps were read and used as routing/limitation evidence; selected local primary sources were compared, not the entire corpus. An empty corpus_refs list means no concrete transferable result was located in this bounded screen, not that none could exist.; verification: Applicability only; no full proof validation, novelty/priority determination, or execution of manuscript/corpus certificate programs. A high rating does not assert that the Ram corpus supplies the missing estimate or a complete solution.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
241/122	Uniform quasipolynomial-time trace reconstruction	L1	–	The claimed uniform deletion-trace decoder rests on multiscale masks, quantitative word separation, and a bounded-cost decoding relaxation. No read Ram result supplies these deletion-channel estimates or their bit complexity. **Boundary:** Finite moment inversion or Fourier coordinates alone do not give trace identifiability at the stated sample budget. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
242/122	A latest-anchor induction with spectrally compact masks for worst-case trace reconstruction	L1	–	The latest-anchor induction controls exact-symbol trace statistics with spectrally compact masks and retention-dependent sample bounds. No concrete corpus result was found for this induction. **Boundary:** This is a sample-complexity theorem, not the uniform decoder of 241; ordinary convolution inversion does not reproduce its mask estimates. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory. Additionally read the first 180 lines of build/part-1.tex, including the sample theorem and anchor-mask induction setup. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
243/122	Quantitative lower bounds for trace reconstruction	L1	–	The lower bound uses nested signed words, likelihood-score cancellation, and a branching construction of hard deletion-channel pairs. The read corpus supplies no corresponding indistinguishability estimate. **Boundary:** Representing likelihoods by moments would not prove the n^{Omega(log log n)} information-theoretic lower bound. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
244/124	A Polynomial-Time Algorithm for Three-Machine Unit-Job Scheduling	L1	–	The scheduling proof relies on bounded descriptions of job intervals, global separator conditions, and pruning sufficient for a polynomial dynamic program. No matching scheduling or bounded-state decomposition theorem was located. **Boundary:** Generic dynamic programming or finite operator representations do not establish completeness of the proposed description family. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
245/125	Single-exponential recovery and bounded-price strictness for metric k-median	L1	–	The k-median recovery and strict-below-two approximation require distinct cluster proxies, exact facility budgets, and a compatible bounded-price execution. No corpus result supports these combinatorial payment and recovery steps. **Boundary:** Optimization coordinates alone do not preserve the exact k budget or improve the approximation factor. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
246/125	The approximation threshold for metric k-median	L1	–	The deterministic 1+2/e threshold argument uses tapered rounding, scalar drift inequalities, and derandomization of a facility-opening process. No concrete matching rounding theorem was found. **Boundary:** This is not the recovery theorem of 245; generic convex or polynomial certificates do not supply its approximation analysis. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
247/126	Exponential PSD rank of positively shifted matching matrices	L1	–	The matching-slack PSD-rank lower bound uses Fourier squares, parity expansion, and control of PSD factorizations. No read corpus theorem yields the exponential rank lower bound. **Boundary:** The existence of Gram or moment representations is not evidence of applicability to proving that all small PSD lifts fail. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
248/127	Average sensitivity of polynomial threshold functions	L1	–	The polynomial-threshold sensitivity bound is driven by Fourier grading, an anticommutator identity, and one-sided coupling. No corpus estimate was identified for degree-dependent hypercube sensitivity. **Boundary:** Fourier transform identities alone do not bound total influence by the claimed O(d sqrt(n)) scale. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
249/128	A Polynomial-Time 2-Approximation for Shortest Common Superstring	L1	–	The shortest-superstring factor-two proof is organized around occurrence counts, periodic layers, and opening linked cycles. No relevant Ram string-overlap or cycle-opening guarantee was located. **Boundary:** Finite encoding of overlaps supplies no approximation guarantee for the global superstring. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
250/129	An exponential state lower bound for two-way nondeterministic complementation	L1	–	The two-way nondeterministic complementation lower bound is built from diagram monoids and rank loss in nested idempotents. No read corpus theorem addresses this automaton-state separation. **Boundary:** Matrix rank or finite inverse formulas do not imply a state-complexity lower bound. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
251/129	An exponential two-way deterministic state lower bound for one-way liveness	L1	–	The one-way nondeterministic to two-way deterministic state lower bound uses matching-diagram corners and their rank loss. No applicable simulation lower-bound theorem was found. **Boundary:** The relevant resource is automaton states, not the arithmetic cost of representing a transition matrix. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
252/130	An explicit power saving for the exact discrete Fourier transform	L2	–	The Ram finite inverse recurrence can check the triangular Toeplitz reciprocal-series subroutine inside the exact local Fourier compiler. This is a concrete but small algebraic component of the claimed all-length power saving. **Boundary:** Ram polynomial inversion does not provide the tensor-network saving, all-length synchronization, or the O(n(log n)^{1-10^{-13}} arithmetic bound. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the opening 240 lines of sections/local.tex, including the local factorization and reciprocal-series construction. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
253/130	Finite tensor savings and exact Fourier circuits	L1	–	The nonuniform subsequence result needs a strict saving in tensor calls, a priced packing argument, and Fourier transfer. No read corpus result establishes any such circuit saving. **Boundary:** Exact Fourier or Schur-coordinate identities are not an operation-count improvement; unlike 252, no specific reciprocal-series bottleneck is identified here. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
254/131	Polynomial mixing of the switch chain for every graphical degree sequence	L1	–	The switch-chain claim needs a uniform conditional-expectation comparison for arbitrary graphical degree sequences. No corpus mixing-time or switch-chain comparison theorem was found. **Boundary:** A finite Markov operator description does not establish the asserted polynomial spectral gap or sampling runtime. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
255/132	A superquadratic separation between sensitivity and block sensitivity	L1	–	The superquadratic sensitivity separation uses recursively labelled tournament predicates and a block-sensitivity recurrence. No concrete corpus construction or inequality applies. **Boundary:** Boolean moment encoding alone does not produce the required separation between sensitivity parameters. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
256/133	Parity lifts and bounded-treewidth witnesses for Weisfeiler-Leman equivalence	L2	–	The Ram refinement report explicitly separates sound WL signatures from complete isomorphism certificates and notes fixed-dimensional WL obstructions. It is relevant to checking what the parity-lift equivalence actually certifies. **Boundary:** The report does not provide the bounded-treewidth homomorphism construction, dimension lower bound, or ETH reduction. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
257/133	The complexity of identifying a graph by Weisfeiler-Leman refinement	L2	–	The Ram complete individualization certificate and its distinction from incomplete refinement are peripheral to the manuscript's WL identification decision problem. **Boundary:** Correctness of an exhaustive canonical search does not prove EXPTIME hardness or membership for variable-dimensional WL identification. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
258/133	Unconditional time lower bounds for Weisfeiler-Leman equivalence	L2	–	The refinement report supplies a soundness boundary for the graph invariants used in WL equivalence. The manuscript's main work is instead an unconditional time-hierarchy reduction with compressed consistency gadgets. **Boundary:** No n^{Omega(k)} running-time lower bound follows from the Ram refinement hierarchy or its factorial exhaustive search. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
259/133	Variable-dimension Weisfeiler-Leman equivalence on general and subcubic graphs	L2	–	The same exact-refinement conventions are relevant to joint WL comparisons in the subcubic gadget encoding. The hardness proof itself depends on affine blocks and exact projection relations absent from the read corpus. **Boundary:** This does not establish EXPTIME completeness, subcubic gadget correctness, or variable-dimension resource bounds. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
260/134	Finite Monoid Computations and a Uniform Generalized Star-Height Bound	L1	–	The generalized star-height-13 construction uses affine update clocks and a finite splitting grammar. No relevant Ram theorem on regular-language star height was located. **Boundary:** Finite automaton or matrix representations do not control nesting of complement and Kleene star. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
261/134	Generalized Star Height at Most Four	L1	–	The star-height-four proof reduces the number of splits using moving periodic boundaries. No read corpus result matches this language-theoretic boundary construction. **Boundary:** The improvement over 260 requires a grammar argument, not generic linearization of a finite monoid. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
262/134	Generalized Star Height at Most Three	L1	–	The star-height-three claim uses prefix episode codes, two splits, and affine correction tables. No concrete Ram star-height or episode-coding result was found. **Boundary:** A finite table can be encoded algebraically without giving the claimed star-height bound. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
263/135	Homogeneous depth-five lower bounds for iterated matrix multiplication	L1	–	The homogeneous depth-five IMM lower bound depends on derivative-multiplication rank, occupation trace moments, and path-type counting. No corpus result controls that circuit-specific rank statistic. **Boundary:** Generic positivity of trace moments is not an arithmetic-circuit size lower bound. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
264/136	The PCP-for-PPAD conjecture: a quasilinear reduction	L1	–	The PPAD result requires a quasilinear PCP reduction, End-of-Line encoding, and robust generalized-circuit gadgets. No matching corpus complexity reduction was identified. **Boundary:** Finite equations for gadgets do not establish gap preservation, completeness, or the reduction's size. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
265/137	Simulating One-Tape Time in Two-Fifths-Power Space	L1	–	The one-tape simulation space bound uses block epochs and finite-field evaluation of a dependency path. No Ram result was found that limits the simulator's writable tape to T^{2/5}. **Boundary:** Algebraic representation of a computation is distinct from a uniform time-space simulation theorem. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
266/138	Subset Sum in Time O(2^{0.49n})	L1	–	The subset-sum 2^{0.49n} algorithm uses checksum filters, frequency selection, and bounded-bit alias extraction. No read corpus result supports that exponential-time improvement. **Boundary:** Convolution identities do not supply the required collision statistics or total bit-cost analysis. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
267/138	A Low-Space Algorithm for Worst-Case Subset Sum	L1	–	The subset-sum time-space result uses compressed sum streams and modular weighted disjointness. No matching corpus streaming or meet-in-the-middle tradeoff theorem was found. **Boundary:** This is a 2^{0.5n}-time, reduced-space claim, distinct from 266; a transformed sum table does not prove the storage bound. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
268/139	Subpolynomial query complexity for well-conditioned log-concave sampling	L2	–	The reverse-heat report gives explicit posterior covariance generators and material-moment jet recurrences. These can inform the conditional-mean derivative calculations in the Gaussian transport construction. **Boundary:** The corpus assumes compact support and its own moment envelopes; the target is unbounded C^2 strongly log-concave sampling. All-split bounds, finite gradient-query implementation, initialization, and subpolynomial query complexity remain separate. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the first 5,000 characters of sections/jets.tex and the corpus setting, covariance-generator proof, and closed-normal-form proof. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
269/140	Memory and precision in noiseless Gaussian regression	L2	–	The joint-Gaussian covariance calculus provides a peripheral exact check for the smoothed difference-vector calculation in the projection proof, specifically the shared first-noise term and independence across rows. **Boundary:** This finite Gaussian-law check does not establish inverse Gram-determinant integrability, the Frostman projection bound, finite-state recurrences, or T >= c_A d log(1/epsilon). Generic Gram positivity is not being counted as a memory lower-bound argument. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read sections/projection.tex lines 1-125 and 161-265, including the exact density theorem and complete Gaussian-difference identity proof. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
270/140	Posterior replicas and conditional information in Gaussian regression	L2	–	The joint-law and conditioning discussion is a peripheral check on how posterior replicas share observations: conditional independence is not unconditional independence. **Boundary:** Total covariance and joint-moment identities do not supply the affine-volume cancellation or the H(W)/t + O(d) information bound. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
271/140	Localization costs and information growth for exact Gaussian observations	L1	–	The information-cost result depends on dyadic localization of spherical/cube priors and dimension-sensitive inverse-distance projection estimates. No read Ram theorem supplies that localization-cost inequality. **Boundary:** Generic localized moments or entropy bookkeeping do not establish the O(dt) information budget; the specific mixed projection and cell-correction estimates remain unmatched. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
272/140	Projection moments, positive cap domination, and Riesz estimates on the sphere	L1	–	The key claims are integrated random-frame density moments, positive domination by cap measures, and Riesz estimates with explicit dimensional margins. No corresponding theorem was found in the inspected corpus. **Boundary:** Positive moment matrices are not positive measure-domination certificates and do not bound singular inverse-volume integrals. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
273/140	Replacing Gaussian observations in memory-constrained inference	L2	–	The corpus's insistence on one joint conditional law is directly pertinent to the distinction between actual selected rows and fresh independent rows. It supplies a limited bookkeeping check. **Boundary:** The corpus does not prove the O(d) information-comparison cost, shared/fresh projection moment, or critical-radius cancellation. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
274/140	Subsphere methods for memory-sample lower bounds in noiseless Gaussian regression	L1	–	The explicit precision constant is obtained from residual subspheres, cap probabilities, stopping budgets, and a beta-mixture comparison. No corpus result was found for these spherical block inequalities. **Boundary:** Conditioning a Gaussian vector on a plane is not the uniform-subsphere law needed here, and cannot justify the 2^{-16} constant. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
275/141	Existential-universal real sentences in the counting hierarchy	L1	–	The counting-hierarchy inclusion for an existential-universal real theory uses controlled polynomial representations and determinant/root labels. No read Ram result addresses alternating real quantifiers or counting-hierarchy reductions. **Boundary:** A finite polynomial inverse or numerical root certificate has neither the universal quantifier nor the complexity guarantee of this theorem. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
276/142	Deterministic Polynomial Factorization over Prime Fields	L1	–	The deterministic finite-field factoring claim depends on a uniform Hecke zero-free input, cyclic covers, and norm equations. No concrete local Ram theorem establishes that arithmetic input. **Boundary:** Finite algebra or transform coordinates cannot substitute for the required zero-free theorem or uniform factoring runtime. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
277/143	Uniform bounds for planar polynomial limit cycles	L5	C3	Direct core match: TR-item-015 explicitly targets a degree-only bound for limit cycles of planar polynomial fields. Its singular-transition and uniformity discussion meets the manuscript's finite passage templates, absolute-zero closure, and projection-count argument. **Boundary:** The corpus supplies fixed-flow certificates and an explicit statement of the missing uniform degeneration theorem, not B(d). L5 measures exact target applicability; the manuscript's absolute-zero and passage-closure proofs have not been validated. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the first 9,500 and last 6,000 characters of sections/11-counting.tex: projection-count hypotheses/proof opening, hyperbolic rotation proof, and final uniform-count deduction. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
278/143	Two limit cycles for quintic Lienard systems	L4	C2	Very relevant specialized application of the polynomial limit-cycle report: the manuscript reduces quintic Lienard cycles to isolated zeros of one matching function, while the corpus supplies careful return-map counting and perturbative root-count contracts. **Boundary:** The Ram report does not prove the quadratic fitting transport inequalities or the global at-most-two theorem. Its hyperbolic and perturbative tests cover only restricted regimes without further work. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read sections/application.tex through all four coefficient cases; supporting fitting/model inequalities were not read in full. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
279/144	A smooth three-torus diffeomorphism with simple Lebesgue spectrum	L5	C3	Direct core match to Banach's probability-preserving simple-Lebesgue-spectrum problem. The corpus separates exact Haar correlations from cyclicity and gives a moving-window transfer condition, precisely the two goals of the manuscript's whitening and diagonal completion. **Boundary:** The corpus constructs no required probability transformation or smooth torus diffeomorphism. Its frame-budget criterion is a comparison tool, not verification of the manuscript's passage and finite-prediction construction. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the first 9,000 and last 7,500 characters of sections/05-completion.tex, including the whitening and main diagonal-limit proof; finite prediction/passage proofs remain only structurally screened. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
280/145	Rokhlin's multiple-mixing problem for one transformation	L5	C3	Direct core match: TR-item-313 has the same ordinary-mixing-to-all-finite-orders target and explicitly analyzes off-diagonal joining laws. Its parity examples test a central danger in the manuscript's ordered-array and conditional-locality route. **Boundary:** The corpus's homoclinic sufficient condition is not implied by ordinary mixing. The target removes standardness by a generated factor; that reduction must be checked separately. Neither report nor this screen validates the all-order theorem. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the complete sections/07-main-proof.tex and exact corpus joining/parity/residual theorem sections; the ordered-array, operator-square, and entropy-cancellation proofs were not read in full. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
281/146	Positive Metric Entropy for the Standard Map at Large Parameters	L5	C3	Direct core match to positive Lebesgue metric entropy for the standard map. The corpus's physical-time expansion and parameter-set contracts apply to the same derivative cocycle and entropy conclusion; the manuscript asserts the stronger all-large-parameters result. **Boundary:** The corpus proves only conditional all-return/positive-area implications and supplies no successful nonzero-parameter certificate. Its positive-measure parameter target is weaker than the manuscript's eventual-every-parameter assertion. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the first 7,000 and last 6,500 characters of sections/geometry.tex, including support exclusions and final positive-entropy deduction; the complete multiscale estimates were not audited. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
282/147	Continuous Phase Foliations Create Analytic Caustic Collars	L5	C3	Direct match to the literal near-boundary continuous invariant phase-foliation target of TR-item-336, with smooth positive curvature and no symmetry or near-ellipse hypothesis. **Boundary:** Rational rotation alone and formal grazing jets do not suffice, as the corpus demonstrates. Its conditional contraction statement is not an unrestricted billiard inverse; the analytic and companion rigidity arguments remain unverified here. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the first 8,000 and last 6,500 characters of sections/geometry.tex, including twist coordinates, rational-leaf closure, and the stationary-chain proof; analytic inversion and the companion proof remain unverified. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
283/147	Rigidity of Smooth Billiards with a Continuous Caustic Collar	L5	C3	Direct core Birkhoff-rigidity match through an explicit, domain-correct bridge: the physical smooth-caustic collar produces a continuous invariant phase-graph collar in lem:rational-periodicity, bringing it into TR-item-336's exact billiard model and ellipse-rigidity question. **Boundary:** The physical hypothesis has smoother individual leaves than the corpus's phase formulation; the bridge must be retained. The Ram report does not supply the meromorphic-wall, critical moment-law, or final sign-test proof. These are proof-completeness limits, not a denial of the core mapping. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the first 6,500 characters of sections/01-introduction.tex and first 6,000 of sections/02-analytic.tex, including the physical-to-phase collar lemma and its proof opening; complex continuation/sign tests were not read in full. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
284/148	The entropy-rate dimension formula for self-similar measures on the line	L1	–	The self-similar dimension formula depends on random-walk entropy, overlap blocks, and a quantitative window entropy gain. No relevant Ram entropy-dimension or overlap theorem was located. **Boundary:** A moment representation of a self-similar measure does not determine its dimension or remove overlap entropy losses. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
285/149	Uniform Permanence in Weakly Reversible Mass-Action Systems	L2	–	The reaction-network chapter directly provides the stoichiometric invariant, quasipositivity, and bounded-class existence checks that precede the proposed class-uniform absorbing-set theorem. **Boundary:** The corpus obtains boundedness only under its bounded-class/conservation hypotheses. It does not derive a uniform absorbing compact subset of every weakly reversible class or exclusion of approach to its boundary. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
286/149	Boundedness and persistence of weakly reversible mass-action systems	L2	–	The same mass-action positivity and conservation contract is relevant to the ODE foundation of the per-initial-condition persistence and boundedness claim. **Boundary:** Unlike 285, the target bound may depend on the initial state. The corpus still does not cover all weakly reversible networks or supply persistence away from the boundary. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
287/150	Weak mixing of triangular billiards with an irrational angle	L3	C3	The polygon-billiard report treats the same full Liouville phase flow and singular cone geometry, including a conditional ergodicity route. This is significant context for the invariant/eigenfunction and singular-set controls in the triangle proof. **Boundary:** The corpus does not prove weak mixing for every irrational triangle, exclude nonzero eigenfrequencies, or eliminate Diophantine restrictions. Its ergodicity criterion is weaker and conditional. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
288/150	Ergodicity of triangular billiards with an irrational angle	L4	C2	Very relevant, domain-correct overlap with the polygon report's full-flow ergodicity problem: both use the Liouville measure on the nonsingular flat-cone phase space and the geodesic/perpendicular/angular operator framework. The triangle proof targets each irrational table rather than a generic polygon family. **Boundary:** The corpus's stated polygon quantifier and conditional Cheeger route differ from the manuscript's every-irrational-triangle theorem. Its operator formulation applies, but it does not prove the source's transport-to-parallel-coefficient passage. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read source/main.tex lines 1-170 and 366-467, including the actual full-flow theorem, strategy, and complete localized-energy and bounded-mode proofs. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
289/151	A C^1 Counterexample to the Entropy Conjecture	L1	–	The C^1 entropy counterexample requires a global clock construction, zero topological entropy, and an induced H_2 eigenvalue of two. No concrete Ram result constructs or controls this low-regularity example. **Boundary:** Finite homology or operator-coordinate transport alone does not connect the required differentiability, dynamics, and induced homology. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
290/152	A zero-entropy system without a smooth positive-volume model	L1	–	The no-smooth-model theorem uses word-window laws and a derivative/volume obstruction for smooth positive-volume realizations. No matching corpus realization obstruction was found. **Boundary:** Neither abstract stationary moments nor a finite spectral representation proves that an ergodic system has no smooth model. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
291/153	Arithmetic classification and non-Pisot singularity for Bernoulli convolutions	L1	–	The Bernoulli-convolution classification is driven by arithmetic unit roots, special algebraic parameters, and lattice characters. No read corpus theorem supplies this algebraic absolute-continuity/singularity classification. **Boundary:** Convolution moments and cumulants do not decide regularity of the infinite measure for the stated arithmetic parameters. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
292/154	Pointwise Multiple Ergodic Averages for Mixing Transformations	L3	C3	Two substantive imported components match the corpus: higher-order mixing and trilinear pointwise convergence. The source explicitly uses these companion inputs before its new selected-law, channel, and tensor-packing argument for arbitrary length. **Boundary:** The Ram reports do not prove the new all-length selected-law theorem. Higher-order correlation limits alone do not imply almost-everywhere convergence of multiple averages. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the first 4,500 characters of sections/inputs.tex, including the exact two companion dependencies; channel/packing proofs were not fully audited. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
293/154	Pointwise convergence of fourfold ergodic averages for mixing transformations	L3	C3	The corpus's trilinear oscillation and approximation criteria match a substantial lower-length input in the fourfold proof. The source expressly needs those inputs on extensions that need not themselves be mixing. **Boundary:** The report treats (n,2n,3n), not all changed triples or the fourfold activation theorem. Its estimates are conditional and do not furnish the needed local analytic bounds. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the first 4,500 characters of sections/07-lower-length.tex, including the changed-coefficient local analytic input and its scope. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
294/154	Triple ergodic averages with distinct integer slopes	L4	C2	Very relevant extension of the corpus's triple-average transference problem to distinct nonzero integer slopes. The source makes the same separation between an oscillation estimate on arbitrary systems and mixing used only to identify the limit. **Boundary:** TR-item-168 fixes slopes 1,2,3; the changed-coefficient local estimate and rational-shift arithmetic must be proved anew. A moment cancellation identity alone does not establish oscillation control. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the first 6,000 characters of sections/transfer.tex, including profile and kernel construction and the transference setup; full local analytic proof not audited. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
295/154	Pointwise convergence of triple ergodic averages for mixing transformations	L5	C3	Direct core trilinear pointwise-convergence match: the averages at n,2n,3n are exactly those in TR-item-168. Although the headline assumes mixing, the source's finite oscillation proposition claims convergence on arbitrary invertible systems before identifying the mixing limit. **Boundary:** The corpus proves the conditional transfer and Cauchy/approximation implications, not the missing trilinear analytic estimate. The large local-form construction and its companion input were not fully audited. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the first 8,000 characters of sections/base.tex and first 10,000 of sections/ergodic.tex, including the finite oscillation proposition, complete Cauchy criterion proof, and norm-limit induction opening. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
296/155	A translational tile with no fully periodic tiling in dimension three	L1	–	The three-dimensional aperiodic translational tile is built from word constraints and stacked finite-coordinate gadgets, followed by a discrete-to-continuous tiling transfer. No Ram tiling or periodicity-obstruction theorem was found. **Boundary:** A finite encoding of tiles does not ensure a tiling exists or rule out every finite-index period lattice. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
297/156	A nine-dimensional counterexample to Borsuk's covering assertion	L1	–	The dimension-nine Borsuk counterexample uses a projective rank-one configuration and a topological obstruction to a ten-label cover. No corresponding Ram covering-diameter obstruction was identified. **Boundary:** Gram realization of the configuration is not the topological lower bound for its covering number. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
298/157	A counterexample to Hadwiger's conjecture	L1	–	The Hadwiger counterexample construction uses characteristic-two moment holes, a graph with independence number at most two, and control of all clique minors. No corpus graph-minor construction supplies this. **Boundary:** Boolean moments or finite low-rank coordinates do not bound the Hadwiger number of the constructed graph. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
299/157	A counterexample to the Colin de Verdiere chromatic conjecture	L1	–	The Colin de Verdiere chromatic counterexample additionally needs a well-signed matrix with a precise nullity and sign pattern. No read corpus theorem provides that invariant calculation or the associated graph construction. **Boundary:** A generic Lorentz/Gram block or spectral compression is not a certificate for the full Colin de Verdiere conditions. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
300/157	A linear list-coloring bound in terms of the Hadwiger number	L1	–	The list-coloring/Hadwiger bound relies on locally sampled palettes, rooted minor density, and recursive weaving. No concrete Ram list-coloring or rooted-minor theorem was found. **Boundary:** The plane-coloring corpus concerns a different graph class and does not transfer to list chromatic number versus clique-minor order. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
301/158	The Euclidean plane is not five-colorable	L5	C3	Direct core match to the ordinary Hadwiger-Nelson lower-bound problem. TR-item-094 explicitly distinguishes arbitrary from measurable colorings, provides finite exact witnesses, and rules out fractional-only routes to six colors, all pertinent to this manuscript's transfer-based proof. **Boundary:** The corpus does not supply a non-five-colorable witness or the source's Haar-rigidity and palette theorems. Its fractional barrier limits one method, not the ordinary-color target. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the first 6,500 characters of sections/transfer.tex, including the continuous-factor/positivity lemma and forward-transfer statement; the full Haar-rigidity and angular-obstruction proofs were not audited. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
302/159	Quasipolynomial Bounds for Arithmetic Progressions	L2	–	TR-item-233 gives the exact translation/common-difference progression-count expression that appears before a density-increment argument. It is a peripheral representation and finite-test tool for the manuscript's all-length progression theorem. **Boundary:** The Ram panel provides no nilmanifold inverse theorem, localized energy gain, or quantitative iteration preventing the losses behind the claimed quasipolynomial bound. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
303/160	Quantitative Superexponential Bounds for van der Waerden Numbers	L3	C3	The progression-hypergraph/Property-B formulation concretely matches the goal of constructing interval colorings with no monochromatic k-progression. It supports a significant finite certificate layer of the cyclic-to-integer coloring transfer. **Boundary:** The corpus's quantitative work is principally on two-color sparse forcing, whereas the manuscript proves superexponential lower bounds for W_r(k). It does not supply the outer-color perturbation or the k^{c k floor(log_2 r)} construction. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
304/161	A counterexample to Sidorenko's conjecture	L1	–	The Sidorenko counterexample uses a particular bipartite graph, finite symplectic/Lagrangian geometry, and a negative kernel-correlation calculation. No read Ram result supplies its homomorphism-density inequality or obstruction. **Boundary:** Generic Boolean gluing or PSD moment feasibility is not a certificate for the claimed negative Sidorenko defect. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
305/162	Balanced counterexamples to Ryser's conjecture at prime orders	L1	–	The balanced prime-order Ryser construction requires an intersecting (q+1)-partite hypergraph with q+1 vertices per part and cover number q+1. Its affine-line deletion and concentration proof has no identified Ram counterpart. **Boundary:** The exact balance and cover-number conditions cannot be inferred from generic incidence or moment coordinates. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
306/162	A counterexample to Ryser's covering conjecture	L1	–	The prime-power Ryser construction uses compatible translations and finite-field incidence covering, with q=s^n and odd n. No corpus construction or cover-number theorem was identified. **Boundary:** This is distinct from the balanced prime-order result of 305; finite-field representation alone does not rule out every size-q transversal. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
307/164	Monochromatic finite sums and products in the positive integers	L1	–	The finite-sums-and-products result depends on divisor-weighted prime scales, a dense model, and face alignment in multiplicative/additive cubes. No relevant Ram simultaneous sum-product Ramsey theorem was found. **Boundary:** The arithmetic-progression panel is not a model of all finite sums and products and does not supply the required monochromatic structure. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
308/165	The crossing number of complete graphs	L3	C2	The crossing report supplies a closely related topological drawing contract, star-order state representation, and parity-sensitive deletion accounting. These are substantive comparison tools for the complete-graph intersection approach. **Boundary:** TR-item-071 targets K_{m,n}, not K_n. Its bipartite rotation-frequency quadratic form does not directly prove the Hill formula or the manuscript's complete-graph rank inequality. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
309/165	The crossing number of complete bipartite graphs	L5	C3	Direct core match to the exact Zarankiewicz-Turan crossing formula for all complete bipartite graphs. The corpus retains precisely the topological, realizability, and odd/even finite-size distinctions used in the source's final deduction. **Boundary:** The corpus uses a rotation-frequency relaxation and does not prove its sharp lower bound or the manuscript's signed-intersection detector. Core applicability is exact despite these unresolved proof obligations. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the complete sections/conclusion.tex, including odd-part rank transfer and even-part deletion; detector and subspace lemmas were not fully audited. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
310/166	The higher-dimensional Erdos distinct-distances conjecture	L1	–	The high-dimensional distinct-distance bound uses polynomial separators, concentration on algebraic sets, and secant/incidence estimates. No read corpus theorem gives the needed incidence exponent. **Boundary:** Euclidean distance or Gram encoding is not an n^{2/d}-scale distinct-distance theorem. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
311/167	The weak pinned planar distance theorem	L1	–	The almost-all-pinned planar distance result requires arithmetic nested partitions and a quantitative variance/fiber argument. No corpus pinned-distance estimate was located. **Boundary:** The Euclidean Ramsey corpus studies monochromatic congruent sets, not the number of distances from most pins. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
312/167	A power saving for planar unit distances	L1	–	The sub-4/3 unit-distance bound is driven by algebraic specialization, height control, and rare-level prediction obstructions. No corpus edge-count estimate at that exponent was identified. **Boundary:** The plane-coloring report concerns chromatic obstructions on the same unit-distance relation, but its certificates do not bound the number of unit-distance pairs. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
313/168	Combinatorial invariance of Kazhdan-Lusztig polynomials	L1	–	The Kazhdan-Lusztig invariance claim uses moment-graph sheaves and rank-two/dihedral comparisons over arbitrary Coxeter systems. No matching Ram sheaf or Bruhat-poset theorem was found. **Boundary:** The word 'moment' here denotes a different geometric construction; ordinary statistical moment coordinates do not imply poset invariance. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
314/169	Elementary positivity of chromatic quasisymmetric functions	L1	–	Natural unit-interval e-positivity is proved through permutation witnesses, shuffle/quantum-torus operations, and triangular wall transport. No concrete Ram symmetric-function positivity result was located. **Boundary:** Triangular coefficient algebra is not positivity in the elementary symmetric-function basis. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
315/170	The sharp logarithmic exponent of r(5,t)	L1	–	The K_5-free Ramsey lower bound uses projective flags, sparse-pair geometry, and entropy/prediction compression. No corpus construction realizes its t^4/log^3(t) scale. **Boundary:** Euclidean Ramsey or arithmetic-progression certificates have different forbidden configurations and do not transfer this graph Ramsey bound. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
316/170	Sharp logarithmic exponents for fixed off-diagonal Ramsey numbers	L1	–	The general off-diagonal Ramsey bound extends the projective-incidence learning construction to fixed clique order s. No matching corpus entropy or incidence estimate was found. **Boundary:** This needs the s-dependent exponent and logarithmic savings, not merely a finite graph encoding or the special K_5 construction. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
317/171	The hypercube Ramsey number has linear order	L1	–	The linear-in-2^n hypercube Ramsey result relies on bias-purity regimes, slice tiling, and late-exposure matching. No relevant Ram hypercube-embedding or Ramsey-host theorem was located. **Boundary:** A cube-shaped coordinate representation is not a monochromatic embedding of Q_n in the required host size. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
318/172	A classification of finite Euclidean Ramsey configurations	L5	C3	Direct core Euclidean-Ramsey match: TR-item-229 pins the same variable-dimension, arbitrary-finite-color congruence property and Graham's spherical-set question. The classification manuscript explicitly claims cyclic-quadrilateral consequences and a spherical nine-point obstruction, so it directly addresses that core question. **Boundary:** The corpus's box/cut-cone criterion is not a criterion for Ramsey-ness. It does not supply the rational tensor classification or prove its claimed nine-point counterexample; this is an exact problem mapping, not endorsement. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the first 7,000 characters of sections/07-consequences.tex, including spherical necessity, finite transitive sets, cyclic quadrilaterals, and the nine-point obstruction statement; the full tensor classification proof was not audited. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
319/173	A proof of Seymour's second-neighborhood conjecture	L5	C3	Direct core specialization of the Seymour-Sullivan report: give every arc of an oriented graph both labels 1 and 2. Rainbow paths have length at most two, no rainbow cycle is possible, and the target becomes |N_1^+(v)|+|N_2^+(v)| >= 2|N_1^+(v)|, exactly the manuscript's second-neighborhood inequality. **Boundary:** This specialization is a direct deduction from the read definitions. The general corpus surplus criterion does not prove existence of a successful vertex and does not establish the manuscript's strengthened subset deficit, pruning, or extremal contradiction. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the first 7,500 characters of source/01-reduction.tex, including the complete subset-deficit proof and blowup setup; pair pruning and extremal contradiction were not fully audited. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
320/174	The strong thin tree conjecture	L5	C3	Direct core match to the strong thin-tree conjecture in TR-item-224: one spanning tree with C/k thinness for all cuts of every k-edge-connected graph. The source follows the report's exact fractional-to-one-tree and cut-versus-spectral distinctions. **Boundary:** The corpus proves fractional and conditional cut-family certificates, not the uniform rounding theorem. It does not supply the source's hierarchy extraction, fixed point, or packing-preserving sparsification estimates. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the complete source/sections/iteration.tex, including the simultaneous-cut product and packing reduction; hierarchy, extraction, and MSS machinery were not fully audited. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
321/174	A polynomial-time construction of strong thin trees	L5	C3	The algorithm constructs the very same C/k-thin tree targeted by TR-item-224, adding deterministic polynomial-bit complexity. Its cut and packing certificates have a direct mathematical mapping to the corpus's simultaneous-cut and Laplacian contracts. **Boundary:** The corpus does not give a polynomial-time signing/extraction routine or compressed-input bit complexity. Those algorithmic obligations are additional to, not reasons to downgrade, the exact thin-tree target match. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the main-theorem input and first 6,500 of 6,547 characters of source/sections/iteration.tex, including explicit-graph sparsification and binary multiplicity reduction; signing and precision proofs were not fully audited. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
322/175	Graph Decompositions at the Integral Expectation Threshold	L1	–	The graph-decomposition expectation-threshold claim uses a discrete convexity theorem with fixed pieces before the host graph varies. No relevant Ram threshold-cover or decomposition theorem was found. **Boundary:** Finite convex representation does not establish the expectation-threshold comparison or its quantifier over host graphs. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
323/175	Integral and fractional expectation thresholds are equivalent	L1	–	The constant fractional-to-integral expectation-threshold comparison depends on a multiscale selector-rounding argument. No read corpus result supplies that rounding or cover-weight control. **Boundary:** A fractional feasible point or generic convex optimization is not an integral threshold certificate at a constant factor. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
324/175	Talagrand's discrete-convexity conjecture	L1	–	The discrete Talagrand convexity claim controls covers and the complement of unions with an explicit exponential-in-k constant. No matching corpus discrete covering theorem was identified. **Boundary:** Ordinary linear convexity or moment positivity does not yield this Boolean-family cover inequality. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
325/176	The second Kahn-Kalai conjecture	L1	–	The second Kahn-Kalai graph threshold bound uses spread resampling, rooted cover trees, and graph-specific counting. No corpus theorem supplies the p_E log(h) scale. **Boundary:** This is not a standard probabilistic moment calculation; the graph-family witness/cover construction remains essential. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
326/177	Bounded-degree coboundary expanders in every dimension	L1	–	The bounded-degree coboundary-expander construction uses congruence/coset complexes, homology vanishing, and quantitative cosystolic inputs. No read Ram theorem supplies that simultaneous degree-and-expansion guarantee. **Boundary:** Exact finite cochain representations preserve algebraic identities but do not produce coboundary expansion constants. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
327/178	Deterministic nonbipartite Ramanujan graphs in every fixed degree	L1	–	The all-even-order deterministic Ramanujan construction requires controlled configuration pencils, trace accounts, and sparse boundary repair. No corpus construction proves the exact degree, size, nonbipartiteness, and spectral edge together. **Boundary:** Finite spectral or Gram certification does not create an all-size Ramanujan graph or control the repair step. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
328/179	The circulant Hadamard conjecture	L1	–	The circulant Hadamard/Barker argument depends on cyclotomic prime-square descent and binary coefficient constraints. No relevant corpus arithmetic descent theorem was identified. **Boundary:** Fourier diagonalization checks the autocorrelation identity but cannot by itself force the asserted length restriction. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
329/180	Paired states and Hamiltonian cycles in cubic bipartite planar graphs	L1	–	The Barnette Hamiltonicity proof uses paired states in the dual graph and compatible spanning-tree transitions. No corpus planar cubic bipartite Hamiltonicity theorem was found. **Boundary:** The thin-tree corpus controls cuts, not the degree-two connected spanning subgraph required for a Hamiltonian cycle. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
330/181	A linear cycle-and-edge decomposition of every graph	L1	–	The linear edge partition into cycles and singletons uses expansion reservoirs, pair folding, and routing induction. No read Ram cycle-decomposition or routing estimate was located. **Boundary:** An incidence-matrix or cycle-space representation does not bound the number of edge-disjoint output cycles. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
331/182	A power saving for intersective polynomial differences with an exponent depending only on the degree	L1	–	The power-saving intersective-polynomial difference theorem uses local auxiliary polynomials, reflection-positive tuples, and signed-kernel induction. No read corpus estimate supports the required polynomial recurrence or density loss. **Boundary:** Arithmetic-progression panels concern a different configuration count and do not supply the singular Fourier or local-root estimates used here. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
332/182	A Power Saving for Polynomial Differences at Prime Arguments	L1	–	The prime-argument extension requires a companion prime-distribution/zero-free input in addition to the intersective-polynomial machinery. No relevant Ram theorem was found for that arithmetic transfer. **Boundary:** Formal convolution or residue coordinates cannot replace the uniform prime estimates or account for exceptional arithmetic moduli. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
333/182	A power saving for square-difference-free sets	L1	–	The squarefree-difference power saving relies on finite-field square-tuple laws, conductor decay, and a signed multilinear-kernel induction. No matching Ram squarefree-difference estimate was located. **Boundary:** A positive finite tuple law does not by itself provide the global signed-kernel cancellation or density exponent. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
334/183	A power saving for planar halving lines	L5	C3	Direct core match to the halving-line/k-set exponent problem treated in TR-item-214. Both formulations count rank switches in planar sweeps and isolate the missing uniform geometric estimate needed to beat the 4/3 scale. **Boundary:** The corpus's sharp telescoping coefficient improves a constant, not the exponent, and its exponent-transfer theorem is conditional. It does not prove the source's spatial-dimension or transverse-foliation packing estimate. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. Additionally read the first 6,500 characters of sections/recurrence.tex, including the complete packing-to-power implication and halving-line transfer; the geometric activity-packing estimate was not fully audited. Relevant corpus sections in corpus_refs were read directly; coverage is selected sections, not the entire corpus. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
335/184	Correspondence coloring graphs with a forbidden clique	L1	–	The clique-forbidden DP-coloring theorem uses local multiplier adjustment, triangle splitting, and dependent palette selection. No read corpus correspondence-coloring theorem supplies its Delta/log(Delta) bound. **Boundary:** The ordinary plane-coloring report has different constraints and cannot replace DP-cover compatibility or palette survival estimates. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
336/184	A logarithmic independence bound for clique-free graphs	L1	–	The clique-free independent-set bound uses entropy-optimal weights, triangle-mass vertex splitting, and a closure argument. No matching Ram independence-number theorem at n log(d)/d was identified. **Boundary:** Moment positivity and fractional weighting do not establish the needed independent-set extraction or closure inequalities. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
337/185	A Counterexample to the Infinite Matroid Packing/Covering Conjecture	L1	–	The infinite matroid packing/covering counterexample is countable, self-dual, and nonfinitary, using ultrafilter and density-zero constructions. No read corpus theorem covers that infinite independence structure. **Boundary:** Finite tree packing or finite matching certificates cannot be transferred to a nonfinitary matroid without new infinite extension arguments. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
338/186	A uniform influence bound for hypergraph properties	L1	–	The hypergraph-property influence bound depends on block restrictions and biased Fourier analysis with the stated uniformity-dependent logarithmic exponent. No corpus influence inequality matches this setting. **Boundary:** An exact Boolean-moment basis does not supply influence growth under hypergraph symmetry. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
339/186	A Sharp Threshold Bound for Monotone Graph Properties	L1	–	The monotone graph-property threshold-width bound uses biased Fourier low-degree blocks and a variance-to-width step. No read Ram sharp-threshold estimate was found. **Boundary:** This needs uniform control across edge probabilities, not only covariance identities at one probability law. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
340/187	Snaky in 21 Maker moves	L1	–	The Snaky winning claim is an explicit finite game strategy backed by reconstructed templates and policy certificates. No applicable Ram positional-game strategy or certificate theorem was found. **Boundary:** The strategy data and checker were not executed. Generic finite verification machinery is not a concrete strategy or a proof of the 21-move bound. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
341/188	The sharp terminal leave in random triangle removal	L1	–	The random triangle-removal terminal count is obtained from an early quasirandom regime, priority unfolding, collision witnesses, and terminal stability/moments. No corpus theorem controls that process to its stopping scale. **Boundary:** General stochastic moment bookkeeping does not establish the terminal n^{3/2}/(2 sqrt(2)) asymptotic or justify stopping-time estimates. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
342/189	Cycle–clique Ramsey numbers	L1	–	The exact cycle-clique Ramsey result combines a minimal-counterexample path analysis with a stated finite case certification. No matching Ram graph Ramsey theorem was found. **Boundary:** The finite-case programs and their 3099-case claim were not rerun; a certificate framework from an unrelated coloring problem is not a mathematical match. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
343/190	Polynomial removal fails for ordered binary matrices	L1	–	The ordered binary-matrix removal counterexample uses a particular 66-by-66 forbidden pattern, tree pinning, and leaf-count distance estimates. No corpus ordered-removal construction was located. **Boundary:** Matrix coordinate changes do not preserve the ordered-pattern quantifier or prove the required removal-distance separation. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
344/191	A power improvement in the Heilbronn triangle lower bound	L1	–	The Heilbronn-triangle lower construction uses integer-lattice determinants, field norms, cap sampling, and deletion of small triangles. No read Ram small-triangle extremal estimate matches it. **Boundary:** Convex-body moments or determinant representation alone do not bound the minimum triangle area in the proposed large point sets. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
345/192	Unbounded Violations of the Square-Root Degree Bound	L1	–	The unbounded Boolean Fourier linear-sum/degree ratio uses variance-reporting functions and a Gaussian-mixture amplification. No corpus theorem constructs or controls this separation. **Boundary:** Wick formulas may check individual Gaussian moments but do not establish the Boolean realization, degree bound, or amplification ratio. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
346/193	Positivity of Serre's Intersection Multiplicity	L1	–	The Serre intersection-multiplicity positivity claim depends on regular local rings, supported complexes, and Tor Euler characteristics. No relevant Ram local-algebra intersection theorem was found. **Boundary:** Finite matrix or coordinate transport does not preserve or prove the required homological positivity and dimension hypotheses. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
347/194	Lech's multiplicity conjecture	L1	–	The Lech multiplicity argument concerns flat local homomorphisms and characteristic-dependent normalized lengths. No read corpus multiplicity-comparison theorem applies. **Boundary:** Generic asymptotic moments or change of variables do not imply a local-ring multiplicity inequality. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
348/195	A Complete Local Domain without a Small Cohen-Macaulay Module	L1	–	The complete normal complex threefold without a maximal Cohen-Macaulay module is built via branched surface/Kummer geometry. No matching Ram existence obstruction for such modules was located. **Boundary:** Finite algebra certificates do not address the completed local ring, depth, or absence of all nonzero maximal Cohen-Macaulay modules. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
349/196	A Torsion-Free Group Algebra with Zero Divisors	L1	–	The torsion-free characteristic-two zero-divisor construction requires labelled random matching, bounded-path topology, and asphericity before the group-algebra identity. No read Ram result supplies that construction. **Boundary:** The finite-group characteristic-zero coordinate report does not transfer to torsion-free F_2 group rings; exact product encoding alone is insufficient. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
350/197	A Torsion-Free Group Algebra That Is Not Directly Finite	L1	–	The characteristic-two one-sided inverse/annihilator construction uses parity-labelled graphs and a torsion-free group realization. No relevant Ram realization or nonzero-annihilator theorem was found. **Boundary:** Checking ab=1 and ac=0 in a proposed encoding would not establish that c survives in the torsion-free group algebra. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
351/197	A Counterexample to Kaplansky's Direct-Finiteness Conjecture in Characteristic Two	L1	–	The direct-finiteness counterexample uses finite-field local groups, incidence embeddings, subgroup averages, and cellular assembly. No corpus theorem supplies the claimed one-sided inverse in the resulting torsion-group algebra. **Boundary:** Finite-dimensional matrix direct finiteness or finite-group coordinates cannot certify the infinite assembly's injectivity and nonzero defect. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
352/197	A Counterexample to the Group-Ring Determinant Conjecture	L1	–	The determinant-conjecture counterexample passes from a group-algebra defect to an integral matrix and a Fuglede-Kadison determinant strictly between zero and one. No read Ram operator-algebra determinant theorem supplies that passage. **Boundary:** Ordinary finite determinants, positive Gram matrices, and spectral truncations are not the von Neumann-algebra trace/determinant needed here. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
353/197	A Counterexample to Kaplansky's Direct-Finiteness Conjecture in Odd Characteristic	L1	–	The odd-characteristic group-ring construction relies on a specified finite field, local idempotents, annihilation ranks, and group-sector gluing. No corresponding Ram characteristic-p construction was located. **Boundary:** The precise field characteristic, survival of the defect, and infinite group realization cannot be replaced by generic algebra-coordinate identities. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
354/198	An algebra of infinite little finitistic dimension	L1	–	The finite algebra with infinite little finitistic dimension is obtained through a noncommutative selection/localization and bimodule tensor simulation. No read corpus theorem controls these projective-dimension constructions. **Boundary:** Finite multiplication tables or terminating polynomial inverses do not bound, or force unboundedness of, finite projective dimensions. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
355/199	An explicit counterexample to the Auslander-Reiten conjecture	L1	–	The Auslander-Reiten counterexample requires complete resolutions, stable actions, cones, and a twisted bimodule lift over the specified characteristic-two algebra. No matching Ram homological-algebra result was found. **Boundary:** An exact algebra representation does not prove simultaneous Ext vanishing or the nonprojectivity required by the counterexample. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
356/199	A counterexample to Tachikawa's second conjecture	L1	–	The Tachikawa counterexample hinges on a particular symmetric algebra, a nonprojective module, and vanishing positive self-Ext groups via stable polynomial twists. No corpus theorem supplies those properties. **Boundary:** Checking the ten-dimensional multiplication table is only a finite preliminary; symmetry and all-degree self-Ext vanishing are separate obligations. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
357/200	The Artinian Lex-Plus-Powers Betti Theorem	L1	–	The Artinian lex-plus-powers Betti claim uses commuting division forms, leading-coefficient resolutions, monomial boxes, and Tor comparison. No matching Ram graded Betti or regular-sequence theorem was located. **Boundary:** A triangular coefficient representation alone does not control minimal resolutions or prove the asserted Betti-number inequalities. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
358/200	Commuting Division-Coefficient Forms and the Artinian Eisenbud–Green–Harris Conjecture	L1	–	The EGH-related construction uses commuting forms over a division algebra, parameter intersections, and index descent in characteristic zero. No relevant Ram division-form or Hilbert-function theorem was found. **Boundary:** Finite jet recovery and moment coordinates do not establish regular-sequence constraints or the required commutative-algebra descent. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
359/201	A Counterexample to Kurosh's Division-Ring Problem	L1	–	The algebraic two-generated infinite-dimensional division ring over a countable center is constructed by successive free-series corrections and a realization schedule. No matching Ram division-ring construction was located. **Boundary:** Finite inverse formulas do not prove algebraicity of every element, preservation of division, or the infinite-dimensional limit over the stated center. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
360/202	The Blockwise Alperin Weight Conjecture	L1	–	The blockwise Alperin weight result depends on modular block/weight transforms, generating tuples, and arithmetic specialization of twisted covers. No concrete Ram modular-character or block correspondence theorem was found. **Boundary:** The local finite-group characteristic-zero induction report has different representation-theoretic hypotheses and does not establish a blockwise modular weight count. **Review:** Screening: catalogue/pack, actual main statement or substantive introduction excerpt, source section inventory, and a proof-opening excerpt. No specialized corpus match established in the local maps and targeted searches; this is not an exhaustive no-relevance theorem. Applicability only; no full proof verification, external source checking, numerical replication, or novelty assessment.
361/203	Donovan's Conjecture over Algebraically Closed Fields	L1	–	The bounded-defect Morita-finiteness claim is driven by block reduction, quasisimple Cartan bounds and tilting/crossed-extension arguments. No inspected Ram theorem supplies those structural finiteness inputs. **Boundary:** Finite coordinates for one algebra do not bound the number of Morita classes uniformly over groups or defect data. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
362/203	Integral Donovan Finiteness over Witt Vectors	L1	–	The integral Donovan statement over Witt vectors needs integral rigidity and descent in addition to the field case. The corpus's real/complex moment and operator inversions do not address this lifting problem. **Boundary:** Do not equate nonsingular coefficient recovery with integral equivalence of orders or uniform integral Morita finiteness. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
363/204	Tensor saturation for even spin groups	L1	–	Even-spin tensor saturation is an integral representation-theoretic statement, with root-lattice admissibility and spin-column/path switching central to the proof. No concrete corpus route to that mechanism was located. **Boundary:** Real convex feasibility or a positive moment matrix does not establish lattice saturation, parity compatibility, or the all-N assertion. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
364/205	Universal Tensor Squares for Symmetric Groups	L2	–	Finite character-coordinate transport can check a proposed tensor square at a fixed symmetric-group rank. This is a peripheral audit of the cyclic-polytabloid construction, not its uniform existence argument. **Boundary:** The cited source concerns a finite induction test for a different group problem. Its exact-coordinate audit pattern transfers; its group theorem does not prove this all-rank tensor-square claim. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
365/205	A Cyclic Polytabloid Proof of Saxl's Conjecture	L2	–	The corpus offers exact finite character/tensor bookkeeping relevant to small staircase tensor-square checks. The core cyclic-polytabloid nonvanishing and row/column alternation argument remains representation-specific. **Boundary:** Small-rank verification cannot replace the uniform polytabloid construction or prove Saxl's conjecture. No novel theorem follows from a character basis change. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
366/206	Finite congruence lattices: characterization and undecidability	L1	–	Finite-congruence-lattice characterization and undecidability require subgroup-interval constructions and computability arguments. No inspected operator, convexity or arithmetic theorem gives a concrete route. **Boundary:** An enumerable collection of finite certificates is not a decision procedure; finite graph coordinates do not settle a representation problem over all finite algebras. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
367/206	A negative solution to the finite lattice representation problem	L1	–	The negative finite-lattice representation claim rests on group-chain rigidity and Boolean gluing obstructions. These are not the weighted-graph or probabilistic gluing problems treated by the corpus. **Boundary:** The corpus's restricted three-edge probabilistic coupling theorem must not be substituted for arbitrary lattice or subgroup-interval gluing. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
368/207	The l1-Bass Conjecture for Discrete Groups	L2	–	The analytic correction of approximate idempotents in a weighted group Banach algebra has a concrete local connection to norm-convergent inverse series. The trace-support and cyclic-chain conclusion has no comparable corpus bridge. **Boundary:** The cited inverse theorem is conditional on a specified bounded operator with norm below one. It neither constructs the needed group-algebra estimates nor proves finite torsion support of the l1 Bass trace. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
369/207	The Bass trace conjecture and the characteristic-zero Kaplansky idempotent conjecture	L1	–	The complex-group-ring Bass trace argument uses cyclic cocycles, geometry and group-algebra K-theory. A finite matrix/idempotent representation would only restate an input, with no concrete route to the central trace vanishing. **Boundary:** Do not promote finite polynomial identities or inverse-coordinate formulas into a Bass or Kaplansky theorem for unrestricted groups. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
370/208	Fiber functors for finite symmetric tensor categories in positive characteristic	L1	–	The positive-characteristic symmetric tensor-category result depends on Verlinde-valued fibre functors and projectivity detection. No matching corpus theorem or domain-correct operator mechanism was located. **Boundary:** Characteristic-zero moment positivity and Gaussian covariance do not transfer to positive-characteristic tensor categories. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
371/209	An integral counterexample to Gersten's conjecture	L1	–	The mixed-characteristic K5 Gersten claim is built around localization, ramified residue calculations and cyclotomic/Bott input for the specified regular local ring. The corpus arithmetic sources do not address this K-theoretic mechanism. **Boundary:** Integer or rational exact arithmetic is not a Gersten exactness theorem; the special ramified ring and residue characteristic cannot be discarded. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
372/209	An integral degree-three Gersten counterexample	L1	–	The K3 Gersten manuscript requires residue-degree and coefficient-lifting arguments in the specified mixed-characteristic ramified ring. No concrete corpus route to those exact-sequence steps was found. **Boundary:** A finite coordinate representation does not certify K-theory localization exactness, and a K5 or unramified comparison cannot silently be substituted. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
373/210	Foulkes' conjecture for the sixth symmetric power	L2	–	The Foulkes-six proof has a substantial finite exact-character verification stage, but the transferable corpus contribution is an audit pattern for those integer certificates, not a theorem proving the uniform plethysm statement. **Boundary:** The manuscript reports finite checks and a uniform reduction; neither was replayed. Integer coefficient nonnegativity here is not continuous polynomial positivity, so Bernstein enclosure is not the missing representation-theoretic argument. **Review:** Read main statement/proof outline and the first 120 lines each of certificates.tex and verification.tex; no character enumeration or certificate execution. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
374/210	Quadratic stabilization of the canonical Foulkes–Howe map	L2	–	The directional differential map on homogeneous bidegree spaces admits exact finite derivative rows of the kind developed in the corpus. This can audit the displayed map, but its injectivity at b >= a(a-1) is the difficult representation-theoretic content. **Boundary:** A banded or exact derivative matrix is not automatically injective. Dimension growth, equivariance and uniform kernel exclusion must come from the manuscript's argument. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
375/211	Random Walks on Critical FK-Ising Maps and Liouville Brownian Motion	L4	C2	The random-walk limit uses the same physical Dirichlet-form, resolvent and semigroup objects treated by the corpus. A source- and measure-preserving representation is very relevant to the operator-to-path portion, beyond generic graph algebra. **Boundary:** The GHP normalization and quantum-map input are assumed or separately proved. Finite spectral convergence alone does not provide path tightness, starting-point control or the correct time normalization. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
376/211	Spectral convergence for critical FK-Ising planar maps	L5	C3	Direct core match: the claimed spectral limit is formulated through a centered Green operator on the limiting quantum measure and finite graph Dirichlet energies. Physical Gram lifting and full-residual/tail control apply to these very operators, not merely to an auxiliary coordinate system. **Boundary:** The singular quantum measure is not Lebesgue measure on a uniformly elliptic domain. The ball bounds, trace inequality, Hilbert-Schmidt limit and heat/eigenvalue tails remain substantive inputs; the corpus does not supply their estimates. **Review:** Read main statement/proof structure, 08-continuum.tex and 09-spectrum.tex through line 205 each, including trace and energy-compactness arguments; no full proof certification. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
377/211	A Linear Clock for Random Walk on Tree-Weighted Planar Maps	L4	C3	The tree-weighted map's electrical-energy and primal/dual reduction steps have a concrete match to source-correct Green-pairing elimination. This is very relevant to identifying the limiting diffusion, though it does not establish the random geometry. **Boundary:** Deleting loaded vertices requires the corresponding source correction. Network identities do not prove conditional path-law convergence or the conformal and metric scaling inputs. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
378/211	Canonical conformal limits of subcritical FK planar maps	L2	–	A finite electrical-network calculation within the FK map construction could be audited by the corpus's Green-pairing identities. The main joint conformal, metric and CLE convergence for q in (0,4) is not an operator-coordinate consequence. **Boundary:** This is peripheral to the conformal coupling and universality claim. Preserve q-dependent constants and do not infer CLE or metric limits from finite graph equivalence. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
379/211	The critical Liouville quantum sphere and geometric limits of FK maps at q=4	L1	–	The q=4 limit depends on critical quantum-surface/Gaussian-multiplicative-chaos limits as gamma approaches two and on metric completion. No inspected Ram result addresses that critical limiting mechanism concretely. **Boundary:** Finite Gaussian moments or smooth-kernel quadrature do not control critical GMC renormalization, field topology or the endpoint metric limit. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
380/211	Metric-measure limits of subcritical FK and spanning-tree planar maps	L1	–	The FK-tree metric-measure limit is a random-word and distance-geometry problem. The inspected corpus graph results concern linear operators and finite reconstruction, not the required random metric identification. **Boundary:** A graph Laplacian encoding does not yield GHP convergence, tightness of graph distances or the asserted continuum tree law. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
381/211	Brownian continuum random tree limits of finite Fortuin-Kasteleyn maps above four	L2	–	The block generating functions and inventory estimates can be checked with analytic-series and exact coefficient methods. This is peripheral to the conditioned branching-process mechanism producing the q>4 CRT limit. **Boundary:** Analytic continuation must avoid singularities; uniqueness of a continuation supplies neither coefficient asymptotics nor a conditioned Galton-Watson invariance principle. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
382/212	No bigeodesics in planar first-passage percolation	L1	–	The no-bigeodesics FPP result is driven by path geometry under atomless weights and its stated moment assumption. No concrete corpus theorem matches the independence, coalescence or geodesic obstruction mechanism. **Boundary:** A finite shortest-path encoding or convex hull does not rule out bi-infinite geodesics; retain the distributional and moment assumptions. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
383/212	Strict convexity and differentiability of the planar exponential first-passage limit shape	L1	–	Strict convexity and C1 regularity of the exponential FPP limit shape require strip changes of measure and rare crossing witnesses. The corpus's moment-cone convexity is a different problem and offers no concrete route to these geometric estimates. **Boundary:** Convexity of a positive-moment cone does not imply strict convexity or differentiability of a passage-time norm. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
384/213	Critical bond and site percolation on the cubic lattice	L1	–	The critical cubic-lattice percolation claim is based on geometric move constructions and dynamic renormalization. No inspected corpus result controls the multiscale crossing events needed for absence of an infinite critical cluster. **Boundary:** Exact finite graph calculations do not settle infinite-volume critical percolation or replace uniform renormalization estimates. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
385/213	No percolation at criticality on quasi-transitive graphs	L1	–	The quasi-transitive critical-percolation argument uses growth alternatives, entropy and exploration geometry. No domain-correct corpus route to that critical infinite-cluster conclusion was found. **Boundary:** Finite transition or Green matrices do not imply a critical percolation theorem on all quasi-transitive graphs satisfying the manuscript's hypotheses. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
386/214	Nonuniqueness of percolation on nonamenable quasi-transitive graphs	L3	C3	The two-point operator T_p on l2 is a genuine operator-norm target for Schur reduction and retained/discarded residual estimates. These methods address a significant analytic component of the p_c < p_2-to-2 comparison. **Boundary:** The row-reducing deformation and critical probabilistic bounds are not supplied by the corpus. A finite norm bound does not prove the infinite-graph critical inequality or a relation to p_u. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
387/215	The canonical massive continuum limit of the two-dimensional O(3) model	L3	C3	Exact blocking and source-normalized Schwinger observables admit moment/remainder and physical-norm checks. This can support significant finite-scale pieces of the O(3) continuum construction, while the central renormalization estimates remain external to the corpus. **Boundary:** Finite-scale moment control does not establish uniform renormalization, all Schwinger axioms, uniqueness of the continuum theory or a mass gap. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
388/215	An Isolated Particle Pole for the Two-Dimensional O(3) Spin Field	L4	C2	The passage from transfer spectral measures to an isolated mass has a close positive-measure and spectral-promotion match. The manuscript explicitly realizes moments as squared projection weights, making the corpus tools directly useful to a major proof component. **Boundary:** Finite moments do not imply an atom or mass gap. Uniform cylinder/plane comparison, field tails, compact momentum control and the state-count estimate remain necessary; a contour certificate requires actual tail resolvent hypotheses. **Review:** Read main statement/proof outline and first 145 lines of 07-mass-support.tex, including the spectral-measure definition and moment-convergence proof; later isolation argument not fully audited. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
389/215	Exact mass asymptotics for the two-dimensional O(4) lattice model	L4	C2	The O(4) mass calculation uses an actual contact Hamiltonian and a weighted spectral integral, not just a formal transform. The corpus's kernel representation and physical spectral-domain discipline are very relevant to that analytic mechanism. **Boundary:** The corpus does not prove completeness of the Bethe resolution, residue cancellation, strong-coupling matching or the exact asymptotic mass constant. Fixed particle number in the inspected lemma must not be confused with uniform many-particle control. **Review:** Read main statement/proof structure and first 145 lines of 04-bethe-line.tex, including the heat-kernel resolution and start of its scattering/residue proof; not the complete mass calculation. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
390/215	Sharp mass bounds for the two-dimensional O(4) model	L5	C3	Direct core match: the full-gap criterion is built from the positive trace-class integral transfer kernel K_w and trace-power ratios. The corpus can represent this same compact positive operator with its physical metric and certify retained/tail spectral information; this is more than generic coordinate algebra. **Boundary:** The corpus does not establish the required small square-box test Delta_beta(n) <= 2^-10 or the renormalization bounds producing it. One fixed-width gap is insufficient for a uniform infinite-volume gap; plane-state uniqueness is a separate premise. **Review:** Read main statement/proof outline and first 145 lines of finite-volume.tex, including positivity/trace-class reasoning, the squaring proof and start of the finite-volume criterion proof. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
391/215	Exponential decay in two-dimensional classical O(n) models	L2	–	Exact derivative/product rows can audit the fourth-rotation response algebra in the O(n) annulus argument. This is a peripheral check on a proof whose force comes from sign-cluster geometry and the competing logarithmic estimates. **Boundary:** A correct derivative table does not establish the annular inequalities or exponential decay for every finite beta and n >= 3. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
392/216	The critical logarithmic correction for the planar XY model	L2	–	Local Gaussian/Fourier representations can check finite annulus observables, but the r^-1/4 (log r)^1/8 result is driven by marginal flow and amplitude-ratio estimates. The corpus contribution is peripheral. **Boundary:** Ordinary transform inversion does not prove the marginal renormalization flow, identify its coefficient or control the logarithmic correction uniformly in scale. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
393/216	Critical Center Magnetization in the Planar XY Model	L2	–	The center-spin asymptotic uses finite annular height ratios plus a companion critical input. The corpus can check finite moment/normalization calculations but does not supply that companion theorem or the scale iteration. **Boundary:** Do not treat an imported critical input as independently proved here, or infer a logarithmic exponent from a finite covariance calculation. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
394/216	The Critical Spin Field of the Planar XY Model	L3	C3	The punctured-domain Gaussian Dirichlet-energy calculations underlying spin moments have a concrete weak-PDE and Green-function match. This addresses significant parts of the imaginary-chaos identification. **Boundary:** Puncture collisions, charge-sector summation, moment determinacy and H^-3_loc tightness are not supplied by a smooth-domain solve. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
395/216	Essential Singularity of the Correlation Length in the Planar XY Model	L3	C3	The analytic finite-scale flow and exit-time initialization have a useful match to nonlinear residual/branch certification and parameter-majorant control. These can test significant deterministic components of the essential-singularity argument. **Boundary:** A branch certificate does not create the RG reduction or prove a uniform positive limit of sqrt(beta_c-beta) log xi. Analytic uniqueness alone gives no stable continuation estimate. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
396/216	The critical correlation exponent of the planar XY model	L3	C3	The mixed Gaussian pinning and radial Dirichlet-flow energy in the critical XY exponent proof are concrete targets for boundary-correct weak operators and Fourier normalization. This is a substantial analytic subproblem. **Boundary:** The endpoint critical identification and radial-flow bounds remain separate. Neither a gauge-fixed matrix nor Fourier inversion proves the exponent 1/4. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
397/216	BKT universality for height and planar spin fields	L3	C3	The effective Gaussian stiffness and neutral/charged boundary sectors can be represented through physical Dirichlet operators and source-correct inverses. This reaches significant finite-scale components of the BKT mechanism. **Boundary:** The corpus does not prove roughening, a BKT transition or uniform renormalization control. The charged problem cannot be replaced by the neutral inverse without source corrections. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
398/217	The low-temperature Sherrington-Kirkpatrick free-energy limiting law	L3	C3	The low-temperature SK fluctuation argument repeatedly evaluates nonlinear hierarchical Gaussian expectations and Laplace windows. Joint-moment/remainder methods can address those specific finite-recursion calculations, a significant part of the analytic work. **Boundary:** The exponential tilt is not a quadratic Gaussian observable; exponential integrability and denominator control need separate bounds. Finite recursion checks do not prove the fluctuation law or its rate. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
399/217	The low-temperature Sherrington-Kirkpatrick fluctuation scale	L3	C3	The recursive map X -> u^-1 log E exp(uX) is a concrete nonlinear integral operation for which moment remainder and derivative auditing are relevant. It addresses a significant component of the stated low-temperature SK scale argument. **Boundary:** Polynomial moment closure is not exact after exponentiation and logarithms. Uniform hierarchy-depth estimates and the n^1/3 variance law remain outside the corpus theorem. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
400/218	Quenched SLE3 limits for general weak random-bond Ising models	L2	–	Conditional moments can audit local environment perturbations in the weak random-bond Ising comparison. The quenched SLE3 limit and critical-temperature shift depend on a separate uniform probabilistic comparison. **Boundary:** Fixed weak disorder is not a vanishing perturbation at every scale. Moment matching alone does not establish quenched interface universality or the stated beta_c expansion. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
401/218	Quenched SLE3 Universality for the Weak Random-Bond Ising Model	L2	–	For the binary random-bond model, exact local joint-law calculations may check the pure/disordered comparison terms. They are peripheral to the full domain/interface convergence argument. **Boundary:** Do not replace the fixed-epsilon quenched problem by an annealed or independent-insertion approximation, or infer SLE3 from a covariance identity. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
402/218	Logarithmic Relative Fluctuations in the Weakly Disordered Planar Ising Model	L2	–	The relative-fluctuation proof has concrete two-insertion environment covariance calculations that can be checked with the corpus's joint-moment formulas. The scale-dependent logarithmic fluctuation estimate remains the main difficulty. **Boundary:** The deterministic reference is assumed critical. Independence must be justified after conditioning; local covariance calculations do not yield the claimed logarithmic exponent. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
403/218	Conformal universality of bulk Ising correlations under weak interactions	L3	C3	The radial conditional-averaging and analytic local recursion for weakly interacting Ising correlations can be handled as nonlinear integral maps with controlled derivatives and remainders. This is a meaningful component of the bulk-correlation analysis. **Boundary:** Finite-scale branch control does not establish conformal covariance, collision asymptotics, the all-order RG bounds or universality of the continuum n-spin/m-energy family. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
404/218	SLE3 universality for weak finite-range Ising interactions	L2	–	The weak-interaction SLE3 proof imports a conditional finite-graph comparison and then establishes interface convergence. The corpus provides peripheral conditional-moment and error bookkeeping, not that comparison theorem. **Boundary:** The companion input must remain explicit. Moment accuracy alone does not control changing Jordan domains, boundary pins or tightness of the entire interface. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
405/218	Buffered comparison and stopping-band resolution in critical Ising	L2	–	The finite-graph likelihood-ratio and two-spin covariance calculations are concrete but peripheral targets for exact joint-law auditing. The ordered coupling and bad-event bounds are not consequences of generic moment transport. **Boundary:** Preserve FKG/monotonicity and conditioning assumptions. The corpus's restricted graph-gluing theorem is not a substitute for the manuscript's Ising coupling or its total-variation estimate. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
406/219	Fixed-energy universality for weak Anderson disorder on random regular graphs	L3	C3	The tree resolvent recursion is exactly a Schur-complement equation, and its law-stability step is a nonlinear residual problem. This is a substantial concrete route into the proof, but not the GOE universality argument as a whole. **Boundary:** The proof uses a multi-step/one-leaf mechanism, not an unconditional one-step contraction. Small second moments alone do not control near-zero denominators or establish the finite-graph GOE replacement. **Review:** Read main statement/proof structure and first 152 lines of 02-tree.tex, including the Schur recursion and law-stability setup; no full universality audit. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
407/219	GOE bulk universality for fixed-degree random regular graphs	L2	–	The Gaussian edge-mark construction has exact finite joint moments that the corpus can audit. The spectral universality claim is driven by the manuscript's entropy and representation arguments, so the match is peripheral. **Boundary:** Correct Gaussian moments do not establish a GOE point process, local laws or replacement uniformly in graph size. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
408/220	A directional zero-one law for finite-range-dependent random environments	L1	–	The finite-range-dependent RWRE zero-one law depends on regeneration and fresh-row conditioning under a dependent environment. No concrete corpus result controls that directional event or its conditioning structure. **Boundary:** A finite transition matrix or independent-sampling theorem does not apply to future-conditioned environment exploration; keep d >= 3 and uniform ellipticity explicit. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
409/220	A directional zero-one law under strict ellipticity	L1	–	The iid strictly elliptic RWRE zero-one argument uses fresh-row products and regeneration geometry. No inspected Ram theorem supplies the required directional-event argument. **Boundary:** Strict ellipticity here must not be silently upgraded to uniform ellipticity. Finite-state Markov inverses do not settle the infinite random-environment zero-one law. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
410/220	Directional transience implies ballisticity	L1	–	The directional-transience-to-ballisticity result requires strong bad-exit and regeneration-duration tails. The corpus's finite Markov or moment identities provide no concrete route to those estimates. **Boundary:** Directional transience is not itself a finite-moment certificate; iid uniform ellipticity and the manuscript's quantitative exit estimate remain essential. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
411/221	The Mezard-Parisi formula for diluted spin glasses	L3	C3	The diluted-spin-glass variational functional is assembled from finite-depth random tree expectations. Controlled cubature, joint-law moments and nonlinear integral derivatives can help evaluate and audit significant pieces of that functional. **Boundary:** Numerical trial upper/lower values are not the Mezard-Parisi formula. Conditional independence, tail control, the overlap mechanism and the limiting infimum must be proved separately. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
412/222	The free energy of the Ising random perceptron	L4	C3	The variational core explicitly uses Gaussian heat kernels, exponential tilts and their nonlinear generator. These are very relevant same-operator targets for the corpus's integral derivatives, finite-moment remainders and branch control. **Boundary:** The smooth compact-potential calculation does not automatically extend to bounded Borel potentials by Taylor expansion. The variational equality, disorder limit and approximation arguments remain substantive. **Review:** Read main/model statements and proof route, plus first 171 lines of variational.tex and additional labelled-statement excerpts; inspected derivative/generator calculations, not the full free-energy proof. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
413/222	Microscopic jamming in the negative spherical perceptron	L5	C3	Direct core match: the exponent certificate is an explicit nonlinear boundary ODE, rational polynomial-panel residual calculation and tail/eigenfunction selection problem. These are exactly the corpus's differential-operator and rigorous residual-certificate objects, even though the earlier jamming theory is not supplied. **Boundary:** Panel residuals alone do not prove a global solution or select the correct eigenfunction. The numerical margins and rational certificate suite were not replayed; pressure/critical-similarity and mixture-universality inputs were not fully audited. **Review:** Read main statement/proof structure and main.tex lines 8872-9211 covering the certificate ODE, panel error, Riccati tails and shooting arguments; no computation replay or certification of the full jamming proof. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
414/222	The spherical perceptron with bi-orthogonally invariant disorder	L3	C3	The spectral-block variational functional for the biorthogonally invariant perceptron offers a concrete route for measure-aware spectral integration and nonlinear expectation evaluation. This covers significant parts of the angular/radial functional calculation. **Boundary:** Spectral moments without tail/no-outlier control do not justify the limit. Haar geometry, cavity estimates and the variational equality are not consequences of coordinate transport. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
415/222	The free energy of the spherical random perceptron	L4	C2	The spherical perceptron's variational functional is built from nonlinear Gaussian integral recursions of the same type treated in the corpus. A domain-correct representation is very relevant to the main analytic functional, without claiming the corpus proves the formula. **Boundary:** The comparison/interpolation proof and infinite-hierarchy limit remain separate. Smoothness, exponential-integrability and invariant-ball hypotheses must be checked for any proposed numerical certificate. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
416/223	Square-lattice FK interfaces and nested loops for 1 <= q < 4	L2	–	Finite observable/Fourier normalization can be audited locally, but the FK q in [1,4) SLE/CLE and Cardy conclusions require probabilistic tightness and conformal identification not present in the corpus. **Boundary:** The continuum elliptic theorem does not automatically apply to a discrete complex observable. Constants are not stated uniformly up to q=4; transform identities do not prove SLE/CLE convergence. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
417/223	Self-dual random-cluster interfaces below one	L2	–	The q<1 open-cap construction has finite complex-weight observable identities that can receive algebraic/transform checks. Their role is peripheral to the domain Markov and interface-limit argument. **Boundary:** Complex weights are not a positive probability measure or Gram weight. Do not invoke FKG, positivity or covariance inequalities unavailable in the q<1 argument. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
418/223	Quenched SLE Universality for Weakly Disordered FK-Ising Interfaces	L2	–	Local disorder-conditioned comparisons in weakly disordered FK-Ising can be checked through joint moments and explicit remainder bounds. This is peripheral to the quenched SLE16/3 identification. **Boundary:** Independent environment moments cannot replace quenched conditioning or provide uniform estimates in evolving slit domains. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
419/223	Thermal FK-Ising interfaces and massive SLE	L4	C2	The massive observable is an actual nonlinear boundary-value problem, Delta h=-4m|grad h|, followed by a moving-domain drift formula. Boundary-correct Green formulations and nonlinear residual control are very relevant to this central PDE mechanism. **Boundary:** Rough prime-end boundary jumps need not admit the global H1 lift required by the cited elliptic theorem. Bounded-drift uniqueness, passage to rough domains, stochastic drift integrability and interface tightness require their own arguments. **Review:** Read model.tex and energy.tex through line 240 each, including the BVP, bounded-drift uniqueness route and energy estimate; not a complete massive-SLE proof audit. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
420/223	Natural Occupation Measures for Critical Square-Lattice FK Interfaces	L2	–	Finite spectral/electrical normalizations may be checked for the occupation observable, but the core natural-parameter limit needs arm probabilities and random-curve measure convergence. The corpus route is peripheral. **Boundary:** An exact finite Green or Gram representation does not identify Minkowski content or provide the multi-point arm and tightness estimates needed for natural parametrization. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
421/223	Conformal Limits of Critical Square-Lattice Random-Cluster Interfaces	L2	–	The extension to q=4 includes sign/loop-multiplicity and rotation bookkeeping that can receive finite exact-coordinate checks. The endpoint GFF and conformal interface conclusions are not supplied by these checks. **Boundary:** Neither a complex transform identity nor an exact finite polynomial representation supplies the q=4 endpoint estimates, positive loop law or GFF identification. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
422/224	From critical crossings to quenched near-critical universality in Voronoi percolation	L2	–	The near-critical Voronoi statement is conditional on a Cardy input and uses pivotal intensities. The corpus can check finite conditional observables but has only a peripheral role in the near-critical construction. **Boundary:** Moment consistency does not provide pivotal scaling, spatial decorrelation or the near-critical continuum limit; the conditional premise must not be reported as proved. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
423/224	A Pivotal Amplitude for Voronoi Percolation from Cardy's Formula	L3	C3	The rank-one annulus propagation and decay of complementary modes form a concrete spectral approximation problem. Physical projection and omitted-space estimates can address a significant component of the pivotal-amplitude argument. **Boundary:** The Cardy premise and scale-uniform fast-mode decay must be supplied. A numerically dominant finite eigenvector does not certify rank-one asymptotics or pivotal normalization. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
424/224	Cardy's formula for critical Poisson-Voronoi percolation	L2	–	The discrete holomorphic transport calculation offers a local transform/operator audit, but the Voronoi Cardy proof is principally a random-geometry and boundary-limit problem. **Boundary:** A continuum Green representation does not justify interchanging those limits or prove tightness and conformal identification of the random observable. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
425/225	The Gaussian free field limit of the balanced six-vertex model with variance multiplier $1/\arcsin(c/2)$	L3	C2	The six-vertex analysis uses commuting transfer rows and resolvent factorization, so physical spectral representations are relevant to significant analytic steps. The GFF conclusion and exact stiffness require additional control. **Boundary:** No complete transfer-domain or factorization proof was audited here. Finite spectra do not establish all-c GFF convergence, the coefficient, or endpoint uniformity. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
426/226	The curve scaling limit of half-plane double dimers	L3	C3	The doubledimer proof's one-particle sine-mode transfer and exterior-power construction provide a concrete finite spectral mechanism. Corpus Gram and source-preserving operator checks can support that part of the argument. **Boundary:** A correct sine basis or determinant formula is not CLE4 convergence. The random-curve tightness and excursion-to-loop identification remain separate. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
427/227	Universality of critical quench autocorrelations in the Sherrington-Kirkpatrick model	L3	C3	The critical SK quench observables are Gibbs-weighted semigroup pairings, with a concrete physical inner product and source functions. Gram/residual methods address significant finite-mode aspects of these quantities. **Boundary:** The entrance law, critical scaling and functional convergence are not obtained from a finite covariance match. Gibbs weighting cannot be dropped. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
428/227	Functional universality of critical SK autocorrelations	L5	C3	Direct core match: the main autocorrelation limit is expressed through the resolvent trace of the limiting Dirichlet operator H_Pi. The corpus's physical Gram, variational inverse and full-tail framework applies to this very source-resolvent observable. **Boundary:** The sources need not lie in an ordinary square-summable coordinate space; their stated dual-norm decay must be used. Mosco/resolvent convergence, the correctors, stochastic edge input and tightness remain manuscript-specific hard estimates. **Review:** Read main statement/proof route and first 145 lines of 07-trace.tex, including the resolvent representation, variational dual-norm argument and trace-convergence setup; not the entire universality proof. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
429/227	A spectral gap throughout the high-temperature Sherrington-Kirkpatrick phase	L3	C2	The Gaussian-observation localization argument has concrete posterior covariance and residual-matrix quantities that joint-moment and physical-norm methods can audit. These are significant pieces of the terminal Poincare argument. **Boundary:** The localization stopping/growth estimates and high-temperature SK bound do not follow from covariance positivity. The two posterior laws must not be conflated. **Review:** Read main statement/proof-route excerpts and labelled theorem/proof-input excerpts from sections/introduction.tex; no full localization or Poincare proof audit. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
430/227	Cutoff throughout the high-temperature Sherrington-Kirkpatrick phase	L4	C2	The exact slow sector is a weighted self-adjoint operator with two specified Hilbert completions. This is a very relevant match to physical Gram geometry, domain-aware spectra and full-residual control. **Boundary:** The corpus does not establish the absence of harmonic defects, isolation or cutoff. Do not replace the exact spectral quantity by (1-beta)^2 without proof, or lose the factor n in the clock. **Review:** Read corrected 01-introduction.tex statement/proof-route excerpts and labelled statements with nearby argument from 08-exact.tex; not the full exact-sector or cutoff proof. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
431/227	Critical slowing down in the Sherrington-Kirkpatrick model	L3	C3	The covariance/linear-Rayleigh calculation and stationary increment bound are concrete finite physical-norm targets. They support a significant test-function part of the claimed critical SK scale. **Boundary:** A linear test function can bound one direction of a gap estimate; it does not certify all-observable Poincare behavior, high-probability edge control or a matching mixing theorem. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
432/227	Stretched-exponential barriers for typical SK initial states	L2	–	Finite constrained-overlap observables can receive joint-moment checks, but the low-temperature SK barrier proof is driven by a constrained free-energy action and rare exit estimates. **Boundary:** Moment positivity cannot supply an exponentially small conductance or the claimed exp(n^(1/10000)) barrier. Typical Gibbs-start and worst-case statements are different. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
433/227	A typical-start upper bound for low-temperature SK Glauber dynamics	L3	C3	The scalar Gaussian proposal and conditional log-concave calculation in the typical-start mixing argument have a concrete nonlinear expectation/covariance route. These are significant analytic components, not a proof of the claimed running time. **Boundary:** The strong-log-concavity and entropy comparison premises must be proved for the conditional problem. The subexponential bound is not automatically a worst-case mixing guarantee. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
434/227	Critical mixing in the Sherrington-Kirkpatrick model	L3	C2	The all-observable critical SK mixing route uses Gaussian localization covariance and operator inequalities, making physical-metric and joint-moment auditing useful to substantial intermediate steps. **Boundary:** Covariance positivity and a finite linear-mode calculation do not prove a uniform all-observable functional inequality or cutoff/mixing for the full dynamics. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
435/228	A continuum temperature singularity for a radial pair potential	L2	–	Finite pressure-branch and cluster-energy calculations can be expressed exactly or enclosed polynomially. This is a peripheral route into a continuum phase-coexistence argument whose main issue is thermodynamic control. **Boundary:** A finite polynomial certificate does not prove stable thermodynamic phases at fixed density. Preserve radial-potential stability and distinguish finite cluster convexity from pressure coexistence. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
436/228	A radial continuum phase transition with algebraic decay	L2	–	The algebraically decaying pair potential admits finite-range tail and local energy checks using integral remainders. Such checks are peripheral to the phase-coexistence construction. **Boundary:** Finite truncation need not preserve stability or the thermodynamic phase diagram. A local quadrature bound cannot establish coexistence in the full continuum model. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
437/229	The Reconstruction Threshold for the Ferromagnetic Four-State Potts Model	L5	C3	Direct core match: the four-state reconstruction threshold is reduced to explicit posterior-polynomial inequalities and exact finite subdivision certificates. Exact moment coordinates and Bernstein/positive-basis enclosures act on the same posterior cone and branch-saving polynomials. **Boundary:** The corpus's one-variable Bernstein statement needs the correct tensor or simplex extension; an arbitrary basis change does not prove positivity. Boundary zeros need exact treatment, and depth exhaustion is not success. The reported 4479/1935-cell checks were not replayed. **Review:** Read the main statement/proof route and the complete 05-polynomial-certificates.tex and 06-finite-verification.tex files. Inspected the symbolic inequality/subdivision arguments; did not run the certificate suite or certify the probabilistic reduction. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
438/229	A Capacity Criterion for Four-State Potts Reconstruction on Trees	L4	C3	The capacity criterion uses the same four-state posterior moments and cubic branch saving as the directly matched certificate problem. Exact joint-moment transport and polynomial error control are very relevant to its nonlinear information/flow comparison. **Boundary:** The nonlinear L3 capacity is not ordinary electrical L2 capacity. The companion saving inequality, telescoping flow construction and arbitrary bounded-tree accumulation bounds remain essential; exact moments alone do not prove the equivalence. **Review:** Read thm:main, section headings and proof-strategy excerpts in main.tex, plus the companion manuscript 437's polynomial/certificate files; did not read the full capacity proof. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
439/229	The exact reconstruction threshold for the three-state symmetric channel	L5	C3	Direct core match: the three-state threshold argument uses exact rational posterior polynomial inequalities, barycentric approximation and finite certified bounds. The corpus's exact polynomial representation, positivity enclosure and shared-law moment accounting target the same central inequalities. **Boundary:** Keep symmetric-channel and conditional-independence assumptions, including the coloring endpoint and denominator-zero limiting cases. A grid sample alone is not a global certificate. The reported 469-point/H=72 and rational table computations were not replayed. **Review:** Read main statement/proof route, first 260 lines of 06-grid.tex and first 260 lines of appendix-certificates.tex (including the appendix's certificate prose); no numerical or rational-code replay. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
440/230	An exact Hausdorff gauge for SLE	L2	–	The SLE Hausdorff-gauge proof uses multi-point Green moments and their order-dependent growth. The corpus can help enclose a specified finite integral but offers only a peripheral route to the all-order random covering argument. **Boundary:** Smooth finite-panel remainders do not cover singular collisions or uniform growth in moment order. They do not identify an exact Hausdorff gauge or an almost-sure measure. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
441/230	An explicit exact Hausdorff gauge for SLE	L2	–	Stopped Green-law integrals at a fixed selected disk are possible local integral-audit targets. The explicit log-log gauge over all times requires multiscale stopping and random geometry beyond the corpus. **Boundary:** A deterministic transform identity is not a stopping theorem. Do not infer all-time control, exceptional-set bounds or the exact gauge from a finite collection of moments. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
442/231	The free uniform spanning forest is a factor of IID	L4	C2	The factor construction contains a concrete causal inverse y=b+integral drift(y), proved by a factorial Picard estimate. This is a very relevant Volterra mechanism, not generic coordinate algebra, with a clear route from the corpus's causal Lipschitz integral theory. **Boundary:** The corpus theorem is stated for finite-dimensional continuous-state kernels; countably many coordinates and Borel time dependence require the manuscript's explicit extension, not a smooth jet expansion. Forest-law identification and posterior innovations are additional arguments. **Review:** Read main statement/proof route and first 245 lines of decoder.tex, including the causal inverse/Picard and Brownian-sampler arguments; not the full forest-law/factor proof. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
443/232	Gaussian free-field limits of weighted integer Lipschitz heights	L3	C3	The integer-Lipschitz surface proof's conditional Dirichlet covariance and Gaussian moment calculations have a concrete boundary-aware Green-operator match. These are significant analytic pieces of the GFF identification. **Boundary:** Conditioning on geometry alone may lose the spin information used by the manuscript. Finite Gaussian covariance agreement does not establish the conditional law or distributional tightness. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
444/232	The Gaussian free field limit of integer Lipschitz heights with two-arc boundary data	L3	C3	The two-arc height problem explicitly combines harmonic means, Green covariances and boundary traces. Boundary-lift and physical-energy certificates address substantial deterministic components. **Boundary:** A regular-domain elliptic solve does not automatically control singular boundary data, moment collisions or the complete GFF convergence. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
445/232	Uniform real Lipschitz surfaces on the triangular lattice	L3	C3	The real-Lipschitz surface argument uses a stiffness cell-flux calculation and conditional Dirichlet covariances. These provide concrete PDE/physical-metric targets within a much larger SLE4/GFF proof. **Boundary:** The corpus does not establish local-set structure, random-domain tightness or SLE4 identification. A deterministic stiffness calculation alone is insufficient. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
446/233	The joint scaling limit of critical Ashkin-Teller currents	L3	C2	The Ashkin-Teller joint limit contains a concrete height-field covariance/Green component. Moment and weak-operator representations can support this significant part while leaving cluster identification separate. **Boundary:** Marginal covariance agreement does not establish a joint local-set coupling or the cluster topology; current constraints and shared conditioning must be retained. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
447/234	All-temperature pressure for orthogonally invariant Ising spin glasses	L3	C3	The invariant Ising pressure functional depends explicitly on the limiting spectrum and nonlinear finite-alphabet overlap averages. Measure-aware spectral integration and controlled expectation evaluation address significant functional components. **Boundary:** Spectral moment matching is insufficient without support and tail control. Haar rotation estimates, overlap optimization and the all-temperature variational equality are not proved by a finite operator representation. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
448/235	A Limiting Satisfiability Threshold for Every Fixed Clause Size	L1	–	The fixed-k SAT threshold argument is based on a capped hitting process, deletion delays and formula combinatorics. No inspected corpus theorem supplies a concrete route to the central threshold mechanism. **Boundary:** Generic moment or finite Boolean encodings do not locate a satisfiability threshold or prove its sharpness for every fixed k >= 3. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
449/235	Linear Variance of the Random 3-SAT Hitting Time	L1	–	The linear-variance claim for the 3SAT hitting index uses clause-puncture occupation and an Efron-Stein argument. No concrete corpus result controls the required combinatorial influence and occupation estimates. **Boundary:** Covariance positivity or a finite moment representation does not prove a Theta(n) lower and upper variance bound for this adaptive hitting statistic. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
450/235	Variance of the Random k-SAT Hitting Time	L1	–	The capped kSAT hitting-variance dichotomy relies on deletion/occupation estimates specific to random clauses. No matching operator or moment theorem was found that advances those core estimates. **Boundary:** The cap and conditioning assumptions cannot be dropped. Generic concentration or exact finite counting does not establish the claimed n log n versus n scaling. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
451/235	Computing the Random 3-SAT Threshold	L3	C3	The random-CSP certificate paper explicitly separates finite trial-functional evaluation from the variational theorem that makes the search complete. Controlled integration and exact coefficient/error checks can support significant parts of the finite certificate layer. **Boundary:** A strict-margin semidecision is not a total decision algorithm or an efficient threshold solver. Conditional sampling must retain its actual law, and cubature requires its total-variation/error hypotheses. No trial value or certificate was computed here. **Review:** Read the main statement/proof route and the complete certificates.tex and proof-map.tex files; inspected the certificate/completeness distinction but did not execute a search or certify the variational inputs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
452/236	The sharp factor-of-IID threshold for the free Ising model on regular trees	L3	C3	The free-Ising factor construction on a tree contains an innovation integral equation with a meaningful Volterra/Picard route. It can support a substantial deterministic construction step while leaving the sharp spatial threshold to the manuscript. **Boundary:** Temporal causal invertibility does not imply the needed spatial decay or factor property. The critical endpoint requires cancellation estimates, not an unconditional contraction assertion. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
453/237	Radial transfer estimates and polygon length laws for honeycomb walks	L2	–	The radial-transfer calculation contains finite normalized moment/spectral quantities that can be audited in the correct metric. The asserted annular exponents and scale-uniform polygon moments are not delivered by that finite representation. **Boundary:** Finite transfer spectra or moment identities do not establish annular scaling, uniform radial limits or the necessary geometry-to-transfer comparison. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
454/237	Critical honeycomb chords with prescribed boundary endpoints	L2	–	Finite endpoint-kernel and stencil calculations offer a peripheral exact-operator check in the prescribed-endpoint chord argument. The length exponent and terminal attachment cost require uniform geometric estimates. **Boundary:** A local kernel expansion cannot supply the R^(-5/4) endpoint cost, a uniform tail bound or a conditioned chord-length law. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
455/237	Cylinder loop weights and planar nesting	L3	C3	The cylinder calculation combines an exact vacuum polynomial and particle contour integrals. Domain-aware kernel coordinates and analytic-majorant checks are relevant to substantial analytic steps of that representation. **Boundary:** Complex contour weights are not a positive Gram measure. The full particle-number sum, pole residues and uniform asymptotics require estimates beyond finite polynomial exactness. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
456/237	Mass and covering exponents for fixed-length honeycomb walks	L2	–	The fixed-length walk statement mainly converts companion analytic inputs into geometric mass-cover estimates. A finite moment/truncation audit is only peripheral to its all-n high-probability conclusion. **Boundary:** Do not replace all-n control by density-one or subsequential information imported from another paper. A finite moment approximation does not imply the n^(3/4) geometry or a high-probability cover bound. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
457/237	Signed cylinder propagation and marked polygons on the honeycomb lattice	L3	C3	The signed-cylinder proof uses finite polynomial interpolation, a transfer projector and interaction estimates in a negative Sobolev norm. Exact reconstruction and full-residual methods can address meaningful parts of those operator calculations. **Boundary:** Signed amplitudes cannot be treated as probabilities, and finite interpolation does not control the negative-Sobolev interaction tail or give uniform cylinder asymptotics. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
458/237	Cylinder amplitudes and logarithmic bridge-length windows on the honeycomb lattice	L3	C3	The amplitude calculation turns Pfaffian/determinant expressions into a positive Fourier norm. Exact polynomial and physical Gram methods can audit this significant nonvanishing mechanism. **Boundary:** A positive semidefinite Gram form can have a kernel. It does not by itself prove nonzero amplitude or a size-uniform leading coefficient. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
459/237	Marked polygon correlations and one-arc bounds	L3	C3	The marked-polygon analysis contains an explicit Cauchy interpolation/Pfaffian calculation and a conversion from complex contours to physical positive weights. These are substantive targets for exact reconstruction and analytic-domain auditing. **Boundary:** A complex contour formula is not automatically a positive measure. Finite interpolation cannot justify a uniform denominator bound or exchanging contour, size and particle-number limits. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
460/237	Disk transfer representations and confined bridge mass	L5	C3	Direct core match: the boundary deficit is an inverse pairing of a concrete moment matrix, reduced to a polynomial-evaluation norm in a positive weight. Exact moment coordinates and physical Gram normalization act on the same operator that yields the key boundary exponent. **Boundary:** Only the scalar inverse pairing is asserted to continue at coincident rapidities, not the original inverse matrix. Gram representation does not itself supply the N^(3/4) evaluation-norm estimate, spatial comparisons or bridge-mass theorem. **Review:** Read main statement/proof outline; boundary.tex lines 1-145, 261-336 and 559-673, including the inverse-pairing statement and homogeneous Gram/evaluation-norm argument. Also inspected labelled analytic.tex excerpts; not the full disk-transfer or corridor proof. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
461/237	Polynomial vacuum representations and bridge mass for honeycomb walks	L3	C3	The vacuum-polynomial calculation combines exact finite transfer algebra with a positive Fourier-period norm. The corpus can address significant reconstruction and positivity/normalization components. **Boundary:** A finite positive norm does not give the asserted large-size amplitude. Omitting the zero mode or treating a complex pre-limit contour as positive would invalidate the comparison. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
462/237	Renewal and changes of law for critical honeycomb walks	L2	–	The renewal identity H=U/q and geometric-series changes of law are concrete finite normalization targets. The corpus can audit these algebraic steps, but not the uniform combinatorial estimates behind the walk laws. **Boundary:** A geometric inverse requires a valid scalar/operator convergence bound. The inspected unrestricted exact-length law is density-one, not an all-length result; no transform makes that quantifier stronger. **Review:** Read the screening entry and main-source headings, then actual theorem/proof excerpts from thermal.tex and uniform.tex, including the renewal identity and density-one qualification. Did not read every renewal proof. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
463/237	Critical strip-crossing mass on the honeycomb lattice	L4	C3	The strip-crossing mass is explicitly represented by a Cauchy bimoment matrix and a positive multiple integral. This is a very relevant same-kernel moment problem, with a concrete domain-correct representation route beyond generic coordinate algebra. **Boundary:** The bimoment matrix is not automatically a symmetric Hankel Gram matrix; its determinant positivity uses the Cauchy/Vandermonde argument. Finite moment reconstruction does not establish the N-dependent mass asymptotic or uniform inverse conditioning. **Review:** Read main statement/proof route and first 210 lines of 05_positive_integral.tex, including the Cauchy bimoment and positive-integral derivation; no full asymptotic or determinant proof certification. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
464/237	Uniform marked-polygon estimates and sharp finite bridge moments	L3	C3	The finite bridge comparison uses a polynomial double-row transfer, positive interpolation and current identities. Exact polynomial reconstruction and domain-aware operator checks address significant finite analytic steps. **Boundary:** A finite identity does not imply constants uniform in N and d, or the sharp bridge comparison. Interpolation nodes and positivity of the physical law must be justified. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
465/237	Cap-selected amplitudes and triangle chords for honeycomb walks	L3	C3	The cap-amplitude proof is a concrete spectral-projector and source-overlap problem. Physical Gram/residual and spectral-promotion gates are relevant to a significant part of establishing a nonzero selected amplitude. **Boundary:** A nonzero finite compression overlap does not establish the full amplitude or exclude competing modes. The promotion gate needs an actual tail resolvent and separation hypotheses. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
466/238	Optimal-order mixing of the Thorp shuffle	L2	–	The Thorp argument's conditional weighted mean-square calculation is a concrete local joint-law audit target. The dimension-uniform contraction and optimal mixing scale remain representation/combinatorics results. **Boundary:** Correct finite conditional moments do not prove the 1600d mixing bound. Keep the preceding-sweep conditioning distinct from independence of all future randomness. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
467/238	Random coordinate frames and partial permutation laws	L2	–	The random-coordinate frame analysis can be checked in its finite physical metric and joint law. The decisive adaptation argument for the Thorp process is not supplied by a generic frame identity. **Boundary:** A deterministic frame bound does not automatically survive conditioning on future choices; no corpus frame theorem establishes the claimed global mixing time. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
468/238	From partial permutation information to Fourier bounds	L2	–	The Fourier-coset estimate admits exact finite character/projection auditing. The key cap bound depends on representation multiplicities and type counting, not on the existence of a Fourier transform. **Boundary:** The corpus's group example does not prove this symmetric-group multiplicity estimate. Fourier inversion alone yields neither the dimension gain nor the optimized cap bound. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
469/238	Conditional information under deterministic coordinate sweeps	L2	–	The partial-list Thorp proof has finite conditional-information quantities that can be audited exactly. This is peripheral to the manuscript's multi-sweep combinatorial contraction. **Boundary:** A covariance or information table for one stage cannot establish the full chain's total-variation bound; list conditioning and stage dependence must be retained. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
470/238	Conditional permutations in a revealed switching environment	L2	–	The conditional free-half-deck calculation contains a concrete two-copy collision law. Joint-moment auditing can check this finite ingredient, while representation contraction supplies the main mixing argument. **Boundary:** The manuscript separately states 16,040,400d physical shuffles; preserve and check its conversion rather than substituting the auxiliary J=40000 sweep parameter. Finite collision moments do not prove the global contraction. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
471/238	Routing densities and representation contraction for Thorp sweeps	L2	–	The weighted routing-forest factorial moments provide specific finite joint-law quantities for exact moment checks. The representation-degree entropy gain and routing-density estimate are separate combinatorial inputs. **Boundary:** The corpus's shared-law requirement is useful precisely because the forest heights are dependent. Exact moments do not establish the required uniform entropy or dimension gain. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
472/238	Row-column symmetry and contraction of coordinate sweeps	L2	–	The weighted Schatten-moment calculation can be checked as a finite physical trace identity. Its effective weight range and overlap counting are the difficult representation-specific parts. **Boundary:** A physical metric does not prove the combinatorial overlap estimate or uniform weighted entropy bound. Schatten and operator norms cannot be interchanged without their dimension factors. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
473/238	Random-subspace tests and trace smoothing for coordinate sweeps	L3	C2	The regular-trace-to-mixing step is a concrete operator-power problem in which nonnormality matters. The corpus's full-residual and pseudospectral cautions address a significant verification point beyond generic matrix notation. **Boundary:** For nonnormal Q, (Q*Q)^v is not generally (Q^v)*Q^v. A spectral radius or a finite eigenvalue plot alone cannot prove the stated trace-to-mixing estimate; the combinatorial trace bound itself remains unverified. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
474/238	Compatibility entropy and the spectrum of a Thorp sweep	L2	–	The compatibility-entropy proof contains finite weighted positive-trace and Holder calculations that the corpus's physical Gram discipline can audit. This is peripheral to the entropy counting that creates the gain. **Boundary:** Noncommuting factors cannot be rearranged as scalar weights. A correct finite trace inequality does not supply the compatibility-entropy estimate or uniform mixing contraction. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
475/238	Signed tensor densities and diagram budgets for the Thorp shuffle	L2	–	The signed tensor-density argument uses unitary averages and inverse square-root cell Gram matrices. Physical metric conditioning is a concrete peripheral audit target. **Boundary:** Entrywise-small Gram errors do not control inverse square roots near rank loss. This metric check does not establish the signed-density cancellation or the type-entropy theorem. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
476/238	Conditional coordinate sweeps and analytic transfer	L3	C3	The analytic transfer from a small real parameter window to the binary endpoint requires precisely the branch/domain and locally uniform majorant discipline recorded in the corpus. This is a substantial analytic step, not just a formal polynomial substitution. **Boundary:** The identity theorem gives uniqueness, not stable continuation or an endpoint bound. Small-parameter control without a uniform complex-domain majorant cannot justify the binary endpoint. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
477/238	A strict four-row permanent inequality and permutation moments	L3	C3	The near-uniform four-row permanent inequality has a concrete joint-moment expansion and strict quadratic margin. Exact covariance and polynomial remainder enclosures can support a significant quantitative part of its local proof. **Boundary:** The result concerns a near-uniform law with exact marginals, not arbitrary permutation laws. The first variation can fail without those marginals; the p=2 complement gap and the choice of p0>4/3 remain necessary. **Review:** Read main statement/proof route and the complete 02-permanent.tex proof; no numerical optimization of constants or independent verification of external permanent inequalities. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
478/239	The sharp exponential rate of singularity for symmetric Bernoulli matrices	L2	–	The symmetric Bernoulli singularity proof uses a Schur reduction that can be audited exactly. The sharp exponential probability is an arithmetic/entropy statement about random rows, so the operator match is peripheral. **Boundary:** A Schur identity does not bound the rare arithmetic extension events or prove 2^(-n+o(n)) singularity. Gaussian covariance substitution loses the discrete arithmetic mechanism. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
479/239	The sharp singularity rate for biased symmetric sign matrices	L2	–	For biased symmetric Bernoulli matrices, exact Schur/source bookkeeping can check the finite reduction. The claimed sharp rate comes from conditioned bit entropy and low-rank mean handling, not generic spectral coordinates. **Boundary:** The fixed-p hypothesis and exclusion of p=1/2 matter. Independent or Gaussian moment matching does not prove the rate p^2+(1-p)^2 or the needed rare-extension estimates. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
480/240	A CH obstruction to a prescribed categoricity threshold	L1	–	The CH-dependent AEC threshold failure is a model-theoretic construction using countable languages, fibre torsors and almost-equality colorings at large cardinals. No concrete corpus theorem addresses its core mechanism. **Boundary:** Finite Boolean, lattice or moment encodings do not transfer to categoricity thresholds beth_{omega_2} and beth_{(2^{aleph_1})^+}. The continuum hypothesis, language size and cardinal quantifiers must remain explicit. **Review:** Read the screening-pack entry, actual main-result excerpts and section headings; read selected proof-route narrative where present, not the full manuscript or all proofs. All three corpus maps were read. Corpus references may be map-mediated; only originals listed in evidence_files are claimed as directly inspected. Applicability is not validity, novelty or proof success; no source computations were replayed.
481/240	Eventual categoricity for abstract elementary classes	L1	–	The uniform eventual-categoricity statement is organized through AEC cardinal transfer and support/subcube arguments. These are model-theoretic structures, not a specified integral operator or positive moment problem. **Boundary:** No concrete Ram route to categoricity or the required cardinal transfer was identified; finite coefficient encodings would not address the core implication. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
482/241	Rigidity of the Turing degrees	L1	–	The rigidity claim for Turing degrees uses definability, Borel lifts and recovery on a comeager set. The inspected rigidity proof does not expose an analytic inverse or Hilbert-space reconstruction problem. **Boundary:** Topological or descriptive-set recovery here is not frame reconstruction; no applicable domain-correct corpus operator was identified. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
483/242	Single-fold Diophantine representations	L1	–	The singlefold Diophantine representation claim requires exactly one auxiliary integer witness for arbitrary recursively enumerable data. The witness-ledger construction, not bounded real-polynomial positivity, is the decisive mechanism. **Boundary:** Bernstein/SOS certificates on compact real boxes cannot establish unbounded integer existence, uniqueness or recursive-enumeration equivalence. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
484/243	Choiceless polynomial time with counting does not capture polynomial time	L1	–	The noncapture result for choiceless polynomial time compares counting and finite-field rank logics. The linear-algebraic-logics section concerns logical expressibility rather than a numerically approximated linear inverse. **Boundary:** Real or complex positive-Gram machinery does not transfer to finite-field rank definability or prove the separation. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
485/243	Witnessed symmetric choice is strictly stronger than choiceless polynomial time with counting	L1	–	The witnessed-choice result proceeds through CPT syntax and evaluation traces with polynomial hereditary bounds. The trace lemma is combinatorial execution bookkeeping, not an operator trace or transform identity. **Boundary:** No concrete analytic, moment or residual subproblem was located in the statements and proof excerpts reviewed. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
486/244	The Partition Principle does not imply Choice	L1	–	The partition-principle/choice comparison relies on a forcing quotient and a local quotient bound. Its quotient is set-theoretic, without the norms and linearity needed for the corpus quotient or frame constructions. **Boundary:** No route from the corpus's analytic inverses to the forcing or choice argument was identified. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
487/245	Weak and strong normalization in pure type systems	L1	–	The normalization claim for pure type systems is proved through labelled reduction and an encoding/metatheory argument. Normalization refers to terms and reduction sequences, not analytic normalization of a kernel. **Boundary:** Finite jets and contraction estimates do not supply termination or typed reduction lifting. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
488/246	A Modulus Proof of Cannon's Conjecture	L1	–	The Cannon-modulus argument uses packing pairs and an exclusion of subexponential records. The inspected record-counting proposition has no explicit polynomial positivity or coercive PDE certificate. **Boundary:** The word 'modulus' does not establish a transform bridge; no concrete corpus operator was found at this depth. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
489/247	An infinite finitely presented residually finite 2-group	L1	–	The finitely presented residually finite infinite 2-group construction uses a Steinberg strip and algebra over characteristic two. Its finite algebra is not an ordered real/complex Gram system. **Boundary:** Positive-definite moment arguments do not establish infinitude, residual finiteness or finite presentation in this setting. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
490/247	An infinite finitely presented periodic group	L1	–	The periodic-group construction uses a nil algebra and Steinberg stability, including a finite-field linear-algebra lemma. This is algebraic stability of a group construction, not a bounded analytic inverse. **Boundary:** No corpus theorem was identified that supplies periodicity, infinitude or the required group-theoretic stabilization. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
491/248	Thompson's group F is nonamenable	L1	–	The Thompson-group nonamenability statement is supported by recursive dyadic partitions and eventual coloring arguments. Those combinatorial objects are not a specified frame analysis map. **Boundary:** An arbitrary encoding of the partitions would be peripheral and would not establish the nonamenability mechanism. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
492/249	A finitely generated counterexample to the Eilenberg-Ganea conjecture	L1	–	The cohomological/geometric-dimension separation uses small-label transport and holonomy in a constructed complex. The obstruction concerns existence of a low-dimensional classifying space. **Boundary:** Finite cochain coordinate equivalence cannot replace the global geometric-dimension obstruction or supply the asserted classifying space. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
493/250	Finite algebraic envelopes and the Boone-Higman conjecture	L1	–	The Boone-Higman envelope construction is driven by HNN extensions and group-input embedding data. No analytic kernel or certificate inequality appears in the reviewed group-input lemma. **Boundary:** Ram transforms do not give the required algorithmic group embedding or simplicity theorem merely by representing its relations. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
494/250	Simple Finfinity overgroups of groups with decidable word problem	L1	–	The simple F-infinity overgroup result uses highly transitive actions and noncommutative matrix-group constructions. Its matrices encode group operations rather than self-adjoint forms with a physical Gram metric. **Boundary:** No concrete positivity, spectral or nonlinear-IE route to the action and finiteness properties was identified. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
495/250	A universal group of type Finfinity	L1	–	The universal F-infinity construction rests on HNN injection and preservation of group-theoretic finiteness properties. The inspected proof structure is categorical and combinatorial. **Boundary:** Coordinate transforms do not establish the HNN embedding or the required universal group property. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
496/251	Unitarizability implies amenability for discrete groups	L2	–	The finite random-sign frames in weights.tex provide a genuine, but auxiliary, Gram-norm target: selected subfamilies of unit vectors must have uniformly bounded synthesis norm. The corpus's finite frame and metric identities apply to those actual vectors. **Boundary:** This checks the finite frame lemma or instances of it; it does not establish the infinite-group unitarizability/amenability implication or its uniform construction. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
497/251	Strong Ulam Stability Characterizes Amenability	L1	–	The strong-Ulam-stability characterization proceeds through boundary flags and strong proximality. The inspected argument does not reduce the core group-stability assertion to a specified invertible Banach derivative. **Boundary:** Small representation defects alone are not the corpus's root-certificate hypotheses; no concrete core map was located. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
498/252	A torsion-free hyperbolic group that is not residually finite	L1	–	The hyperbolic non-residually-finite construction uses thin-triangle and path-stability geometry. The replacement read of geometry-thinness.tex confirms a geometric small-cancellation route rather than an integral or Gram lift. **Boundary:** Analytic stability estimates do not supply torsion-freeness, hyperbolicity or failure of finite-quotient separation. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
499/253	An infinite finitely presented simple amenable group	L1	–	The finitely presented simple amenable group construction uses transport slots and combinatorial frames. These frames are organizational data for a group action, not Hilbert frames with an analysis bound. **Boundary:** No operator-domain or positive-measure interpretation supporting a concrete corpus theorem was identified. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
500/254	Harmonic heights and the Artin K(pi,1) conjecture	L1	–	The Artin K(pi,1) argument uses harmonic heights, minimal bimodule complexes and a two-point candidate-uniqueness statement. 'Harmonic' here does not identify a Laplacian boundary-value problem. **Boundary:** Finite differential coordinates cannot establish asphericity or the candidate-uniqueness/topological comparison. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
501/254	An Artin group with no geometric CAT(0) action	L1	–	The non-CAT(0) Artin-group example is driven by a triangle obstruction for sector length ratios. The reviewed obstruction is metric-geometric rather than a positivity statement for a displayed polynomial family. **Boundary:** No concrete core certificate map was identified; merely encoding the length constraints would be peripheral. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
502/254	Parabolic intersections in Artin groups	L1	–	The parabolic-intersection theorem for Artin groups uses stabilizer and conjugation arguments. 'Parabolic' denotes subgroups, not a parabolic PDE. **Boundary:** The corpus heat and Green operators do not address subgroup intersections or conjugation. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
503/255	Quasi-isometric recognition of virtually polycyclic groups	L1	–	The quasi-isometric polycyclic recognition argument uses bounded-height Lie-group charts and suffix concentration. The inspected probability/geometry lemma is not a finite kernel reconstruction problem. **Boundary:** Generic concentration or coordinate changes would not give quasi-isometric recognition or its uniform chart argument. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
504/256	Nonsingular systems of equations over arbitrary groups	L1	–	The nonsingular-equations-over-groups result uses an exponent-sum rank condition and planar/unitary phase-counting arguments to construct an overgroup. The rank condition is input to an algebraic existence theorem. **Boundary:** Checking that one rational matrix has full rank is incidental; the corpus does not supply the overgroup solution or the planar obstruction. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
505/256	The Kervaire theorem for groups	L1	–	The unimodular-word/Kervaire argument proceeds through planar topology and normal closures. It does not expose a Banach nonlinear equation with a normed-domain inverse. **Boundary:** A formal analogy between equations over groups and nonlinear IE solving is not an applicable theorem. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
506/257	A hyperbolic group with no geometric CAT(0) action	L1	–	The hyperbolic non-CAT(0) construction uses random complexes, linear filling and diagonal completion parameters. No finite analytic certificate controlling the CAT(0) obstruction was located. **Boundary:** A numerical or coordinate representation of the complexes would not supply their geometric obstruction. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
507/258	Baumslag-Solitar-free one-relator groups are hyperbolic	L1	–	The one-relator hyperbolicity claim uses stationary currents and atomic axes. The needed stationarity and group-action conclusions are not consequences of finite positive moment matrices. **Boundary:** Measure terminology alone does not identify the corpus's finite moment recovery hypotheses or settle the group dynamics. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
508/258	Virtual compact specialness of hyperbolic one-relator groups	L1	–	The virtually special one-relator result is organized around group decompositions and free-by-cyclic consequences. The reviewed proof does not identify a core analytic operator or positive Gram certificate. **Boundary:** No concrete route from the corpus to virtual specialness or the required group embeddings was found. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
509/259	A group without fixed price	L1	–	The fixed-price counterexample uses rank surgery on group presentations. The rank in the inspected rank:main argument is combinatorial presentation data, not a matrix-rank or frame-conditioning problem. **Boundary:** Finite numerical rank certificates do not establish cost, fixed price or the rank-surgery group construction. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
510/260	The spacetime Penrose inequality with charge and original-data rigidity	L4	C2	The core deformation has explicit uniformly elliptic linearized operators and source terms in negative Sobolev spaces. Those admit domain-correct weak/Green representations on bounded exhaustion pieces, making the corpus very relevant to the PDE part of the charged Penrose argument. **Boundary:** The corpus's bounded coercive theorem does not prove the manuscript's rough nondivergence estimates, exterior uniformity, mass monotonicity or original-data rigidity. This is applicability, not verification. **Review:** Main statements plus selected elliptic theorem/proof excerpts and the baseline variation module; no complete Penrose proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
511/260	A Charged Reduction of the Spacetime Penrose Inequality in Spatial Dimensions at Least Four	L4	C2	The charged reduction explicitly uses a neutral companion theorem and an elliptic deformation with a natural-growth source. The corpus can represent and certify the bounded-domain linearized/weak solves central to this reduction without changing their boundary conditions. **Boundary:** No neutral Penrose theorem, uniform exhaustion estimate or charge/area monotonicity follows merely from the representation. The nonlinear derivative and source spaces must be verified. **Review:** Main and neutral-input statements plus selected elliptic estimates/proofs; no full geometric proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
512/260	Spacetime Penrose inequalities: enclosing area, charge, and rigidity	L4	C2	This composite manuscript contains the same concrete elliptic deformation and charged upper-branch mechanisms, so the PDE/residual corpus is very relevant. The directly inspected charged-extension module also exposes exact algebraic branch conditions. **Boundary:** Only selected modules of the large composite source were inspected. No assessment of every extension or of compatibility among all imported companion results is claimed. **Review:** Main statement, selected static-rigidity excerpts and charged-extensions statements/proof excerpts; not all 47 source files or proofs. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
513/260	The Kerr-Newman Penrose Inequality for Axisymmetric Electrovacuum Exteriors	L4	C2	The trace-solve result gives a concrete quasilinear Dirichlet problem in the axisymmetric Einstein-Maxwell deformation. Exact weak/Green representations and full nonlinear variation are relevant to the equation actually solved, not just to the manuscript's use of spectral language. **Boundary:** A local or bounded-domain solve does not establish the global Kerr-Newman inequality. A contraction, uniform exterior bounds and geometric comparison cannot be assumed from the corpus. **Review:** Main statement and selected trace-solve theorem/proof-structure excerpts; no complete Einstein-Maxwell argument audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
514/260	Electromagnetic tails and the Kerr-Newman Penrose inequality	L5	C4	The central conformal construction is already an explicit nonlinear integral fixed point on bounded continuous functions on r >= 1: T(v)=R[p(u0+v)^(-7)+ell(u0+v)^(-3)]. The reflected Newton Green operator enforces the Robin boundary condition, and the manuscript supplies a large-L contraction estimate. This is a direct core match to the corpus's Banach/nonlinear-IE certificate. **Boundary:** The direct match uses the manuscript's exterior Green operator, not the corpus's bounded-Dirichlet inverse. Positivity, tail derivatives and later geometric counterexample properties remain independent obligations; no fixed point or counterexample was numerically certified here. **Review:** Entire 02-elliptic.tex read, with main statement and selected later proof structure; no numerical replay or full geometric proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
515/260	The nonmaximal anti-de Sitter Penrose Inequality and original-data rigidity	L4	C3	The nonmaximal asymptotically AdS argument contains a concrete mixed-boundary linear elliptic estimate. Weak operator/trace representations and physically scaled residuals are very relevant to its deformation analysis. **Boundary:** The cited coercive corpus theorem does not itself supply the manuscript's drift, corner, exterior or uniform W^{2,p} estimates. Nor does it yield the global mass-area inequality. **Review:** Main statement and selected linear-estimates theorem/proof excerpts; other nonlinear and geometric stages not fully read. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
516/260	The Penrose inequality for maximal asymptotically hyperbolic initial data	L3	C2	The maximal asymptotically hyperbolic argument uses conformal approximation and an elliptic deformation to compare mass and area. These are meaningful PDE subproblems for domain-aware weak inverses and residual accounting. **Boundary:** No uniform AH-end estimate, stress comparison or global Penrose conclusion is supplied by a bounded-domain Green representation. **Review:** Main theorem and selected completion proof-structure excerpts; no full conformal/stress argument audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
517/260	A local Penrose inequality for conformal perturbations of Schwarzschild-anti-de Sitter data	L5	C3	The local Penrose problem is governed by explicit quadratic and quartic positivity on a conformal solution branch. Its core identity expresses the quadratic coefficient as a weighted squared norm after solving L=Delta-3 with the stated boundary operator. This is a concrete physical-Gram/jet certificate target, not an arbitrary coordinate recoding. **Boundary:** The corpus does not identify this TT identity with its unrelated hypercircle theorem. Exterior weights, boundary cancellation and a remainder for the actual solution branch must be established; finite jets alone are not the nonlinear theorem. **Review:** Setting plus selected quadratic identity/proof and quartic statement/proof excerpts; no complete exterior analytic or symbolic replay. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
518/260	The spacetime Penrose inequality and enclosing area	L3	C3	The enclosing-area argument uses weak conformal deformation and limiting area comparisons. The weak PDE/source and boundary bookkeeping have a concrete corpus counterpart, but are only part of the spacetime-geometric claim. **Boundary:** The corpus does not supply all-dimensional singular-set control, decay thresholds or enclosing-area monotonicity. **Review:** Main statements and selected weak4 theorem/proof-structure excerpts; no full weak-flow or singular-set audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
519/260	Conformal flow and the Riemannian Penrose inequality with minimizing frontiers	L3	C3	The conformal-flow proof uses harmonic velocities on regular patches and simultaneous regularity/separation of the moving frontier. Local harmonic Green representations and flux residuals apply on those actual regular pieces. **Boundary:** The local mapping does not cover the singular frontier, detachment arguments or the global conformal-flow mass law; it must not silently smooth away the singular set. **Review:** Main theorem and selected regularity/separation proof excerpts; singular and global flow arguments not fully audited. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
520/260	Equality and rigidity in the spacetime Penrose inequality	L3	C3	The rigidity analysis produces static scalar-flat conformal metrics and elliptic conditions whose boundary/decay identities can be represented and checked. That is a significant analytic part of the equality argument. **Boundary:** The corpus does not establish positive-mass rigidity, global Schwarzschild identification or causal embedding from a small local residual. **Review:** Main statement and selected static classification proof-structure excerpts; no complete rigidity proof. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
521/260	Boundary graph deformations for the spacetime Penrose inequality	L4	C2	The boundary-graph deformation is a coupled nonlinear elliptic existence problem with explicitly tracked weak natural-growth terms and flux jumps. The corpus's full nonlinear derivative, boundary weak forms and PDE-first mapping apply to major parts of this same model. **Boundary:** Leray-Schauder degree and exhaustion are not Banach contraction. The representation does not supply the Cordes estimate, uniform a priori bounds or admissibility needed for existence. **Review:** Main statement plus selected degree, natural-growth and boundary-flux theorem/proof excerpts; no full continuation or exhaustion verification. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
522/260	Area-controlled end replacement and the Bondi Penrose inequality in the CKS class	L2	–	The Bondi/end-replacement argument has an explicit quantitative metric comparison and charge error budget. A physical-norm residual ledger could check those finite-end comparison inequalities, but the main mechanism is geometric replacement. **Boundary:** No direct domain-correct Ram model for the full Bondi construction was identified at this depth; the proposed use is peripheral error accounting, not a Bondi mass theorem. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
523/261	Absolutely Continuous Spectrum for Weak-Disorder Anderson Models in Dimensions at Least Three	L5	C3	The Anderson argument's committed Schur/Ward states are concrete resolvent matrices for the original finite-volume random Hamiltonian. The identities g=(M-t diag r)^(-1), F_ij=|sqrt(t)g_ij|^2 and the weighted Dirichlet form identify a direct core Schur/physical-metric certificate, even though the probabilistic and infinite-volume estimates remain difficult. **Boundary:** Classical Schur/Ward identities do not prove adaptive freshness, probability bounds uniform in volume/regularization or absolute continuity. Finite matrices are not substitutes for the infinite-volume resolvent limit. **Review:** Core finite-state, assembly and spectral-averaging statements plus selected proofs read; no multiscale probability proof or computational replay. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
524/261	Pure-Point Spectrum for the Two-Dimensional Anderson Model at Every Positive Disorder	L5	C3	The two-dimensional Anderson proof uses the original resolvent through passive block matrices, Im G=G* W G and analytic Schur closure. These are explicit core operator representations to which the corpus's Schur residual and domain-aware inverse framework directly applies. **Boundary:** A finite analytic Schur certificate does not establish the two-dimensional bootstrap or the uniform fractional-moment estimate. Analytic continuation cannot bypass poles or replace the infinite-volume spectral argument. **Review:** Matrix identities/closure and spectral-column statements with selected proof excerpts; not the entire localization proof. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
525/262	Equality cases in the sharp one-dimensional matrix Lieb-Thirring inequality	L3	C3	The matrix Lieb-Thirring equality analysis includes H1 trial-state approximation and Gram orthonormalization for a matrix Schroedinger form. The corpus gives a precise physical metric and full-residual interpretation for this spectral subproblem. **Boundary:** The map does not establish the sharp constant, commutation or the classification by constant-unitary conjugates of sech-squared potentials. Trial-space spectral data alone cannot prove equality rigidity. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
526/262	Sharp one-dimensional Lieb-Thirring inequalities for matrix potentials	L5	C3	The sharp matrix trace step is built from the coercive matrix pencil (sI-B)^2+V and its integrated resolvent. Componentwise Fourier/RamCS realizes exactly the corresponding constant-coefficient matrix differential operator on the whole line; physical-Gram and resolvent certificates therefore address the core spectral model, not merely an incidental basis change. **Boundary:** The classical Fourier representation does not by itself prove the sharp trace inequality or the full Lieb-Thirring bound. Positivity must hold for the whole real parameter line, and trace/integral exchanges need the stated bounds. **Review:** Main statement and selected trace-theorem/proof-structure excerpts; no complete sharp-inequality proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
527/262	Sharp one-dimensional Lieb-Thirring constants	L3	C3	The scalar Lieb-Thirring proof uses the H1 quadratic form on the line, Dirichlet exhaustion and negative eigenvalues. The corpus applies concretely to variational spectral certificates on each interval and to full-residual/domain checks. **Boundary:** The sharp action inequality and constant are not supplied by a Gram representation, and accumulation at zero requires an independent limiting argument. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
528/263	Uniform excess charge for Coulomb molecules and the outer radius of neutral atoms	L3	C3	The common localization module gives an exact Coulomb-energy identity D(f)=||grad(K*f)||^2/(8 pi) on the homogeneous Sobolev domain. This is a concrete positive-energy Gram target and a significant part of the quantum comparison, with signed compact sources explicitly allowed. **Boundary:** The corpus hypercircle theorem applies only after its own domain conditions are met, not directly to the whole-space identity. Uniform excess charge, conditional quantum localization and all-sector energy comparison remain outside this finite energy certificate. **Review:** Main statements, interface excerpts and the Coulomb-duality statement/proof in the localization module; no full molecular comparison audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
529/263	Generalized ionization energies for full Coulomb atoms	L4	C2	The universal ionization coefficient is identified through a concrete semilinear limiting field on punctured R3. On compact annuli its Poisson equation has an exact Green/nonlinear-IE representation, while log-radius coordinates can organize the singular and far-field scalings. This is very relevant to the limiting-profile part of the core argument. **Boundary:** The corpus does not identify the limiting field with the many-electron model or supply the barrier, charge tightness and singular endpoint estimates. A global Mellin line cannot be assumed for a singular profile. **Review:** Main statement, selected fixed-sector proof structure and limiting-field selection/compactness excerpts; not the full many-body asymptotic proof. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
530/263	Generalized outer-electron radii of neutral Coulomb atoms	L4	C3	The outer-radius constant is governed by a scale-invariant semilinear Poisson profile, not only by an abstract density tail. The inspected compactness argument gives an explicit local Green representation and dilation law F(x)->D^4 F(Dx), which match the corpus's nonlinear PDE and logarithmic-scale machinery. **Boundary:** The profile representation does not prove the positive barrier, radial rigidity, quantum-to-profile limit or inversion of electron-mass tails into radii. Uniform-in-scale estimates remain separate. **Review:** Main theorem, interface excerpts and the profile compactness statement/proof read; later rigidity/tail proof not fully audited. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
531/264	Generic Future Inextendibility with Square-Integrable Connection Near a Fixed Kerr Spacetime	L1	–	The inspected near-Kerr C0/W1,2 inextendibility material focuses on causal homotopy, weak exits and compact-diamond obstructions. No explicit core integral or finite residual certificate was identified in those excerpts. **Boundary:** This is a limited-depth no-route finding, not a claim that Einstein evolution has no analytic representation. A PDE applicability upgrade would require reading and mapping its actual evolution estimates. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
532/264	Generic C1 Future Inextendibility Near Rotating Subextremal Kerr Spacetimes	L1	–	The inspected C1 near-Kerr statement and trivial-isometry/normal-graph supplement concern geometric genericity and extension obstructions. They do not supply a concrete normed operator and certificate to which the corpus theorem can be applied. **Boundary:** No applicability claim is inferred from 'evolution' alone; the main nonlinear evolution modules were not deeply reviewed. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
533/264	Quantitative Near-Kerr Evolution and Generic C2 Future Inextendibility	L3	C3	The finite-signal packet lemma exposes a normalized wave operator, bounded coefficients and Hamiltonian/Gronwall control on a finite window. Full derivative jets and remainder accounting can support that actual local signal construction. **Boundary:** The corpus elliptic Green theorem is not a hyperbolic propagator theorem. Local packet certificates do not establish near-Kerr evolution, generic blue shift or C2 inextendibility. **Review:** Main statement plus selected packet theorem/proof excerpts; history-only material excluded from mathematical evidence; not the global evolution proof. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
534/265	A two-dimensional area law from a global spectral gap	L3	C3	The positive-constraint construction uses actual Heisenberg evolution and rapidly decaying graph-local tails of a smooth spectral filter. Componentwise Fourier filtering and quantitative omitted-space/error accounting apply to a significant spectral-localization stage of the area-law proof. **Boundary:** The global gap is an assumption, not a product of the transform. Entropy comparison, replica transport and volume-independent amplification are not proved by the spectral representation. **Review:** Main theorem, baseline entropy/geometry excerpts and the quasilocal propagation statement/proof read; later filter and entropy arguments not fully audited. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
535/265	Polynomial PEPS approximation of gapped square-grid ground states	L4	C2	The PEPS construction has an explicit partial-isometry encoding K with K* K=P and a quantitative loss on the ground vector. The corpus frame/range and leakage identities map directly to this central compression stage, while the tensor truncation has a physical norm rather than merely coefficient error. **Boundary:** An 'encoded frame' is not automatically a frame for the entire many-body Hilbert space. The corpus does not supply patch projectors, ownership geometry, polynomial resource counts or positivity of the compressed density network. **Review:** Core encoding, tensor-bound and rewrite excerpts plus the pair-effect lemma/proof read; not all compression/assembly modules. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
536/266	The maximum number of mutually unbiased bases in dimension six	L5	C3	The main dimension-six MUB exclusion is an explicit finite phase-cover/Gram certificate problem. The corpus's exact coordinate conversion, validated Taylor remainder and physical-Gram bounds directly address the core covering and incompatibility certificates, regardless of whether the complete computation succeeds. **Boundary:** The rectangular 36-by-30 correction map does not meet an automatic square Banach-root theorem. Trigonometric panels require rigorous remainders before Bernstein use. No compiled certificate, full cover or search was replayed. **Review:** Main statement and core phase-cover, refinement and exclusion statement/proof excerpts read; source-level screening only, not computational verification. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
537/266	Exact Fourier certificates for complex Hadamard matrices of order six	L5	C3	The decisive contradiction is an exact Fourier-moment Gram identity with positive rational blocks and exact moment elimination. This is a direct core match to classical positive-measure Gram/SOS certificates expressed in the corpus's physical coordinates. **Boundary:** The univariate Hankel theorem is not a Hadamard-specific theorem; the function-family Gram identity is its elementary positive-measure analogue. Exact block data, elimination and the full MUB-to-moment reduction still require checking; no table/script replay occurred. **Review:** Entire 04-mixed.tex and 05-verification.tex read, plus main statements; exact arithmetic and external data not recomputed. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
538/267	Bose-Einstein condensation at positive temperature in the dilute hard-sphere gas	L2	–	The hard-sphere finite-temperature argument uses spatial localization with p^2+q^2=1. The corpus's squared partition-of-unity/Parseval mechanism can represent this particular localization and expose the lost component, but it is auxiliary to the thermodynamic argument. **Boundary:** Frame norm identities do not establish Bose condensation, entropy bounds or uniform thermodynamic control, and hard-core configuration-space boundary conditions remain essential. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
539/267	Quantum Depletion and Momentum Distribution in the Dilute Hard-Sphere Bose Gas	L2	–	The depletion statement concerns the physical momentum distribution of a hard-sphere many-body state on an exclusion configuration domain. Physical Gram norms can check a specified retained momentum component and its omitted mass, but no core correlation estimate is supplied by the corpus. **Boundary:** Finite momentum data do not establish the thermodynamic depletion law. A Fourier basis that ignores hard-core traces is not a domain-correct Hamiltonian model. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
540/267	Quantum Depletion for Fixed Bounded Repulsive Potentials	L2	–	For bounded repulsive interactions the depletion proof still requires companion density/correlation inputs. The concrete peripheral use is a norm-preserving representation of measured momentum components with a separate missing-density tail. **Boundary:** A transform of the density does not prove the imported many-body bounds or the universal depletion coefficient; no independent core route was identified. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
541/267	A density-uniform condensate bound for dilute Bose gases	L2	–	The density-uniform condensation argument uses Brownian-bridge/heat-kernel slab operations. Classical heat-semigroup convolution provides a concrete representation for a free segment, but the interacting path conditioning is an additional part of the proof. **Boundary:** The bounded-domain heat theorem is not automatically the interacting many-body bridge measure. Positivity of a free kernel does not prove uniform condensation or control interaction/path tails. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
542/267	Ground-state condensation in the dilute hard-sphere gas	L2	–	The hard-sphere ground-state proof uses local cube energy and ray/Cauchy-Schwarz estimates with mixed physical boundaries. A physical quadratic-form representation can check those local trial inequalities while preserving the exclusion traces. **Boundary:** This auxiliary variational check does not establish the dilute-limit condensation theorem or its volume-uniform estimates. No unrestricted periodic model may replace the stated domain. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
543/268	The periodic spin-one Haldane gap	L5	C3	The spin-one gap proof has a concrete Hilbert-Schmidt transfer operator and certified short-chain thermal moments. Both the same physical finite Hamiltonian's polynomial exponential approximation and the transfer-moment inequalities are direct core targets for the corpus's physical-Gram and positive-moment certificates. **Boundary:** The corpus does not prove the transfer construction, long-chain bootstrap or final gap from an arbitrary finite spectrum. The printed integer totals and remaining thermal inputs were not independently replayed. **Review:** Transfer statement/proof excerpts and thermal72 calculation substantially read, including matrix, rounding, total and enclosure formulas; other initialization/thermal/assembly proofs only selectively inspected. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
544/268	A boundary-field gap for the spin-one Heisenberg chain	L5	C3	The boundary-field gap uses the same explicit self-adjoint transfer operator with endpoint vector, and a polynomial moment bound ||P_*(X) Pi v||^2. The exact thermal recurrence uses a nonorthogonal integer coordinate system with a known physical metric. These are directly core Gram/moment certificate problems. **Boundary:** The nonsymmetric integer D is not Euclidean-self-adjoint or contractive. These classical certificates do not replace the boundary transfer construction or long-chain gap argument; no integer table was rerun. **Review:** Entire thermal.tex read and selected boundary-transfer/trial statements and proofs reviewed; no full gap proof or arithmetic replay. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
545/269	Uniform Stability of the Spherical Laughlin Gap	L4	C3	The disorder-stability proof uses concrete retained-block positivity and a relative quadratic-form bound against the parent Hamiltonian. The corpus is very relevant to the exact local Gram certificates and their physical error budgets. **Boundary:** The finite retained-block check does not imply uniform stability in particle number, flux or disorder. That extension needs the manuscript's locality and relative-bound proof. **Review:** Main theorem plus retained-block and stability statement/proof excerpts; bundled pgf utility file excluded; no table replay or complete disorder proof. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
546/269	A Fock-space inequality and the Laughlin spectral gap	L5	C3	The Laughlin-gap proof's core finite positivity certificates are explicit rational Gram congruences, coupled to an explicit local-to-full-Fock comparison. The corpus's exact polynomial and PSD identity framework directly applies to those certificates, with the singular Gram range handled correctly. **Boundary:** The determinant-coefficient PSD test uses real symmetric matrices and the exact polynomial, not floating eigenvalue signs. The local-to-global comparison, spectator modes and tail bounds are distinct proofs; none is supplied automatically by coordinate equivalence. **Review:** Entire 04-comparison.tex, 05-transfer.tex and 08-arithmetic.tex read; exact certificate tables/code not replayed and the full gap proof not certified. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
547/270	The unique threshold bound state of the SU(N) BFSS model	L3	C2	The BFSS threshold-state argument has a concrete closed Hamiltonian, massive approximants and annular tightness estimates. The corpus's full-residual and graph-domain conditions give a significant framework for judging whether trial states survive the noncompact massless limit. **Boundary:** A small finite residual near zero does not prove an L2 threshold eigenstate, uniqueness or the required index/gauge induction. Compactness and tails must be established independently. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
548/270	Positive eigenvalues of the relative SU(2) BFSS Hamiltonian	L3	C3	The relative SU(2) BFSS spectral problem is defined by a closed form and symmetry/gauge sectors. Domain-aware Gram and variational certificates apply to the actual sector Hamiltonian, making the corpus relevant to significant spectral steps. **Boundary:** Finite Ritz levels do not prove embedded positive eigenvalues or exclude escape through noncompact channels; the sector confinement and infinite spectral sequence remain independent. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
549/271	Bloch's Law for Finite-Range Heisenberg Ferromagnets in Three Dimensions	L2	–	The Bloch-law coefficient contains a concrete long-wave quadratic dispersion and a determinant/heat-kernel factor. The corpus Fourier and local-moment machinery applies to this asymptotic coefficient calculation, but only peripherally to the interacting magnetization proof. **Boundary:** Computing a quadratic dispersion integral does not establish the interacting pressure comparison, uniform limit or spontaneous magnetization law. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
550/271	The first lattice correction to Bloch's law	L3	C3	The lattice correction is sensitive to higher terms of the dispersion and to a quantitative capacity comparison. Finite Taylor/moment identities with a remainder can address the specific small-momentum coefficient, a significant analytic component of the correction. **Boundary:** A formal Taylor coefficient is not the thermodynamic correction theorem. Bernstein can bound polynomial panels only after nonpolynomial remainders and tails are enclosed. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
551/271	The spherical magnetization law for the three-dimensional quantum Heisenberg ferromagnet	L2	–	The spherical magnetization law has a concrete scalar distribution with transform $\sinh(m\|t\|)/(m\|t\|)$. Positive moment/Gram checks can test finite moment consequences for a fixed directional observable, but the core assertion is a thermodynamic distributional limit. **Boundary:** Finite moment positivity neither identifies the full limiting law without determinacy/tail control nor proves the interacting magnetization amplitude or ergodic limit. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
552/271	Spontaneous magnetization in the quantum Heisenberg ferromagnet	L2	–	The spontaneous-magnetization proof uses pinned-loop pressure and an analytic-strip limiting argument. The corpus's holomorphy/continuation conditions are relevant to the explicit limit exchange, but only as an auxiliary analytic check. **Boundary:** Analytic continuation supplies uniqueness on a declared connected domain, not existence of KMS states, pressure bounds or spontaneous symmetry breaking. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
553/272	Entanglement with zero distillable secret key in local dimension ten	L5	C3	The dimension-ten construction has explicit integer matrix pencils whose Gram identities certify PPT positivity and whose rank-loss directions drive the obstruction. These finite polynomial/Gram certificates are directly relevant to the core constructed example. **Boundary:** Gram positivity does not establish the projective ideal computation or the all-copy/all-protocol key obstruction. Those are additional core proofs, not consequences of a Ram basis; no integer or ideal calculation was rerun. **Review:** Main construction and protocol excerpts plus selected pencil/PPT and projective-certificate statements/proofs; no full quantum-protocol or exact-algebra replay. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
554/273	The entropy photon-number inequality	L2	–	The entropy-photon-number inequality concerns bosonic states with finite energy. A domain-aware number-state truncation can certify retained positive matrices and the discarded mass/energy, a concrete but peripheral computational component. **Boundary:** A finite PSD density matrix or classical heat transform does not prove the noncommutative entropy inequality, tensorization or the channel-capacity consequences. Entropy continuity needs its own energy-dependent bound. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
555/274	Product-projection localization and the QAC0 parity lower bound	L3	C2	The QAC0 parity argument contains a specific root-product/Chebyshev approximating polynomial with quantitative degree and coefficient bounds. Exact polynomial-coordinate identities and panel enclosures apply to this substantial approximation step. **Boundary:** A finite polynomial certificate does not establish the uniform asymptotic parameter choice, gate pruning or the circuit lower bound. Discrete-grid validity must not be replaced by an unrelated interval estimate. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
556/274	Regular trajectories, pruning and quantum parity	L2	–	The regular-trajectory pruning argument has a concrete accumulated operator-error budget. Physical norm and projection-leakage identities can audit that budget for specified circuit states, but the decisive trajectory and parity arguments are combinatorial. **Boundary:** This is an auxiliary error check, not a route to the regular-trajectory construction or a QAC0 lower bound. State-specific error does not imply a uniform circuit operator bound. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
557/275	Continuum Coulomb hardness with binary nuclear charges	L5	C3	The continuum Coulomb hardness reduction explicitly constructs U:C^m->L2(R3), G=U*U and J0=U G^(-1/2), then compares the actual shifted Hamiltonian in those modes. This is exactly the physical-Gram/full-residual model in the corpus and is central to controlling YES/NO energies. **Boundary:** A finite variational upper bound is not the NO-instance lower bound. Uniform well/tunneling estimates, the entire orthogonal complement, many-electron coupling and polynomial promise precision remain independent obligations. **Review:** Main theorem, completion excerpts and substantial one-particle Gram/residual statement/proof in 04-continuum.tex read; the full complement/many-body reduction not audited. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
558/275	QMA-hardness of continuum Coulomb energy with unit nuclear charges	L5	C3	The unit-charge construction has an explicit whole-plane Green resolvent phi=(-Delta_r+1)^(-1)f, physical localized modes and a Coulomb Gram matrix controlling occupancy defects. Those operators and forms directly match the corpus's Fourier/Green and physical-Gram representations at a core stage of the reduction. **Boundary:** The whole-plane resolvent is not the bounded-Dirichlet theorem. Local mode/Gram certificates do not by themselves implement rational unit nuclei, control the continuum complement or establish QMA-hardness; those later constructions and promise bounds remain. **Review:** Main statement and substantial wells/local-mode/occupancy excerpts read; later nuclear implementation and full many-body reduction not audited. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
559/276	Classical capacity and entropy inequalities for generalized amplitude damping	L3	C3	The generalized-amplitude-damping capacity proof uses explicit positive thermal blocks built from XX*, YY* and XY*, with a coefficient relation alpha delta-beta eta=K^2. Physical Gram/congruence and constrained polynomial positivity are relevant to that significant block comparison. **Boundary:** A fixed block PSD check does not establish tensor-power additivity or the channel capacity. Bernstein applies only to polynomial pieces or rigorously bounded approximants, not directly to entropy logs. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
560/277	Threshold parallel repetition for finite-dimensional entangled games	L2	–	For a fixed finite-dimensional entangled strategy, game correlations have explicit Gram/PSD constraints that can be independently checked. This gives a peripheral certificate route for the strategy data in the threshold-repetition setting. **Boundary:** Finite strategy feasibility or an SDP bound at one level is not a dimension-uniform repeated-game theorem. The postselection and dependence estimates remain outside the mapped certificate. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
561/278	A Coulomb ground-state density without Kohn-Sham ensemble representation	L4	C3	The non-Kohn-Sham density construction contains an explicit weighted ground-state form and a spectral-gap certificate for the candidate ensemble. The corpus's physical metric, domain and full-residual conditions are very relevant to the operator that controls this obstruction. **Boundary:** A formal positive ground-state identity does not establish closure, the weighted gap or representability by an admissible Kohn-Sham potential. The ensemble obstruction and explicit large-parameter bounds require their own proof. **Review:** Main statement and selected weighted-form/ensemble and potential-identification statements/proofs read; no full obstruction proof or explicit-constant verification. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
562/279	Exact quantum factoring over a fixed finite gate set	L1	–	The exact fixed-gate factoring construction is organized around quantum arithmetic and primality/polynomial congruence checks. No actual Mellin, positive-Gram or nonlinear-IE core operator was identified in the inspected proof structure. **Boundary:** Classical polynomial-coordinate equivalence does not supply modular arithmetic, exact quantum compilation or the factoring guarantee. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
563/280	Strongly rational unitary vertex operator algebras and conformal nets	L2	–	The VOA-to-conformal-net argument has unbounded fields with polynomial energy bounds. Domain transport and physical-norm truncation can audit a specified mode approximation, but only peripherally to strong locality and representation theory. **Boundary:** Finite-mode boundedness does not establish strong commutativity/locality of closed fields or the conformal-net and tensor-category conclusions. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
564/281	QAOA attains the SK ground-state energy in the thermodynamic-first limit	L3	C3	The classical-synthesis theorem is formulated in weighted Fock-space smooth norms with an explicit insertion-vector error. That supplies a concrete domain/norm target for finite-jet and full-residual bookkeeping in a significant synthesis stage. **Boundary:** A finite coefficient model does not establish uniform smooth bounds or the helper contraction. The thermodynamic-first order of limits, seed removal and SK optimum identification remain separate. **Review:** Main statement, MaxCut consequence and classical-synthesis statement/domain excerpts read; the long helper-contraction and limiting proof not fully audited. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
565/281	Full support of the zero-temperature Sherrington-Kirkpatrick order parameter	L5	C3	The SK full-support argument has a decisive exact rational polynomial identity: a divergence term plus positive products/squares yields a strict coercive lower bound. The corpus's exact coefficient-equivalence and SOS/Gram certificate machinery directly targets this core inequality, while the Parisi/heat representation supplies its analytic context. **Boundary:** This is classical polynomial positivity plus integration by parts, not a new transform theorem. The finite identity does not establish the Parisi regularity, variation formula or vanishing boundary term; those remain required. No symbolic certificate replay occurred. **Review:** Entire certificate.tex read, with main/variation excerpts and selected regularity statements/proofs; no complete Parisi PDE or full-support proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
566/282	Scale and conformal symmetry in four-dimensional operational quantum field theory	L2	–	The scale-to-conformal QFT argument uses Fourier-Laplace boundary values and positivity for field distributions. The corpus can help audit transform conventions and regularized pairings, but its weighted L1/L2 inversion is not itself a Wightman distribution theorem. **Boundary:** No direct implication to conformal improvement, distributional positivity or removal of field-theoretic contact terms is supplied. Analytic continuation requires its own domain and domination argument. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
567/283	Polynomial-Time Unitary Synthesis from a Boolean Oracle	L4	C3	The synthesis reduction deliberately produces an internal unitary block D with ||D|| <= 3/4 so its geometric series converges. The actual block operators, Walsh/sign transformations and moment bound provide a strong concrete match to Schur/Neumann and physical-Gram certificates at a central recursive step. **Boundary:** The corpus's analytic-jet inverse theorem is only a Neumann-series analogue here; this finite matrix uses its own norm bound. A certificate for a chosen block does not prove uniform programmable-slot synthesis or efficient Boolean-oracle access. **Review:** Main statement, compilation excerpts and contracting-block statement/proof substantially read; no full oracle/circuit complexity proof. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
568/284	A Nearly Quartic Separation Between Randomized and Quantum Query Complexity	L2	–	The query-separation argument can expose finite adversary/correlation matrices for specified input sizes. Their Gram feasibility and norm bounds are a concrete peripheral certificate target, distinct from the asymptotic lower-bound construction. **Boundary:** Finite matrix checks do not prove the quartic asymptotic separation or the uniform adversary family. An arbitrary coordinate representation would not upgrade applicability. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
569/285	A torsion-free counterexample to reduced Baum-Connes injectivity	L1	–	The torsion-free Baum-Connes injectivity assertion is organized around assembly and K-theoretic constructions. No specified analytic finite certificate connecting the corpus to the assembly obstruction was identified in the inspected source. **Boundary:** A finite cochain or Gram encoding does not establish injectivity of the global assembly map. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
570/285	A torsion-free counterexample to the Kadison-Kaplansky projection conjecture	L4	C3	The projection construction has an explicit self-adjoint cover operator, a separated spectral band and a quantitative compression/contour perturbation argument. The corpus's leakage, physical metric and omitted-space framework is very relevant to this central band-transfer mechanism. **Boundary:** These estimates do not construct the group/voltage cover or prove torsion-freeness, uniform approximation and equivariant trace identities. A spectral projection certificate is not by itself a counterexample to the group-algebra conjecture. **Review:** Main statement, tower excerpts and the compression proof plus contour-perturbation statement read; no full cover construction or group-algebra proof. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
571/285	An irrational-trace counterexample to reduced Baum-Connes	L2	–	The irrational-trace construction uses component spectral projections followed by an infinite trace sum. The corpus can audit finite spectral filters and the need for a uniform omitted-space gap, but the irrational trace and group-algebra membership involve additional structure. **Boundary:** Finite component traces do not prove irrationality of the infinite sum, convergence in the required algebra or the Baum-Connes obstruction. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
572/286	Arithmeticity of twisted finite correspondences for lattices over local fields	L1	–	The twisted-lattice correspondence argument uses arithmetic exhaustion and measurable boundary/operator-algebra correspondences. The inspected structures do not furnish a finite physical Gram problem controlling the classification. **Boundary:** A formal Hilbert-space representation does not replace the arithmetic or measurable correspondence argument; no concrete corpus route was found. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
573/286	The arithmetic category and stable recovery of lattice factors	L1	–	The stable lattice-factor result relies on canonical recovery, cocycles and index data. 'Stable' concerns operator-algebra equivalence, not a numerical inverse or perturbation estimate of the corpus type. **Boundary:** No explicit domain-correct analytic/certificate map to the core classification was identified at the reviewed depth. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
574/287	An isomorphism of the free group factors	L2	–	The free-group-factor isomorphism argument uses Haar-word orthonormality and recovery of a full Hilbert-space representation. The corpus's complete-frame versus retained-bank distinction is relevant to checking that particular completeness claim. **Boundary:** A unitary identification of L2 spaces does not imply a von Neumann algebra isomorphism. Freeness, generation and multiplicative/normal extension remain the decisive independent obligations. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
575/288	Kadison's similarity theorem through uniform derivation estimates	L2	–	The uniform derivation/similarity argument contains concrete Fourier/Hankel manipulations in a dihedral corner. Fourier norm conventions and bounded-similarity domain transport are useful auxiliary checks, but the main universal completely bounded estimate is not a finite moment problem. **Boundary:** This Hankel operator is not automatically a positive-measure Hankel moment matrix. The corpus does not construct a universal similarity constant or control all matrix amplifications. **Review:** Main statement plus selected global and Hankel theorem/proof excerpts; Unicode helper file excluded; no full uniform-derivation proof. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
576/289	Universal strong Kadison-Kastler stability	L2	–	The strong Kadison-Kastler argument contains a concrete small-unitary norm comparison under amplification. Unitary/physical-metric transport can audit that estimate once the implementing unitary is available, but this is auxiliary to constructing it. **Boundary:** The corpus does not construct a spatial implementer from near inclusion or supply a universal constant over all von Neumann algebras. Finite matrix agreement is not uniform control of the unit ball. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
577/289	Near Inclusions of von Neumann Algebras Without Small Spatial Embeddings	L3	C2	The near-inclusion obstruction uses explicit clock/shift unitaries and an integer trace-log invariant. Exact finite-matrix relations and a verified spectral margin from the logarithm branch cut give a concrete certificate route for a significant obstruction step. **Boundary:** The analytic-continuation theorem is only a branch/domain discipline here, not the operator-algebra obstruction theorem. Finite Bott-type data do not establish the infinite near-inclusion construction or nonexistence of small spatial embeddings. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
578/289	Close Separable C*-Algebras Without Spatial Conjugacy	L1	–	The close-separable-C*-algebra counterexample uses sequence-algebra and deformation/embedding structure. In the inspected single-file proof excerpts, no specific finite operator certificate was found that controls spatial conjugacy itself. **Boundary:** Checking a few operator norms is not control of Kadison-Kastler distance over entire unit balls, and arbitrary coordinate algebra would be peripheral rather than a core route. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
579/290	Expected amenable subalgebras preserving core commutants	L2	–	The expected-amenable-subalgebra construction has modular-energy and averaging components. The concrete auxiliary match is spectral filtering in the state's Hilbert space, with operator-norm bounds kept separate from state-L2 estimates. **Boundary:** A bounded spectral filter does not construct an expected amenable subalgebra or preserve the required core commutants. State-L2 approximation cannot be substituted for operator-norm or algebraic inclusion claims. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
580/290	Bounded recovery for modular spectral averages	L5	C3	The central modular-localization/recovery model is explicit: D=log Delta_phi on the standard Hilbert space and y_f=int g(t) sigma_t(y)dt with y_f xi=f(D)y xi. This is a domain-correct Fourier representation of the same core operator, with an L1 kernel norm controlling the algebra element. The hard bounded-recovery estimates do not reduce its direct applicability. **Boundary:** The classical Fourier/spectral representation does not establish bounded recovery, polar truncation or either norm ergodic theorem. A small state-L2 error is not a small operator-norm error, and no norm continuity of the modular orbit is assumed. **Review:** Main theorem, selected bounded-recovery statements/proofs and the modular-averaging localization/predual/ergodic proofs read; later fourth-moment and recovery assembly not fully audited. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
581/291	Equivariant Jiang-Su Stability for Amenable Actions in the Unital Stably Finite Case	L1	–	Equivariant Jiang-Su stability is established through central-sequence/ultrapower splitting for amenable actions. The inspected splitting argument has no concrete core integral operator or finite positivity certificate of the corpus type. **Boundary:** Finite tests on a central sequence do not construct the required equivariant order-zero maps or their uniform ultrapower relations. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
582/291	Cuntz comparison and Jiang-Su absorption	L2	–	The tracial step includes an explicit spectral-block formula for the L2 norm of a commutator with f(H). A physical trace-Gram calculation can audit that auxiliary estimate and its Lipschitz constant. **Boundary:** The resulting L2 inequality is not an operator-norm commutator bound and does not construct Cuntz comparisons, central order-zero maps or Jiang-Su absorption. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
583/291	Nuclear dimension and Jiang-Su stability without elementary subquotients	L1	–	The nonunital nuclear-dimension/Jiang-Su result is driven by covering and central order-zero-map constructions. The inspected central module is not a numerical rank or analytic inverse problem. **Boundary:** No concrete corpus route to the required C*-algebra comparison and absorption structures was identified. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
584/291	Tracial projection methods and uniform property Gamma	L2	–	Uniform property Gamma involves finite tracial projection relations whose defects can be expressed in a positive L2 Gram matrix at a fixed trace. This provides an auxiliary consistency check for the proposed projections. **Boundary:** One trace or finitely many samples cannot establish uniformity over the trace space, centrality in the required quotient or the nonunital absorption result. Positive-measure moments apply only to genuinely linear positive traces. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
585/292	An explicit obstruction to nuclear norm-ultrapower embeddings	L2	–	The nuclear norm-ultrapower obstruction uses a sequence of growing finite matrix models and a gap/approximation contradiction. Exact Gram or norm checks can validate a chosen model's gap, a peripheral component of the sequence argument. **Boundary:** A verified gap in selected dimensions does not establish a uniform family, nonembedding into the norm ultrapower or the nuclearity obstruction. Strong/pointwise convergence is not the required uniform norm estimate. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
586/293	Invariant-projection counterexamples for every irrational rotation	L2	–	For the irrational-rotation weighted operator U f(V), there is a concrete auxiliary route through its iterated weight products and physical norms. This can audit explicitly bounded finite products while retaining zeros and the singular logarithmic behavior. **Boundary:** The integral of log f may be minus infinity; a Mellin/Laplace inverse or bounded similarity must not be assumed. Finite product bounds do not prove absence of invariant projections. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
587/293	Backward intertwiners and a transitive commutant	L3	C3	The backward-intertwiner construction has explicit weighted-shift products with rapid norm decay. This gives a concrete operator-tail and sequence-norm target for a significant part of the bounded intertwiner construction. **Boundary:** The cited l1 convolution model is a tail-accounting analogue, not a theorem about this direct-integral shift. Transitivity of the commutant and exclusion of invariant subspaces require the separate fiber/ergodic proof. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
588/294	A counterexample to Kaplansky's quasitrace conjecture and failure of tensor-product stable finiteness	L1	–	The quasitrace counterexample hinges on nonadditivity and its consequences for tensor-product stable finiteness. The positive-moment corpus assumes a linear positive functional, precisely a property that cannot be silently imposed here. **Boundary:** No Gram moment lift using quasitrace as though it were a trace is justified. The inspected consequence module provides no alternative concrete corpus route. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
589/295	Vanishing of higher bounded Hochschild cohomology	L1	–	The Hochschild primitive argument uses normality, continuity and averaging over a von Neumann algebra. The required uniformly bounded infinite-dimensional homotopy is not supplied by a finite cochain coordinate identity. **Boundary:** No concrete kernel or finite physical model controlling the full bounded cohomology problem was identified; finite-dimensional homotopy analogies are insufficient. **Review:** Main statement and selected 05-cohomology.tex statements/proofs read; typography helper excluded; no complete cohomology proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
590/296	Relative generation and the generator problem for finite factors	L2	–	The relative-generation argument contains a concrete free-sum estimate using orthogonal ranges and an l2 coefficient norm. The corpus frame/Bessel identity applies to this auxiliary norm estimate. **Boundary:** A correct Bessel-type estimate does not prove von Neumann generation, density of generating pairs or the Baire-category conclusion. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
591/297	A counterexample to Naimark's problem in ZFC	L1	–	The nonseparable Naimark construction uses pure-state excision and transfinite/algebraic organization. The inspected source offers no concrete core finite operator or analytic certificate to which the corpus can be applied. **Boundary:** Finite Hilbert-space representations do not settle equivalence of irreducible representations or the nonseparable construction. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
592/298	A finite-entropy separation of microstates and nonmicrostates free entropy	L2	–	The entropy-separation proof uses Gaussian matrix blocks and exact conditional second-moment calculations. Positive Gram/covariance identities can check those finite calculations and their dimension factors, but the core result compares asymptotic entropy quantities. **Boundary:** Finite covariance positivity does not establish microstates-volume asymptotics, Fisher-information flow or a strict finite-entropy separation. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
593/299	The Kirchberg-Rrdam character criterion	L1	–	The Kirchberg-Rordam character criterion relies on tensor independence and structural C*-algebra arguments. The inspected tensor module does not identify a finite positive polynomial or normed IE problem controlling the criterion. **Boundary:** No corpus transform implication to characters, cone embeddings or the required tensor-product structure was found. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
594/300	Approximation Paving over Arbitrary Maximal Abelian Subalgebras	L5	C3	The paving mechanism has an explicit expected characteristic polynomial and a rooted arrow-matrix/Stieltjes representation. Positive spectral measures, exact polynomial identities and Schur quotients are therefore a direct core model for its finite pinning rule, even though measured/global transfer still needs proof. **Boundary:** The corpus does not prove real stability or the size-independent buffered-pinning bound from an arbitrary moment list. Local certificates do not establish measurable coherent coloring or approximation paving in an arbitrary MASA. **Review:** Main statement, slab excerpts and finite-polynomial stability proof plus rooted spectral-data statement read; later pinning and measured transfer not fully audited. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
595/300	Quadratic Strong-Operator Paving over Arbitrary Maximal Abelian Subalgebras	L5	C3	The strong-operator paving proof likewise has a concrete core mixed-determinantal polynomial with an explicit Hermitian arrow realization and positive rooted measures. The corpus's exact polynomial and moment/Gram certificates directly apply to this finite color-assignment model. **Boundary:** The real-rootedness/mixed-bound theorem, dimension-uniform assignment cost and measured-to-strong-operator transfer remain independent proof obligations. Finite PSD checks alone do not prove paving over every MASA. **Review:** Main statement, transfer excerpts and mixed-bound statement/proof plus polynomial-quotient statement/proof excerpts read; no full coloring or completion audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
596/301	The trace cone classifies Razak-Jacelon stabilizations	L1	–	The trace-cone classification is implemented by pointwise/approximate intertwining for Razak-Jacelon stabilizations. A trace cone in this classification is not a finite scalar moment cone with the corpus's atomic recovery hypotheses. **Boundary:** No finite Gram or transform certificate was identified that constructs the C*-algebra intertwining or proves classification. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
597/302	Filtered products and boundary-preserving compression in complex cobordism	L1	–	The filtered-complex-cobordism argument uses boundary-preserving compression, finite CW/stable-homotopy data and torus-bundle characteristic classes. Its compression is topological, not a Hilbert projection. **Boundary:** Finite cochain coordinates do not establish the required MU nullhomotopy, filtered product or boundary-preserving topological construction. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
598/302	Radius of comparison equals half the mean dimension	L1	–	The radius-of-comparison/mean-dimension theorem uses dynamical towers, regularity and Cuntz comparison. The 'radius' is an operator-algebra invariant, not a numerical conditioning or contraction radius. **Boundary:** No concrete domain-correct corpus operator linking the dynamical lower bound and Cuntz-comparison upper bound was found. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
599/303	Weak pure infiniteness and O-infinity absorption	L1	–	The weak-pure-infiniteness argument constructs algebraic separation and partial-isometry structures to obtain O-infinity absorption. Its row-norm bounds are supporting estimates, not the core existence mechanism. **Boundary:** A generic matrix inverse or norm representation cannot supply the required C*-algebra relations and absorption map; no concrete Ram route was identified. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.
600/304	The Hilbert-Smith conjecture in every finite dimension	L1	–	The all-dimensional Hilbert-Smith claim excludes p-adic actions using sheaf/cohomological character classes and derived-limit structure. The inspected character-class module has no analytic integral or positive-Gram core problem. **Boundary:** A finite Fourier representation of an action would not prove the sheaf/cohomological obstruction or exclude arbitrary continuous p-adic actions. **Review:** Main statement/introduction and selected theorem/proof-structure excerpts in the listed manuscript sources; not every theorem or whole file. No full proof audit. All three local corpus maps read; cited corpus definitions/theorem excerpts inspected, not independently certified. Level measures applicability, not validity, novelty or demonstrated success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
601/305	A boundary-only obstruction to four-dimensional disk embedding	L1	–	The boundary-only disk obstruction depends on formal connections, Frobenius-Koszul obstruction and four-dimensional realization; no corpus theorem bridges those steps. **Boundary:** Finite polynomial or chain calculations alone do not establish locally flat disk nonembedding. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
602/305	A marked tensor obstruction to four-dimensional disk embedding	L2	–	Finite tensor transport can help audit the marked tensor obstruction, but the essential comparison is with geometric holonomy data. **Boundary:** Projective tensor transport is not the relative trace comparison, formal-potential algebraicity, or disk-embedding obstruction. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
603/305	A PD4 group without an aspherical manifold model	L1	–	The chamber, controlled homology removal and marked stable realization require geometric group topology outside the identified corpus results. **Boundary:** Finite cohomology preservation does not produce a PD4 group without an aspherical manifold model. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
604/306	Purely cosmetic surgery on knots in the three-sphere	L2	–	The harmonic-background and local-energy subargument permits a limited comparison with coercive residual estimates. **Boundary:** No corpus result supplies instanton compactification, transversality, spinor indices or the nonzero pairing needed for cosmetic surgery. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
605/307	Failure of rational injectivity for maximal coarse assembly	L2	–	Finite congruence fibers offer a place to preserve chain classes under compression; the maximal Roe-algebra transfer remains external to the corpus. **Boundary:** Finite-stage cohomology is not persistence at every coarse scale or rational injectivity failure of maximal assembly. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
606/307	A counterexample to the coarse Novikov conjecture	L2	–	Finite-disc and congruence-fiber bookkeeping has a peripheral chain-complex application. **Boundary:** A finite homotopy certificate does not establish the Roe homomorphism, infinite-order coarse class or vanishing boundary index. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
607/308	Finite Smith-Toda Complexes at Varying Primes	L1	–	The varying-prime argument uses ultraproduct gradings, BP coactions and connectivity uniform in the prime, with no identified corpus substitute. **Boundary:** Real/complex polynomial inversion does not control varying-prime stable homotopy or ultraproduct realization. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
608/308	A finite Smith-Toda complex V(4) at the prime 1009	L2	–	The explicit filtered cobar calculation admits peripheral finite-complex checks. **Boundary:** The corpus finite-complex identities require adaptation to this ring; they do not prove Adams-Novikov survival or realization of V(4). **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
609/309	The Kervaire invariant problem at the prime three	L2	–	The finite recurrence and Adams-Novikov calculation are specific places for exact finite algebra verification. **Boundary:** This does not identify the topological Tate detector, justify order-nine ascent or establish the 322-stem element. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
610/310	Rational homology and Quillen's conjecture	L3	C3	Explicit decorated cycles and restrictions make chain-level preservation relevant to a substantial, checkable part of the proof. **Boundary:** A checked finite component does not discharge all finite-group reduction cases or prove Quillen's conjecture uniformly. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
611/311	Stabilizer orbits and thick tensor ideals of dualizable K(n)-local spectra	L1	–	Formal orbit density and invariant ideals of Lubin-Tate deformation rings have no concrete route from the identified analytic/frame results. **Boundary:** The shared words 'frame' or 'operator' do not connect numerical frames to thick tensor ideals of K(n)-local spectra. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
612/312	The Grothendieck homotopy hypothesis via elementary expansions	L1	–	The proof concerns globular models, elementary expansions and semi-model structures, not the corpus's finite numerical cochain complexes. **Boundary:** A finite chain homotopy is insufficient for a homotopy hypothesis or an equivalence of higher categorical models. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
613/313	Finite generation for the K(n)-local sphere	L1	–	Completed towers, integral coefficient cohomology and Kummer support arguments require arithmetic and homotopical inputs not supplied by the corpus. **Boundary:** Sequence-space analytic convergence does not imply integral finite generation for the K(n)-local sphere. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
614/314	Cyclic length and chromatic fixed-point loss	L2	–	Finite character and congruence calculations are a narrow algebraic checkpoint. **Boundary:** Tensor bookkeeping does not supply procyclic-length classification, profinite passage or chromatic gluing. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
615/315	The Singer conjecture in dimension four	L2	–	The finite-energy-function construction can use physical energy norms when examining finite approximations. **Boundary:** Frame energy estimates do not construct the continuum boundary or prove its topological contradiction and Singer conclusion. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
616/316	The Stable Hurewicz Image of the Sphere at Two	L1	–	Steenrod module extensions and suspension-weight obstructions have no identified theorem-level application from the corpus. **Boundary:** Polynomial jet truncation is not Steenrod algebra or stable Hurewicz detection. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
617/317	Thomason Model Structures in Every Strict Higher Dimension	L1	–	The decisive relative-prism and algebraic-pushout statements are categorical rather than finite analytic operator results. **Boundary:** Numerical cochain preservation does not establish couniversality, cofibration axioms or Quillen equivalence. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
618/318	Filtered chromatic splitting at generic primes	L1	–	The finite frame rings here are arithmetic marking rings, not Hilbert-space frames of the corpus. **Boundary:** Frame stability and moment inversion do not prove integral primitive classes or filtered chromatic splitting. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
619/318	The height-three chromatic overlap: an explicit filtration and its attachments	L1	–	The filtration and attachments depend on Tate coordinates, removal of completion and exact Morava descent. **Boundary:** An explicit finite number of cofibers does not make the spectrum-level attachments a finite numerical inverse problem. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
620/318	A rational obstruction to strong chromatic splitting at height three	L1	–	Transport of the degree-three primitive through Witt coefficients is outside the corpus's analytic transport results. **Boundary:** Moment transport does not justify arithmetic descent, realization or the rational wedge obstruction. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
621/318	Failure of finite assembly for a chromatic overlap at the prime three	L1	–	Deformation jets in this manuscript are adic/arithmetic comparison data, not finite Taylor jets of smooth integral operators. **Boundary:** Finite smooth-jet recovery cannot be substituted for comparison at all deformation orders or Morava descent. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
622/318	Counterexamples to weak chromatic splitting: sphere kernels and descent exponents	L1	–	Sphere products, canonical kernels and descent exponents need homotopy-theoretic arguments absent from the mapped corpus. **Boundary:** A finite numerical nullspace does not establish a null map of spectra or a uniform descent exponent. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
623/319	Counterexamples to the Hahn-Wilson conjecture at height two	L2	–	The complete-operator model offers a limited opportunity to distinguish a bounded inverse from a formal coefficient solve. **Boundary:** A bounded Neumann inverse requires a strict norm bound in the actual space; it does not supply the gluing lattice or topological transfer. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
624/320	Nonhomeomorphic closed aspherical four-manifolds with the same homotopy type	L2	–	The graph tensor obstruction admits finite tensor bookkeeping, peripheral to the manifold construction. **Boundary:** This does not establish reflection realization, hyperbolicity, outer actions or nonhomeomorphism of aspherical manifolds. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
625/321	A Counterexample to Wall's D(2) Problem	L1	–	A tower obstruction to D(2) and a finite plus construction are not consequences of finite cochain compression. **Boundary:** Preserving ordinary finite cohomology is weaker than deciding two-dimensional homotopy type. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
626/322	A positive solution to Tingley's problem	L2	–	Residual fixed-point certification is relevant only to auxiliary genuinely contractive instances, not to the asserted sphere-isometry extension in general. **Boundary:** The corpus requires q<1 and a self-mapping ball; neither follows from sphere isometry or condensing behavior. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
627/323	Relative independence of the separable quotient problem	L1	–	The main work is set-theoretic consistency and coded norming functionals; no concrete corpus theorem addresses it. **Boundary:** Finite numerical models cannot establish relative independence, CH consequences or measurable-cardinal consistency bounds. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
628/324	Lipschitz Equivalent Separable Banach Spaces Need Not Be Linearly Isomorphic	L2	–	The explicit weighted graph criterion has bounded linear maps and norm estimates that can be checked, but the corpus does not supply this Lipschitz nonlinear construction. **Boundary:** This is peripheral norm bookkeeping. The corpus's C1 contraction-ball hypotheses need not hold for a Lipschitz map, and the free-space/complete-continuity obstruction remains separate. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/03-triangular.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
629/324	Bi-Lipschitz Absorption of c0 Without a Linear Copy of c0	L2	–	The frozen shift is an explicit bounded bijection in a c0 block space, allowing peripheral full-tail and inverse-norm checks. **Boundary:** The state-dependent onto argument uses terminal scalars and Brouwer, not an automatically available strict contraction. Frozen linear inversion does not prove the nonlinear absorption or the no-c0 detector. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/04-shift.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
630/325	A direct proof of the complete Crouzeix inequality	L4	C3	The Faber coefficient estimate and positive double-layer resolvent kernel closely match the corpus's frame, physical-Gram and kernel norm tools. **Boundary:** These tools do not prove the sharp constant two; conformal collar positivity, amplification and limiting domains still need the manuscript's arguments. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in main.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
631/325	The complete Crouzeix theorem: optimal similarity and a common positive boundary representation	L4	C3	The minimizing metric SDP and common positive matrix density provide a specific operator-inequality application. **Boundary:** Rational SDP repair is a supplied-candidate test, not a proof that kappa<=2 for every matrix; finite tests do not certify all amplification sizes or the Hilbert-space limit. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in main.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
632/326	The cotype-cotype conjecture under the approximation property	L2	–	The finite Walsh operators can be checked in their actual weighted norms, but generic frame or moment algebra does not supply the dimension-uniform Banach-space argument. **Boundary:** Hilbert orthogonality does not give dimension-independent cotype bounds for arbitrary G and F; compact factorization and the approximation-property conclusion remain separate. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in preliminaries.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
633/327	Nontrivial Markov Type Forces Superreflexivity	L2	–	Finite reversible-chain energy calculations can be represented exactly; the route from Markov type to superreflexive geometry is not supplied. **Boundary:** A spectral or frame estimate for one chain does not give uniform Markov-type constants or a superreflexive renorming. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
634/328	Fixed Points of Nonexpansive Maps in Reflexive Banach Spaces	L2	–	The corpus supplies a useful boundary check on fixed-point methods, but its strict contraction theorem is weaker than the manuscript's nonexpansive claim. **Boundary:** A nonexpansive map has factor one; the residual-ball theorem cannot establish the general reflexive-space fixed-point assertion. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
635/329	Counterexamples to the duality conjecture for metric entropy	L2	–	The partition compression is an explicit finite calculation, but the corpus's Hilbert rank estimates do not directly give the required entropy-norm compression. **Boundary:** This is peripheral finite checking, not a theorem about metric entropy duality. Singular-value control alone does not preserve the relevant covering and packing quantities. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/03-compression.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
636/330	A uniformly discrete counterexample to bounded approximation in Lipschitz-free spaces	L2	–	The finite-support reduction of an operator can use a quantitative truncation budget, peripheral to the signed-measure/graph obstruction. **Boundary:** The norm is the Lipschitz-free Banach norm, not a default Euclidean coefficient norm. Finite operator approximation does not establish the limiting AP-without-BAP assertion. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/operators.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
637/331	Asymptotic midpoint uniform convexity and unbounded diamond distortion in a reflexive tree space	L2	–	The segment-norm and paired-test calculations permit finite energy bookkeeping, but the sharp diamond-tree obstruction is not a corpus theorem. **Boundary:** Finite quadratic energy identities do not imply the claimed universal midpoint-convexity modulus. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
638/331	Midpoint lenses in segment spaces	L3	C3	Finite convex atoms, common-head cancellation and a joint quadratic budget are specific constrained algebraic estimates that can be independently certified. **Boundary:** The reduction from arbitrary Banach-space geometry to those finite constraints is separate; an SOS search need not find an exact sharp certificate. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
639/331	Distortion of countably branching diamonds from midpoint and tree energies	L3	C3	The recursive path/segment energy transfer is a substantive finite inequality subproblem. **Boundary:** Depth-uniform sharp distortion requires the manuscript's recursion, not merely validity of a finite test case. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
640/331	Exact asymptotic moduli in a Daugavet subspace of L1	L2	–	Compact clipping and finite-measure approximation are peripheral places for quantitative truncation estimates. **Boundary:** Hilbert-frame estimates do not automatically transfer to L1 or determine exact Daugavet moduli. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
641/331	Midpoint convexity from bounded tree potentials and path costs	L3	C3	The four potential norms and inward tests reduce the claimed modulus to explicit linear and quadratic path-cost constraints. **Boundary:** Finite compatible marginals or PSD tests do not by themselves justify selection of infinite tail tests or completion of the tree space. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/constrained.tex, sections/introduction.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
642/331	Independent products in real L1: asymptotic midpoint convexity without AUC renormings	L3	C3	The exponential and Gaussian-square laws give actual conditional integral kernels and scalar excess integrals that can be represented with quantitative tails. **Boundary:** The L1 absolute-value integrand is nonsmooth at zero and needs splitting or a justified remainder; finite moments do not prove the no-AUC-renorming obstruction. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
643/331	Midpoint convexity from two recursive potentials	L3	C3	The proof's root-defect to boundary-energy recursion is an explicit quadratic budget that can be checked stage by stage. **Boundary:** Finite-stage inequalities do not establish infinite stopping convergence or the cubic modulus unless their constants remain uniform. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/foundations.tex, sections/introduction.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
644/332	Metric Markov Cotype Two of l1	L3	C3	The finite binary-cut reconstruction and quartic flat-map inequality are concrete weighted quadratic/polynomial steps, followed by geometric/Cesaro walk estimates. **Boundary:** A certificate for a fixed finite cut system does not prove the uniform constant across all chains; the stopped-martingale construction and ell1 reconstruction remain essential. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in main.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
645/333	Smooth isometric immersions of closed surfaces into Euclidean four-space	L3	C3	The exact smooth correction of primitive metric increments is a substantial nonlinear residual problem. **Boundary:** A bounded inverse with no derivative loss and an invariant ball must be established; ordinary finite Newton convergence is not smooth isometric immersion or finite assembly. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
646/334	A Smooth Metric with No Local Isometric Immersion into Three-Space	L3	C3	The height equation on an oscillatory model patch is a specific variable-coefficient differential calculation. **Boundary:** Local smooth-jet formulas do not prove nonexistence for every immersion; the patch obstruction and smooth accumulation at the origin remain essential. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
647/335	An integral scalar curvature bound for real simplicial volume	L2	–	The weighted graph deformation allows a peripheral energy-preserving finite representation. **Boundary:** No corpus result turns that energy estimate into a measured degree bound or the bounded-cohomology detector for real simplicial volume. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
648/335	Positive scalar curvature forces rational inessentiality	L3	C3	Solving the nonlinear graph equation on a cover is a substantive PDE step with a potential inverse/residual formulation. **Boundary:** Coercivity on compact panels is not uniform solvability on the cover; rational inessentiality still requires shortening and the torical obstruction. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
649/336	Spectral scalar curvature and uniform Urysohn width	L3	C3	The spectral lower bound and quadratic reserve give an explicit positive quadratic-form problem central to the stretching estimates. **Boundary:** The corpus does not construct compatible partitions or a uniform Urysohn-width map on every complete manifold; finite spectral tests alone cannot prove the assumed global lower bound. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
650/336	Spectral scalar curvature and Urysohn width in dimension three	L2	–	Weighted stability forms on the cutting surfaces have a limited energy-estimation application. **Boundary:** Generic conforming energy estimates do not supply separating cuts, distance-slab topology or the dimension-three width constant without bounded geometry. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
651/336	Positive scalar curvature and uniform codimension-two width	L2	–	Quadratic pressure budgets and finite cut data admit local finite inequality checks, peripheral to singular shielding and countable reconstruction. **Boundary:** Convex panel convergence does not cover singular minimizing cuts, compatible countable partitions or global codimension-two width. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
652/337	Generalized Cartan-Hadamard isoperimetry and Euclidean equality rigidity	L2	–	The spectral and torsional comparison consequences have a concrete variational residual application, but this is downstream of the principal isoperimetric claim. **Boundary:** This does not supply the sharp hypersurface Sobolev inequality, all-dimensional isoperimetry or equality rigidity; the spectral corpus's Euclidean convex-domain hypotheses cannot be silently transferred. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
653/337	Sharp integral fillings in CAT(0) spaces	L2	–	Finite Euclidean current charts permit peripheral integration and energy checks. **Boundary:** Smooth Taylor assumptions do not hold for an arbitrary CAT(0) current; no finite kernel approximation proves sharp integral filling or existence of a minimizing current. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
654/338	Uniformization of complete Kahler manifolds with positive bisectional curvature	L3	C3	Polynomial chart models and nonuniform contraction create a specific coordinate-composition and error-propagation subproblem. **Boundary:** A local contraction certificate does not prove global Kahler-Ricci flow, Harnack estimates or surjectivity of the eventual holomorphic coordinates. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
655/339	Entropy equality and local symmetry in negative curvature	L2	–	Contracting plaque jets offer finite differentiated-row bookkeeping, but the formal-to-smooth rigidity is a different issue. **Boundary:** Finite Taylor identities do not realize coherent formal fields smoothly or prove that entropy equality forces local symmetry. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
656/340	A counterexample to the nearby Lagrangian conjecture	L1	–	Stable tubes, a K-theory kernel class and exact Lagrangian embedding require symplectic-topological inputs not identified in the corpus. **Boundary:** A transform or finite cochain model does not establish the obstruction to Hamiltonian isotopy. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
657/341	Deforming hypersymplectic four-manifolds to hyperkahler triples	L3	C3	The collapsing local model and auxiliary prescribed-volume equation provide a substantial nonlinear elliptic application. **Boundary:** No uniform inverse, continuation to the endpoint or preservation of the cohomology classes follows merely from a local residual representation. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
658/342	Taming implies compatibility on four-manifolds	L2	–	Regularization and angular energy of currents admit local smoothing-error accounting, but the needed density restriction is not covered by a smooth-kernel theorem. **Boundary:** Pointwise Taylor remainders require smooth fields and absolute moments; they do not prove planar tangent densities, closedness of a restricted current or compatibility. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
659/343	Symplectic Ball Packings in Higher Dimensions	L2	–	Finite toric Hamiltonian comparison offers a narrow trajectory and local residual check. **Boundary:** Finite-time flow error control is not existence of the required symplectic embedding; algebraic-curve and global packing obstructions remain separate. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
660/344	The metric Blaschke theorem	L3	C3	The Jacobi/Lagrangian comparison along a geodesic is an actual matrix differential boundary problem suitable for full-jet and boundary compatibility analysis. **Boundary:** A finite BVP solve does not prove the sharp comparison in every geodesic or the incidence/characteristic-class identities needed for Blaschke classification. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
661/345	Infinitely many closed geodesic images on every Riemannian sphere	L2	–	Finite-cut and filtered cut-join calculations can use chain-level preservation checks at a fixed action and degree. **Boundary:** Unfiltered finite cohomology preservation does not establish global equivariant classes or infinitely many distinct geodesic images. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
662/346	A Codimension-One Bound for the Singular Set of a Stationary Integral Varifold	L2	–	Finite observations of linearized height measures are a possible diagnostic, but the codimension-one singular-set theorem is driven by blow-up and dimension reduction. **Boundary:** Finite-rank observations do not detect all singular centers or justify weak measure limits; smooth Taylor models are not automatically valid on the varifold. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
663/346	Almost-everywhere regularity of stationary integral varifolds	L3	C3	Heat-potential contact and low-frequency barriers use a concrete smoothing operator where exact semigroup identities and bias bounds matter. **Boundary:** This does not establish the signed-excess estimate, multiplicity control or almost-everywhere varifold regularity from a finite heat computation. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
664/347	Hamiltonian Fixed Points Below the Stable Morse Number in Dimension Twenty-Two	L2	–	The explicit integral counts permit a peripheral exact chain and homology audit. **Boundary:** Real or complex ranks lose torsion. Even an integral chain verification does not realize handle cancellation or the 22-dimensional Hamiltonian dynamics. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
665/347	Sharpness of the Cyclic Integral Floer Bound Below the Stable Morse Number	L2	–	The balanced torsion profiles and stated integer deficit give finite bookkeeping targets. **Boundary:** The finite-complex criterion must be used integrally; it neither constructs the Kahler manifold nor realizes the prescribed nondegenerate fixed points. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
666/347	Hamiltonian Fixed Points Below the Stable Morse Number	L2	–	The explicit two-prime homology counts allow an exact finite-complex consistency check. **Boundary:** An integer count does not establish geometric handle attainment, Hamiltonian perturbation or absence of other fixed points. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
667/347	A nondegenerate counterexample to the Morse-number Arnold bound	L2	–	Augmentation-ideal and relative Morse presentations offer limited finite algebra checks. **Boundary:** The corpus supplies no group-generator lower bound or geometric Hamiltonian closing theorem; rational ranks alone are insufficient. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
668/347	Three fixed points on the symplectic quadric threefold	L1	–	The quadric involution and degenerate three-critical-point construction have no concrete theorem-level route identified in the corpus. **Boundary:** Nondegenerate residual or event certificates cannot certify this construction's degenerate fixed point, critical number and global fixed-point count. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
669/348	Zero-Plane Rigidity for Einstein Four-Manifolds	L5	C3	Exact polynomial positivity is directly part of the core Weyl-block coupling argument. The corpus supplies the same rational power-to-Bernstein conversion and rigorous Gram/residual certificate machinery. **Boundary:** The interval Bernstein theorem must be extended with the manuscript's proved triangular formula; matrix entries need PSD certificates, not entrywise positivity. No certificate was rerun, and the weighted Hessian/equality-splitting arguments remain separate. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/07-certificates.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
670/348	An L2 Einstein Gap for Nonnegatively Curved Four-Manifolds	L3	C3	The L2 Einstein-defect quantity is directly suited to physical residual and energy accounting in the compactness/smoothing part of the argument. **Boundary:** The main theorem explicitly assumes Classification Assumption ass:classification. This screening does not remove it, infer a universal epsilon0 or certify the singular compactness/smoothing argument. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
671/348	Positively curved Einstein four-manifolds	L5	C3	The classification proof centrally uses explicit scalar and matrix polynomial certificates on the Weyl spectral domain, matching the corpus's exact positivity tools. **Boundary:** Positivity verification is a core application, not proof of the global classification. The half-conformal-flatness argument, topological reduction and all triangular/matrix hypotheses remain obligations. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/07-certificates.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
672/349	The Solomon-Yau least-volume theorem	L2	–	The curvature/index quadratic forms admit peripheral physical-Gram calculations. **Boundary:** Finite index estimates do not prove low-density cone attainment or the cohomological min-max input and sharp least-volume bound. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
673/350	Sharp nodal length on smooth surfaces	L3	C2	Wavelength-scale equations and affine improvement under subdivision provide a concrete multi-scale differential/remainder application. **Boundary:** Smooth coefficient jets alone do not control zero-set length, persistence or the sharp square-root eigenvalue rate uniformly over eigenspaces. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
674/350	Smooth counterexamples to Yau's nodal upper bound in dimensions three and four	L4	C3	The scalar exactification is an explicit residual-to-coefficient correction with a nonvanishing first-jet denominator, closely aligned with full differentiated residual accounting. **Boundary:** A small eigen-residual alone gives neither nodal-area growth nor a fixed limiting metric. Noncancellation, positivity and the separate dimension-four construction were not fully checked. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/s3-exactification.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
675/350	Power-law violations of Yau's nodal upper bound	L4	C3	The quantitative packet construction has explicit finite jet covariance, noncancellation and exact coefficient-correction steps. **Boundary:** Finite covariance positivity does not prove the probability limit, the power-law sign witnesses or preservation of all eigenfunctions in one smooth metric. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/global.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
676/351	A closed Ricci flow with bounded scalar curvature and finite-time curvature blowup	L3	C3	The warped reaction matrix and barrier comparison give a specific local parabolic/operator-estimate application. **Boundary:** The energy-level heat theorem does not supply the needed derivative-free parabolic Calderon-Zygmund estimate; the construction input and two-scale singular evolution remain independent. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/05-reaction.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
677/351	Bounded scalar curvature and smooth extension of four-dimensional Ricci flow	L4	C2	The collar extension uses a genuine gauged elliptic inverse and analytic correction with weighted residual bounds, strongly relevant to corpus inverse and nonlinear certificates. **Boundary:** A local inverse does not prove Ricci-flow extension; concentration estimates and the imported static inequality/path selection are essential. Uniformity in the geometric degeneration must be proved, not assumed. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/06-collar.tex, sections/01-introduction.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
678/351	Path selection and an elliptic inequality on degenerating Ricci-flat trees	L5	C2	The manuscript's core is an augmented elliptic inverse with finite tests/sources and controlled infinite end modes. Corpus bounded inverse, low-rank correction and nonlinear residual tools apply directly to this operator architecture. **Boundary:** No finite matrix check proves Fredholmness or length-uniform estimates on noncompact ends. Indicial gaps, all partial disconnections, complex contours and the subsequent path-selection argument remain to be established. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/04-family.tex, sections/01-introduction.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
679/352	A finite-time singularity of Calabi flow on projective space	L5	C3	Finite shooting is a core existence step using exactly the kind of jet recurrence, positive denominator, propagated inverse error and nonlinear remainder certification developed in the corpus. **Boundary:** The source specifies its own integer-interval verifier; it was not run here. Finite shooting positivity alone does not prove the ancient inverse, compact parabolic gluing or finite-time Calabi singularity. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/certificate.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
680/353	A Smooth Nonquadratic Entire Affine Maximal Graph in Dimension Ten	L4	C4	The ten-dimensional ansatz reduces the affine-maximal equation to an explicit radial nonlinear system; derivative identities, local series and invariant-box inequalities are concrete corpus applications. **Boundary:** Pointwise positive Hessian is not a uniform global lower bound; no affine-metric completeness is asserted. The global invariant-region proof is not supplied merely by local Newton solvability. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in source/main.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
681/353	The affine Bernstein theorem in dimensions three through nine	L3	C3	Convex sections and the affine-area comparison create explicit Hessian/determinant and integral-inequality subproblems. **Boundary:** Local positive Hessian and finitely many section moments do not prove the all-scale rigidity in dimensions three through nine; degenerating sections need separate bounds. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
682/354	The Isoperimetric Conjecture for the Cubic Flat Three-Torus	L5	C2	The core elimination of the remaining torus candidates uses explicitly certified scalar integrals and between-knot curvature bounds, a direct rigorous-integration application. **Boundary:** The functions are not globally polynomial: Bernstein certification applies only after a justified polynomial enclosure. No arithmetic was rerun, and the geometric candidate reduction is not certified by these scalar checks. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/numerics.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
683/355	Uniqueness of tangent flows at the first singular time of embedded surface mean-curvature flow	L4	C3	The normalized augmented inverse on finite contours has a concrete block/inverse residual application in the main analytic chain. **Boundary:** A nonzero drift divisor is not accretivity, and a one-sided residual is not surjectivity. Tangent-flow uniqueness still needs the finite-jet, localized-gradient and action-jump arguments. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/finite-linear.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
684/356	Gigli's distributional curvature characterization of Alexandrov spaces	L3	C2	Heat/Hessian quadratic forms and distributional curvature testing offer a substantive weak-operator comparison. **Boundary:** The smooth-domain corpus theorems require an RCD-specific extension; finite weak tests do not imply Alexandrov comparison along all geodesics. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
685/356	Weak Hessian bounds along every geodesic in RCD spaces	L3	C2	The bounded-drift generator and weighted measure are concrete operator data for the weak-Hessian propagation argument. **Boundary:** Energy semigroup equivalence does not promote an almost-everywhere Hessian statement to every geodesic; localization and nonsmooth geometric calculus remain necessary. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
686/357	Bi-Lipschitz Coordinates at Regular Points of Noncollapsed RCD Spaces	L3	C2	Harmonic coordinate maps and summable pinching permit quantitative finite-probe conditioning and coordinate-error accumulation. **Boundary:** A finite probe rank condition does not prove full-dimensional RCD regularity, bi-Lipschitz bounds or control at every limiting scale. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
687/358	A Three-Manifold Without Conjugate Points and Without a Nonpositively Curved Metric	L3	C2	The construction's Jacobi/index estimates and concave cometric are concrete matrix differential and positivity problems. **Boundary:** Checking one segment or parameter box is not exclusion of conjugate points on all complete geodesics, and does not supply the three-manifold topology. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
688/359	A negatively pinched Kahler threefold without bounded holomorphic coordinates	L4	C3	Positive representing densities with exact phase moments and uniform derivative estimates are a strong, model-specific moment/kernel application. **Boundary:** Finite Hankel/Gram positivity alone does not construct a positive representing density or ensure geometric curvature pinching; the Hartogs construction and bounded-map obstruction remain separate. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/04-densities-estimates.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
689/359	One-sided negative sectional curvature and the holomorphic Liouville property	L2	–	The Legendre/radial metric construction has finite differentiated identities that can be checked, but the Liouville mechanism is not supplied by the corpus. **Boundary:** This is peripheral algebra, not a proof of the required global curvature sign, completeness or harmonic-function growth/constancy. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
690/360	Global Support and Convex Injectivity Domains under Weak MTW	L3	C3	Jacobi fields and branch Hessians give a substantial matrix boundary-value subproblem. **Boundary:** Finite inverse tests do not prove global support, injectivity-domain convexity or the degree argument under weak MTW. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
691/360	Uniform Bi-Holder Transport from Weak MTW	L3	C3	The lower-jet and determinant comparison offers a specific finite conditioning and Hessian-inequality application. **Boundary:** Finite jets do not establish a uniform Holder modulus through conjugate or cut-locus limits; the weak-MTW geometry supplies additional obligations. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
692/361	A counterexample to integer-degree harmonic dimension comparison	L4	C3	The harmonic transmission is an explicit finite matrix Riccati/transfer system, with uniform damping and full-period correction, not merely an abstract coordinate analogy. **Boundary:** The dimension-16 arithmetic is supplementary and does not construct the control word or metric. All preceding histories, radial tails, Ricci bounds and global harmonic growth still require the analytic proof. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/finite-selection.tex, sections/transmission.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
693/361	A Three-Dimensional Counterexample to Integer-Degree Harmonic Dimension Comparison	L4	C3	The proof explicitly separates a finite angular frame from the full harmonic restriction operator, matching finite-frame plus discarded-mode error analysis. **Boundary:** A sharper error on finitely many vectors is not a full operator-norm bound. All crossing choices, infinite compatible traces and every-radius growth remain obligations. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/transfer.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
694/362	Global classical solutions of the three-dimensional relativistic Vlasov-Maxwell system	L3	C2	The signed retarded-force identity exposes a precise weak derivative-transfer and residual bookkeeping application in a major PDE estimate. **Boundary:** A smooth finite jet cannot replace the singular angular/retarded kernel estimates. TR813 is a finite recovery/diagnostic result, not a global Vlasov-Maxwell theorem; characteristic occupation and bootstrap closure remain unproved here. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/cancellation.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
695/363	Nonuniqueness with local conservation for the hard-sphere Boltzmann equation	L4	C2	The Carleman gain, Gaussian parameter sheets and normalized collision tails supply a concrete kinetic kernel model for exact weak moments and rigorously truncated observation banks. **Boundary:** Ordinary smooth Taylor formulas cannot be applied across singular sheets without justification. Local conservation, branching separation and compactness are not consequences of finite moment recovery; no nonuniqueness proof was certified. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in paper/03-kernel.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
696/363	Nonuniqueness for the periodic hard-sphere Boltzmann equation	L4	C3	The periodic construction uses the same explicit collision operator but additionally needs quantitative axis and nonconcentration control. **Boundary:** Finite weighted rank does not imply the uniform non-atomicity limits or separation of periodic Boltzmann solutions. The corpus's recorded kinetic validation is synthetic, not proof of this theorem. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/03-kernel.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
697/364	The Boltzmann-Grad limit for stable radial potentials on regular kinetic intervals	L3	C2	The exact finite collision-history expansions and layer centering provide substantial finite integral and cancellation calculations. **Boundary:** Local moment identities do not control the recollision set, hierarchy tails or Boltzmann-Grad limit uniformly in particle number and time. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
698/364	Hard-sphere fluctuations on the regular Boltzmann lifespan	L3	C3	The backward mild fluctuation equation and collision covariance permit a concrete semigroup/covariance representation and finite cumulant checks. **Boundary:** The OU report is not a theorem for this collision generator, and Gaussian covariance calculations alone do not establish cumulant decay, tightness or the fluctuation limit. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
699/365	Determination of a metric and a unitary connection from one boundary patch	L5	C2	Finite harmonic jet spanning and first/second moment remainders are central to the metric/connection extension argument and directly match corpus moment and rank-stability results. **Boundary:** Apply the moment identity entrywise to the actual sphere measure and retain the uniform density bound; no commutation of matrix factors is allowed. Finite jet recovery does not prove continuation, gauge equivalence or global boundary uniqueness. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/06-matrix.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
700/365	Smooth anisotropic uniqueness in the Calderon problem from one boundary patch	L5	C2	The affine-reproducing shrinking-sphere functional and its two-power moment bound are directly core to the Calderon bootstrap, with a domain-correct finite-moment representation. **Boundary:** The moment representation is directly applicable but does not establish the decisive angular coercivity or product continuation. Generic symmetric coercive PDE results cannot silently replace the moving trace operator or prove global uniqueness. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/05-angular-estimate.tex, sections/06-bootstrap.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
701/365	Nonuniqueness for bounded measurable scalar conductivities in three dimensions	L3	C3	Exact scalarization of two Cauchy pairs has genuine flux/constraint-preservation and finite laminate algebra subproblems. **Boundary:** Two Cauchy-pair matches are not the entire DN map without the branching argument. Convex variational convergence does not supply the Baire strong limit or this nonconvex scalarization theorem. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/scalarization.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
702/366	Interior regularity of planar Mumford-Shah minimizers	L3	C2	The harmonic pieces in Mumford-Shah admit exact local Green/energy error accounting on a fixed crack configuration. **Boundary:** The full free-discontinuity functional is not a convex fixed-domain panel problem. Crack classification, rectifiability and changes of topology remain outside this application. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
703/367	The critical dimension for one-phase Bernoulli minimizers	L5	C3	The core free-boundary argument depends on a finite tensor-polynomial positivity certificate with exact contraction rules and a coercive margin. **Boundary:** No coefficient table or verifier was rerun here. A valid algebraic certificate still requires the geometric flux identity, cutoff passage and minimizer/dimension-reduction arguments. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/certificate-proof.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
704/368	Strong diffeomorphic approximation in three dimensions for 1 <= p <= 2	L2	–	Local smoothing and trace budgets offer peripheral approximation-error accounting for Sobolev maps. **Boundary:** Unconstrained conforming approximation does not preserve injectivity, orientation, topology or diffeomorphism; the corpus does not supply the required topological approximation theorem. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
705/368	Strong diffeomorphic approximation in three dimensions for p>2	L3	C2	Finite rank data, exact carriers and constrained routing create substantive local algebra and approximation subproblems. **Boundary:** Finite full rank is not global one-to-one behavior. Standard convex Gamma-convergence does not ensure diffeomorphic approximants or control all high-p trace/topological constraints. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
706/369	Strict hot spots and absence of interior critical points on smooth simply connected planar domains	L5	C3	The subcritical Green resolvent and finite rank-one updates form a core kernel argument with a concrete Neumann/low-rank representation under the stated d<1 form bound. **Boundary:** Entrywise reciprocal positivity is not ordinary inverse positivity and is not furnished by Woodbury alone. The boundary limit, conditional negative semidefiniteness and Hilbert-multiplier argument still need the manuscript's estimates. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/05-reciprocal-kernels.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
707/370	The Subcritical Henon-Lane-Emden Conjecture	L3	C2	The Newton kernel and explicit weighted interval-pair comparison are concrete integral/derivative-transfer subproblems in the central energy estimate. **Boundary:** Finite smooth-kernel moments do not control the Newton singularity or nonintegrable tails automatically. The all-scale energy bound and subcritical scaling contradiction remain separate. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/04-intervals.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
708/371	Stable self-similar blowup for a supercritical defocusing Schrodinger equation on the torus	L5	C2	The profile and outgoing-spectrum argument contains core exact recurrence, coefficient-enclosure, positivity and spectral-count calculations matching the corpus's strongest certificate machinery. **Boundary:** The corpus spectral bridge is explicitly conditional: residuals alone are pseudospectral, and contour holomorphy, outgoing tails and all errors are required. No verifier was run; nonlinear stability and torus gluing are not certified. The corpus's radial mass-critical biharmonic NLS result is a different equation. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/certificates.tex, sections/spectrum.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
709/372	Global Uniqueness for the Smooth Isotropic Elasticity Inverse Problem	L4	C3	The Cauchy-transform frame construction is an explicit small-norm integral equation, while physical amplitude and coefficient recovery provide concrete full-rank inverse checks. **Boundary:** Local Neumann inversion does not supply global holomorphic trivialization, physical CGO realization or large-parameter uniqueness for arbitrary smooth Lame coefficients. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/physical-amplitudes.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
710/373	A counterexample to the Monge ansatz for the three-marginal Coulomb cost	L4	C3	The five-component supporting inequality and contact branches are directly relevant to corpus finite coupling/duality and nonlinear correction tools. **Boundary:** The finite marginal-coupling theorem must be extended to the manuscript's continuous branch measures; discrete feasibility alone does not prove the Coulomb supporting inequality or absence of a deterministic Monge optimizer. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/certificate.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
711/374	Sharp One-Third Stability of Brenier Maps	L4	C3	The stationary-site moment inequality concerns actual convex cells, masses and first-moment derivatives, a strong quantitative moment application. **Boundary:** The corpus's static body-moment theorem does not prove the derivative inequality with C0=162 or the sharp one-third exponent. Differentiating cell indicators is a shape-calculus step, not ordinary smooth Taylor expansion. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in source/sections/moments.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
712/375	A positive resolution of De Giorgi's conjecture in dimension eight	L4	C3	The diffuse stability and normal-average interaction equations provide a specific elliptic Green/moment and weighted-energy application. **Boundary:** Uniformity in the stable solution and interface separation is not provided by a finite panel solve. Dimension-eight rigidity still depends on density, fourth-order growth and radial-threshold arguments. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in sections/04-interface.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
713/376	Finite Instructions and Solenoidal Shear Flows	L3	C3	The transverse shear stages have an explicit solenoidal velocity and force residual, giving substantive derivative and finite-time flow checks. **Boundary:** The constructed force is tailored to a prescribed smooth U; this is not arbitrary-force or unforced 3D regularity. Corpus event-error bounds need a separation margin and cannot decide halting or certify an exact infinite-time detector. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
714/376	A Fixed Particle Test for Computation in a Forced Viscous Flow	L3	C3	The fixed-particle construction adds time-integrable derivative budgets and a solenoidal projection to an explicit forced flow. **Boundary:** The comparison class and time-decay assumptions are essential. Finite flow error control cannot establish the exact halting equivalence or general Navier-Stokes global regularity. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
715/376	Universal Computation with Eventually Stationary Navier-Stokes Forcing	L3	C2	The autonomous/spatial-clock construction has a concrete stationary forcing residual and smooth startup check. **Boundary:** Eventually stationary forcing is specially constructed, not arbitrary stationary forcing. Local residual identities do not prove the global processor itinerary or a robustness margin for halting detection. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
716/376	Geometric Programs for Solenoidal Forcing	L3	C3	The one-component-at-a-time transverse pulses make zero self-advection and exact residual checking particularly concrete. **Boundary:** Symbolic interpreter correctness and all-real-time detection remain separate. A finite residual or clock estimate does not turn this special forced solution into a general regularity theorem. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. No deeper manuscript proof-body review. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
717/376	Prefix Instructions and Incompressible Flows	L3	C3	Prefix instructions require normal compensation for area-changing planar maps, a specific divergence and local flow-construction problem. **Boundary:** Planar area change cannot be ignored in an incompressible lift. The force is a general prescribed body force; no arbitrary-data regularity or finite-precision halting certificate follows. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in principal-theorem.tex, introduction.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
718/376	Computation under Rapidly Vanishing Navier-Stokes Forcing	L3	C2	The compact moving curls and rapidly decreasing scales give an explicit derivative/tail budget for an exact forced solution. **Boundary:** The detector signal shrinks and is exact, not uniformly noise-separated. Rapid forcing decay and a prescribed solution do not decide halting numerically or establish general global regularity. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in addressed.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
719/376	Velocity-Field Detection of Computation in Forced Navier-Stokes Flows	L4	C3	The velocity-field detector couples transport to an actual diffusion/observation problem, giving a more direct PDE-semigroup application than trajectory bookkeeping alone. **Boundary:** The torus construction allows derivative bounds to grow with the time horizon; the whole-space case has different support/comparison hypotheses. Semigroup representation does not prove the encoded computation or a general Navier-Stokes theorem. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in main.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
720/376	Scalar Potentials and Slow Clocks for Forced Fluid Computation	L3	C3	Three scalar potentials and an unbounded slow clock yield concrete curl, composition and decay estimates. **Boundary:** Clock reparameterization must also transform the force; decay does not imply finite computational horizon. The corpus's conditional NS criteria do not prove arbitrary-data regularity or exact detector correctness. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in main-result.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

M/F	Manuscript	App.	Contr.	Applicability, boundary, and review depth
721/376	Incompressible Box Transport and Finite Computation	L3	C3	Whole-box affine transport has explicit determinant-one, curl-extension and neighborhood-margin formulas suitable for substantial local checks. **Boundary:** Finite routing checks do not validate the entire compiler/history invariant or all-time observation. Prescribed forced smooth solutions are not a resolution of general 3D Navier-Stokes. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in geometry.tex, introduction.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.
722/377	Uniform Interior C^{1,alpha} Estimates for Infinity-Harmonic Functions	L2	–	The discrete propagation section has finite quadratic-envelope optimization that can be checked, but the decisive infinity-Laplacian regularity mechanism is not covered by the corpus's coercive PDE results. **Boundary:** These are peripheral finite inequalities. The degenerate viscosity equation, contact localization, limiting rigidity and uniform C1,alpha rate do not follow from a smooth coercive inverse or generic Bellman contraction. **Review:** Bounded source screening: main-statement/introductory excerpt and the first two section/subsection headings per indexed source, not full proofs. Additional selected statement/proof or introductory passages inspected in manuscript/sections/06-discrete-propagation.tex. All three corpus maps read. Corpus originals listed in evidence_files had selected theorem/limitation passages inspected; other corpus_refs are map-based leads only. No numerical/formal verifier run and no exhaustive corpus or manuscript proof checking. Levels: 5 direct core relevance; 4 very relevant; 3 significant parts; 2 peripheral; 1 no concrete route identified. This is applicability, not validity, novelty or proof success.

6 Sources and Next Stage

Public release: `https://openai.com/index/sharing-ai-progress-in-mathematics/`. Manuscript catalogue and original sources: `https://github.com/openai/math`. Local evidence maps are in `../_audit/`; row-specific complete references are in `applicability.json`. The next stage rederives selected steps with explicit assumptions, error budgets, proofs or reproducible certificates, comparisons, and research directions. Any stronger headline result requires a valid new proof, not a high relevance score.